



Real Estate – Individual Properties: The Definitive Guide

This Real Estate Financial Modeling course covers both **individual properties** and **real estate investment trusts (REITs)**. These two topics are related because you must know about individual properties if you want to understand REITs.

However, they are also different in terms of the analysis and financial modeling, and so we have separated them into two guides. This one covers **individual properties**.

There are dozens of hours of video tutorials in the course, so it's impossible to cover everything in a single PDF, but this guide summarizes the key points:

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Property Types and Strategies

“Real estate” in this course means **land and buildings that generate revenue or have the potential to do so.**

The industry exists because humans need spaces to live in, and businesses need spaces where they can operate, produce goods and services, and sell those goods and services.

“Tenants” – individuals and businesses who need space – pay **rent** to property owners to use this space.

They may also have to pay for expenses such as utilities, property taxes, insurance, and others, depending on the terms of the lease. Some tenants even pay a percentage of sales to the owners!

When you conduct a financial analysis of a property, you typically do so **from the perspective of an Equity Investor or Debt Investor (Lender) in the property.**

For example, if you acquire a “multifamily” property (i.e., an apartment building) for \$100 million and hold it for 5 years, could you earn a 12% annualized return on your investment?

If you identify the most important assumptions and set up your analysis correctly, real estate financial modeling helps you answer these questions.

As with all financial modeling, it is more important to be “roughly correct” than “precisely wrong.”

You can’t know if a property will generate an 11% or 13% annualized return, but a decent analysis should tell you whether or not that *range of returns* – 10% to 15% – is plausible.

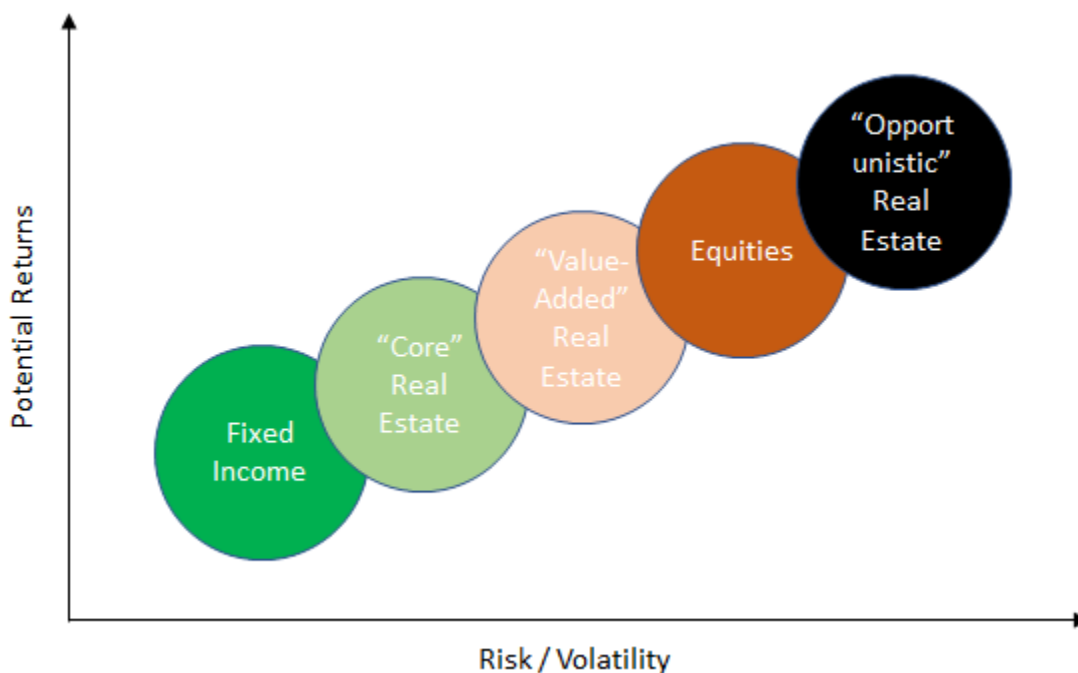
For investors, real estate combines elements of equities and fixed income and can offer a risk / potential return profile that is somewhere in between them.

For example, a “Core” real estate deal for an existing, stabilized property might offer risk and potential returns closer to those of an investment-grade corporate bond.

On the other hand, a “Value-Added” deal for a property with a lower occupancy rate that needs renovations might offer risk and potential return profile closer to that of stocks.

And an “Opportunistic” deal with significant development or redevelopment needs might offer even higher risk than stocks, but also higher potential returns.

Here's how you can think about these strategies:



And here's a summary of the main property types:

	Office / Retail / Industrial	Multifamily	Condominiums	Hotels
Customers	▪ Businesses	▪ Individuals	▪ Individuals	▪ Individuals
Lease Type	▪ Long-term (several years); details vary	▪ Medium (1 year?); very similar leases	▪ N / A – Ownership	▪ Short (days); very similar terms
Timing of Lease or Sale	▪ After building is done	▪ After building is done	▪ Pre-sold	▪ After building is done
Purpose	▪ Business	▪ Residential	▪ Residential	▪ Business or leisure

Office, retail, and industrial properties are used for different purposes, but the **lease terms** tend to be similar, and so do the customers (businesses).

For example, leases are almost always multi-year, and they might require tenants to pay for some portion of the operating expenses and property taxes.



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If a developer builds a new office, retail, or industrial property, it will almost always aim to complete the property, lease it up to a high occupancy rate, and then sell it.

Multifamily and condominium properties are places for individuals to live in. “Multifamily” refers to apartment buildings where you can **rent** a unit for a year at a time, while “condominiums” are units that you **own**.

In a multifamily development, the aim is to complete the property, lease it up to a high occupancy rate, such as ~95%, and then sell it.

In a condo development, by contrast, the developer aims to sell individual units *in advance* – as the construction is taking place – to reduce risk and provide funding for the construction.

Finally, hotels are also living spaces for individuals, but they are primarily for short stays of a few days or less.

If a developer constructs a new hotel, it will finish the building, aim to reach a stabilized, predictable occupancy rate, and then sell it.

Another big difference between these property types is the **granularity** of the revenue and expenses.

Specifically, office/retail/industrial properties tend to have relatively few tenants, and each tenant’s lease may have different terms, so the financial modeling tends to be more granular.

One tenant might have a 5-year lease with 6 months of free rent and rental escalations of 2.0% per year, while another might have a 7-year lease with 9 months of free rent and 1.5% rental escalations.

You might have to pay a 3% commission to a real estate broker to find a tenant for 5,000 square feet of space, but only a 2% commission for 1,000 square feet of space.

As a result, you usually project each tenant’s rent, expenses, reimbursements, incentives, commissions, and other items separately.

By contrast, financial modeling for multifamily, condo, and hotel properties tends to be higher-level **because there are many units and the lease/deal terms are quite similar**.

If an apartment building has 400 units and each lease is nearly the same, there is no point in modeling all 400 tenants separately – just use the average rent and expenses across all the units.

You can think of the spectrum like this:

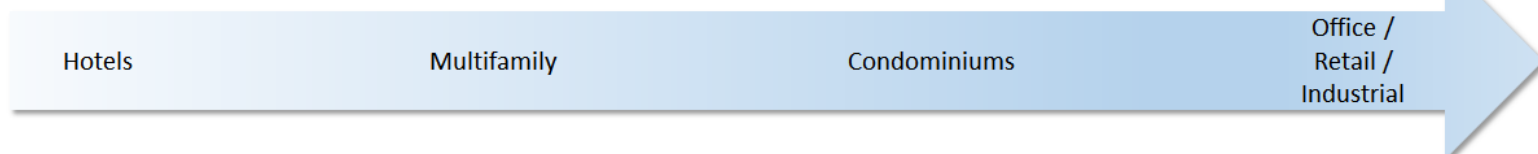


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Closer to “normal” (non-real-estate) companies

VERY different due to timing and revenue/expense detail



Other property types, such as storage, student housing, nursing, and data centers are all variations of the ones above.

Think about the number of tenants, the average lease length, the similarity of lease terms, and the granularity of the revenue and expenses to pick the most appropriate comparison.

For example, student housing is nearly the same as multifamily because leases are typically for a year or less, there are dozens or hundreds of students in each building, and the lease terms are very similar.

On the other hand, data centers are closer to industrial/office/retail properties because there tend to be fewer tenants, leases are longer-term (e.g., 3-7 years), and individual lease terms vary greatly.

Once you understand the property types, you need to acquaint yourself with the main **deal types**:

- **Strategy #1: Develop a Property and Sell It Upon Completion (“Opportunistic”)** – This one is common for all the property types above except for condos.
- **Strategy #2: Develop a Property and Pre-Sell It Before Completion (“Opportunistic”)** – This one applies primarily to condos. Other property types may pre-lease, but pre-sales are rare.
- **Strategy #3: Acquire an Existing Property and Then Sell It (“Core” and “Core Plus”)** – This one applies to most property types except for condos. If nothing changes, it’s “Core”; if there are minor upgrades, it might be labeled “Core Plus.”
- **Strategy #4: Acquire an Existing Property, Renovate It, and Then Sell It (“Value-Added”)** – This one also applies to everything but condos. With this strategy, a common goal is to boost a low occupancy rate – say, 65% – up to the 80-90% range. The sponsor may also attempt to bring below-market rents up to market rates.

Here’s a summary of the property and deal types:

	Office/Retail/Ind.	Multifamily	Condominiums	Hotels
Develop & Sell Upon Completion				
Develop and Pre-Sell Before Completion				
Acquire Existing & Do Nothing				
Acquire Existing & Renovate				
	 = Common	 = Less Common	 = Rare or Not Possible	

In this course, we focus on **the most common property and deal combinations**, which means those in green above.

In most of these deals, you need to understand something about property valuation before you can complete the analysis and make an investment decision.

We cover this topic in detail in [the Valuation section](#), but the short version is that you typically make purchase and exit assumptions based on **Cap Rates** (also called “Yields”) and **Net Operating Income**, or NOI.

Cap Rates in real estate are the reciprocals of valuation multiples; if a company’s EV / EBITDA is 10x, then its Cap Rate is 1 / 10, or 10%.

Net Operating Income is similar to, but **not** the same as, EBITDA for normal companies: it measures the property’s cash flow from operations on a capital structure-neutral basis before most of the “capital costs” (e.g., capital expenditures, tenant improvements, and leasing commissions).



At a high level, Net Operating Income equals the property's revenue (from rental income, expense reimbursements, and other sources) minus its operating expenses and property taxes.

The Cap Rate for a property is defined as Net Operating Income (NOI) / Property Price. You should ideally use the forward, stabilized NOI, and the "Property Price" could be either the asking price or the actual price paid for the property.

Sometimes you *calculate* the Cap Rate, and sometimes you *select* the Cap Rate to estimate the property's price if you're considering an acquisition or sale.

You use Cap Rates rather than multiples because many investors view real estate as more of a fixed-income investment (e.g., a corporate bond) than an equity investment (a stock).

In reality, certain properties might be closer to equities in terms of risk and potential returns.

But you still need to pick Cap Rates to decide what to pay for a property and how much you might be able to sell it for in a few years.

You often assume that the property's cash flow stabilizes at a certain level over several years to make a reasonable estimate.

A Cap Rate reflects the property's location, desirability, growth potential, and risk.

For example, a luxury apartment building in the center of Manhattan might sell for a 4% Cap Rate (equivalent to a 25x multiple), while a run-down, second-tier apartment building in The Middle of Nowhere, Iowa, might sell for a 10% Cap Rate (equivalent to a 10x multiple).

The major real estate brokerage firms publish Cap Rate data for different cities and regions in the world, and you need to review this data to make the appropriate assumptions in your analysis.

If an office building down the street recently sold for a 6% Cap Rate, that doesn't necessarily tell you anything about *the office building you are currently valuing* because it could be quite different:

- Is it the same "class" (e.g., Class A, Class B, etc.) property with similar amenities and conditions?
- How similar are the properties' locations? The one that's in a safe, wealthy suburb will be a lot more desirable than the one in an area with a high crime rate and low-quality schools.
- What is the occupancy rate of each building? Are the tenants high quality (e.g., blue-chip, Fortune 500 companies), or do they represent more risk (e.g., small businesses)?



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- How stable are the cash flows? Is there a substantial chance that many of the leases will expire in 2-3 years, or are the tenants on longer-term leases with well-staggered lease expirations?

Just as with valuation multiples for comparable public companies, Cap Rates for properties are meaningful only if they're based on *properties that are truly comparable*.

Similar to leveraged buyouts for traditional companies, nearly all acquisition and development deals in real estate involve a significant amount of **Debt** – often 50-70%+ of the property's value.

Real estate investors use leverage for the same reason that private equity firms use leverage: **to amplify their returns**. Leverage does not “boost” returns – it *amplifies* them so that positive returns become even more positive, and negative returns become even more negative.

Also, leverage makes it possible to acquire and develop more properties and diversify because investors do not have to pay for the entire price upfront – just the smaller Equity portion.

Investors use a variety of loans to fund real estate deals, but the most important categories include:

- **Construction Loans:** These are used to fund new developments. Interest is capitalized when the property is still under construction and not yet generating revenue, and the loan is refinanced and replaced with a Permanent Loan once tenants move in and the property stabilizes.
- **Bridge/Temporary Loans:** These are often used for acquisition and renovation scenarios in value-added deals. Interest rates are higher to compensate investors for the added risk during the renovation period, and these loans are refinanced once the property stabilizes.
- **Permanent Loans:** This category is split into Senior Loans, which may represent up to 60-70% of the property's value, and Mezzanine, which might increase the “Loan to Value” (LTV) Ratio to the 80-90% range. **These are just arbitrary examples – the acceptable percentages, interest rates, and other terms vary based on the region and market conditions.** These loans are longer-term than the first two categories and tend to carry lower interest rates (at least, the Senior Loans carry lower interest rates; Mezzanine is significantly higher).

If you're used to analyzing traditional companies, you probably have two questions now:

QUESTION #1: How can 60%, 70%, or 80% leverage possibly work? Won't the interest and principal repayments exceed the property's cash flow?



ANSWER: These high leverage levels work because real estate loans often have interest-only periods, and even when principal amortization begins, the loan often amortizes over 20-30 years rather than the 5-10 years common with normal companies.

Here's a simple example to illustrate the math:

- **Property Purchase Price:** \$10 million
- **Debt Used:** \$7 million (70% LTV) with a 4% fixed interest rate, 10-year maturity, and 30-year amortization
- **Property Net Operating Income ("NOI"):** \$600K / year (i.e., a 6% Yield or "Cap Rate" on the Purchase Price)
- **Cash Flow Available to Service Debt:** \$550K / year due to Capital Expenditures and other capital costs below the Net Operating Income line
- **Annual Interest Expense:** \$7 million * 4% = \$280K
- **Annual Amortization:** \$7 million / 30 = \$233K

So, the Debt Service in Year 1 is \$280K + \$233K = \$513K, while the Year 1 Cash Flow is \$550K.

It's a bit close, but the sponsor can use 70% leverage to fund this deal – if all goes according to plan, and the NOI keeps increasing as the Debt Service decreases due to a declining Debt balance.

QUESTION #2: Why do investors bother to refinance loans?

ANSWER: For the same reason that PE firms pursue **dividend recapitalizations** (or "dividend recaps") in LBOs: to boost their returns by making the company or property take on more Debt and then distributing the net proceeds to themselves.

Sometimes investors also refinance to take advantage of lower interest rates if rates have been falling.

Continuing with the example above, let's say that a real estate investor acquires a \$10 million property using \$3 million of Equity and \$7 million of Debt.

By Year 3 of this acquisition, the property's NOI has grown from \$600K / year to \$650K / year.



Market conditions have stayed about the same, and there have been minor unit upgrades to offset the fact that the property is now older, so it's now worth $\$650K / 6\% = \10.8 million.

If the investor now refinances at a 70% LTV, he/she can use $\$10.8 \text{ million} * 70\% = \7.6 million of Debt.

By Year 3, there have been 3 years of amortization, so the initial Debt balance has decreased by $\$233K * 3 = \sim \$700K$.

Therefore, the investor can use $\$7.6$ million of new Debt, but he/she has to repay only $\$6.3$ million of existing Debt.

That “extra” \$1.3 million goes directly to the investor as Equity Proceeds in Year 3, which boosts the investor’s returns in this deal. It’s like a “free bonus” before the exit.

If the property’s value has fallen by Year 3, a refinancing is unlikely. And if interest rates have risen significantly, a refinancing is also unlikely.

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Office, Retail, and Industrial Properties

We group together office, retail, and industrial properties because they share characteristics that set them apart from other property types:

- **Number of Tenants:** Relatively low compared with *most* multifamily properties; some buildings have only a single tenant, while others might have 5-10 up to a few dozen.
- **Length of Leases:** Often 5-10 years; sometimes as short as 2-3 years, but almost always longer than 1 year.
- **Variability of Lease Terms:** High. If an office building has 3 tenants, each occupying a different amount of space, the chances are that the terms of each lease will be quite different.
- **Key Lease Terms:** Here are the key ones for *financial modeling purposes*:
 - **Start Date and End Date:** How many years and months does the lease last?



- **Monthly Rate and Step-Ups:** How much is the tenant paying per month for its space? This rate is usually quoted on a per-square-foot or per-square-meter basis, and it often increases by a small percentage each year, but other variations are possible.
- **Months of Free Rent:** When a tenant first signs a lease or renews an existing lease, the owner usually provides “incentives” such as free months of rent to encourage the tenant. For example, if the tenant signs a 7-year lease, it might receive 6 rent-free months at the beginning, so it effectively pays for only 6.5 years of rent.
- **Expense Reimbursements:** Which expenses are the tenant’s responsibility? For example, must the tenant pay for its share of property taxes, insurance, and maintenance/utilities? Or just one or two of those? Or none of those?
- **Percentage Rent:** Must the tenant also pay a percentage of its monthly retail sales to the owner? If so, what is this percentage, and what volume of sales does it apply to? This term **only** applies to retail properties for obvious reasons.
- **Tenant Improvements (TIs):** Tenants often want to customize their space, especially in office and retail properties. So, as another incentive, landlords pay for certain customizations called “Tenant Improvements” (e.g., partitioning space differently, additional lighting, or installation of retail shelving) when the tenant first moves in or renews its lease. These are quoted on a per-square-foot or per-square-meter basis.

Office, industrial, and retail properties also tend to have higher **capital costs** – the Tenant Improvements described above, commissions paid to brokers who find the tenants (Leasing Commissions), and normal capital expenditures (CapEx) than property types such as multifamily.

If you search online, you’ll find sources that describe various types of leases. For example, you’ll often see the following descriptions:

- **Triple Net (NNN) Lease:** Tenant must pay for “everything,” i.e., the proportion of property taxes, insurance, and maintenance/utilities that correspond to its occupied space.
- **Double Net (NN) Lease:** Tenant must pay for its proportion of property taxes and insurance, but not maintenance/utilities.
- **Single Net (N) Lease:** Tenant must pay for its proportion of property taxes, but not insurance or maintenance/utilities.



- **Full Service (FS) Lease:** Tenant does **not** pay for its proportion of property taxes, insurance, or maintenance/utilities.

There's no such thing as a free lunch because tenants also pay the **highest rent** with FS leases and the **lowest rent** with NNN leases.

Many owners and landlords *prefer* Full Service Leases because they do not have to disclose the property's expenses and can, therefore, earn a higher margin.

When it comes to leases, read the fine print closely because no two agreements are the same.

The descriptions above work as *rough* categories, but in practice, there are always variations, exceptions, and special cases.

As one simple example, a tenant with a NNN lease does not necessarily pay for "all" the expenses associated with a property because some expenses may not be specified in the lease.

For example, fees paid to the 3rd party firm in charge of day-to-day property management are usually the owner's responsibility.

Also, if the owner sets aside funds for **reserves** to pay for future Capital Expenditures, Tenant Improvements (TIs), or Leasing Commissions (LCs), those are the owner's responsibility as well.

Do **not** assume anything unless it is stated explicitly in the case study instructions or lease documents.

The Pro-Forma for Office, Retail, and Industrial Properties

Properties are simpler than companies, so you do not need to build full Income Statement, Balance Sheet, and Cash Flow Statement projections for them.

Instead, you just need one statement: **The "Pro-Forma"** (you will also build supporting schedules, but they're not full financial statements).

The Pro-Forma is sufficient because it is, effectively, **a combined Income Statement and Cash Flow Statement**.

And if you need to track Balance Sheet line items, such as Cash, Debt, and Equity, you can do that at the bottom of this Pro-Forma.

It's easy to simplify the financial statements of a property because:



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- **You Use Cash Accounting, Not Accrual Accounting** – Financial analysis of properties is based on Cash Accounting, so you do not need Working Capital line items, CapEx *and* Depreciation, and the other line items necessary for Accrual Accounting.
- **You Typically Ignore Income Taxes** – Most properties are owned by pass-through entities such as LLCs, Partnerships, S Corporations, and REITs that do not pay corporate income taxes. Instead, the earnings “pass through” to the individual shareholders, who then pay taxes as individuals. Therefore, you almost always ignore income taxes in property models.
- **“Other Activities” Are Minimal** – Properties do not spend excess cash flow on financial investments, acquisitions, or non-core activities.... *for the most part*. That removes the need for much of the Cash Flow from Investing and Cash Flow from Financing sections on the CFS.
- **Debt and Equity Issuances and Debt Service Are Handled on the Pro-Forma and in the Returns Calculations** – When investors develop properties, they draw on Debt and Equity over time. But you can represent this activity directly on the Pro-Forma, and if a refinancing takes place, you can also show it directly on the Pro-Forma and in the Returns schedule.

Here’s a simple, annual Pro-Forma for an office property:

Property Pro-Forma:	Units:	Historical:	Projected:					Stabilized
		FY17	FY18	FY19	FY20	FY21	FY22	Year:
Revenue:								
(+) Base Rental Income:	\$	\$ 9,666,931	\$ 10,246,947	\$ 10,759,294	\$ 11,189,666	\$ 11,525,356	\$ 11,871,117	\$ 12,227,250
(-) Absorption & Turnover Vacancy:	\$	-	(512,347)	(268,982)	(559,483)	(288,134)	(593,556)	(305,681)
(-) Concessions & Free Rent:	\$	-	(256,174)	(134,491)	(279,742)	(144,067)	(296,778)	(152,841)
(+) Expense Reimbursements:	\$	62,936	64,824	66,769	68,772	70,835	72,960	75,149
Potential Gross Revenue:	\$	9,729,867	9,543,250	10,422,589	10,419,213	11,163,990	11,053,743	11,843,877
(-) General Vacancy:	\$	(1,450,040)	(1,434,573)	(1,398,708)	(1,342,760)	(1,267,789)	(1,187,112)	(1,222,725)
Effective Gross Income (EGI):	\$	8,279,827	8,108,677	9,023,881	9,076,453	9,896,201	9,866,631	10,621,152
Expenses:								
(-) Property Management Fees:	\$	(248,395)	(243,260)	(270,716)	(272,294)	(296,886)	(295,999)	(318,635)
(-) Operating Expenses:	\$	(1,258,715)	(1,296,476)	(1,335,371)	(1,375,432)	(1,416,695)	(1,459,196)	(1,502,972)
(-) Real Estate & Property Taxes:	\$	(902,247)	(920,292)	(938,698)	(957,472)	(976,621)	(996,153)	(1,016,077)
(-) CapEx, TI, and LC Reserves:	\$	(100,697)	(103,718)	(106,830)	(110,035)	(113,336)	(116,736)	(120,238)
Total Expenses:	\$	(2,510,054)	(2,563,747)	(2,651,615)	(2,715,232)	(2,803,538)	(2,868,084)	(2,957,920)
Net Operating Income (NOI):	\$	5,769,773	5,544,931	6,372,267	6,361,221	7,092,663	6,998,547	7,663,232
NOI Margin:	%	69.7%	68.4%	70.6%	70.1%	71.7%	70.9%	72.2%
(-) CapEx, TIs, and LCs:	\$	-	(181,507)	(93,476)	(192,560)	(99,169)	(204,287)	(105,208)
(+) Capital Costs Paid from Reserves:	\$	-	181,507	93,476	146,297	99,169	130,903	105,208
Adjusted Net Operating Income:	\$	5,769,773	5,544,931	6,372,267	6,314,958	7,092,663	6,925,163	7,663,232
Adjusted NOI Margin:	%	69.7%	68.4%	70.6%	69.6%	71.7%	70.2%	72.2%
(-) Interest Expense on Debt:	\$		(1,832,689)	(1,797,187)	(1,760,443)	(1,722,413)	(1,683,052)	
(-) Debt Principal Repayment:	\$		(1,014,332)	(1,049,834)	(1,086,578)	(1,124,608)	(1,163,970)	
Cash Flow to Equity Investors:	\$		2,697,909	3,525,245	3,467,936	4,245,642	4,078,141	
Ending Debt Balance:	\$	52,362,544	51,348,212	50,298,378	49,211,800	48,087,192	46,923,222	

NOI is similar to EBITDA for normal companies because it's a capital structure-neutral measure of core-business cash flow. But unlike EBITDA, it partially reflects some capital costs due to the Reserves!

This section is sort of like the "Cash Flow Statement" since it reflects capital costs; Adjusted NOI is similar to Unlevered FCF.

We deduct Debt Service to calculate "Cash Flow to Equity Investors," which is similar to Levered FCF.



Many of these line items correspond to the key lease terms described above, but some are new, so let's go through each one:

- **Base Rental Income:** This represents the "Potential" Rental Income of the property if it is 100% occupied and all tenants pay market rates. For example, if the property has 10,000 square feet and the market rate is \$50 per square foot per year, the Base Rental Income is \$500,000.
- **Absorption & Turnover Vacancy:** This represents the rental income lost when an existing tenant leaves and it takes time to find a new tenant. For example, if a tenant is renting 2,000 square feet for \$50 per square foot per year, this tenant leaves, and it takes 6 months to find a new tenant, there will be an Absorption & Turnover Vacancy line item for $2,000 * \$50 * (6 / 12) = \$50,000$. The owner does not "lose money," but the property generates *less than its full potential Rental Income* since this space goes vacant for half a year. Therefore, this item is shown with a negative sign and counts as a deduction from Base Rental Income.
- **Concessions & Free Rent:** This line item corresponds to the "Months of Free Rent" lease term described above. If a new tenant moves into this same 2,000-square-foot space with rent of \$50 per square foot per year and receives 3 rent-free months as a move-in incentive, there will be a Concessions & Free Rent line item of $2,000 * \$50 * (3 / 12) = \$25,000$ on the Pro-Forma.

Again, this item is not an "expense"; it represents *lost potential Rental Income*, so it is shown with a negative sign and counts as a deduction from Base Rental Income.

- **Expense Reimbursements:** This one corresponds to the "Expense Reimbursements" described above and represents the amount of property taxes, insurance, and maintenance/utilities for which the tenants are responsible.

For example, if the tenant occupying these 2,000 square feet of space is responsible only for its share of property taxes, and property taxes for the entire building are \$50,000 per year, then this tenant must pay the owner $(2,000 / 10,000) * \$50,000 = \$10,000$.

This \$10,000 is shown in "Expense Reimbursements" with a positive sign and *increases* the Base Rental Income.

In the Pro-Forma above, the Expense Reimbursements represent only ~3% of Operating Expenses + Property Taxes. This very low percentage means that the tenants are *most likely* on Full Service (FS) or Single Net (N) Leases.



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- **Potential Gross Revenue:** If you sum up everything above, you'll calculate the Potential Gross Revenue or Potential Gross Income for the property. This number represents the cash revenue from all space in the building, **ignoring permanently vacant spaces**.

"Permanently vacant" means there is no tenant in the space currently, and the owner has no plans to find a tenant in the near term. "Permanently vacant" does **not** refer to spaces that are temporarily free because of a lease expiration, tenant move-out, and search for a new tenant – those count toward the Absorption & Turnover Vacancy line item.

- **General Vacancy:** This line item represents potential Rental Income lost to spaces that are permanently vacant (per the definition above). For example, if this same 10,000-square-foot building has 1,000 square feet (10% of the total) that is always vacant, the General Vacancy line item would be $10,000 * 10\% * \$50 = \$50,000$.

Once again, this is not an "expense," but rather *lost potential Rental Income*, so it is shown with a negative sign and counted as a deduction from Base Rental Income.

All landlords and owners want their properties to be 100% occupied, but sometimes there are practical or market reasons that explain this General Vacancy (e.g., the remaining space in the building is too much for small companies in the area but not enough for bigger companies).

In many renovation deals, the goal is to reduce General Vacancy from 25-40% to the 10-20% range, which, if effective, will provide a huge boost to the returns.

- **Effective Gross Income (EGI):** This is similar to "Net Sales" or "Net Revenue" for a normal company, but it's on a **cash basis** rather than an accrual basis. EGI represents **the cash revenue** (rental income + expense reimbursements) that a property generates *after* the adjustments for temporary and permanent vacancy and free months of rent.

It may also include adjustments for above or below-market leases, "percentage rent," and other items that bridge the gap between *potential* revenue and *actual* revenue.

This property's **expenses** are easier to explain because most of them mean what they say:

- **Property Management Fees:** Owners of large properties rarely manage them directly. Instead, they hire 3rd party property management to deal with tenants, collect rent, resolve problems, set up repairs and maintenance, and so on. These fees are usually a percentage of EGI.
- **Operating Expenses:** These include building insurance, maintenance and repairs, utilities, and other miscellaneous items. If the property is big enough, this category may also include staff



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payroll, sales & marketing, janitorial, landscaping, and security services. These expenses are usually quoted on a per-square-foot or per-square-meter basis.

- **Real Estate & Property Taxes:** Nearly all *local* governments (cities, counties, and districts, sometimes up to the state or province-level) charge a tax on properties to fund local governments, school systems, and infrastructure. Often, this tax is a small percentage (1-4%) of the property's assessed value, and it increases by a small percentage each year.

NOTE: Tax schemes worldwide vary tremendously. In some places, such as Australia and the U.K., there's a "Stamp Duty" tax charged on property sales, which may exist along with or in place of other property taxes, depending on the region. In other countries, taxes may be tied to the property's annual rental income or the owner's income or other metrics. The statement above is a *general guideline* that is most applicable to properties in the U.S.

- **CapEx, TI, and LC Reserves:** This item exists to "smooth out" the property's spending on capital costs and ensure that its Adjusted NOI does not diverge too far from its NOI. It's easiest to illustrate this concept with an example:

Let's say that the property owner expects to spend \$0 to attract new tenants in Year 1, \$200K in Year 2, \$0 in Year 3, \$400K in Year 4, and \$500K in Year 5.

This spending will be in the form of Tenant Improvements ("TIs") to customize each tenant's space as well as Leasing Commissions ("LCs") paid to brokers who find these new tenants. There will also be Capital Expenditures ("CapEx") to improve the *entire* building's condition (e.g., fixing the roof, installing new flooring, or adding an elevator).

If there are no Reserves, the property owners deduct these costs from the NOI in each period, resulting in an Adjusted NOI that is significantly lower than the normal NOI in Years 2, 4, and 5, but the same in Years 1 and 3.

Not only does that make the property's performance "lumpier," but it also makes it trickier to service Debt. For example, what happens if the property's capital costs are so high one year that it cannot afford Interest Expense and Principal Repayment?

To get around these issues, properties use "Reserves." Each year, the owner allocates some smaller amount to the Reserves based on expected future costs.

In this example, the owner might allocate \$200K per year based on expected future capital costs of \$1,100K over 5 years.



Then, the owner withdraws funds from these Reserves when they are needed to pay for the capital costs:

Year 1: Allocate \$200K; withdraw \$0; Ending Reserves of \$200K.

Year 2: Allocate \$200K; withdraw \$200K; Ending Reserves of \$200K.

Year 3: Allocate \$200K; withdraw \$0; Ending Reserves of \$400K.

Year 4: Allocate \$200K; withdraw \$400K; Ending Reserves of \$200K.

Year 5: Allocate \$200K; withdraw \$400K; spend \$100K additional cash; Ending Reserves of \$0.

As a result of these Reserves, the NOI and Adjusted NOI are the same in the first four years, and the property never faces a “cash crunch.”

The Adjusted NOI is \$100K lower than the NOI in Year 5 because of the \$500K in capital costs vs. the \$400K in Reserves, but that will happen from time to time, even with healthy reserving.

NOTE: There is considerable debate about whether or not additions to these Reserves should be deducted to calculate Net Operating Income (NOI).

We believe they should be counted as expenses that reduce NOI because if you do *not* do this, NOI does not reflect the true costs of owning and maintaining the property. Many lenders also deduct these Reserve additions to get a more conservative view of the property’s financials.

However, other people argue that these items are not true cash expenses, so they should only be counted in between the NOI and Adjusted NOI line items as additional capital costs.

Throughout this course, we always deduct Reserve Additions to calculate NOI. But you will see different treatments in real life, and there isn’t one “correct” way to reflect the Reserves.

- **Net Operating Income (NOI):** This item is *similar* to EBITDA for normal companies – a **capital structure-neutral approximation of core-business cash flow** – but it’s not the same. For example, if you deduct the Reserve Additions, as we do, then NOI partially reflects capital costs.

By contrast, EBITDA never reflects capital costs because it excludes both CapEx and Depreciation.

NOI is critical for valuation purposes because properties are often valued based on their projected NOI divided by an appropriate “Capitalization Rate” (Cap Rate) or “Yield,” which is the reciprocal of a valuation multiple.

For example, if a property’s projected NOI is \$5 million and Cap Rates for similar properties in the area are 5%, this property should be worth \$5 million / 5% = \$100 million.

- **CapEx, TIs, and LCs and Capital Costs Paid from Reserves:** We went through the mechanics of these items above, so please refer to that explanation.

These line items, netted together, produce **0** if the Reserves are sufficient to pay for the CapEx, TIs, and LCs in a given year.

If they are not, then the net of these two numbers will be a negative figure, indicating that the property must spend some of its cash flow on capital costs:

Property Pro-Forma:	Units:	Historical:	Projected:					Stabilized
		FY17	FY18	FY19	FY20	FY21	FY22	Year:
Net Operating Income (NOI):	\$	5,769,773	5,544,931	6,372,267	6,361,221	7,092,663	6,998,547	7,663,232
NOI Margin:	%	69.7%	68.4%	70.6%	70.1%	71.7%	70.9%	72.2%
(-) CapEx, TIs, and LCs:	\$	-	(181,507)	(93,476)	(192,560)	(99,169)	(204,287)	(105,208)
(+) Capital Costs Paid from Reserves:	\$	-	181,507	93,476	146,297	99,169	130,903	105,208
Adjusted Net Operating Income:	\$	5,769,773	5,544,931	6,372,267	6,314,958	7,092,663	6,925,163	7,663,232
Adjusted NOI Margin:	%	69.7%	68.4%	70.6%	69.6%	71.7%	70.2%	72.2%

In Year 5, the Net of CapEx, TIs, LCs, and Capital Costs Paid from Reserves is finally a negative, indicating the the Reserves are insufficient and that NOI and Adjusted NOI will differ.

- **Adjusted Net Operating Income:** This is what’s shown above: the property’s NOI minus the capital costs *not* covered by the Reserves. It’s a similar concept to Unlevered FCF for normal companies: **the cash flow generated by core-business operations, ignoring capital structure.**

...but it’s *not* the same because Adjusted NOI excludes income taxes. Also, the Change in Working Capital is a key component of Unlevered FCF for companies, but it does not exist in quite the same way for properties.

- **Interest Expense on Debt:** Exactly what it sounds like. In real estate, you may use the IPMT function to calculate this so that the total Interest Expense + Principal Repayment stays the same each year. Or, you could calculate it manually based on Interest Rate * Debt Balance.



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- **Debt Principal Repayment:** See above. You might use the PPMT function, the counterpart of IPMT, to calculate this one. **Since you ignore income taxes in most real estate models, both Interest Expense and Principal Repayments reduce the property's Cash Flow to Equity Investors, and there's no tax benefit for the Interest.**
- **Cash Flow to Equity Investors:** The bottom line. This item is similar to the "Net Change in Cash" at the bottom of the Cash Flow Statement for normal companies, and it represents what's available for distribution to the equity investors in the property ("the owners").

Unlike companies, which might avoid distributions (i.e., Dividends) and accumulate large Cash balances over time, most properties distribute their available annual cash flow.

The Simple Way to Project Line Items on the Pro-Forma

You could use simple percentage growth rates for many of the key assumptions, such as the average annual increases for rent, expense reimbursements, and property taxes, and you could link other items to the Effective Gross Income or per-square-foot numbers:

Operating Assumptions:		Units:					Units:		
Building / Construction Name:	Name	Culver Drive Tower	Real Estate Taxes % Property Value:	%	1.12%				
Location:	Name	Irvine, CA	Real Estate Taxes Annual Growth Rate:	%	2.0%				
Months in Year:	#	12	Replacement Reserve per SF per Year:	\$ / sq. ft. / Yr	\$ 0.40				
Rentable Square Feet (RSF):	sq. ft.	251,743 sq. ft.	Replacement Reserve Annual Growth Rate:	%	3.0%				
Average Annual Rent per Square Foot:	\$ / sq. ft. / Yr	\$ 38.40	Capital Costs per SF for New Tenants:	\$ / sq. ft. / Yr	7.00				
Expense Reimbursements per Square Foot:	\$ / sq. ft. / Yr	0.25	Capital Costs Annual Growth Rate:	%	3.0%				
Expense Reimbursements Annual Growth Rate:	%	3.0%							
Property Management Fees % EGI:	%	3.0%							
Operating Expenses per SF per Year:	\$ / sq. ft. / Yr	5.00							
Operating Expenses Annual Growth Rate:	%	3.0%							
Operating Assumptions:		Units:	Historical:	Projected:				Stabilized	
			FY17	FY18	FY19	FY20	FY21	FY22	Year:
Expiring Leases - % Rentable Square Feet:	%		-	10.0%	5.0%	10.0%	5.0%	10.0%	5.0%
Fraction of Year Required to Find New Tenants:	Year Frac.		0.50	0.50	0.50	0.50	0.50	0.50	0.50
Fraction of Year of Free Rent for New Tenants:	Year Frac.		0.25	0.25	0.25	0.25	0.25	0.25	0.25
Rental Income Growth Rate:	%			6.0%	5.0%	4.0%	3.0%	3.0%	3.0%
General Vacancy % Base Rent:	%		15.0%	14.0%	13.0%	12.0%	11.0%	10.0%	10.0%
New Tenant Capital Costs per SF:	\$ / sq. ft. / Yr	\$ 7.00	\$ 7.21	\$ 7.43	\$ 7.65	\$ 7.88	\$ 8.11	\$ 8.36	
Annual Capital Costs:	\$	-	(181,507)	(93,476)	(192,560)	(99,169)	(204,287)	(105,208)	
Replacement Reserve Amount:	\$	100,697	22,909	36,262	-	14,167	-	15,030	



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To calculate the **Absorption & Turnover Vacancy**, you could multiply the Base Rental Income by the percentage of leased Rentable Square Feet that expire in the year and then multiply by the percentage of the year required to find new tenants to occupy that space.

The **Free Rent** calculation is similar, but the percentage of the year will be different.

There's nothing "wrong" with this method, but it is better to use tenant-by-tenant projections whenever possible because that makes it easier to **justify** your numbers.

More Complex Line Items on the Pro-Forma

Let's now look at a more complex Pro-Forma, with new items circled in red:



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Property Pro-Forma:	Units:	Historical:	Projected:				Stabilized:
		FY18	FY19	FY20	FY21	FY22	FY23
Revenue:							
(+) Base Rental Income:	\$	\$ 2,805,084	\$ 2,942,642	\$ 3,044,289	\$ 3,135,618	\$ 3,221,150	\$ 3,301,678
(-) Loss to Lease:	\$	(252,957)	(296,740)	(295,303)	(178,965)	(98,434)	(39,238)
(-) Absorption & Turnover Vacancy:	\$	-	(55,228)	(48,771)	(252,211)	(63,248)	(79,547)
(-) Concessions & Free Rent:	\$	(61,763)	(27,389)	(193,336)	(182,115)	(109,447)	(79,630)
(+) Expense Reimbursements:	\$	695,031	701,479	721,686	675,469	745,972	752,454
Potential Gross Revenue:	\$	3,185,396	3,264,764	3,228,565	3,197,795	3,695,993	3,855,718
(-) General Vacancy:	\$	(578,017)	(606,363)	(441,442)	(266,052)	(273,310)	(280,142)
(+) Percentage Rent:	\$	111,360	114,701	118,142	90,175	124,123	126,605
Effective Gross Income (EGI):	\$	2,718,738	2,773,103	2,905,264	3,021,918	3,546,807	3,702,181
EGI Growth Rate:	%	N/A	2.0%	4.8%	4.0%	17.4%	4.4%
Operating Expenses:							
(-) Property Management Fees:	\$	(81,562)	(83,193)	(87,158)	(90,658)	(106,404)	(111,065)
(-) Common Area Maintenance (CAM):	\$	(140,206)	(145,049)	(148,610)	(151,582)	(154,202)	(156,515)
(-) Common Area Utilities:	\$	(112,165)	(116,039)	(118,888)	(121,266)	(123,362)	(125,212)
(-) Insurance:	\$	(56,082)	(58,020)	(59,444)	(60,633)	(61,681)	(62,606)
(-) Real Estate & Property Taxes:	\$	(534,008)	(552,455)	(566,018)	(577,338)	(587,315)	(596,125)
(-) CapEx, TI, and LC Reserves:	\$	(168,247)	(174,059)	(178,332)	(181,899)	(185,042)	(187,818)
Total Operating Expenses:	\$	(1,092,271)	(1,128,814)	(1,158,450)	(1,183,376)	(1,218,006)	(1,239,341)
Net Operating Income (NOI):	\$	1,626,468	1,644,288	1,746,814	1,838,542	2,328,801	2,462,839
NOI Margin:	%	59.8%	59.3%	60.1%	60.8%	65.7%	66.5%
(-) Capital Expenditures (CapEx):	\$	-	(1,260,750)	(1,014,750)	(799,500)	-	-
(-) Tenant Improvements (TIs):	\$	-	(69,707)	(553,404)	(527,757)	(386,765)	(234,219)
(-) Leasing Commissions (LCs):	\$	-	(26,174)	(221,208)	(195,693)	(137,471)	(87,100)
(+) Capital Costs Paid from Reserves:	\$	-	1,303,056	1,730,371	949,110	76,766	217,496
Adjusted Net Operating Income:	\$	1,626,468	1,590,713	1,687,824	1,264,702	1,881,331	2,359,017
Adjusted NOI Margin:	%	59.8%	57.4%	58.1%	41.9%	53.0%	63.7%
(-) Cash Interest Expense on Senior Loan:	\$		(911,963)	(958,334)	(1,004,705)	(1,016,040)	
(-) Cash Interest Expense on Mezzanine:	\$		(71,340)	(75,620)	(80,158)	(84,967)	
(-) Senior Loan Principal Repayment:	\$		-	-	(515,233)	(515,233)	
Cash Flow to Equity Investors:	\$		607,410	653,870	(335,394)	265,090	
PIK Interest on Mezzanine:	\$		(142,680)	(151,241)	(160,315)	(169,934)	
Additional Draw from Senior Loan TI/LC Holdback:	\$		960,750	1,921,500	397,750	-	
(+) Ending Senior Loan Balance:	\$	15,457,000	15,457,000	15,457,000	14,941,767	14,426,533	
(+) Ending Mezzanine Balance:	\$	2,378,000	2,520,680	2,671,921	2,832,236	3,002,170	
Ending Debt Balance:	\$	17,835,000	17,977,680	18,128,921	17,774,003	17,428,704	
LIBOR:	%	2.60%	2.90%	3.20%	3.50%	3.80%	
Interest Rate on Senior Loan:	%		5.90%	6.20%	6.50%	6.80%	
Senior Loan Balance - Basis to Lenders:	\$	12,177,000	13,137,750	15,059,250	14,941,767	14,426,533	
Replacement Reserves:	\$		-	369,461	-	108,276	

Here's a summary of these new items, along with a few differences for the items described in the previous section:

- **Loss to Lease:** If certain tenants are paying below-market rates, this item represents the difference between in-place rents and market rents. For example, if the market rent is \$50 per

square foot per year, but a tenant occupying 5,000 square feet is paying \$45 per square foot per year, the Loss to Lease would be $(\$50 - \$45) * 5,000$, or \$25,000.

- **Percentage Rent:** This line item represents the rent that retail tenants pay as a percentage of their monthly gross sales. This one can fluctuate wildly based on the exact lease terms and economic conditions, but retailers might agree to it if it means lower fixed rent for them.
- **Operating Expenses:** Nothing here is new, but these are broken out into more detail – Maintenance, Utilities, and Insurance are shown separately, for example. We did that because in this property, each tenant is responsible for slightly different expenses.
- **Cash Flow and Debt Line Items:** Again, nothing here is new, but the items are shown in more detail, including separate lines for Tenant Commissions, Leasing Commissions, and CapEx, as well as separate Interest Expense for each tranche of Debt (Senior Loan and Mezzanine). The Senior Loan also has an Interest-Only Period for the first 2 years.
- **PIK Interest on Mezzanine:** “Paid-in-Kind” or PIK Interest, also known as Capitalized Interest or Accrued Interest, means that it accrues to the loan principal instead of being paid in Cash. The advantage is that the property owners pay less in Cash, but the disadvantage is that they will have to repay more Debt principal upon maturity or exit. This term is common with Mezzanine, and interest is often split into Cash and PIK portions (e.g., 6% Cash Interest and 4% PIK Interest).
- **Additional Draw from Senior Loan TI/LC Holdback:** Often, lenders “hold back” a portion of the funds, especially in renovation and other value-added deals, to reduce risk and ensure that the project is proceeding as planned. The property owners still must pay Interest based on the full amount of Debt, but they cannot *use* that full amount until it is released.

For example, the lenders might fund a \$20 million, 5% Senior Loan but “hold back” \$5 million of it and say that it will be released gradually as the property incurs capital costs.

When this loan is first issued, the owners must pay interest on all \$20 million of it – so, $\$20 \text{ million} * 5\% = \1 million of Cash Interest per year.

However, they can only access \$15 million of these funds initially. An additional amount, such as \$1 million per year, might be released over time as project milestones are met.

If the property is sold at the end of Year 3, when \$18 million of the Loan has been released, then the owners must repay only the \$18 million available at that point – even though Interest has been based on the full \$20 million the entire time.



- **Other Items:** We also track more Balance Sheet items here, including the Senior Loan and Mezzanine balances and the Senior Loan “face value” (what the Interest is based on) and basis to lenders (the initial amount, plus the cumulative amounts released from holdback, minus cumulative amortization).

How to Project the Pro-Forma Correctly Based on Individual Tenants

You do **not** necessarily have to model individual tenants to create a Pro-Forma, but it is often a requirement in case studies and modeling tests of 2-3+ hours.

If you do this, you must always include Renewal and Non-Renewal Cases for existing tenants.

Projecting Base Rental Income for each tenant is easy; the hard part is assigning values for everything else once a lease comes up for renewal.

The industry convention is to assign **probabilities** to the Renewal and Non-Renewal cases, multiply the relevant items by those probabilities, and add up everything.

You can start with the assumptions for the average lease term, the renewal probability, the months of “downtime” it takes to find a new tenant if an existing one does not renew, and the months of free rent and TIs and LCs for new and renewal tenants:

Property-Wide Operating Assumptions:

New Lease Term (Years):	# Years	10
Renewal Probability:	%	70.0%
# Months of Downtime for Non-Renewal:	#	6

10 years is on the long side, but plausible; renewal probabilities often range from 50% to 70%. The Months of Downtime typically range from 6 to 12 depending on the location and property type.

Free Rent and Capital Costs:

		New:	Renewal:
# Months of Free Rent:	#	6	3
Tenant Improvements (TIs) per RSF:	\$ / sq. ft. / Yr	\$ 10.00	3.00
Leasing Commissions (LCs) % Total Lease Value:	%	3.0%	1.0%

More incentives are needed to win *new* tenants than to keep existing tenants.

Once you have those, you can project the Base Rental Income for the tenant based on an annual percentage growth rate (typically specified in the lease):



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Rent Roll & Operating Assumptions:		Units:	Historical:	Projected:					Stabilized Year:
			FY17	FY18	FY19	FY20	FY21	FY22	
Office Tenant #1 - Full Service (FS) Lease:									
Rentable Square Feet Occupied:	sq. ft.	10,000 sq. ft.							
Lease Expiration Date:	Date	2019-12-31							
Baseline Rent per Square Foot:	\$ / sq. ft. / Yr		\$ 120.00	\$ 124.80	\$ 129.79	\$ 134.98	\$ 139.03	\$ 143.20	\$ 147.50
Rental Growth Rate:	%			4.0%	4.0%	4.0%	3.0%	3.0%	3.0%
(+) Base Rental Income:	\$		1,200,000	1,248,000	1,297,920	1,349,837	1,390,332	1,432,042	1,475,003
(-) Absorption & Turnover Vacancy:	\$		-	-	-	(202,476)	-	-	-
(-) Concessions & Free Rent:	\$		-	-	-	(438,697)	-	-	-
(-) Tenant Improvements (TIs):	\$		-	-	-	(51,000)	-	-	-
(-) Leasing Commissions (LCs):	\$		-	-	-	(215,974)	-	-	-

This tenant's lease expires at the end of Year 2, so this space might be vacant for part of Year 3. To calculate the Absorption & Turnover Vacancy, you multiply the Year 3 Base Rental Income by the probability of a non-renewal by the percentage of the year in which the space might be vacant.

This line item counts *only if the tenant does **not** renew*. Here's the formula:

$$= -\text{Base Rental Income in Year 3} * (1 - \text{Renewal Probability}) * (\text{Downtime Months} / \text{Months in Year})$$

$$= -\$1,349,837 * (1 - 70\%) * (6 / 12) = -\$202,476$$

You do **not** consider the Renewal Case because if the tenant renews, there is no downtime. The Concessions & Free Rent line is more complicated because you **must** consider both cases there:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
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77														
78														

Rent Roll & Operating Assumptions:		Units:	Historical:	Projected:					Stabilized
			FY17	FY18	FY19	FY20	FY21	FY22	Year:
Property-Wide Operating Assumptions:									
New Lease Term (Years):	# Years	10							
Renewal Probability:	%	70.0%							
# Months of Downtime for Non-Renewal:	#	6							
Free Rent and Capital Costs:									
			New:	Renewal:					
# Months of Free Rent:	#	6	3						
Tenant Improvements (TIs) per RSF:	\$ / sq. ft. / Yr	\$ 10.00	3.00						
Leasing Commissions (LCs) % Total Lease Value:	%	3.0%	1.0%						
Office Tenant #1 - Full Service (FS) Lease:									
Rentable Square Feet Occupied:	sq. ft.	10,000 sq. ft.							
Lease Expiration Date:	Date	2019-12-31							
Baseline Rent per Square Foot:	\$ / sq. ft. / Yr	\$ 120.00	\$ 124.80	\$ 129.79	\$ 134.98	\$ 139.03	\$ 143.20	\$ 147.50	
Rental Growth Rate:	%		4.0%	4.0%	4.0%	3.0%	3.0%	3.0%	
(+) Base Rental Income:	\$	1,200,000	1,248,000	1,297,920	1,349,837	1,390,332	1,432,042	1,475,003	
(-) Absorption & Turnover Vacancy:	\$	-	-	-	(202,476)	-	-	-	
(-) Concessions & Free Rent:	\$	-	-	-	=174*(1-Renewal_Probability)*(New_Free_Rent_Months/Months)-	-	-	-	
(-) Tenant Improvements (TIs):	\$	-	-	-	74*Renewal_Probability*(Renewal_Free_Rent_Months/Months)	-	-	-	
(-) Leasing Commissions (LCs):	\$	-	-	-	(215,974)	-	-	-	



That formula may look complicated, but it's simple if you break it into smaller components:

$$\begin{aligned} &= -\text{Base Rental Income in Year 3} * (1 - \text{Renewal Probability}) * (\text{New Free-Rent Months} / 12) \\ &-\text{Base Rental Income in Year 3} * \text{Renewal Probability} * (\text{Renewal Free-Rent Months} / 12) \\ &= -\$1,349,837 * (1 - 70\%) * (6 / 12) - \$1,349,837 * 70\% * (3 / 12) \\ &= -\$438,697 \end{aligned}$$

The first line handles the **non-renewal case** – for 6 months of free rent – and the second line handles the **renewal case** – for 3 months of free rent.

The formulas for the Tenant Improvements and Leasing Commissions use similar principles, but the calculations are different since the TIs are linked to the tenant's rentable square feet, and the LCs are linked to the tenant's total lease value.

Here is the formula for the Year 3 TIs:

$$\begin{aligned} &= -\text{Rentable Square Feet} * (1 - \text{Renewal Probability}) * \text{New TIs per Square Foot} \\ &-\text{Rentable Square Feet} * \text{Renewal Probability} * \text{Renewal TIs per Square Foot} \\ &= -10,000 * (1 - 70\%) * \$10.00 - 10,000 * 70\% * \$3.00 \\ &= -\$51,000 \end{aligned}$$

To calculate the Leasing Commissions, you must determine the total lease value over 10 years and then multiply by the commission percentages and probabilities.

Technically, you should adjust this lease value for the annual rental escalations because the broker who referred the tenant will expect those to be factored in.

To simplify the calculation, though, we left this out and based the lease value on 10 Years * Rentable Square Feet * Rent per Square Foot in Year 3, which made the formula:

$$\begin{aligned} &= -(1 - \text{Renewal Probability}) * \text{Rent per SF} * \text{Rentable Square Feet} * \text{New Lease Years} * \text{New LC Pct} \\ &-\text{Renewal Probability} * \text{Rent per SF} * \text{Rentable Square Feet} * \text{New Lease Years} * \text{Renewal LC Pct} \\ &= -(1 - 70\%) * \$134.98 * 10,000 * 10 * 3\% - 70\% * \$134.98 * 10,000 * 10 * 1\% \\ &= -\$215,974 \end{aligned}$$

The calculations here are simple because the rent for new tenants and renewal tenants is the same, and so are the years in the lease; only the commission percentages differ.



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You complete this process for each tenant and then sum up everything.

This tenant has a **Full Service (FS) lease**, which makes things even easier because there are no Expense Reimbursements.

If this tenant had a lease with Expense Reimbursements, you would project the expenses and then calculate the Reimbursements based on the tenant's percentage of total rentable square feet:

Office Tenant #2 - Single Net (N) Lease:								
Rentable Square Feet Occupied:	sq. ft.	7,000 sq. ft.						
Lease Expiration Date:	Date	2020-12-31						
Baseline Rent per Square Foot:	\$ / sq. ft. / Yr	\$ 105.00	\$ 110.25	\$ 115.76	\$ 120.39	\$ 125.21	\$ 128.96	\$ 132.83
Rental Growth Rate:	%	5.0%	5.0%	4.0%	4.0%	3.0%	3.0%	
(+) Base Rental Income:	\$	735,000	771,750	810,338	842,751	876,461	902,755	929,838
(-) Absorption & Turnover Vacancy:	\$	-	-	-	-	(131,469)	-	-
(-) Concessions & Free Rent:	\$	-	-	-	-	(284,850)	-	-
(-) Tenant Improvements (TIs):	\$	-	-	-	-	(35,700)	-	-
(-) Leasing Commissions (LCs):	\$	-	-	-	-	(140,234)	-	-
(+) Expense Reimbursements:	\$	280,000	285,600	291,312	=-132*(581/Rentable_SF)	309,143	315,325	
Numerical Year:	Year	2017	2018	2019	2020	2021	2022	2023
Property Pro-Forma:	Units:	Historical: FY17	FY18	FY19	Projected: FY20	FY21	FY22	Stabilized Year:
Revenue:								
(+) Base Rental Income:	\$	\$ 2,715,000	\$ 2,841,750	\$ 2,968,862	\$ 3,090,621	\$ 3,198,048	\$ 3,293,989	\$ 3,392,809
(-) Absorption & Turnover Vacancy:	\$	-	-	-	(202,476)	(131,469)	-	-
(-) Concessions & Free Rent:	\$	-	-	-	(438,697)	(284,850)	-	-
(+) Expense Reimbursements:	\$	580,000	592,290	604,848	617,681	585,333	644,196	657,891
Potential Gross Revenue:	\$	3,295,000	3,434,040	3,573,710	3,067,130	3,367,062	3,938,186	4,050,700
(-) General Vacancy:	\$	(240,000)	(249,600)	(259,584)	(269,967)	(278,066)	(286,408)	(295,001)
Effective Gross Income (EGI):	\$	3,055,000	3,184,440	3,314,126	2,797,163	3,088,996	3,651,777	3,755,700
Expenses:								
(-) Property Management Fees:	\$	(91,650)	(95,533)	(99,424)	(83,915)	(92,670)	(109,553)	(112,671)
(-) Common Area Maintenance (CAM):	\$	(125,000)	(129,375)	(133,903)	(138,590)	(143,440)	(148,461)	(153,657)
(-) Common Area Utilities:	\$	(75,000)	(77,250)	(79,568)	(81,955)	(84,413)	(86,946)	(89,554)
(-) Insurance:	\$	(50,000)	(51,250)	(52,531)	(53,845)	(55,191)	(56,570)	(57,985)
(-) Real Estate & Property Taxes:	\$	(1,000,000)	(1,020,000)	(1,040,400)	(1,061,208)	(1,082,432)	(1,104,081)	(1,126,162)
(-) CapEx, TI, and LC Reserves:	\$	(62,500)	(64,375)	(66,306)	(68,295)	(70,344)	(72,455)	(74,628)
Total Expenses:	\$	(1,404,150)	(1,437,783)	(1,472,132)	(1,487,807)	(1,528,491)	(1,578,066)	(1,614,657)

The formula shown at the top of the screenshot above works in most years, **but you must also consider the possible vacancy in Year 4 when this tenant's lease expires**. The formula there becomes:

$$= -\text{Annual Property Taxes} * (\text{Tenant Rentable SF} / \text{Total Rentable SF})$$

$$+ \text{Annual Property Taxes} * (\text{Tenant Rentable SF} / \text{Total Rentable SF}) * (1 - \text{Renewal Probability}) * (\text{Downtime Months} / \text{Months})$$



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The first line handles the case where the tenant renews and pays its proportional share of Property Taxes in full.

The second line handles the case where the tenant does **not** renew. In that case, the space goes vacant for a few months, during which we receive no reimbursements since there is no tenant.

We adjust *down* the total Expense Reimbursements based on this number. Expense Reimbursements are positive because Annual Property Taxes are negative, so by *adding* the second line, we make the Expense Reimbursements less positive.

If this tenant were responsible for other expenses, such as Maintenance, Utilities, or Insurance, those expenses would also be referenced in this formula.

If you are modeling a development, where *all* the tenants are in the “New” category initially, you can use similar formulas with one key difference: **When the leases first begin, you will use the “New” numbers for everything and not consider the Renewal vs. Non-Renewal cases at all.**

After that, when the tenant’s lease comes up for renewal, you can follow the approach above and use probability-weighting to calculate all the numbers. Here’s an example:

Rent Roll & Operating Assumptions:		Units:	Construction:	Operational Years:							Stabilized:
			FY18	FY19	FY20	FY21	FY22	FY23	FY24	FY25	FY26
Tenant #1 - Triple Net (NNN) Lease:											
% Rentable Square Feet Occupied:	%	65.0%									
Rentable Square Feet Occupied:	sq. ft.	208,193 sq. ft.									
Lease Start Date:	Date	2018-12-31									
Lease Expiration Date:	Date	2022-12-31									
Baseline Rent per Square Foot:	\$ / sq. ft. / Yr		\$ 7.50	\$ 7.73	\$ 7.96	\$ 8.20	\$ 8.44	\$ 8.69	\$ 8.96	\$ 9.22	
Rental Growth Rate:	%			3.0%	3.0%	3.0%	3.0%	3.0%	3.0%	3.0%	
(+) Base Rental Income:	\$		\$ 1,561,446	\$ 1,608,290	\$ 1,656,538	\$ 1,706,235	\$ 1,757,422	\$ 1,810,144	\$ 1,864,449	\$ 1,920,382	
(-) Absorption & Turnover Vacancy:	\$		-	-	-	-	(351,484)	-	-	-	
(-) Concessions & Free Rent:	\$		(520,482)	-	-	-	(410,065)	-	-	-	
(-) Tenant Improvements (TIs):	\$		(312,289)	-	-	-	(249,831)	-	-	-	
(-) Leasing Commissions (LCs):	\$		(195,975)	-	-	-	(167,947)	-	-	-	
(+) Expense Reimbursements:	\$		489,253	503,931	519,049	534,620	440,527	567,179	584,194	601,720	

This year corresponds to a traditional lease expiration, where we consider both the Renewal and Non-Renewal cases and probability-weight the numbers.

No Absorption & Turnover Vacancy here because the tenant's lease is just starting. We use the "New" numbers for Free Rent, TIs, and LCs.

These projections are for an **industrial property**, so the construction period is short (1 year), the lease term is 4-5 years rather than 10 years, and the renewal probability is slightly lower (60%).

Industrial properties are often faster and cheaper to build since they tend to be simple warehouse-like structures, but they’re also less “defensible” than complex office buildings.

The Monthly Pro-Forma and the Annual Roll-Up



Since the goal of financial modeling is to be “roughly correct” rather than “precisely wrong,” annual projections for tenants work well in many cases.

However, there are some problems with the annual approach:

- What if a tenant’s lease starts or ends in the **middle of the year** rather than December 31?
- What if the Downtime Months between tenants span 2 years (e.g., a lease expires in October, and it takes 6 months to find a new tenant)?
- What if the Free Rent Months when a new tenant signs up or an existing tenant renews span 2 years (e.g., a sign-up occurs in November and the tenant gets 3 months of free rent)?
- What if the lease does not last for a whole number of years (e.g., 6.5 years rather than 6 or 7 years)?

For these reasons, you usually create **monthly projections** for individual tenants if you’re doing a serious analysis of the property.

In real life, this gets so complicated that you will use ARGUS rather than Excel, especially when the lease terms are highly variable and there are dozens of tenants.

However, if you constrain the assumptions to include only one lease expiration in the holding period for each tenant, it is possible to create reasonable monthly projections in Excel for properties with relatively few tenants (e.g., 3-5 rather than 25).

After you finish the monthly projections, you will create an annual summary because no one wants to look at 50+ columns of an Excel file; they want the high-level overview.

There are a few examples of monthly models in this course, but here we’ll focus on the one used in the office/retail acquisition & renovation deal (45 Milk Street).

You should **start** by projecting the monthly expenses for this property because you’ll need them for the Expense Reimbursements later on.

We prefer to use an “Expense Inflation Multiplier” to determine the monthly numbers.

For example, if expenses are expected to grow at 4% per year, the Monthly Expense Inflation Rate would be $(1 + 4\%)^{(1 / 12)} - 1 = 0.33\%$.



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Then, the Expense Inflation Multiplier in the first month would be $1.00 * (1 + 0.33\%) = 1.00$, and in the second month it would be $1.00 * (1 + 0.33\%) * (1 + 0.33\%) = 1.01$, and so on.

Expenses such as Utilities, Insurance, and Maintenance are still based on per-square-foot values times the number of square feet, but you must divide each one by 12 and then multiply by this Expense Inflation Multiplier to get the correct monthly value:

Rent Roll:	Units:	Jan-18	Feb-18	Mar-18	Apr-18	May-18
Operating Expenses:						
(-) Property Management Fees:	\$	(5,729)	(5,822)	(6,873)	(6,869)	(6,933)
(-) Common Area Maintenance (CAM):	\$	(11,475)	(11,513)	(11,550)	(11,588)	(11,626)
(-) Common Area Utilities:	\$	(9,180)	(9,210)	(9,240)	(9,270)	(9,301)
(-) Insurance:	\$	(4,590)	=Insurance_per_SF*Rentable_Sq_Ft/Months*G\$15			(4,650)
(-) Real Estate & Property Taxes:	\$	(43,705)	(43,848)	(43,992)	(44,136)	(44,280)
(-) CapEx, TI, and LC Reserves:	\$	(13,770)	(13,815)	(13,860)	(13,906)	(13,951)
Total Operating Expenses:	\$	(88,449)	(88,812)	(90,136)	(90,404)	(90,741)

The G\$15 cell in the screenshot above is the Expense Inflation Multiplier for the month.

Property Management Fees are a simple percentage of Effective Gross Income, and Property Taxes are initially a percentage of the purchase price and then grow by a small percentage each year.

This property has 5 existing tenants and 2 “potential” tenants that may move in, depending on the scenario, and each tenant has different lease terms.

To avoid repetition, we’ll walk through the calculations for the most complex one: Tenant #5 (Trader Joe’s), which has a Triple Net Lease with Percentage Rent.

We will cover the main ideas across 3 steps: 1) Market Rents and Escalated Rents, 2) Other Items, and 3) Percentage Rent.

Step 1: Market Rents and Escalated Rents

Projecting the rent is trickier than it sounds because it usually starts at one level, stays the same for a year, and increases by a fixed percentage the next year – and that increase could happen in different months, depending on whether the existing tenant renews or cancels.

If the tenant cancels, you’ll have to assume a Downtime period while you search for a new tenant.

The first step here is to pull in all the lease information from the assumptions provided to us:

	A	B	C	D	E	F	G
1							
2			Monthly Cash Flow Projections				
3			(\$ in USD as Stated)				
4							
5			Rent Roll:	Units:		Jan-18	Feb-18
175							
176			Tenant #5 - Triple Net (NNN) Lease with Percentage Rent:				
177			Tenant Name:	Name	Trader Joe's		
178			% Rentable Square Feet Occupied:	%	20.0%		
179			Rentable Square Feet Occupied:	sq. ft.	10,980 sq. ft.		
180			Lease Start Date:	Date	2016-06-30		
181			Initial Lease Term (Years):	# Years	5		
182			Lease Expiration Date:	Date	2021-06-30		
183			Initial Lease - Annual Rental Escalation:	%	3.00%	-	-
184			Initial Lease - Rent-Free Months:	#	6		
185							
186			Renewal Lease Start Date:	Date	2021-06-30		
187			Renewal Lease Term (Years):	# Years	5		
188			Renewal Lease Expiration Date:	Date	2026-06-30		
189			Renewal Lease - Annual Rental Escalation:	%	2.50%	-	-
190							
191			New Lease Start Date:	Date	2021-12-31		
192			New Lease Term (Years):	# Years	5		
193			New Lease Expiration Date:	Date	2026-12-31		
194			New Lease - Annual Rental Escalation:	%	=INDEX(Annual!	-	-
195					\$F\$65:\$K\$92,16,		
196			Market Rent per Square Foot per Year:	\$ / sq. ft. / Yr.	INDEX(Monthly!	\$ 48.23	\$ 4
197			Escalated Rent Paid by Initial Tenant:	\$ / sq. ft. / Yr.	\$F\$5:\$BY\$9,5,	42.00	4
198			Escalated Rent Paid by Initial Tenant - Renewal:	\$ / sq. ft. / Yr.	MATCH(E191,	-	-
199			Escalated Rent Paid by New Tenant:	\$ / sq. ft. / Yr.	\$F\$5:\$BY\$5,0)))	-	-

Then, we check the current month against the lease start date, and if it matches, we assume a rental escalation:

	A	B	C	D	E	H	I	J	K	L
1										
2			Monthly Cash Flow Projections							
3			(\$ in USD as Stated)							
4										
5			Rent Roll:	Units:		Mar-18	Apr-18	May-18	Jun-18	Jul-18
175										
176			Tenant #5 - Triple Net (NNN) Lease with Percentage Rent:							
177			Tenant Name:	Name	Trader Joe's					
178			% Rentable Square Feet Occupied:	%	20.0%					
179			Rentable Square Feet Occupied:	sq. ft.	10,980 sq. ft.					
180			Lease Start Date:	Date	2016-06-30					
181			Initial Lease Term (Years):	# Years	5					
182			Lease Expiration Date:	Date	2021-06-30					
183			Initial Lease - Annual Rental Escalation:	%	3.00%	=IF(AND(H\$5>\$E180,I\$5<\$E182,MONTH(H\$5)=				3.00%
184			Initial Lease - Rent-Free Months:	#	6	MONTH(\$E180)), \$E183,0)				
185										

We divide this tenant into the Initial, Renewal, and Non-Renewal cases because those are the main possibilities during the term of this lease.



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Then, we project the rents on a per-square-foot-per-year basis:

Rent Roll:	Units:		May-21	Jun-21	Jul-21	Aug-21	Sep-21	Oct-21
Initial Lease Term (Years):	# Years	5						
Lease Expiration Date:	Date	2021-06-30						
Initial Lease - Annual Rental Escalation:	%	3.00%	-	-	-	-	-	-
Initial Lease - Rent-Free Months:	#	6						
Renewal Lease Start Date:	Date	2021-06-30						
Renewal Lease Term (Years):	# Years	5						
Renewal Lease Expiration Date:	Date	2026-06-30						
Renewal Lease - Annual Rental Escalation:	%	2.50%	-	-	-	-	-	-
New Lease Start Date:	Date	2021-12-31						
New Lease Term (Years):	# Years	5						
New Lease Expiration Date:	Date	2026-12-31						
New Lease - Annual Rental Escalation:	%	2.50%	-	-	-	-	-	-
Market Rent per Square Foot per Year:	\$ / sq. ft. / Yr.	\$ 48.00	\$ 55.18	\$ 55.31	\$ 55.45	\$ 55.59	\$ 55.72	\$ 55.86
Escalated Rent Paid by Initial Tenant:	\$ / sq. ft. / Yr.	42.00	45.89	45.89	-	-	-	-
Escalated Rent Paid by Initial Tenant - Renewal:	\$ / sq. ft. / Yr.		-	-	55.45	55.45	55.45	55.45
Escalated Rent Paid by New Tenant:	\$ / sq. ft. / Yr.		-	-	55.45	55.59	55.72	55.86

Market Rent changes each month based on market conditions and the monthly growth and decline rates.

But the Escalated Rents for this initial tenant increase only on the anniversaries of the leases. When a new lease starts, it's based on the Market Rent in that starting month.

The same applies to the Non-Renewal Case Rent, but in the Downtime Period before the new tenant moves in, we set it equal to the Market Rent because the rate is not set until the new tenant moves in and the lease officially begins.

Next, we calculate the lump-sum rent paid in these Initial, Renewal, and Non-Renewal cases, and we probability weight the latter two:

	A	B	C	D	E	AT	AU	AV	AW	AX
1										
2			Monthly Cash Flow Projections							
3			(\$ in USD as Stated)							
4										
5			Rent Roll:	Units:		May-21	Jun-21	Jul-21	Aug-21	Sep-21
175										
176			Tenant #5 - Triple Net (NNN) Lease with Percentage Rent:							
177			Tenant Name:	Name	Trader Joe's					
178			% Rentable Square Feet Occupied:	%	20.0%					
179			Rentable Square Feet Occupied:	sq. ft.	10,980 sq. ft.					
180			Lease Start Date:	Date	2016-06-30					
181			Initial Lease Term (Years):	# Years	5					
182			Lease Expiration Date:	Date	2021-06-30					
183			Initial Lease - Annual Rental Escalation:	%	3.00%	-	-	-	-	-
184			Initial Lease - Rent-Free Months:	#	6					
185										
196			Market Rent per Square Foot per Year:	\$ / sq. ft. / Yr.	\$ 48.00	\$ 55.18	\$ 55.31	\$ 55.45	\$ 55.59	\$ 55.72
197			Escalated Rent Paid by Initial Tenant:	\$ / sq. ft. / Yr.	42.00	45.89	45.89	-	-	-
198			Escalated Rent Paid by Initial Tenant - Renewal:	\$ / sq. ft. / Yr.		-	-	55.45	55.45	55.45
199			Escalated Rent Paid by New Tenant:	\$ / sq. ft. / Yr.		-	-	55.45	55.59	55.72
200										
201			(+) Base Rental Income:	\$		\$ 50,488	\$ 50,612	\$ 50,737	\$ 50,862	\$ 50,988
202			In-Place Rental Income - Initial Tenant:	\$		41,993	41,993	-	-	-
203			In-Place Rental Income - Initial Tenant Renewal:	\$		-	-	=+AV198*\$E179/Months*Renewal_Probability	-	-
204			In-Place Rental Income - New Tenant:	\$		-	-	20,295	20,345	20,395



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Finally, we calculate the “Loss to Lease” line item by summing up the In-Place Rental Income and then subtracting the Base Rental Income:

	A	B	C	D	E	AT	AU	AV	AW	AX
1										
2			Monthly Cash Flow Projections							
3			(\$ in USD as Stated)							
4										
5			Rent Roll:	Units:		May-21	Jun-21	Jul-21	Aug-21	Sep-21
175										
176			Tenant #5 - Triple Net (NNN) Lease with Percentage Rent:							
177			Tenant Name:	Name	Trader Joe's					
178			% Rentable Square Feet Occupied:	%	20.0%					
179			Rentable Square Feet Occupied:	sq. ft.	10,980 sq. ft.					
180			Lease Start Date:	Date	2016-06-30					
181			Initial Lease Term (Years):	# Years	5					
182			Lease Expiration Date:	Date	2021-06-30					
183			Initial Lease - Annual Rental Escalation:	%	3.00%	-	-	-	-	-
184			Initial Lease - Rent-Free Months:	#	6					
185										
196			Market Rent per Square Foot per Year:	\$ / sq. ft. / Yr.	\$ 48.00	\$ 55.18	\$ 55.31	\$ 55.45	\$ 55.59	\$ 55.72
197			Escalated Rent Paid by Initial Tenant:	\$ / sq. ft. / Yr.	42.00	45.89	45.89	-	-	-
198			Escalated Rent Paid by Initial Tenant - Renewal:	\$ / sq. ft. / Yr.		-	-	55.45	55.45	55.45
199			Escalated Rent Paid by New Tenant:	\$ / sq. ft. / Yr.		-	-	55.45	55.59	55.72
200										
201			(+) Base Rental Income:	\$		\$ 50,488	\$ 50,612	\$ 50,737	\$ 50,862	\$ 50,988
202			In-Place Rental Income - Initial Tenant:	\$		41,993	41,993	-	-	-
203			In-Place Rental Income - Initial Tenant Renewal:	\$		-	-	30,442	30,442	30,442
204			In-Place Rental Income - New Tenant:	\$		-	-	20,295	20,345	20,395
205										
206			(-) Loss to Lease:	\$		(8,494)	(8,619)	=IF(AV\$5>\$E180,SUM(AV202:AV204)-AV201,0)		

On the Pro-Forma, we'll show the Base Rental Income and the Loss to Lease to represent the probability-weighted In-Place Rent paid by tenants across all the cases.

Step 2: Other Items (Absorption & Turnover Vacancy, Concessions & Free Rent, Tenant Improvements, Leasing Commissions, Expense Reimbursements, and General Vacancy)

The other line items for this tenant follow the same basic logic – Initial, Renewal, and Non-Renewal cases with slightly different numbers – but some are simpler than others.

For example, the Absorption & Turnover Vacancy is easy to calculate because we just need to check if the current month is in between the initial lease expiration date and the new lease start date, and if so, reverse the in-place rental income paid by the new tenant:



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	A	B	C	D	E	AT	AU	AV	AW	AX	AY
1											
2			Monthly Cash Flow Projections								
3			(\$ in USD as Stated)								
4											
5			Rent Roll:	Units:		May-21	Jun-21	Jul-21	Aug-21	Sep-21	Oct-21
175											
176			Tenant #5 - Triple Net (NNN) Lease with Percentage Rent:								
177			Tenant Name:	Name	Trader Joe's						
178			% Rentable Square Feet Occupied:	%	20.0%						
179			Rentable Square Feet Occupied:	sq. ft.	10,980 sq. ft.						
180			Lease Start Date:	Date	2016-06-30						
181			Initial Lease Term (Years):	# Years	5						
182			Lease Expiration Date:	Date	2021-06-30						
183			Initial Lease - Annual Rental Escalation:	%	3.00%	-	-	-	-	-	-
184			Initial Lease - Rent-Free Months:	#	6						
185											
191			New Lease Start Date:	Date	2021-12-31						
192			New Lease Term (Years):	# Years	5						
193			New Lease Expiration Date:	Date	2026-12-31						
194			New Lease - Annual Rental Escalation:	%	2.50%	-	-	-	-	-	-
195											
201			(+) Base Rental Income:	\$		\$ 50,488	\$ 50,612	\$ 50,737	\$ 50,862	\$ 50,988	\$ 51,114
202			In-Place Rental Income - Initial Tenant:	\$		41,993	41,993	-	-	-	-
203			In-Place Rental Income - Initial Tenant Renewal:	\$		-	-	30,442	30,442	30,442	30,442
204			In-Place Rental Income - New Tenant:	\$		-	-	20,295	20,345	20,395	20,445
205											
206			(-) Loss to Lease:	\$		(8,494)	(8,619)	-	(75)	(150)	(226)
207			(-) Absorption & Turnover Vacancy:	\$		-	-	(20,295)	=IF(AND(AW\$5>\$E182,AW\$5<=\$E191),-AW204,0)		
208			(-) Concessions & Free Rent:	\$		-	-	(30,442)	(30,442)	-	-
209			(-) Tenant Improvements (TIs):	\$		-	-	(145,637)	-	-	-
210			(-) Leasing Commissions (LCs):	\$		-	-	(57,605)	-	-	-

The next few line items are more complicated because they incorporate all 3 cases. With the TIs and LCs, we check if the current month matches a lease start date, and if it does, we record the appropriate amount.

For the LCs, we use the FV (Future Value) function in Excel to calculate the total value of the lease over its term, escalated by the appropriate percentage each year, and then we multiply that by the assumed leasing commission percentage.

We probability weight both these line items in the Renewal and Non-Renewal cases as well.

The Concessions & Free Rent formula is more complex because in each case, we need to check *a range of dates*: the lease start date to the end of the free months of rent following that lease start.

Depending on the case and period we're in, we reverse the in-place rental income for the initial tenant, renewal tenant, or new tenant:



Real Estate & REIT Modeling

Quick Reference Guide for Individual Properties

	A	B	C	D	E	AU	AV	AW	AX	AY	AZ	BA	BB	BC	BD
1															
2			Monthly Cash Flow Projections												
3			(\$ in USD as Stated)												
4															
5			Rent Roll:	Units:		Jun-21	Jul-21	Aug-21	Sep-21	Oct-21	Nov-21	Dec-21	Jan-22	Feb-22	Mar-22
176			Tenant #5 - Triple Net (NNN) Lease with Percentage Rent:												
177			Tenant Name:	Name	Trader Joe's										
178			% Rentable Square Feet Occupied:	%	20.0%										
179			Rentable Square Feet Occupied:	sq. ft.	10,980 sq. ft.										
180			Lease Start Date:	Date	2016-06-30										
181			Initial Lease Term (Years):	# Years	5										
182			Lease Expiration Date:	Date	2021-06-30										
183			Initial Lease - Annual Rental Escalation:	%	3.00%										
184			Initial Lease - Rent-Free Months:	#	6										
185															
191			New Lease Start Date:	Date	2021-12-31										
192			New Lease Term (Years):	# Years	5										
193			New Lease Expiration Date:	Date	2026-12-31										
194			New Lease - Annual Rental Escalation:	%	2.50%										
195															
201			(+) Base Rental Income:	\$		\$ 50,612	\$ 50,737	\$ 50,862	\$ 50,988	\$ 51,114	\$ 51,240	\$ 51,366	\$ 51,492	\$ 51,578	\$ 51,684
202			In-Place Rental Income - Initial Tenant:	\$		41,993	-	-	-	-	-	-	-	-	-
203			In-Place Rental Income - Initial Tenant Renewal:	\$		-	30,442	30,442	30,442	30,442	30,442	30,442	30,442	30,442	30,442
204			In-Place Rental Income - New Tenant:	\$		-	20,295	20,345	20,395	20,445	20,496	20,546	20,589	20,589	20,589
205															
206			(-) Loss to Lease:	\$		(8,619)	-	(75)	(150)	(226)	(301)	(377)	(441)	(547)	(653)
207			(-) Absorption & Turnover Vacancy:	\$		-	(20,295)	(20,345)	(20,395)	(20,445)	(20,496)	(20,546)	-	-	-
208			(-) Concessions & Free Rent:	\$		-	=IF(AND(AV\$5>\$E180,AV\$5<=EOMONTH(\$E180,\$E184)),-								
209			(-) Tenant Improvements (TIs):	\$		-	AV202,0)+IF(AND(AV\$5>\$E182,AV\$5<=EOMONTH(\$E182, Renewal_Free_Rent_Months)),-								

That formula is:

=IF(AND(AV\$5>\$E180,AV\$5<=EOMONTH(\$E180,\$E184)),-
AV202,0)+IF(AND(AV\$5>\$E182,AV\$5<=EOMONTH(\$E182, Renewal_Free_Rent_Months)),-
AV203,0)+IF(AND(AV\$5>\$E191,AV\$5<=EOMONTH(\$E191, New_Free_Rent_Months)),-AV204,0)

The Expense Reimbursements are simple because we just need to check that the initial lease has begun; if it has, we take the applicable expenses and multiply by the tenant's proportional share (20% here), and if not, it's 0.

Then, we also check if we're in the Downtime Period in between the initial lease ending and the new lease starting, and if so, we multiply the Expense Reimbursements by the Renewal Probability (60% here); if not, we multiply by 1.

That reduces the Expense Reimbursements by the *Non-Renewal Probability* in this Downtime Period.

Finally, General Vacancy is also simple: we reverse the Market Rent and show it as a negative if the initial lease has not yet begun. Otherwise, it's 0.

In this model, the General Vacancy line item exists only for the 2 "potential tenants" at first since the other 5 tenants all have active leases as of the first month.

Step 3: Percentage Rent



Percentage Rent, i.e., charging a tenant a percentage of its monthly sales in addition to fixed monthly rent, is **not** specific to monthly models.

However, we're explaining it here because it is a new feature in this office/retail acquisition & renovation monthly model.

Percentage Rent is limited to retail tenants because it does not make sense to charge an office or industrial tenant in this way since those spaces do not generate sales directly.

The two main methods for calculating Percentage Rent are **Artificial Breakpoints** and **Natural Breakpoints**.

With Artificial Breakpoints, the tenant and landlord might agree that if the tenant's sales exceed \$100,000 per month (for example), the tenant will pay 5% of sales above that level to the landlord.

So, if monthly sales are \$120,000, the tenant will owe $5\% * (\$120,000 - \$100,000) = \$1,000$ in Percentage Rent that month.

With Natural Breakpoints, the owner divides the tenant's Annual Rent by an agreed-upon "Set Percentage" and charges that percentage for anything above this number.

For example, if the Annual Rent is \$300,000 and the Set Percentage is 5%, then $\$300,000 / 5\% = \6 million.

The tenant will pay 5% of sales above this "Natural Breakpoint" in Percentage Rent. On a monthly basis, that is $\$6 \text{ million} / 12 = \$500,000$.

So, if the tenant's monthly sales are \$600,000, the tenant will owe $(\$600,000 - \$500,000) * 5\% = \$5,000$. But if the tenant's monthly sales are \$450,000, it will owe nothing.

The lease terms might also adjust these numbers over time to reflect inflation or growth in the tenant's business.

Retail tenants often agree to Percentage Rent if it means their normal, fixed rent will be lower – it reduces the risk in recessions but also costs the company more when the economy is strong.

To set up the formula for Percentage Rent in Excel, we check to make sure that the initial lease has begun, and if it has, we take the MAX between 0 and (Monthly Retail Sales – Breakpoint * Percentage of Rent Beyond Breakpoint). We may also adjust with an inflation or retail sales growth multiplier.

We also multiply this figure by the Renewal Probability if we're in the Downtime Period to reflect that the space will not generate any Percentage Rent if it does not have a tenant:

Rent Roll:	Units:		Apr-21	May-21	Jun-21	Jul-21	Aug-21	Sep-21	Oct-21
New Lease - Annual Rental Escalation:	%	2.50%	-	-	-	-	-	-	-
Market Rent per Square Foot per Year:	\$ / sq. ft. / Yr.	\$ 48.00	\$ 55.04	\$ 55.18	\$ 55.31	\$ 55.45	\$ 55.59	\$ 55.72	\$ 55.86
Escalated Rent Paid by Initial Tenant:	\$ / sq. ft. / Yr.	42.00	45.89	45.89	45.89	-	-	-	-
Escalated Rent Paid by Initial Tenant - Renewal:	\$ / sq. ft. / Yr.		-	-	-	55.45	55.45	55.45	55.45
Escalated Rent Paid by New Tenant:	\$ / sq. ft. / Yr.		-	-	-	55.45	55.59	55.72	55.86
(+) Base Rental Income:	\$		\$ 50,364	\$ 50,488	\$ 50,612	\$ 50,737	\$ 50,862	\$ 50,988	\$ 51,114
In-Place Rental Income - Initial Tenant:	\$		41,993	41,993	41,993	-	-	-	-
In-Place Rental Income - Initial Tenant Renewal:	\$		-	-	-	30,442	30,442	30,442	30,442
In-Place Rental Income - New Tenant:	\$		-	-	-	20,295	20,345	20,395	20,445
(-) Loss to Lease:	\$		(8,370)	(8,494)	(8,619)	-	(75)	(150)	(226)
(-) Absorption & Turnover Vacancy:	\$		-	-	-	(20,295)	(20,345)	(20,395)	(20,445)
(-) Concessions & Free Rent:	\$		-	-	-	(30,442)	(30,442)	-	-
(-) Tenant Improvements (TIs):	\$		-	-	-	(145,637)	-	-	-
(-) Leasing Commissions (LCs):	\$		-	-	-	(57,605)	-	-	-
(+) Expense Reimbursements:	\$		15,118	15,143	15,168	9,116	9,131	9,146	9,161
(-) General Vacancy:	\$		-	-	-	-	-	-	-
Initial Retail Sales per Square Foot per Year:	\$ / sq. ft. / Yr.	\$ 800.00							
Initial Average Monthly Sales:	\$	\$ 732,000							
Initial Artificial Breakpoint:	\$	\$ 500,000							
Percentage Rent Beyond Breakpoint:	%	4.0%							
Monthly Distribution of Retail Sales:	%		8.0%	8.6%	7.8%	7.9%	8.9%	7.4%	8.0%
Monthly Retail Sales:	\$		\$ 766,704	\$ 819,305	\$ 740,403	\$ 749,967	\$ 850,388	\$ 704,539	\$ 759,531
(+) Percentage Rent:	\$		8,920	11,024	7,868	4,950	7,360	3,860	5,180

Monthly Sales exceed the Artificial Breakpoint, so Percentage Rent is owed.

But as soon as the initial lease expires, Percentage Rent decreases because we now multiply it by the Renewal Probability in these Downtime Months.

Strengths and Weaknesses of Office, Industrial, and Retail Properties

The key feature that sets office, industrial, and retail properties apart are the **long-term leases** that lock in rent and annual rental escalations regardless of market conditions.

A hotel might benefit greatly when the economy is roaring, but its revenue will also drop immediately when there's a recession because business and leisure travel will plummet.

By contrast, rent from a tenant on a 10-year lease will keep going to the owner of an office property even if there's a recession – unless the tenant goes bankrupt in that recession, of course.

But that's also a weakness of these properties: since the lease terms are set far in advance, the owners cannot take advantage of a high-growth market to increase rent more rapidly.

Within this category, office properties have had the most volatile *historical* returns, indicating the highest risk, primarily because economic downturns impact them the most.

Also, they take more time and money to construct, which is a major risk factor in development deals.



However, it's hard to say if office properties in the modern environment are truly the riskiest in this category because retailers are facing major headwinds due to the rise of e-commerce (read: Amazon).

That same trend has benefited industrial properties, many of which are warehouses used by companies such as FedEx, at the expense of retail properties.

On the other hand, industrial properties often have the fewest number of tenants and take the least amount of time and money to build, so from that perspective, they are less defensible and incur more lease-expiration risk.

So, offices have been the most volatile historically and industrial and retail properties have had similar volatility to each other, but we hesitate to say that will be the case going forward.

If you get a question about this topic in interviews, it's best to describe these trade-offs between the property types rather than proclaiming that one has the highest or lowest risk and potential returns.

One final point: although office, industrial, and retail properties benefit from their long-term leases, they're also hurt by the fact that the terms of these leases vary substantially, and that there tend to be fewer tenants than in property types such as [multifamily](#).

Also, the high capital costs in the form of TIs, LCs, and CapEx count against office/industrial/retail properties because they make the company's cash flow "lumpier"; these items either do not exist or are much smaller for other types of properties.

So, even though properties in this class offer some advantages, [multifamily](#) still has the lowest long-term historical volatility.

The sector is simply less sensitive to economic conditions because people *always* need somewhere to live, and that's a major benefit that outweighs the drawbacks of shorter-term leases.

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Condominiums

Condominiums ("condos"), or properties that are divided into units that are purchased outright, are quite different from all the other property types here.



The primary difference is that condo developers never plan to hold the units or rent them out – they always aim to sell them to individuals as soon as possible.

In most cases, they attempt to “pre-sell” these units before construction is finished so they can reduce risk and assess market demand as construction progresses.

Often, the new owners pay an upfront deposit for a portion of the unit price, another deposit upon construction end, and the final deposit when they move in.

There are dozens of variations, ranging from monthly deposits spread out over the entire construction duration to an initial deposit and final deposit upon move-in.

There’s no such thing as a condominium Pro-Forma because there is no “stabilized state.”

Instead, you project the cash flows associated with the development of the condos, including the Hard Costs, Soft Costs, FF&E and Move-In Costs, and Land Acquisition Costs, and the Deposits paid by the owners.

Condo developers, like all real estate developers, use a combination of Debt and Equity to fund their projects, and if cash flow is insufficient to pay for interest, the interest will be capitalized.

Usually, there are restrictions on Debt draws – for example, a certain percentage or amount of Equity must be drawn first, or a certain percentage of units must be sold.

There’s no point modeling acquisitions or renovations of condominiums because it’s logistically difficult to acquire entire condos when there are dozens or hundreds of different owners.

Some private equity firms do specialize in acquisitions of individual condo units, but this strategy is **not** a common case study or interview topic, so we do not focus on it here.

The **trickiest part** of condominium models is getting the trigger conditions for sales and deposits correct.

You normally split the construction timeline into pre-construction, construction, and post-construction phases, and you also divide the condo sales into multiple phases.

It’s not always straightforward to determine when one phase ends and the next one begins, especially if deposits are paid over an entire project phase rather than just at the beginning or end.

Here’s an example of the basic Assumptions for a simple condo development & pre-sale:



Real Estate & REIT Modeling

Quick Reference Guide for Individual Properties

Construction Assumptions:			Units:			Units:		
Development Name:	Name	Island Garden	Number of Flats:	#	470	Plot Ratio (PR):	#	5.0
Address:	Name	33 Chai Wan Road	Car Parks:	#	N/A	Lot Square Meters:	sq. m.	8,502 sq. m.
Location:	Name	Hong Kong	Number of Floors:	#	25	Land Price per Lot Square Meter:	HK\$ / sq. m.	\$ 170,000
Development Start Date:	Date	2015-01-31				Lot Asking Price:	HK\$	\$ 1,445,420,898
Months in Year:	#	12				Lot Upfront Payment %:	%	50.0%
Days in Year:	#	365				Month # for Upfront Payment:	Month #	1
Units:	#	1,000				Month # for Remainder of Payment:	Month #	6
Developer Name:	Name	Nan Fung Group				Flat Selling Price per Rentable Square Meter:	HK\$ / sq. m.	\$ 250,000
Investor Name:	Name	Jade Sea Partners				Upfront Client Payment:	%	30.0%
Rentable Square Meters:	sq. m.	34,010 sq. m.				Client Payment Upon Construction End:	%	30.0%
Rentable to Gross Square Meters %:	%	80.0%				Client Payment Upon Move-In:	%	40.0%
Gross Square Meters:	sq. m.	42,512 sq. m.				# Months Before Construction Completion Pre-Sales Begin:	# Months	30
Hard Costs per Gross Square Meter:	HK\$ / sq. m.	\$ 55,000				Sales Velocity per Month:	# Units	14
Soft Costs per Gross Square Meter:	HK\$ / sq. m.	\$ 15,000				Annual Sales Price Inflation:	%	5.0%
FF&E and Move-In Costs % Monthly Sales:	%	10.0%						
Annual Expense Inflation:	%	7.0%						

The most important assumptions, circled in red above, are the size of the property, the number of flats (units) to be sold, the land and construction costs, the anticipated selling costs per square meter, the payment percentages, and the number of units to be sold each month ("Sales Velocity per Month").

You might assume that Phase I of the sales process begins after the land is acquired, or a few years before construction ends.

Then, Phase II might begin once a certain number or percentage of units have been sold in Phase I, or once a certain number of months have passed.

The assumptions vary based on region and local zoning regulations, but here's a simple example:



Real Estate & REIT Modeling

Quick Reference Guide for Individual Properties

Construction Timeline & Assumptions:		Units:	Units:
Construction Phases:			
Pre-Construction Start Month #:	Month #	1	Pre-Construction: # 1
Pre-Construction Start Date:	Date	Jan 15	Construction: # 2
Pre-Construction # of Months:	# Months	6	Post-Construction: # 3
Construction Start Month #:	Month #	7	Post-Construction Start Month #: Month # 49
Construction Start Date:	Date	Jul 15	Post-Construction Start Date: Date Jan 19
Construction # of Months:	# Months	42	Post-Construction # of Months: # Months 6
Construction End Month #:	Month #	48	Post-Construction End Month: Month # 54
Construction End Date:	Date	Dec 18	Post-Construction End Date: Date Jun 19
Earliest Month # for Pre-Sales:	Month #	18	
Earliest Date for Pre-Sales:	Date	Jun 16	
Phase I: # Months: Month #: 18 Phase I Start Month: Month # 18 Sales Phase I - # Units To Be Sold: # Units 148 Sales Phase I - Average Unit Size: sq. m. 50 sq. m.			
Phase II: # Months: Month #: 24 Phase II Starting Delay: # Months 6 Phase II Start Month: Month # 24 Sales Phase II - # Units To Be Sold: # Units 322 Sales Phase II - Average Unit Size: sq. m. 83 sq. m.			

In Hong Kong, pre-sales can begin, at most, 30 months before construction ends.

In this case, the Phase I start date is based on local regulations in Hong Kong, and Phase II starts 6 months after Phase I.

Fewer units are sold in Phase I because the developer plans to “test the market” and assess demand. If all goes well, they plan to sell more units, at a greater average size, in Phase II.

Once you’ve set up the basic assumptions, you can project the number of units sold and the value of units sold in each month:



We assume Sales Prices and Expenses increase on a *monthly basis*.

Each month, we compare the assumed units sold to the total remaining units to be sold in this phase, and take the lesser of those two. This is why we get 8 in Apr 17 - only 8 units are left in Phase I by this point.

Total Value of Flats Sold keeps increasing because of the Sales Inflation Multiplier.

With the unit projections in place, you can forecast the Initial, Construction-End, and Final Deposits in each phase:

Based on the value of Flats Sold in this month * the upfront payment percentage - as long as Phase I has already begun.

Based on everything sold in Phase I
* the percentage due upon
construction completion; only
triggered in the very final month of
construction.

Based on the total value of Flats Sold in Phase I minus the total Initial and Construction-End Deposits paid so far; only triggered in the last month of the model.

This model assumes that the Construction-End and Final Deposits all arrive in the same month in both phases, which you can see below:

Monthly Cash Flows:	Units:	Nov 18	Dec 18	Jan 19	Feb 19	Mar 19	Apr 19	May 19	Jun 19
		End of Construction				End of Project			
Monthly Gross Sales:									
(+) Phase I - Initial Deposits:	HK\$	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
(+) Phase I - Construction-End Deposits:	HK\$	-	608,949,738	-	-	-	-	-	-
(+) Phase I - Final Deposits:	HK\$	-	-	-	-	-	-	-	811,932,984
Phase I Total Sales:	HK\$	-	608,949,738	-	-	-	-	-	811,932,984
(+) Phase II - Initial Deposits:	HK\$	-	-	-	-	-	-	-	-
(+) Phase II - Construction-End Deposits:	HK\$	-	2,301,784,279	-	-	-	-	-	-
(+) Phase II - Final Deposits:	HK\$	-	-	-	-	-	-	-	3,069,045,705
Phase II Total Sales:	HK\$	-	2,301,784,279	-	-	-	-	-	3,069,045,705
Total Monthly Gross Sales:	HK\$	-	2,910,734,017	-	-	-	-	-	3,880,978,689

Next, you have to think about the expenses associated with this development.

The main ones here are **Hard Costs** (buying the materials and paying for the construction), **Soft Costs** (legal and architectural fees, permits, and other expenses owed to service providers), **FF&E and Move-In Costs** (FF&E is "Furniture, Fixtures & Equipment), and **Land Acquisition Costs**.

Hard Costs and Soft Costs tend to be distributed irregularly, so you might use monthly percentages based on spending patterns from prior projects, or you might use a Normal or other Distribution:

Construction Timeline:	Units:	Aug 16	Sep 16	Oct 16	Nov 16	Dec 16	Jan 17	Feb 17	Mar 17
Status (1 = Pre, 2 = Constr, 3 = Post-Constr):	#	2	2	2	2	2	2	2	2
% of Hard Costs Incurred:	%	=IF(AB32=Constr,NORM.DIST(AB46-Pre Constr Months,(Final Constr Month-Pre Constr Months)/2,5,FALSE),0)							
% of Soft Costs Incurred:	%	2.0%	1.0%	4.5%	1.0%	1.5%	6.5%	1.0%	2.5%

The Hard Cost percentages follow a Normal Distribution based on the middle month of the Construction Period as the mid-point of the curve.

Random, fairly low percentages each month.

FF&E and Move-In Costs should be linked to buyer deposits because they are only incurred once the new owners purchase their units and then move in.

Land Acquisition Costs are normally paid in the first month of the project with an initial deposit, and then again within the first year with the final deposit (around the time construction begins).

Here's our schedule across several months of Phases I and II:



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Quick Reference Guide for Individual Properties

Monthly Cash Flows:	Units:	Oct 16	Nov 16	Dec 16	Jan 17
Monthly Gross Sales:					
(+) Phase I - Initial Deposits:	HK\$	\$ 57,412,486	\$ 57,646,391	\$ 57,881,250	\$ 58,117,065
(+) Phase I - Construction-End Deposits:	HK\$	-	-	-	-
(+) Phase I - Final Deposits:	HK\$	-	-	-	-
Phase I Total Sales:	HK\$	57,412,486	57,646,391	57,881,250	58,117,065
(+) Phase II - Initial Deposits:	HK\$	-	-	95,665,495	96,055,248
(+) Phase II - Construction-End Deposits:	HK\$	-	-	-	-
(+) Phase II - Final Deposits:	HK\$	-	-	-	-
Phase II Total Sales:	HK\$	-	-	95,665,495	96,055,248
Total Monthly Gross Sales:	HK\$	57,412,486	57,646,391	153,546,745	154,172,314
Monthly Expenses:					
(-) Hard Costs:	HK\$	(128,097,656)	(154,227,861)	(178,407,337)	(198,285,432)
(-) Soft Costs:	HK\$	(32,485,492)	(7,259,815)	(10,951,295)	(47,723,933)
(-) FF&E and Move-In Costs:	HK\$	(5,741,249)	(5,764,639)	=-AC75*FFE_Move_In_Pct	
Total Monthly Expenses:	HK\$	(166,324,396)	(167,252,316)	(204,713,307)	(261,426,597)
(-) Land Acquisition Costs:	HK\$	-	-	-	-
Gross Income:	HK\$	(108,911,910)	(109,605,924)	(51,166,562)	(107,254,283)

Hard Costs and Soft Costs are based on \$ per Gross Square Meter figures * Total Gross Square Meters * Expense Inflation Multiplier * % Incurred.

Once you've projected these expenses, you can move to the Debt and Equity Draws, Interest Expense, and Debt Principal Repayments.

In *most* real estate developments, a minimum amount of Equity must fund the project first before lenders step in to fund the Construction Loan.

The Debt Draws may begin once a certain percentage of units have been sold; in other cases, they begin once a certain amount of Equity has been drawn. We use the latter approach here:

Monthly Cash Flows:	Units:	Jan 15	Feb 15	Mar 15	Apr 15
Monthly Gross Sales:					
(+) Phase I - Initial Deposits:	HK\$	\$ -	\$ -	\$ -	\$ -
(+) Phase I - Construction-End Deposits:	HK\$	-	-	-	-
(+) Phase I - Final Deposits:	HK\$	-	-	-	-
Phase I Total Sales:	HK\$	-	-	-	-
(+) Phase II - Initial Deposits:	HK\$	-	-	-	-
(+) Phase II - Construction-End Deposits:	HK\$	-	-	-	-
(+) Phase II - Final Deposits:	HK\$	-	-	-	-
Phase II Total Sales:	HK\$	-	-	-	-
Total Monthly Gross Sales:	HK\$	-	-	-	-
Monthly Expenses:					
(-) Hard Costs:	HK\$	-	-	-	-
(-) Soft Costs:	HK\$	(32,064,563)	(6,449,172)	(3,242,818)	(3,261,154)
(-) FF&E and Move-In Costs:	HK\$	-	-	-	-
Total Monthly Expenses:	HK\$	(32,064,563)	(6,449,172)	(3,242,818)	(3,261,154)
(-) Land Acquisition Costs:	HK\$	(722,710,449)	-	-	-
Gross Income:	HK\$	(754,775,012)	(6,449,172)	(3,242,818)	(3,261,154)

We add up ALL the negative Gross Incomes over the entire project duration using SUMIF and flip the sign, and that becomes the "Total Operating Deficit." We plan to fund this with 50% Debt and 50% Equity, drawing on the Equity first.

Total Operating Deficit:	HK\$	\$ 3,067,823,692
% Equity for Total Operating Deficit:	%	50.0%
Maximum Equity Draw:	HK\$	1,533,911,846
Maximum Construction Loan Draw:	HK\$	1,533,911,846
Construction Loan Interest Rate:	%	10.00%

Then, you project the Debt and Equity Draws based on these assumptions:

Debt & Equity Schedules:	Units:	Jan 15	Feb 15	Mar 15	Apr 15	May 15	Jun 15	Jul 15	Aug 15	Sep 15
Monthly Interest Rate on Construction Loan:	%	0.80%	0.80%	0.80%	0.80%	0.80%	0.80%	0.80%	0.80%	0.80%
Interest on Construction Loan:	HK\$	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 313,883	\$ 423,917
Cash Interest on Construction Loan:	HK\$	-	-	-	-	-	-	-	-	-
Interest Accrued to Loan Principal:	HK\$	-	-	-	-	-	-	-	313,883	423,917
Equity & Debt Draws:										
(+) Equity:	HK\$	754,775,012	6,449,172	3,242,818	3,261,154	9,838,778	729,306,721	27,038,191	-	-
(+) Construction Loan:	HK\$	-	-	-	-	-	-	39,362,596	13,484,975	7,009,837
Total Equity & Debt Draws:	HK\$	754,775,012	6,449,172	3,242,818	3,261,154	9,838,778	729,306,721	66,400,787	13,484,975	7,009,837
Cash Generated After Equity & Debt Draws:	HK\$	-	-	-	-	-	-	-	-	-
Equity & Debt Balances:										
(+) Equity:	HK\$	754,775,012	761,224,184	764,467,002	767,728,156	777,566,934	1,506,873,655	1,533,911,846	1,533,911,846	1,533,911,846
(+) Construction Loan:	HK\$	-	-	-	-	-	-	39,362,596	53,161,454	60,595,208
Total Equity & Debt Balances:	HK\$	754,775,012	761,224,184	764,467,002	767,728,156	777,566,934	1,506,873,655	1,573,274,442	1,587,073,300	1,594,507,054
Construction Loan Repayments:	HK\$	-	-	-	-	-	-	-	-	-
Cash Flow After Draws & Debt Repayment:	HK\$	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -

We hit the Max Equity in Month #7 and then switch to Debt for all the funding. At that point, we also start paying Interest on this Debt.

However, since this property is under construction and has no cash flow yet, this Interest simply accrues to the Construction Loan principal instead of being paid in cash.

The Debt has an annual interest rate of 10%, which you can convert to a monthly rate with $(1 + 10\%)^{\frac{1}{12}} - 1$.

Here is the formula used for the Equity Draws in this model:

$=\text{MIN}(\text{Max_Equity_Draw}-\text{L105}, \text{MAX}(0, -\text{M85}))+\text{IF}(\text{M\$46}=\text{Post_Constr_End_Month}, \text{MAX}(\text{L\$106}+\text{M\$95}-\text{M\$87}, 0), 0)$

The $\text{MIN}(\text{Max_Equity_Draw}-\text{L105}, \text{MAX}(0, -\text{M85}))$ part compares the total remaining amount of Equity available and the Gross Income for the month with the sign flipped.

So, if Gross Income is (\$50,000) and the maximum Equity left to draw on is \$500,000, then the Equity Draw will be \$50,000 since \$50,000 is the minimum between \$50,000 and \$500,000.

But if Gross Income is \$50,000, nothing happens because the maximum between \$0 and (\$50,000) is \$0.

If Gross Income is (\$50,000) and the maximum Equity left to draw on is \$30,000, then the Equity Draw will be \$30,000 since \$30,000 is the minimum between \$30,000 and \$50,000.

The second part of this formula – $\text{IF}(\text{M\$46}=\text{Post_Constr_End_Month}, \text{MAX}(\text{L\$106}+\text{M\$95}-\text{M\$87}, 0), 0)$ – checks to see if there is any Debt remaining in the final month, and if so, draws on Equity to repay it.

The logic is that if the project is at an end, there cannot be any remaining Debt – the lenders must be repaid in full. If there is insufficient cash flow to do that, then the Equity Investors must pay for this.

L106 is the Construction Loan balance from the previous month, M95 is the Accrued Interest from the current month, and M87 is the Cash Flow After Cash Interest, Before Equity/Debt Draws.

So, this entire term of the formula – $L\$106 + M\$95 - M\$87$ – just represents *the Construction Loan balance as of the current month*.

In addition to the Equity and Debt Draws, there may also be Construction Loan Repayments while development is taking place.

These Construction Loan Repayments only happen when there is sufficient positive cash flow, so they only become a factor when the Total Deposits across both phases exceed the Cash Flow After Cash Interest, Before Equity/Debt Draws:

		AL	AM	AN	AO	AP	AQ
Debt & Equity Schedules:	Units:	Sep 17	Oct 17	Nov 17	Dec 17	Jan 18	Feb 18
Monthly Interest Rate on Construction Loan:	%	0.80%	0.80%	0.80%	0.80%	0.80%	0.80%
Interest on Construction Loan:	HK\$	\$ 11,603,327	\$ 12,070,269	\$ 12,158,574	\$ 12,072,389	\$ 11,870,714	\$ 11,552,224
Cash Interest on Construction Loan:	HK\$	-	996,342	12,158,574	12,072,389	11,870,714	11,552,224
Interest Accrued to Loan Principal:	HK\$	11,603,327	11,073,927	-	-	-	-
Equity & Debt Draws:							
(+) Equity:	HK\$	-	-	-	-	-	-
(+) Construction Loan:	HK\$	46,953,651	-	-	-	-	-
Total Equity & Debt Draws:	HK\$	46,953,651	-	-	-	-	-
Cash Generated After Equity & Debt Draws:	HK\$	-	-	10,808,052	25,291,054	39,940,475	43,235,828
Equity & Debt Balances:							
(+) Equity:	HK\$	1,533,911,846	1,533,911,846	1,533,911,846	1,533,911,846	1,533,911,846	1,533,911,846
(+) Construction Loan:	HK\$	1,513,676,475	1,524,750,401	1,513,942,349	1,488,651,295	1,448,710,819	1,405,474,991
Total Equity & Debt Balances:	HK\$	3,047,588,321	3,058,662,247	3,047,854,195	3,022,563,140	2,982,622,665	2,939,386,837
Construction Loan Repayments:	HK\$	-	-	=-MIN(AM106+AN95,AN102)	(39,940,475)	(43,235,828)	-
Cash Flow After Draws & Debt Repayment:	HK\$	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -

Repay the MIN between (Prior Month Construction Loan balance + Accrued Interest) and Cash Generated After Equity & Debt Draws and Cash Interest.

Cash Flow at the bottom equals Cash Generated After Equity & Debt Draws minus Construction Loan Repayments.

In most pre-sold condo development models, “Cash Flow After Draws & Debt Repayment” will be \$0 until the final deposits arrive upon construction completion or the end of the project.

In the earlier months, Gross Income is negative, creating the need to draw on Debt and Equity; in later months, as the deposits arrive, the positive cash flows from the deposits are used to repay the Construction Loan.

Here’s the model in the final 8-month period when all remaining deposits from all phases come due:



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Debt & Equity Schedules:	Units:	Nov 18	Dec 18	Jan 19	Feb 19	Mar 19	Apr 19	May 19	Jun 19
Monthly Interest Rate on Construction Loan:	%	0.80%	0.80%	0.80%	0.80%	0.80%	0.80%	0.80%	0.80%
Interest on Construction Loan:	HK\$	\$ 6,744,612	\$ 6,865,324	\$ -	\$ 502,731	\$ 574,150	\$ 612,623	\$ 787,944	\$ 931,347
Cash Interest on Construction Loan:	HK\$	-	6,865,324	-	-	-	-	-	931,347
Interest Accrued to Loan Principal:	HK\$	6,744,612	-	-	502,731	574,150	612,623	787,944	-
Equity & Debt Draws:									
(+) Equity:	HK\$	-	-	-	-	-	-	-	-
(+) Construction Loan:	HK\$	8,393,336	-	63,045,152	8,453,549	4,250,673	21,373,536	17,195,508	-
Total Equity & Debt Draws:	HK\$	8,393,336	-	63,045,152	8,453,549	4,250,673	21,373,536	17,195,508	-
Cash Generated After Equity & Debt Draws:	HK\$	-	2,604,400,402	-	-	-	-	-	3,474,656,739
Equity & Debt Balances:									
(+) Equity:	HK\$	1,533,911,846	1,533,911,846	1,533,911,846	1,533,911,846	1,533,911,846	1,533,911,846	1,533,911,846	1,533,911,846
(+) Construction Loan:	HK\$	860,948,449	-	63,045,152	72,001,432	76,826,254	98,812,414	116,795,867	-
Total Equity & Debt Balances:	HK\$	2,394,860,295	1,533,911,846	1,596,956,998	1,605,913,278	1,610,738,100	1,632,724,260	1,650,707,712	1,533,911,846
Construction Loan Repayments:	HK\$	-	(860,948,449)	-	-	-	-	-	(116,795,867)
Cash Flow After Draws & Debt Repayment:	HK\$	\$ -	\$ 1,743,451,953	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 3,357,860,873

The cash flows aren't always this lopsided; in some cases, the final deposits come due in different months or are distributed over several months. However, these types of developments *generally* generate zero-to-minimal positive cash flow until the final stages.

At this point, you can calculate the **unleveraged and leveraged IRR and multiples** to get a rough sense of whether or not the deal works:

Debt & Equity Schedules:	Units:	Jan 15	Feb 15	Mar 15	Apr 15	May 15	Jun 15
Monthly Interest Rate on Construction Loan:	%	0.80%	0.80%	0.80%	0.80%	0.80%	0.80%
Interest on Construction Loan:	HK\$	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Cash Interest on Construction Loan:	HK\$	-	-	-	-	-	-
Interest Accrued to Loan Principal:	HK\$	-	-	-	-	-	-
Equity & Debt Draws:							
(+) Equity:	HK\$	754,775,012	6,449,172	3,242,818	3,261,154	9,838,778	729,306,721
(+) Construction Loan:	HK\$	-	-	-	-	-	-
Total Equity & Debt Draws:	HK\$	754,775,012	6,449,172	3,242,818	3,261,154	9,838,778	729,306,721
Cash Generated After Equity & Debt Draws:	HK\$	-	-	-	-	-	-
Equity & Debt Balances:							
(+) Equity:	HK\$	754,775,012	761,224,184	764,467,002	767,728,156	777,566,934	1,506,873,655
(+) Construction Loan:	HK\$	-	-	-	-	-	-
Total Equity & Debt Balances:	HK\$	754,775,012	761,224,184	764,467,002	767,728,156	777,566,934	1,506,873,655
Construction Loan Repayments:	HK\$	-	-	-	-	-	-
Cash Flow After Draws & Debt Repayment:	HK\$	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Waterfall Returns Schedule:	Units:	Jan 15	Feb 15	Mar 15	Apr 15	May 15	Jun 15
Cash Flow to Equity Investors:							
(-) Invested Equity:	HK\$	\$ (754,775,012)	\$ (6,449,172)	\$ (3,242,818)	\$ (3,261,154)	\$ (9,838,778)	\$ (729,306,721)
(+) Net Cash Flow After Debt Service:	HK\$	-	-	-	-	-	-
Total Cash Flow to Equity Investors:	HK\$	(754,775,012)	(6,449,172)	(3,242,818)	(3,261,154)	(9,838,778)	(729,306,721)

The unleveraged numbers are easy because they are based on the Gross Income each month, which is before Interest, Debt/Equity Draws, and Debt Repayments.

For the leveraged numbers, you need to show the **Invested Equity** (i.e., all the Equity Draws) as negatives, show the **Net Cash Flow After Debt Service** as positive figures, and sum those to determine the Total Cash Flow to Equity Investors, as in the screenshot above.

Once you have this set up, you can use the XIRR function to calculate the IRR for each type of cash flow and sum up all the positive cash flows and divide them by all the negative cash flows to calculate the multiples:

Formula Bar: `=XIRR(F160:BG160,$F$152:$BG$152)`

	Units:	Jan 15	Feb 15	Mar 15	Apr 15	May 15	Jun 15	Jul 15	Aug 15	Sep 15
Construction Timeline:										
Returns Summary:										
Unleveraged Cash Flows:										
Internal Rate of Return (IRR):	%	30.1%								
Positive Cash Flows:	HK\$	6,889,613,381								
Invested Amounts:	HK\$	3,067,823,692								
Cash-on-Cash Multiple:	x	2.25 x								
Total Cash Flows to Equity Investors:										
Internal Rate of Return (IRR):	%	34.7%								
Positive Cash Flows:	HK\$	5,101,312,826								
Invested Equity:	HK\$	1,533,911,846								
Cash-on-Cash Multiple:	x	3.33 x								

Annual Cash Flows (F160:BG160):

	Jan 15	Feb 15	Mar 15	Apr 15	May 15	Jun 15	Jul 15	Aug 15	Sep 15
Gross Income	(754,775,012)	(6,449,172)	(3,242,818)	(3,261,154)	(9,838,778)	(729,306,721)	(66,400,787)	(13,484,975)	(7,009,837)
Total Cash Flow to Equity Investors	(754,775,012)	(6,449,172)	(3,242,818)	(3,261,154)	(9,838,778)	(729,306,721)	(27,038,191)	-	-

Annotations:

- Red arrow from Formula Bar to IRR cell (34.7%): **Gross Income each year since it is capital structure-neutral.**
- Red arrow from Total Cash Flow to Equity Investors row to IRR cell: **Total Cash Flow to Equity Investors, calculated above.**

If it's a simple model, you might stop at this point; you have the IRR and multiples, and you can look at them under different assumptions to judge the merits of the deal.

One additional step, however, is to split up the returns differently to Investors vs. Developers in a **waterfall returns schedule**.

We cover this concept in more depth in the section on [Waterfall Returns Schedules](#), but here's the basic idea:

In most real estate deals, there's an "Investor," who contributes most of the Equity, and a "Developer," who contributes a smaller portion of the Equity and does the actual construction work.

Rather than distributing the deal proceeds proportionally to the Investor and Developer, resulting in a 35% IRR for each one (as in the example above), you could **incentivize** the Developer to perform better by granting it additional proceeds if the deal goes well.

For example, here's the structure for this deal:

IRR and Equity Multiple Hurdles:		Low:	High:	
IRR / Multiple Between:	%		20.0%	The Investor contributes 80% of the Equity, and the Developer contributes 20%; in this first tier, up to a 20% IRR, they receive proceeds in that 80 / 20 proportion.
Jade Sea Partners (Investors):	%		80.0%	
Nan Fung Group (Developers):	%		20.0%	
IRR / Multiple Between:	%	20.0%	3.00 x	But... if the IRR is above 20%, the split shifts to 70 / 30 for the portion of returns between a 20% IRR and a 3x multiple.
Jade Sea Partners (Investors):	%		70.0%	
Nan Fung Group (Developers):	%		30.0%	
IRR / Multiple Between:	%	3.00 x		And if the multiple is over 3x, the portion of returns over 3x get distributed in a 60 / 40 split.
Jade Sea Partners (Investors):	%		60.0%	
Nan Fung Group (Developers):	%		40.0%	

You can then use this information to build Tier 1 through Tier 3 of this “waterfall schedule.”

Here are the beginning few months of Tier 1:

Waterfall Returns Schedule:		Units:	Jan 15	Feb 15	Mar 15	Apr 15	May 15
Cash Flow to Equity Investors:							
(-) Invested Equity:	HK\$		\$ (754,775,012)	\$ (6,449,172)	\$ (3,242,818)	\$ (3,261,154)	\$ (9,838,778)
(+) Net Cash Flow After Debt Service:	HK\$		-	-	-	-	-
Total Cash Flow to Equity Investors:	HK\$		(754,775,012)	(6,449,172)	(3,242,818)	(3,261,154)	(9,838,778)
Tier 1 - Up to 20.0% IRR:							
(+) Beginning Balance:	HK\$		-	754,775,012	772,779,390	787,853,051	803,175,818
(+) Investor Injections:	HK\$		754,775,012	6,449,172	3,242,818	3,261,154	9,838,778
(+) Investor Accruals:	HK\$	20.0%	-	=G121*((1+\$E123)^(1/Months)-1)		12,061,613	12,296,196
(-) Tier 1 Accrual Distribution:	HK\$		-	-	-	-	-
Ending Balance:	HK\$		754,775,012	772,779,390	787,853,051	803,175,818	825,310,793
Investor Cash Flow:	HK\$	80.0%	(603,820,009)	(5,159,338)	(2,594,255)	(2,608,923)	(7,871,023)
Developer Cash Flow:	HK\$	20.0%	(150,955,002)	(1,289,834)	(648,564)	(652,231)	(1,967,756)
Cash Flow Available for Tier 2 Distribution:	HK\$		-	-	-	-	-

Represents what the equity investors are "owed" each month to reach a 20% IRR; it's a monthly model, so we convert the 20% IRR into a monthly rate.

Distributing the cash flows according to this 80 / 20 split in this tier.

The “Investor Injections” line item represents the total amount of Equity that both the Investors and the Developers contributed in the month.

The “Investor Accruals” line item represents the amount they are owed on a *monthly basis* based on their cumulative contributions – to reach a 20% IRR.

The “Tier 1 Accrual Distribution” line is blank in these beginning months, but once there are finally positive cash flows near the end, it will represent the distributions up to this 20% IRR level.

Here's this same Tier 1 in the final few months of this project:

Waterfall Returns Schedule:	Units:	Nov 18	Dec 18	Jan 19	Feb 19	Mar 19	Apr 19	May 19	Jun 19
Cash Flow to Equity Investors:									
(-) Invested Equity:	HK\$	\$	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
(+) Net Cash Flow After Debt Service:	HK\$	-	1,743,451,953	-	-	-	-	-	3,357,860,873
Total Cash Flow to Equity Investors:	HK\$	-	1,743,451,953	-	-	-	-	-	3,357,860,873
Tier 1 - Up to 20.0% IRR:									
(+) Beginning Balance:	HK\$	2,926,806,065	2,971,613,917	1,273,655,799	1,293,154,795	1,312,952,310	1,333,052,915	1,353,461,249	1,374,182,024
(+) Investor Injections:	HK\$	-	-	-	-	-	-	-	-
(+) Investor Accruals:	HK\$ 20.0%	44,807,851	45,493,836	19,498,996	19,797,515	20,100,605	20,408,334	20,720,772	21,037,999
(-) Tier 1 Accrual Distribution:	HK\$	-	(1,743,451,953)	-	-	-	-	-	(1,395,220,023)
Ending Balance:	HK\$	2,971,613,917	1,273,655,799	1,293,154,795	1,312,952,310	1,333,052,915	1,353,461,249	1,374,182,024	-
Investor Cash Flow:	HK\$ 80.0%	-	1,394,761,562	-	-	-	-	-	1,116,176,019
Developer Cash Flow:	HK\$ 20.0%	-	348,690,391	-	-	-	-	-	279,044,005
Cash Flow Available for Tier 2 Distribution:	HK\$	-	-	-	-	-	-	-	1,962,640,849

This \$1.7 billion positive cash flow still represents *less than* a 20% IRR on the cumulative Equity contributions - so we simply distribute it 80 / 20, and nothing is left for Tier 2 since we're not yet at a 20% IRR.

But this \$3.4 billion now gives us enough to go *beyond* that 20% IRR. Therefore, we distribute the \$1.4 billion that corresponds to a 20% IRR and leave the rest for the next tiers.

The Tier 1 Accrual Distribution formula looks tricky, but the concept is not difficult. Here it is in the final month of this model, Jun 19:

$$=\text{MIN}(0, \text{MAX}(-(\text{BG121}+\text{BG122}+\text{BG123}), -\text{BG118}))$$

The $-(\text{BG121}+\text{BG122}+\text{BG123})$ sums up the Beginning Balance, Investor Injections, and Investor Accruals to get the amount that Equity Investors are owed for this 20% IRR as of this month – and then it flips the sign to a negative.

The $-\text{BG118}$ part represents the Total Cash Flow to Equity Investors in this month, with the sign flipped to a negative.

We then take the MAX between these two negatives, so the function returns the *least negative* number, which, in this case, is the Beginning Balance + Investor Injections + Investor Accruals one.

We then distribute that amount in an 80 / 20 split to the Investors and Developers in this tier.

The remaining cash flow – the \$3.4 billion minus this \$1.4 billion – is then made available for distribution in Tiers 2 and 3.

Tier 2 is similar, but it is based on a **multiple** rather than an IRR target:

Waterfall Returns Schedule:	Units:	Jan 15	Feb 15	Mar 15	Apr 15
Cash Flow Available for Tier 2 Distribution:	HK\$	-	-	-	-
Tier 2 - 20.0% IRR to 3.0x Multiple:					
(+) Beginning Balance:	HK\$	-	2,264,325,035	2,283,672,551	2,293,401,006
(+) Investor Injections:	HK\$	754,775,012	6,449,172	3,242,818	3,261,154
(+) Investor Accruals:	HK\$	3.00 x	1,509,550,023	=+G134*(\$E135-1)	6,522,307
(-) Tier 1 Accrual Distribution:	HK\$	-	-	-	-
(-) Tier 2 Accrual Distribution:	HK\$	-	-	-	-
Ending Balance:	HK\$	2,264,325,035	2,283,672,551	2,293,401,006	2,303,184,468
Investor Cash Flow:	HK\$	70.0%	-	-	-
Developer Cash Flow:	HK\$	30.0%	-	-	-
Cash Flow Available for Tier 3 Distribution:	HK\$	-	-	-	-
Tier 3 - Above 3.0x Multiple:					
Investor Cash Flow:	HK\$	60.0%	-	-	-
Developer Cash Flow:	HK\$	40.0%	-	-	-

The Investor Accruals here are different because they're based on (Multiple - 1) * Investor Injections.

If the Equity Investors contribute \$5 million, to earn 3x on that, they must earn back the \$5 million *and* get an additional \$10 million.

Tier 3 is a straightforward 60 / 40 split based on the Cash Flow Available for Tier 3 Distribution - nothing tricky since there are no bounds here.

The Tier 1 Accrual Distribution here is linked to the same line item in the Tier 1 schedule.

The Tier 2 Accrual Distribution here is based on a similar formula to the Tier 1 Accrual Distribution one: distribute *the lesser* of all the cash flow available or the total amount that Equity Investors are owed in this tier.

It's just that now, we must also factor the Tier 1 Accrual Distribution into "the total amount that Equity Investors are owed in this tier."

Tier 3 is straightforward since it is a simple percentage split based on the remaining cash flows available for distribution *after* Tiers 1 and 2:

Waterfall Returns Schedule:	Units:	Oct 18	Nov 18	Dec 18	Jan 19	Feb 19	Mar 19	Apr 19	May 19	Jun 19
Cash Flow Available for Tier 2 Distribution:	HK\$	-	-	-	-	-	-	-	-	1,962,640,849
Tier 2 - 20.0% IRR to 3.0x Multiple:										
(+) Beginning Balance:	HK\$	4,601,735,538	4,601,735,538	4,601,735,538	2,858,283,585	2,858,283,585	2,858,283,585	2,858,283,585	2,858,283,585	2,858,283,585
(+) Investor Injections:	HK\$	-	-	-	-	-	-	-	-	-
(+) Investor Accruals:	HK\$	3.00 x	-	-	-	-	-	-	-	-
(-) Tier 1 Accrual Distribution:	HK\$	-	-	(1,743,451,953)	-	-	-	-	-	(1,395,220,023)
(-) Tier 2 Accrual Distribution:	HK\$	-	-	-	-	-	-	-	-	(1,463,063,562)
Ending Balance:	HK\$	4,601,735,538	4,601,735,538	2,858,283,585	2,858,283,585	2,858,283,585	2,858,283,585	2,858,283,585	2,858,283,585	-
Investor Cash Flow:	HK\$	70.0%	-	-	-	-	-	-	-	1,024,144,493
Developer Cash Flow:	HK\$	30.0%	-	-	-	-	-	-	-	438,919,068
Cash Flow Available for Tier 3 Distribution:	HK\$	-	-	-	-	-	-	-	-	499,577,288
Tier 3 - Above 3.0x Multiple:										
Investor Cash Flow:	HK\$	60.0%	-	-	-	-	-	-	-	299,746,373
Developer Cash Flow:	HK\$	40.0%	-	-	-	-	-	-	-	199,830,915

Then, you re-assemble all the cash flows to calculate the IRR and multiple to the Investors and Developers:



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Returns Summary:	Units:		Oct 18	Nov 18	Dec 18	Jan 19	Feb 19	Mar 19	Apr 19	May 19	Jun 19
Unleveraged Cash Flows:			85,713,343	(8,393,336)	2,611,265,726	(63,045,152)	(8,453,549)	(4,250,673)	(21,373,536)	(17,195,508)	3,475,588,086
Internal Rate of Return (IRR):	%	30.1%									
Positive Cash Flows:	HK\$	6,889,613,381									
Invested Amounts:	HK\$	3,067,823,692									
Cash-on-Cash Multiple:	x	2.25 x									
Total Cash Flows to Equity Investors:			-	-	1,743,451,953	-	-	-	-	-	3,357,860,873
Internal Rate of Return (IRR):	%	34.7%									
Positive Cash Flows:	HK\$	5,101,312,826									
Invested Equity:	HK\$	1,533,911,846									
Cash-on-Cash Multiple:	x	3.33 x									
Investor Cash Flow:			-	-	1,394,761,562	-	-	-	-	-	2,440,066,884
Internal Rate of Return (IRR):	%	32.7%									
Positive Cash Flows:	HK\$	3,834,828,447									
Invested Equity:	HK\$	1,227,129,477									
Cash-on-Cash Multiple:	x	3.13 x									
Developer Cash Flow:			-	-	348,690,391	-	-	-	-	-	917,793,988
Internal Rate of Return (IRR):	%	41.7%									
Positive Cash Flows:	HK\$	1,266,484,379									
Invested Equity:	HK\$	306,782,369									
Cash-on-Cash Multiple:	x	4.13 x									

The Developers earn a significantly higher IRR and multiple because of this waterfall returns schedule and incentive fees.

It's tough to say whether or not this deal is a good one because we don't have a great sense of the proper assumptions and the different possible outcomes.

For example, if the condo selling price assumptions here are 50% above market rates, then these IRRs are inflated, and the deal might be a disaster in real life.

But if the prices and expenses are reasonable and the timing is plausible, then this deal looks promising because a 30-40% IRR is well above the typical target of 20-25% in many development deals.

In real life, you would look at sensitivity tables and assess the deal's performance in different scenarios and cases to "stress test" it and see what happens if prices fall, or expenses rise faster than expected, or construction takes longer than expected (for example).

Strengths and Weaknesses of Pre-Sold Condo Developments

Like all real estate developments, pre-sold condo developments offer higher risk but also higher potential returns than acquisitions of stabilized properties ("core deals") or acquisition/renovation projects ("value-added deals").

If the market is on fire and residential housing prices skyrocket, you could earn IRRs that are impossible with a stabilized multifamily acquisition.

On the other hand, real estate developments almost always get delayed or go over-budget because it's the nature of the industry, and those both present major risk factors.

If things go *really* poorly, the Equity Investors could easily lose money with this type of project.



Pre-selling units does **reduce the risk for investors** because it lets them test the market in small increments rather than attempting to sell everything at the end.

If it's a disaster, the investors can cut their losses early and cancel or re-tool the project; if it goes well, they can expand and sell more units.

That business model offers advantages over developing an entire office complex, shopping mall, or multifamily property and then attempting to sell it after several years.

The biggest weakness of pre-sold condo developments is that they are heavily linked to the residential housing market since they *are*, effectively, the residential housing market – and that market fluctuates wildly based on economic conditions.

Regardless of the economy, individuals always need *somewhere* to live, so multifamily properties tend to be resilient.

And while office, industrial, and retail properties are more sensitive to the economy, they also have longer-term leases, so the risk of losing many tenants in a single year isn't high with a properly staggered rent roll.

By contrast, when people lose their jobs or the housing market crashes, they *stop* buying homes and condo units (just look at housing prices and unit sales in any previous recession).

In short, pre-sold condo developments offer risk and potential returns that are in-line with those of other property developments, with one factor that reduces risk (pre-selling the units) and another factor that increases the risk (the link to the residential housing market).

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Multifamily and Student Housing

You can think of multifamily properties as **simplified versions** of office, industrial, and retail properties with a few major differences:

- 1) The **lease terms** in a single property tend to be **nearly identical** except for the amount of monthly rent, which varies based on the unit size.
- 2) The leases are almost always for **one (1) year**.

- 3) **Turnover** tends to be very high because so many tenants move in and move out each year; it's often above 50% for larger properties.
- 4) **Tenant Improvements** and **Leasing Commissions** may still factor in, especially in downturns when it's tougher to find tenants, but they tend to be less significant.
- 5) The **line items** on the Pro-Forma are similar to those on an office, industrial, or retail Pro-Forma, but there are a few new items, such as "Bad Debt" (i.e., tenants simply not paying rent), Parking Income, Laundry Income, and Utility Reimbursements.

Everything above applies to student housing as well, so we group together multifamily and student housing in this guide. The main difference is that student housing is often **counter-cyclical** since university enrollments tend to rise when there's a recession.

Here's a relatively simple Pro-Forma for a multifamily property in Arizona:

Property Pro-Forma:	Historical:		Projected:					Stabilized
	FY17	FY18	FY19	FY20	FY21	FY22	Year:	
Revenue:								
(+) Base Rental Income:	\$ 929,024	\$ 961,540	\$ 995,194	\$ 1,030,026	\$ 1,060,926	\$ 1,092,754	\$ 1,125,537	
(-) Loss to Lease:	(92,902)	(72,115)	(49,760)	(25,751)	(10,609)	(10,928)	(11,255)	
(-) Bad Debt & Concessions:	(25,084)	(26,683)	(18,909)	(20,085)	(10,503)	(10,818)	(11,143)	
(+) Parking Income:	51,528	53,331	55,198	57,130	58,844	60,609	62,428	
(+) Utility Reimbursements:	62,016	66,272	70,727	75,390	80,269	85,374	87,935	
Potential Gross Revenue:	924,582	982,345	1,052,451	1,116,710	1,178,927	1,216,991	1,253,501	
(-) General Vacancy:	-	(9,823)	(21,049)	(33,501)	(35,368)	(36,510)	(37,605)	
Effective Gross Income (EGI):	924,582	972,521	1,031,402	1,083,208	1,143,559	1,180,481	1,215,896	
EGI Growth Rate:	N/A	5.2%	6.1%	5.0%	5.6%	3.2%	3.0%	
Expenses:								
(-) Property Taxes:	(66,470)	(67,799)	(69,155)	(70,538)	(71,949)	(73,388)	(74,856)	
(-) Insurance:	(21,774)	(22,427)	(23,100)	(23,793)	(24,507)	(25,242)	(25,999)	
(-) Utilities:	(77,520)	(79,846)	(82,241)	(84,708)	(87,249)	(89,867)	(92,563)	
(-) Property Management Fee:	(27,737)	(29,176)	(30,942)	(32,496)	(34,307)	(35,414)	(36,477)	
(-) Sales, Marketing & Administrative:	(175,671)	(186,724)	(200,092)	(212,309)	(226,425)	(236,096)	(243,179)	
(-) Replacement Reserves:	(19,000)	(19,570)	(20,157)	(20,762)	(21,385)	(22,026)	(22,687)	
Total Expenses:	(388,172)	(405,542)	(425,687)	(444,607)	(465,822)	(482,034)	(495,761)	
Net Operating Income (NOI):	536,410	566,980	605,714	638,602	677,738	698,447	720,135	
NOI Margin:	58.0%	58.3%	58.7%	59.0%	59.3%	59.2%	59.2%	
(-) Capital Expenditures:	-	(38,000)	(76,000)	(57,000)	-	(15,200)	-	
(+) Capital Costs Paid from Reserves:	-	38,000	39,727	20,762	-	15,200	-	
Adjusted Net Operating Income:	536,410	566,980	569,441	602,364	677,738	698,447	720,135	
Adjusted NOI Margin:	58.0%	58.3%	55.2%	55.6%	59.3%	59.2%	59.2%	

"Concessions" still refer to Free Months of Rent, which exist but are rarer for multifamily properties. "Bad Debt" means "tenants who do not pay or do not pay on time."

Many properties charge for parking and require tenants to pay a portion of their utilities.

These categories are similar to those of office, industrial, and retail properties, but the percentages are a bit different; there's also more of a need for sales & marketing to keep attracting new tenants.

CapEx is standard. It's unusual not to list anything for TIs or LCs here - they tend to be smaller for multifamily properties because each tenant is smaller and has less negotiating power, and brokers are often optional, but most properties have at least *something* in these categories.

Here are the key drivers for this Pro-Forma, with a few explanations on the right-hand side:



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Operating Assumptions:	Units:	Historical:	Projected:					Stabilized
		FY17	FY18	FY19	FY20	FY21	FY22	Year:
Market Rent per RSF per Month:	\$ / RSF / Mo.	\$ 1.78	\$ 1.84	\$ 1.90	\$ 1.97	\$ 2.03	\$ 2.09	\$ 2.15
In-Place Rent per RSF per Month:	\$ / RSF / Mo.	1.60	1.70	1.81	1.92	2.01	2.07	2.13
Rental & Parking Income Growth Rate:	%		3.5%	3.5%	3.5%	3.0%	3.0%	3.0%
In-Place Rent Discount to Market Rent:	%	10.0%	7.5%	5.0%	2.5%	1.0%	1.0%	1.0%
Bad Debt & Concessions % Effective Rent:	%	(3.0%)	(3.0%)	(2.0%)	(2.0%)	(1.0%)	(1.0%)	(1.0%)
Parking Fees per Spot per Month:	\$ / Spot / Mo.	50.00	51.75	53.56	55.44	57.10	58.81	60.58
Utility Reimbursements % Utility Expense:	%	80.0%	83.0%	86.0%	89.0%	92.0%	95.0%	95.0%
General Vacancy:	%	-	(1.0%)	(2.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)
Insurance per Gross Square Foot per Year:	\$ / GSF / Yr.	0.40	0.41	0.42	0.44	0.45	0.46	0.48
Utilities per Unit per Month:	\$ / Unit / Mo.	85.00	87.55	90.18	92.88	95.67	98.54	101.49
Sales, Marketing & Administrative % EGI:	%	19.0%	19.2%	19.4%	19.6%	19.8%	20.0%	20.0%
Replacement Reserves per Unit per Year:	\$ / Unit / Yr.	250.00	257.50	265.23	273.18	281.38	289.82	298.51
Property Tax Growth Rate:	%		2.0%	2.0%	2.0%	2.0%	2.0%	2.0%
Operating Expense Growth Rate:	%		3.0%	3.0%	3.0%	3.0%	3.0%	3.0%
Capital Expenditures per Unit per Year:	\$ / Unit / Yr.	-	500.00	1,000.00	750.00	-	200.00	-
Replacement Reserves:	\$	-	38,000	19,570	-	21,385	28,211	50,898

Both Rent and Parking Fees grow at low single-digit percentages, in-line with inflation, and In-Place Rents move closer to Market Rents over time due to minor renovations.

Multifamily tenants are never responsible for Property Taxes, Insurance, or Maintenance - just some of the Utilities. General Vacancy increases as a result of the In-Place Rent increases.

Notice how some of these are tied to the property size (Gross Square Feet), while others are linked to individual Units or EGI.

Reserves work the same way as always; CapEx spending is highly irregular, making the Reserves quite useful here.

Here are the assumptions used in this model:

Arcadia Gardens - Investment Analysis Model

(\$ USD in \$ as Stated, Unless Otherwise Noted)

Property Assumptions:		Units:		Units:
Building / Construction Name:	Name	Arcadia Gardens	Property Taxes % Property Value:	% 0.68%
Location:	Name	Phoenix, AZ	Property Management Fee % EGI:	% 3.00%
Months in Year:	#	12	Rentable Square Footage (RSF):	sq. ft. 43,548 sq. ft.
Number of Apartment Units:	#	76	Rentable to Gross Square Feet Ratio:	% 80.0%
Average Rentable Square Feet per Unit:	sq. ft.	573 sq. ft.	Gross Square Footage (GSF):	sq. ft. 54,435 sq. ft.
Parking Spots per Unit:	#	1.13		
Parking Spots:	#	86		

If the property has only 5-10 tenants, then you might go into more granular detail and set up something [similar to the office/retail/industrial models in the previous section](#).

But even if the property is “relatively small” for a multifamily – say, 20-30 tenants – it’s rarely worth the effort to do that, given the similarities of the lease terms.

Quarterly or monthly models for multifamily properties tend not to be that useful for similar reasons: with so many tenants on annual leases, “lumpy” lease expirations are not that common.

Therefore, the *most useful* way to make a multifamily model more complex is to add multiple scenarios that represent different market outcomes; you could also calculate line items like TIIs, LCs, and CapEx in more detail and link them to the Turnover Percentage.



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Scenarios are arguably more important in multifamily models than in office, industrial, or retail models because **the leases are much shorter term**.

If all 500 tenants of a large property are on annual leases and the real estate market crashes next year, it's very likely that all 500 tenants' rents will fall.

But if that happens with an office complex, rent might not even change if the average lease is 6-7 years, there are pre-defined annual escalations, and there are no lease expirations in the next year.

Here's a more complex multifamily Pro-Forma for a property in Seattle:

Property Pro-Forma:	Units:	Historical:	Projected:					Stabilized
		FY18	FY19	FY20	FY21	FY22	FY23	Year:
Revenue:								
(+) Base Rental Income:	\$	\$ 7,020,000	\$ 7,371,000	\$ 7,739,550	\$ 8,087,830	\$ 8,451,782	\$ 8,789,853	\$ 9,097,498
(-) Loss to Lease:	\$	(561,600)	(552,825)	(386,978)	(202,196)	(84,518)	(87,899)	(90,975)
(-) Bad Debt & Concessions:	\$	(193,752)	(204,545)	(220,577)	(236,569)	(251,018)	(261,059)	(270,196)
(+) Parking Income:	\$	648,648	681,080	715,134	747,315	780,945	812,182	840,609
(+) Utility Reimbursements:	\$	310,284	326,492	343,500	359,609	376,424	392,074	401,875
Potential Gross Revenue:	\$	7,223,580	7,621,202	8,190,629	8,755,989	9,273,615	9,645,152	9,978,812
(-) General Vacancy:	\$	(216,707)	(228,636)	(245,719)	(262,680)	(278,208)	(289,355)	(299,364)
Effective Gross Income (EGI):	\$	7,006,873	7,392,566	7,944,911	8,493,309	8,995,407	9,355,798	9,679,447
EGI Growth Rate:	%	N/A	5.5%	7.5%	6.9%	5.9%	4.0%	3.5%
Expenses:								
(-) Property Taxes:	\$	(1,068,128)	(1,110,853)	(1,155,287)	(1,195,722)	(1,237,572)	(1,274,700)	(1,306,567)
(-) Insurance:	\$	(128,023)	(133,144)	(138,469)	(143,316)	(148,332)	(152,782)	(156,601)
(-) Utilities:	\$	(365,040)	(379,642)	(394,827)	(408,646)	(422,949)	(435,637)	(446,528)
(-) Property Management Fee:	\$	(210,206)	(221,777)	(238,347)	(254,799)	(269,862)	(280,674)	(290,383)
(-) Sales, Marketing & Administrative:	\$	(700,687)	(746,649)	(810,381)	(874,811)	(935,522)	(982,359)	(1,016,342)
(-) Replacement Reserves:	\$	(93,600)	(97,344)	(101,238)	(104,781)	(108,448)	(111,702)	(114,494)
Total Expenses:	\$	(2,565,684)	(2,689,408)	(2,838,550)	(2,982,076)	(3,122,686)	(3,237,853)	(3,330,917)
Net Operating Income (NOI):	\$	4,441,188	4,703,157	5,106,361	5,511,234	5,872,721	6,117,944	6,348,531
NOI Margin:	%	63.4%	63.6%	64.3%	64.9%	65.3%	65.4%	65.6%
(-) Tenant Improvements (TIs):	\$	(18,720)	(23,363)	(28,347)	(33,530)	(39,041)	(44,681)	(45,798)
(-) Leasing Commissions (LCs):	\$	(18,794)	(23,809)	(29,954)	(36,716)	(43,828)	(50,645)	(52,418)
(-) Capital Expenditures:	\$	-	(234,000)	(351,000)	(175,500)	-	(46,800)	-
(+) Capital Costs Paid from Reserves:	\$	37,514	281,172	409,301	168,976	82,869	137,281	98,216
Adjusted Net Operating Income:	\$	4,441,188	4,703,157	5,106,361	5,434,465	5,872,721	6,113,099	6,348,531
Adjusted NOI Margin:	%	63.4%	63.6%	64.3%	64.0%	65.3%	65.3%	65.6%

The same general idea - Loss to Lease decreases over time, boosting EGI, but here, the General Vacancy percentage doesn't change by much.

CapEx is on a per-unit-per-year basis and doesn't follow a clear trend, but spending is highest in the first two years when a minor renovation takes place.

TIs are linked to "Turnover" (the % of units with new tenants each year) and the \$ per unit required.

LCs are a percentage of the Effective Rent * Turnover % each year.

We model the impact of scenarios such as Base and Downside Cases via the assumptions and drivers.

In the Base Case, everything continues "as is," with market rents increasing in the low-single-digit percentages, expense growth slightly below that, and stable percentages for the General Vacancy and Bad Debt & Concessions.



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In the Downside Case, rents initially decline due to a recession, expenses also fall, and the General Vacancy and Bad Debt & Concession percentages both become more negative (i.e., fewer tenants occupy the property and pay their rent on time, and more concessions are needed).

Tenant Improvements and Leasing Commissions also increase because in a poor market, fewer tenants want to move in, so owners need to pay more to secure tenants.

After the recession in these first few years, there's a recovery period in which the percentages and growth rates move back to their normal levels.

We also build an Extreme Downside Case into this model because it's for a credit analysis; this case is similar to the Downside Case, but with even more pessimistic numbers.

If we had an Upside Case here, it would have been the opposite of the Downside Case, with elevated rent and expense growth in the first few years, lower General Vacancy and Concession percentages, and an eventual return to the norm by the end of the 5-year period.

Here's a portion of the assumptions in this Pro-Forma:

Operating Assumptions:	Units:	Historical:	Projected:					Stabilized
		FY18	FY19	FY20	FY21	FY22	FY23	Year:
Property Management Fee % EGI:	%	3.0%						
Market Rent per Unit per Month:	\$ / RSF / Mo.	\$ 2,500	\$ 2,625	\$ 2,756	\$ 2,880	\$ 3,010	\$ 3,130	\$ 3,240
In-Place Rent per Unit per Month:	\$ / RSF / Mo.	2,300	2,428	2,618	2,808	2,980	3,099	3,207
In-Place Rent Discount to Market Rent:	%	8.0%	7.5%	5.0%	2.5%	1.0%	1.0%	1.0%
Parking Fees per Spot per Month:	\$ / Spot / Mo.	150.00	157.50	165.38	172.82	180.59	187.82	194.39
Utility Reimbursements % Utility Expense:	%	85.0%	86.0%	87.0%	88.0%	89.0%	90.0%	90.0%
Income Growth Rate:								
Base	%	5.0%	5.0%	5.0%	4.5%	4.5%	4.0%	3.5%
Downside	%	5.0%	5.0%	(3.0%)	(1.0%)	6.0%	4.5%	3.5%
Extreme Downside	%	5.0%	5.0%	(6.0%)	(3.0%)	5.0%	4.0%	3.0%
Selected Income Growth Rate:	%	5.0%	5.0%	5.0%	4.5%	4.5%	4.0%	3.5%
General Vacancy:								
Base	%	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)
Downside	%	(3.0%)	(3.0%)	(6.0%)	(4.0%)	(3.5%)	(3.0%)	(3.0%)
Extreme Downside	%	(3.0%)	(3.0%)	(8.0%)	(6.0%)	(4.0%)	(3.0%)	(3.0%)
Selected General Vacancy:	%	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)

This percentage drives both Market Rent and Parking Fees.

In recessions, the Vacancy Rate rises at least a bit because unemployment increases and tenants often relocate to cheaper locations, move back home, etc.

To come up with these numbers in real life, you would find historical data for similar properties and check how the percentages and growth rates changed under different market conditions.

For example, if rents fell by 10% in the worst recession of the past 30 years, then your Downside Case should not assume a 20% drop.

A rental decrease in the 5-10% range might be appropriate, depending on the numbers from the other, milder recessions.



Likewise, if the General Vacancy Rate only rose to 7-10% in past recessions, your model should use something in that range rather than 20% or 25%.

When in doubt, think “roughly correct” rather than “precisely wrong.”

Strengths and Weaknesses of Multifamily and Student Housing Properties

The greatest strength of multifamily and student housing properties is also their greatest weakness: there are no long-term leases with terms set far in advance, so rent and expenses fully reflect market conditions – in good times and bad.

Unlike the single-family owned-home market, though, recessions are *very rarely disastrous* for multifamily properties.

In large cities in the U.S., multifamily occupancy rates often remain in the 90-95% range regardless of the economy because people always need somewhere to live.

If you compare multifamily properties to office/industrial/retail properties, some factors *reduce* risk and other factors *increase* risk:

- **Reduce Risk:** Lease terms are very similar, and there are often dozens or hundreds of tenants, so there’s little risk of an “anchor tenant” leaving. There are usually limits to how much rents and occupancy rates can fall, at least in major cities, and capital costs are much lower.
- **Increase Risk:** Lease terms are much shorter, and individuals are less reliable than companies when it comes to paying rent on time. Rents change far more quickly in response to market conditions.

On balance, however, multifamily properties with high numbers of tenants are **less risky** than office, industrial, and retail properties simply because they are less sensitive to economic conditions.

Historically, multifamily properties have also delivered the highest risk-adjusted returns of the major property types for the same reason.

Compared with pre-sold condo developments, multifamily properties under any strategy (new developments, renovations, or stabilized acquisitions) almost certainly offer lower risk and lower potential returns.

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Hotels

Hotels are an even more extreme version of multifamily properties since the ‘leases’ are even shorter (a few days rather than 1 year) and their occupancy rates and room rates swing wildly in response to market conditions.

If you look up hotel REITs – companies that own many hotel properties – and compare them to multifamily, office, or industrial REITs, you’ll find that the average **Beta** for hotel REITs is much higher.

“Beta” measures a stock’s correlation to the broader stock market; if Beta is 1.0, then the stock increases by 10% when the market increases by 10%. If Beta is 1.5, then the stock increases by 15% when the market increases by 10%.

A quick search of several U.S.-based hotel REITs at the time this guide was written showed that almost every company in the sector had a Beta over 1.00, with most between 1.20 and 1.40.

By contrast, the average Beta for large, U.S.-based multifamily REITs in the same period was between 0.30 and 0.50, with some as low as 0.20.

Beta for hotel REITs is much higher than it is for REITs specializing in the other property types because hotels are more sensitive to the broader economy.

Besides the much shorter-term “leases,” hotels also have **Pro-Formas** that look quite different from those of multifamily, office, industrial, and retail properties.

The hotel Pro-Forma is closer to the **Income Statement** of a normal company, where expenses are split into categories such as Cost of Goods Sold (COGS) and Operating Expenses.

Hotels are the closest to “normal companies” out of all the property types, so their financial statements are also the closest to the statements of normal companies.

Their key drivers are also similar to those of normal companies, where you might project revenue based on Units Sold * Average Selling Price and then assume margins for each revenue category.

Instead of showing you the entire Pro-Forma for a hotel, we’ll break it into segments because it’s easier to explain each section separately.

Here’s the Revenue part of the Pro-Forma and the key assumptions/drivers, taken from a hotel acquisition & renovation model:



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Property Pro-Forma:	Units:	Historical:	Projected:					Stabilised:
		FY17	FY18	FY19	FY20	FY21	FY22	FY23
Revenue:								
(+) Rooms:	\$	\$ 9,657,900	\$ 8,900,721	\$ 8,480,409	\$ 4,944,261	\$ 5,489,793	\$ 11,715,360	\$ 12,219,565
(+) Food & Beverage:	\$	1,149,750	1,136,873	1,138,452	621,504	664,281	1,403,962	1,464,386
(+) Telecommunications:	\$	91,980	90,950	91,076	49,720	53,142	112,317	117,151
(+) Apartment Management Fees:	\$	266,000	273,980	282,199	290,665	299,385	308,367	317,618
(+) Parking:	\$	291,375	288,112	288,512	157,074	168,345	355,799	371,111
Total Revenue:	\$	11,457,005	10,690,635	10,280,648	6,063,224	6,674,947	13,895,805	14,489,831

These first three assumptions are based on an average Fee per Night * Occupied Room Nights. Hotels only make money with these when guests are paying to stay there!

Revenue and Expense Assumptions:	Units:	Historical:	Projected:					Stabilised:
		FY17	FY18	FY19	FY20	FY21	FY22	FY23
Renovation Factor:	%	-	-	-	50.0%	50.0%	-	-
Average Daily Rate (ADR):	\$ / Room Night	\$ 210.00	\$ 201.60	\$ 197.57	\$ 217.32	\$ 232.54	\$ 241.84	\$ 249.09
ADR Growth %:	%		(4.0%)	(2.0%)	10.0%	7.0%	4.0%	3.0%
Occupancy Rate:	%	75.0%	72.0%	70.0%	74.0%	77.0%	79.0%	80.0%
Days in Period:	# Days	365	365	365	366	365	365	365
Revenue per Available Room (RevPAR):	\$ / Room Night	\$ 157.50	\$ 145.15	\$ 138.30	\$ 160.82	\$ 179.05	\$ 191.05	\$ 199.28
Available Room Nights:	# Room Nights	61,320	61,320	61,320	61,488	61,320	61,320	61,320
Occupied Room Nights:	# Room Nights	45,990	44,150	42,924	22,751	23,608	48,443	49,056
Food & Beverage Revenue per Occupied Room Night:	\$ / Room Night	\$ 25.00	\$ 25.75	\$ 26.52	\$ 27.32	\$ 28.14	\$ 28.98	\$ 29.85
Food & Beverage Revenue Growth %:	%		3.0%	3.0%	3.0%	3.0%	3.0%	3.0%
Telecom Revenue per Occupied Room Night:	\$ / Room Night	\$ 2.00	\$ 2.06	\$ 2.12	\$ 2.19	\$ 2.25	\$ 2.32	\$ 2.39
Telecom Revenue Growth %:	%		3.0%	3.0%	3.0%	3.0%	3.0%	3.0%
Management Fees per Apartment:	\$ / Apt	\$ 2,000	\$ 2,060	\$ 2,122	\$ 2,185	\$ 2,251	\$ 2,319	\$ 2,388
Management Fees Growth %:	%		3.0%	3.0%	3.0%	3.0%	3.0%	3.0%
Fees per Occupied Car Park:	\$ / Car Park	\$ 1,500	\$ 1,545	\$ 1,591	\$ 1,639	\$ 1,688	\$ 1,739	\$ 1,791
Car Park Fees Growth %:	%		3.0%	3.0%	3.0%	3.0%	3.0%	3.0%
Occupied Car Parks:	# Car Parks	194	186	181	96	100	205	207

The Average Daily Rate (ADR) and the Occupancy Rate are the key drivers for hotels. A hotel's revenue = ADR * Total # of Rooms * 365 * Occupancy Rate.

Here, we're assuming a market and property decline followed by a renovation, so the ADR and Occupancy Rate fall for ~2 years before recovering.

The Renovation means that 50% of the rooms will be closed off in Years 3-4, so the Occupied Room Nights are cut in half in that period.

These other fees grow at low single-digit percentages, in-line with inflation. Food & Beverage, Telecom, and Parking Revenue are all linked to the Occupancy Rate because it is impossible to generate fees if no guests are staying at the hotel.

The Apartment Management Fees are in a different category because these are paid to hotel management regardless of occupancy. Anyone who owns an apartment in this building pays the hotel a monthly fee to manage it. This is not a "standard" line item, but many hotels do earn income from these sources.

The "Renovation Factor" refers to the percentage of rooms that will be unavailable while this renovation is taking place.

The key drivers, the **Average Daily Rate (ADR)** and **Occupancy Rate**, are what they sound like: the average room rate that hotel guests pay and the average percentage of rooms that are occupied.

Revenue per Available Room, or RevPAR, equals the ADR * the Occupancy Rate, and it's a useful metric that tells you the hotel's income-earning potential, *factoring in occupancy levels*.

Most hotels earn significant revenue from food & beverage sales (e.g., room service, on-premise restaurants, mini-bars, etc.) and smaller amounts from parking and "telecom fees" (e.g., internet access and phone calls). They may also earn revenue from hosting events and conferences.



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The “Apartment Management Fees” line item here is non-standard, but many hotels also have individual apartment units, and the hotel may collect fees for managing these.

Everything in the Revenue section except for this last item depends on the Occupied Room Nights – Occupancy drives everything for a hotel, and it can fluctuate significantly in different environments.

Moving on, the next section of a hotel Pro-Forma is “Departmental Expenses,” which is similar to COGS or Cost of Services for normal companies.

This section includes expenses that correspond *directly* to all the Revenue line items above. For example, it contains the expenses required to operate, clean, and maintain rooms, the cost of food and beverages, and the cost of delivering telecom services to guests.

The margins vary widely because some of these services are much more profitable than others:

Property Pro-Forma:	Units:	Historical:	Projected:						Stabilised:
		FY17	FY18	FY19	FY20	FY21	FY22	FY23	
Revenue:									
(+) Rooms:	\$	\$ 9,657,900	\$ 8,900,721	\$ 8,480,409	\$ 4,944,261	\$ 5,489,793	\$ 11,715,360	\$ 12,219,565	
(+) Food & Beverage:	\$	1,149,750	1,136,873	1,138,452	621,504	664,281	1,403,962	1,464,386	
(+) Telecommunications:	\$	91,980	90,950	91,076	49,720	53,142	112,317	117,151	
(+) Apartment Management Fees:	\$	266,000	273,980	282,199	290,665	299,385	308,367	317,618	
(+) Parking:	\$	291,375	288,112	288,512	157,074	168,345	355,799	371,111	
Total Revenue:	\$	11,457,005	10,690,635	10,280,648	6,063,224	6,674,947	13,895,805	14,489,831	
Departmental Expenses:									
(-) Room Operating Expenses:	\$	(2,173,028)	(1,993,761)	(1,891,131)	(1,097,626)	(1,213,244)	(2,577,379)	(2,688,304)	
(-) Food & Beverage:	\$	(862,313)	(852,655)	(853,839)	(466,128)	(498,211)	(1,052,972)	(1,098,289)	
(-) Telecommunications:	\$	(275,940)	(272,849)	(273,228)	(149,161)	(159,427)	(336,951)	(351,453)	
(-) Apartment Management Fees:	\$	(133,000)	(136,990)	(141,100)	(145,333)	(149,693)	(154,183)	(158,809)	
(-) Parking:	\$	(201,049)	(198,797)	(199,073)	(108,381)	(116,158)	(245,501)	(256,067)	
Total Departmental Expenses:	\$	(3,645,329)	(3,455,052)	(3,358,371)	(1,966,628)	(2,136,733)	(4,366,986)	(4,552,922)	
Gross Operating Income:									
Gross Operating Margin:	%	68.2%	67.7%	67.3%	67.6%	68.0%	68.6%	68.6%	

These are like Cost of Goods Sold or Cost of Services for normal companies, and each expense line item is projected as a simple percentage of the corresponding revenue line item.

Similar to the Gross Profit and Gross Margin for normal companies.

Room Revenue almost always has the highest margins (60-80% range); F&B is lower, and hotels often *lose money* on Telecom and simply offer it as a convenience to guests. Margins in other areas vary, but they're usually significantly lower than Room Margins.

Revenue and Expense Assumptions:	Units:	Historical:	Projected:						Stabilised:
		FY17	FY18	FY19	FY20	FY21	FY22	FY23	
Room Margin:	%	77.5%	77.6%	77.7%	77.8%	77.9%	78.0%	78.0%	
Food & Beverage Margin:	%	25.0%	25.0%	25.0%	25.0%	25.0%	25.0%	25.0%	
Telecommunications Margin:	%	(200.0%)	(200.0%)	(200.0%)	(200.0%)	(200.0%)	(200.0%)	(200.0%)	
Apartment Management Fee Margin:	%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	
Car Park Margin:	%	31.0%	31.0%	31.0%	31.0%	31.0%	31.0%	31.0%	

Even though we are renovating this hotel, we are *not* assuming that these margins change significantly because most renovations aim to improve the ADR and Occupancy Rate.

In practice, it's difficult to boost the Departmental margins by a huge amount because pricier rooms will also be more expensive to maintain, clean, and support, and more expensive food will also cost more in terms of supplies and labor (for example).

The bottom part of the Pro-Forma shows the hotel's **Undistributed Expenses**, i.e., its expenses that trend with Revenue and Occupancy Rates but are *not linked to specific departments*.

These are similar to Operating Expenses for a normal company, but they include only **variable expenses**.

The **fixed expenses** – which do not depend on the property's Occupancy Rate – are shown below in the **Fixed Expenses** section.

Finally, Gross Operating Income minus Undistributed Expenses minus Fixed Expenses equals Net Operating Income (NOI). And NOI minus Capital Costs equals Adjusted NOI.

As shown below, there are also intermediate sections of this Pro-Forma, including "House Profit" (Gross Operating Income – Undistributed Expenses) for showing the hotel's profitability after only *variable expenses*:

Property Pro-Forma:	Units:	Historical:	Projected:					Stabilised:
		FY17	FY18	FY19	FY20	FY21	FY22	FY23
Gross Operating Income:	\$	7,811,676	7,235,582	6,922,277	4,096,596	4,538,213	9,528,818	9,936,909
<i>Gross Operating Margin:</i>	%	68.2%	67.7%	67.3%	67.6%	68.0%	68.6%	68.6%
Undistributed Expenses:								
(-) Administrative & General (A&G):	\$	(572,850)	(534,532)	(514,032)	(303,161)	(333,747)	(694,790)	(724,492)
(-) Sales & Marketing (S&M):	\$	(801,990)	(748,344)	(719,645)	(424,426)	(467,246)	(972,706)	(1,014,288)
(-) Repairs & Maintenance (R&M):	\$	(400,995)	(374,172)	(359,823)	(212,213)	(233,623)	(486,353)	(507,144)
(-) Management Fees:	\$	(343,710)	(320,719)	(308,419)	(181,897)	(200,248)	(416,874)	(434,695)
(-) Utilities:	\$	(91,980)	(90,950)	(91,076)	(49,720)	(53,142)	(112,317)	(117,151)
Total Undistributed Expenses:	\$	(2,211,526)	(2,068,717)	(1,992,996)	(1,171,417)	(1,288,008)	(2,683,041)	(2,797,770)
House Profit:	\$	5,600,150	5,166,865	4,929,281	2,925,179	3,250,206	6,845,778	7,139,140
<i>House Profit Margin:</i>	%	48.9%	48.3%	47.9%	48.2%	48.7%	49.3%	49.3%
Fixed Expenses:								
(-) Insurance:	\$	(75,600)	(77,868)	(80,204)	(82,610)	(85,088)	(87,641)	(90,270)
(-) Property Taxes:	\$	-	-	-	-	-	-	-
(-) Renewal & Replacement Reserves:	\$	(352,800)	(363,384)	(374,286)	(385,514)	(397,080)	(408,992)	(421,262)
(-) Ground Lease:	\$	-	-	-	-	-	-	-
Total Fixed Expenses:	\$	(428,400)	(441,252)	(454,490)	(468,124)	(482,168)	(496,633)	(511,532)
Net Operating Income (NOI):	\$	5,171,750	4,725,613	4,474,791	2,457,055	2,768,038	6,349,145	6,627,608
<i>NOI Margin:</i>	%	45.1%	44.2%	43.5%	40.5%	41.5%	45.7%	45.7%
(-) Renovation Costs:	\$	-	-	-	(3,360,000)	(3,360,000)	-	-
Adjusted Net Operating Income (NOI):	\$	5,171,750	4,725,613	4,474,791	(902,945)	(591,962)	6,349,145	6,627,608
<i>Adjusted NOI Margin:</i>	%	45.1%	44.2%	43.5%	(14.9%)	(8.9%)	45.7%	45.7%

These are all percentages of revenue, except for Utilities, which is based on the Occupied Room Nights (since vacant rooms do not use water or electricity).

No analogue for normal companies; similar to the Operating Margin, but different because Fixed Expenses are excluded.

These expenses all exist regardless of the Occupancy Rate and ADR. Most are linked to per-room figures, though Property Taxes may be based on the initial purchase price and then escalated each year. If the hotel does not own its land, there will be something for the Ground Lease expense, and it could be a significant risk factor.

The lack of Property Taxes here is an unusual quirk of the property system in the Northern Territory of Australia.

Traditional setup for NOI and Adjusted NOI, but there are no TIs or LCs here for obvious reasons: "Tenants" don't stay long enough to justify them. We also skip the business with the Reserve Allocation and Draws to pay for Capital Costs because they're much more predictable for a large property with guests constantly coming and going.



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You can see the assumptions and drivers for this section below:

Revenue and Expense Assumptions:	Units:	Historical:	Projected:					Stabilised:
		FY17	FY18	FY19	FY20	FY21	FY22	FY23
Administrative & General % Revenue:	%	5.0%	5.0%	5.0%	5.0%	5.0%	5.0%	5.0%
Sales & Marketing % Revenue:	%	7.0%	7.0%	7.0%	7.0%	7.0%	7.0%	7.0%
Repairs & Maintenance % Revenue:	%	3.5%	3.5%	3.5%	3.5%	3.5%	3.5%	3.5%
Management Fees % Revenue:	%	3.0%	3.0%	3.0%	3.0%	3.0%	3.0%	3.0%
Utilities per Occupied Room Night:	\$ / Room Night	\$ 2.00	\$ 2.06	\$ 2.12	\$ 2.19	\$ 2.25	\$ 2.32	\$ 2.39
Utilities Growth %:	%		3.0%	3.0%	3.0%	3.0%	3.0%	3.0%
Insurance per Available Room:	\$ / Room	\$ 450	\$ 464	\$ 477	\$ 492	\$ 506	\$ 522	\$ 537
Insurance per Room Growth Rate:	%		3.0%	3.0%	3.0%	3.0%	3.0%	3.0%
Property Taxes per Available Room:	\$ / Room	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Property Taxes per Room Growth Rate:	%		3.0%	3.0%	3.0%	3.0%	3.0%	3.0%
Reserves per Available Room:	\$ / Room	\$ 2,100	\$ 2,163	\$ 2,228	\$ 2,295	\$ 2,364	\$ 2,434	\$ 2,508
Reserves per Room Growth Rate:	%		3.0%	3.0%	3.0%	3.0%	3.0%	3.0%
Ground Lease:	\$	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Renovation Costs:	\$				3,360,000	3,360,000		

Utilities depend on Occupancy, so they are Undistributed Expenses, while Insurance is a Fixed Expense since it doesn't change based on Occupancy.

It's highly unusual to have no Property Taxes here; they are normally linked to a per-room rate or the property's initial value and an annual escalation.

There are no TIs or LCs, and Maintenance CapEx is more predictable and less "lumpy" than it is for other property types, so we skip the full Reserves setup here.

The distinction between Undistributed Expenses and Fixed Expenses is important because the more Fixed Expenses the hotel has, the less room there is for margin improvement.

Also, since a hotel's revenue can change very quickly, high Fixed Expenses represent a much bigger risk factor than they do for multifamily, office, industrial, or retail properties.

More Complex Additions to a Hotel Pro-Forma

You can always list more revenue and expense line items and break out expenses in more detail, but those don't make the Pro-Forma "more complex"; they just make it longer.

Similarly, you could project the hotel's performance on a monthly or quarterly basis, but that isn't necessarily "more complex" – it's just more information. And it's a bit pointless for hotels because most guests only stay for a few days.

The main *useful*, more complex additions are **operating and market scenarios**.

As with multifamily, office, industrial, and retail properties, you can create Base, Upside, and Downside cases for hotels, or focus on the Base, Downside, and Extreme Downside cases in credit deals.

A hotel can also operate **independently** or choose to join a **franchise**, such as Marriott or Hilton.

If the hotel joins a franchise, it will be more expensive to operate and it will owe fees to the franchisor, but it may also be able to boost its Occupancy Rate and ADR.

Finally, the hotel owner may decide to pay an incentive fee to the management team if the hotel's NOI Margin (or another performance metric) exceeds a certain percentage.

Here are the Independent and Franchise scenarios for the hotel property in Dubai in this course:

Selected Market Scenario:	Name	Base
Selected Operating Scenario:	Name	Independent

Operating Scenarios:		Selected:	Independent	Franchise
Property Occupancy % Market Occupancy:	%	95.0%	95.0%	100.0%
Property ADR % Market ADR:	%	90.0%	90.0%	95.0%
Renovation Costs per Key:	\$ / Key	\$ 35,000	\$ 35,000	\$ 40,000
Management Fee % Revenue:	%	2.0%	2.0%	3.0%
Franchise Fee % Revenue:	%	-	-	3.0%
NOI Margin Hurdle for Management Incentive:	%	20.0%	20.0%	30.0%
Incentive Fee % NOI Above Hurdle:	%	30.0%	30.0%	20.0%
Room Margin Through Holding Period:	%	81.0%	81.0%	81.0%
F&B Margin Through Holding Period:	%	25.0%	25.0%	25.0%
Other Revenue Margin Through Holding Period:	%	11.0%	11.0%	11.0%
Administrative & General % Revenue:	%	5.0%	5.0%	6.0%
Sales & Marketing % Revenue:	%	7.0%	7.0%	8.0%
Repairs & Maintenance % Revenue:	%	3.5%	3.5%	3.5%
Initial FF&E Reserves per Available Room:	\$ / Key	\$ 2,100	\$ 2,100	\$ 2,300

The Franchise option may offer a higher Occupancy Rate and ADR...

But expenses, including management and franchise fees, G&A, S&M, R&M, and Capital Costs and Reserves, also tend to be higher.

The screenshot below shows the setup for the market and operating scenarios and how they drive the Occupancy Rate, ADR, and other revenue and expense growth rates for this property:



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Market Data and Scenarios:	Units:	Historical:	Projected:					Stabilized:
		FY16	FY17	FY18	FY19	FY20	FY21	FY22
Market Occupancy Rate:								
Base	%	85.0%	84.0%	82.0%	83.0%	83.5%	84.0%	84.5%
Downside	%	85.0%	83.0%	81.0%	81.5%	82.0%	82.5%	83.0%
Upside	%	85.0%	84.5%	84.0%	85.0%	86.0%	87.0%	87.0%
Selected Market Occupancy Rate:	%	85.0%	84.0%	82.0%	83.0%	83.5%	84.0%	84.5%
Market Average Daily Rate (ADR):								
	\$ / Room Night	\$ 450.00	\$ 432.00	\$ 423.36	\$ 440.29	\$ 457.91	\$ 471.64	\$ 481.08
Market Average Daily Rate (ADR) Growth Rate:								
Base	%	(5.0%)	(4.0%)	(2.0%)	4.0%	4.0%	3.0%	2.0%
Downside	%	(5.0%)	(7.0%)	(5.0%)	4.0%	3.0%	2.0%	2.0%
Upside	%	(5.0%)	(2.0%)	4.0%	3.0%	3.0%	3.0%	2.0%
Selected Market ADR Growth Rate:	%	(5.0%)	(4.0%)	(2.0%)	4.0%	4.0%	3.0%	2.0%
Market RevPAR:								
	\$ / Room Night	382.50	362.88	347.16	365.44	382.35	396.18	406.51
Property's Occupancy Rate:								
	%	78.0%	79.8%	77.9%	78.9%	79.3%	79.8%	80.3%
Property's Average Daily Rate (ADR):								
	\$ / Room Night	350.00	388.80	381.02	396.26	412.12	424.48	432.97
Property's RevPAR:								
	\$ / Room Night	273.00	310.26	296.82	312.45	326.91	338.73	347.57
Other Revenue Growth Rate:								
Base	%	(2.0%)	(2.0%)	(1.0%)	3.0%	2.0%	2.0%	2.0%
Downside	%	(2.0%)	(5.0%)	(3.0%)	2.0%	2.0%	2.0%	2.0%
Upside	%	(2.0%)	-	1.0%	2.0%	2.0%	2.0%	2.0%
Other Revenue Growth Rate:	%	(2.0%)	(2.0%)	(1.0%)	3.0%	2.0%	2.0%	2.0%
Expense Growth Rate:								
Base	%	3.0%	1.0%	1.0%	2.0%	2.0%	2.0%	2.0%
Downside	%	3.0%	0.5%	0.5%	1.0%	1.0%	2.0%	2.0%
Upside	%	3.0%	1.5%	1.5%	2.0%	2.0%	2.0%	2.0%
Selected Expense Growth Rate:	%	3.0%	1.0%	1.0%	2.0%	2.0%	2.0%	2.0%

There's a mild downturn in all the cases here, but it's most severe in the Downside Case and least severe in the Upside Case. We look at market-wide stats first and then assume the property "captures" a certain percentage of these.

Market Occupancy Rate * Market ADR.

These are based on the market figures and the property's "capture" percentages, which depend on the Independent vs. Franchise choice.

Revenue grows more quickly in strong markets and economies, but so do expenses as wages and supply costs rise. No such thing as a free lunch!

There's still a Renovation Factor in this model because it's for an "Acquire & Renovate" deal.

The other line items are similar to those in the simplified example, but this property also has a Ground Lease, which increases risk significantly.

There are still no ongoing property taxes because the UAE only taxes the initial property acquisition. This is an unusual feature that is unlikely in most other parts of the world.

Here's the Pro-Forma in the Base Market Case and Independent Operating Case:



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Property Pro-Forma:	Units:	Historical:	Projected:					Stabilized:
		FY16	FY17	FY18	FY19	FY20	FY21	FY22
Revenue:								
(+) Rooms:	\$	\$ 61,749,324	\$ 52,489,417	\$ 33,476,584	\$ 70,480,457	\$ 73,943,270	\$ 76,408,287	\$ 78,400,361
(+) Food & Beverage:	\$	22,053,330	16,537,947	10,655,163	22,217,314	22,860,637	23,393,386	24,003,285
(+) Other:	\$	4,410,666	3,307,589	2,131,033	4,443,463	4,572,127	4,678,677	4,800,657
Total Revenue:	\$	88,213,320	72,334,953	46,262,779	97,141,233	101,376,034	104,480,350	107,204,302
Departmental Expenses:								
(-) Room Operating Expenses:	\$	(11,732,372)	(9,972,989)	(6,360,551)	(13,391,287)	(14,049,221)	(14,517,575)	(14,896,069)
(-) Food & Beverage:	\$	(16,539,998)	(12,403,460)	(7,991,372)	(16,662,985)	(17,145,477)	(17,545,039)	(18,002,464)
(-) Other:	\$	(3,925,493)	(2,943,754)	(1,896,619)	(3,954,682)	(4,069,193)	(4,164,023)	(4,272,585)
Total Departmental Expenses:	\$	(32,197,862)	(25,320,204)	(16,248,542)	(34,008,954)	(35,263,892)	(36,226,637)	(37,171,117)
Gross Operating Income:	\$	56,015,458	47,014,749	30,014,237	63,132,279	66,112,142	68,253,714	70,033,186
Gross Operating Margin:	%	63.5%	65.0%	64.9%	65.0%	65.2%	65.3%	65.3%
Undistributed Expenses:								
(-) Administrative & General (A&G):	\$	(4,410,666)	(3,616,748)	(2,313,139)	(4,857,062)	(5,068,802)	(5,224,018)	(5,360,215)
(-) Sales & Marketing (S&M):	\$	(6,174,932)	(5,063,447)	(3,238,395)	(6,799,886)	(7,096,322)	(7,313,625)	(7,504,301)
(-) Repairs & Maintenance (R&M):	\$	(3,087,466)	(2,531,723)	(1,619,197)	(3,399,943)	(3,548,161)	(3,656,812)	(3,752,151)
(-) Franchise Fee:	\$	-	-	-	-	-	-	-
(-) Management Fee:	\$	(1,764,266)	(1,446,699)	(925,256)	(1,942,825)	(2,027,521)	(2,089,607)	(2,144,086)
(-) Utilities:	\$	(1,764,266)	(1,363,537)	(896,255)	(1,850,657)	(1,904,245)	(1,948,622)	(1,999,425)
Total Undistributed Expenses:	\$	(17,201,597)	(14,022,154)	(8,992,241)	(18,850,373)	(19,645,051)	(20,232,683)	(20,760,178)
House Profit:	\$	38,813,861	32,992,596	21,021,996	44,281,906	46,467,091	48,021,031	49,273,008
House Profit Margin:	%	44.0%	45.6%	45.4%	45.6%	45.8%	46.0%	46.0%
Fixed Expenses:								
(-) Insurance:	\$	(927,000)	(936,270)	(945,633)	(964,545)	(983,836)	(1,003,513)	(1,023,583)
(-) Property Taxes:	\$	-	-	-	-	-	-	-
(-) FF&E Reserves:	\$	(1,297,800)	(1,310,778)	(1,323,886)	(1,350,363)	(1,377,371)	(1,404,918)	(1,433,017)
(-) Ground Lease:	\$	(15,563,120)	(15,718,751)	(15,875,939)	(16,193,457)	(16,517,326)	(16,847,673)	(17,184,626)
Total Fixed Expenses:	\$	(17,787,920)	(17,965,799)	(18,145,457)	(18,508,366)	(18,878,533)	(19,256,104)	(19,641,226)
Net Operating Income (NOI):	\$	21,025,941	15,026,797	2,876,539	25,773,540	27,588,558	28,764,927	29,631,782
NOI Margin:	%	23.8%	20.8%	6.2%	26.5%	27.2%	27.5%	27.6%
(-) Incentive Fee to Management:	\$	-	-	-	(1,903,588)	(2,194,005)	(2,360,657)	(2,457,276)
(-) Renovation Costs:	\$	-	(10,815,000)	(10,815,000)	-	-	-	-
Adjusted Net Operating Income (NOI):	\$	21,025,941	4,211,797	(7,938,461)	23,869,952	25,394,552	26,404,270	27,174,505
Adjusted NOI Margin:	%	23.8%	5.8%	(17.2%)	24.6%	25.0%	25.3%	25.3%

The NOI Margin of 24-28% here is closer to the normal range for hotels, and so is the Adjusted NOI Margin of 24-25%. These percentages are lower in the two renovation years, but that's normal when Revenue is reduced due to far fewer Occupied Room Nights.

Notice how the Revenue reaches \$107 million in the Stabilized Year in the end, with NOI of \$30 million and Adjusted NOI of \$27 million.

By contrast, here's this same property in the Upside Market Case and Franchise Operating Case:



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Property Pro-Forma:	Units:	Historical:	Projected:					Stabilized:
		FY16	FY17	FY18	FY19	FY20	FY21	FY22
Revenue:								
(+) Rooms:	\$	\$ 61,749,324	\$ 59,890,992	\$ 41,278,714	\$ 86,046,463	\$ 89,916,210	\$ 93,434,615	\$ 95,303,307
(+) Food & Beverage:	\$	22,053,330	17,869,373	11,960,849	24,690,610	25,550,520	26,292,537	26,818,388
(+) Other:	\$	4,410,666	3,573,875	2,392,170	4,938,122	5,110,104	5,258,507	5,363,678
Total Revenue:	\$	88,213,320	81,334,240	55,631,734	115,675,195	120,576,834	124,985,659	127,485,373
Departmental Expenses:								
(-) Room Operating Expenses:	\$	(11,732,372)	(11,379,288)	(7,842,956)	(16,348,828)	(17,084,080)	(17,752,577)	(18,107,628)
(-) Food & Beverage:	\$	(16,539,998)	(13,402,030)	(8,970,637)	(18,517,958)	(19,162,890)	(19,719,403)	(20,113,791)
(-) Other:	\$	(3,925,493)	(3,180,748)	(2,129,031)	(4,394,929)	(4,547,993)	(4,680,072)	(4,773,673)
Total Departmental Expenses:	\$	(32,197,862)	(27,962,067)	(18,942,624)	(39,261,714)	(40,794,962)	(42,152,051)	(42,995,092)
Gross Operating Income:	\$	56,015,458	53,372,173	36,689,110	76,413,481	79,781,872	82,833,608	84,490,280
Gross Operating Margin:	%	63.5%	65.6%	65.9%	66.1%	66.2%	66.3%	66.3%
Undistributed Expenses:								
(-) Administrative & General (A&G):	\$	(4,410,666)	(4,880,054)	(3,337,904)	(6,940,512)	(7,234,610)	(7,499,140)	(7,649,122)
(-) Sales & Marketing (S&M):	\$	(6,174,932)	(6,506,739)	(4,450,539)	(9,254,016)	(9,646,147)	(9,998,853)	(10,198,830)
(-) Repairs & Maintenance (R&M):	\$	(3,087,466)	(2,846,698)	(1,947,111)	(4,048,632)	(4,220,189)	(4,374,498)	(4,461,988)
(-) Franchise Fee:	\$	-	-	-	(3,470,256)	(3,617,305)	(3,749,570)	(3,824,561)
(-) Management Fee:	\$	(1,764,266)	(2,440,027)	(1,668,952)	(3,470,256)	(3,617,305)	(3,749,570)	(3,824,561)
(-) Utilities:	\$	(1,764,266)	(1,450,993)	(976,029)	(2,014,803)	(2,084,973)	(2,145,523)	(2,188,434)
Total Undistributed Expenses:	\$	(17,201,597)	(18,124,512)	(12,380,534)	(29,198,474)	(30,420,529)	(31,517,153)	(32,147,496)
House Profit:	\$	38,813,861	35,247,661	24,308,575	47,215,008	49,361,343	51,316,455	52,342,784
House Profit Margin:	%	44.0%	43.3%	43.7%	40.8%	40.9%	41.1%	41.1%
Fixed Expenses:								
(-) Insurance:	\$	(927,000)	(940,905)	(955,019)	(974,119)	(993,601)	(1,013,473)	(1,033,743)
(-) Property Taxes:	\$	-	-	-	-	-	-	-
(-) FF&E Reserves:	\$	(1,297,800)	(1,442,721)	(1,464,362)	(1,493,649)	(1,523,522)	(1,553,992)	(1,585,072)
(-) Ground Lease:	\$	(15,563,120)	(15,796,567)	(16,033,515)	(16,354,185)	(16,681,269)	(17,014,894)	(17,355,192)
Total Fixed Expenses:	\$	(17,787,920)	(18,180,193)	(18,452,895)	(18,821,953)	(19,198,392)	(19,582,360)	(19,974,008)
Net Operating Income (NOI):	\$	21,025,941	17,067,468	5,855,680	28,393,054	30,162,950	31,734,095	32,368,777
NOI Margin:	%	23.8%	21.0%	10.5%	24.5%	25.0%	25.4%	25.4%
(-) Incentive Fee to Management:	\$	-	-	-	-	-	-	-
(-) Renovation Costs:	\$	-	(12,360,000)	(12,360,000)	-	-	-	-
Adjusted Net Operating Income (NOI):	\$	21,025,941	4,707,468	(6,504,320)	28,393,054	30,162,950	31,734,095	32,368,777
Adjusted NOI Margin:	%	23.8%	5.8%	(11.7%)	24.5%	25.0%	25.4%	25.4%

The Revenue in the Stabilized Year is \$127 million, nearly 20% higher, and the NOI is \$32 million, nearly 10% higher.

That difference illustrates the trade-offs of becoming a franchisee: joining a franchise can boost a hotel's Revenue by giving it access to a built-in client base, but it also makes it more expensive to operate the hotel, so margins might decrease.

In the Downside Market Case and Independent Operating Scenario, this hotel's Revenue reaches only \$97 million in the Stabilized Year, with NOI of only \$25 million.



Outcomes this far apart would be unlikely for an office or industrial property because the rental income and annual escalations are determined long in advance, and the expenses tend to increase at low, predictable rates.

Strengths and Weaknesses of Hotel Properties

Hotels tend to sell for higher Cap Rates than the other property types, meaning *lower* valuations, and for a good reason: **they're significantly riskier due to the lack of long-term leases and the high sensitivity to economic conditions.**

They're the closest to "normal companies" out of everything in this guide, so it's no surprise that hotel REITs often have Betas and valuation multiples that are closer to those of normal companies.

More so than the other property types, hotels are dependent on **the human element**: the management teams that run them.

You could interpret these differences as both strengths and weaknesses: a hotel in a good market with a talented management team will do very well, while a hotel in a bad market with a mediocre management team will perform quite poorly.

By contrast, even if a landlord or owner manages an office building incompetently, *things will probably be OK* as long as the property has reliable, blue-chip tenants with long-term leases that keep paying the rent on time.

Because of hotels' risk profile, acquisitions of stabilized hotels ("Core" deals) are less common than renovations or new developments.

The sponsor usually wants to do *something* to improve the hotel and mitigate the risk factors above; if it acquires a stabilized hotel and changes very little, the risk may not be enough to justify the deal.

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Other Property Types: Storage, Healthcare, Lumber, Data Center, etc.

All the other property types are subsets or variations of the main types discussed above. To figure out which is which, think about **the number of tenants**, the **average lease length and terms**, and the **granularity of the revenue and expenses**.



Most properties fit into the “Multifamily” (M) or “Office / Industrial / Retail” (OIR) categories; relatively few are in the “Hotel” and “Condominium” categories.

Here are a few examples:

- **Storage:** These are similar to industrial properties, but the leases tend to be shorter term if individuals rather than companies rent the units. Also, there are usually many tenants rather than only a handful. So, these are a cross between M and OIR properties.
- **Student Housing:** Discussed above. These are nearly the same as M properties because the leases are 1 year or less, there’s little variation in the lease terms, there are dozens, hundreds, or thousands of tenants, and you don’t project revenue and expenses in granular detail.
- **Data Centers:** These fall into the OIR category because companies rent space here for years at a time and the lease terms vary significantly. There may also be extra fees based on power usage, which is one difference in this sector.
- **Healthcare:** These could be more like M or more like OIR properties. Nursing homes fall into the M category because of the number of tenants, the shorter-term leases, and the similar lease terms. But a lab or pharmaceutical facility rented to companies is in the OIR category due to the longer-term, highly variable leases.
- **Lumber:** These are arguably the closest to Hotels because they earn money from selling products (wood), not from tenant-derived rental income, and they operate more like normal companies, with a mix of fixed and variable expenses.
- **Single-Family Owned Homes:** These are like smaller, unconnected Condominiums, and they’re riskier since the developer builds them one-by-one and then attempts to sell them at the end of the process.

There are also mixed-use properties that combine several property types, such as apartment buildings that have retail space on the ground floor.

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Strategy #1: Develop a Property and Sell It Upon Completion (“Opportunistic”)

Investors and developers can use this strategy for most of the property types described in this guide.



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New property development works best when there is a **supply/demand mismatch** in the market – for example, there's rising demand for warehouses to support e-commerce, but the supply is lagging, and companies cannot find anything centrally located.

We'll cover two examples of new property developments here: a simplified one for an industrial property in Canada, and a more complex one for an office property in the U.K.

We'll focus on the **transaction assumptions, construction projections, debt service, and returns calculations** since we've covered the Pro-Forma in previous sections.

Simplified Model, Part 1 – Transaction Assumptions

First, you need to make assumptions about the construction start date, the plot of land, and the construction costs per gross square foot or gross square meter:

4216 61 Ave SE - Investment Analysis

(\$ in CAD as Stated)

Development Overview:			Units:			Units:		
Project Name:	Name	4216 61 Ave SE	# Square Feet in 1 Acre:	sq. ft.	43,560 sq. ft.			
Property Type:	Name	Industrial	Total Square Feet to Purchase:	sq. ft.	784,080 sq. ft.			
			Percentage Required for Property:	%	43.0%			
Location:	Name	Calgary	Gross Square Feet of Property:	sq. ft.	337,154 sq. ft.			
Construction Start Date:	Date	2018-01-01						
Months in Year:	#	12	Rentable to Gross Square Feet %:	%	95.0%			
			Rentable Square Feet:	sq. ft.	320,297 sq. ft.			
Number of Acres to Purchase:	# Acres	18.0 Acres						
Estimated Price per Acre:	\$ / Acre	\$ 700,000	Construction Costs per Gross SF:	\$ / sq. ft.	\$ 50.00			
Land Acquisition Costs:	\$	\$ 12,600,000	Total Construction Costs:	\$	\$ 16,857,720			
Saleable Excess Land Percentage:	%	57.0%						

Not needed for the property, so the investors may be able to sell this at the end to boost returns.

You'll also have to assume that a **Construction Loan** is used to fund part of the development costs.

Such loans have higher interest rates than Permanent Loans due to the higher risk, and interest is capitalized when the property is under construction due to the lack of cash flows:

Construction, Financing, and Exit Assumptions: Units:

Construction Financing:

Loan-to-Cost (LTC) Ratio:	%	50.0%
Interest Rate - Annual:	%	6.25%
Interest Rate - Monthly:	%	0.506%
Issuance Fees:	%	1.00%
Amortization (Years):	# Years	Interest Only
Maturity:	Date	2018-12-31
Term:	# Years	1 year
Developer Equity Contribution:	%	10.0%
IRR Hurdle 1:	%	10.0%
Developer Cash Flow Above IRR Hurdle 1:	%	20.0%
IRR Hurdle 2:	%	20.0%
Developer Cash Flow Above IRR Hurdle 2:	%	30.0%

So, 50% Equity and 50% Debt funding.

$(1 + \text{Annual Interest Rate})^{(1 / 12)} - 1$

Matches the expected construction time; industrial / warehouse properties are fast to build, while office and retail properties take longer.

Developer contributes 10% of the Equity and earns 10% of the returns up to a 10% IRR; 20% up to a 20% IRR, and 30% above a 20% IRR.

Then, you'll assume an Equity contribution from the Developer and the Investor and set up a **Sources & Uses schedule** to show where the funding is coming from and where it's going to:

Sources & Uses of Funds:

Sources of Funds:	
Construction Loan:	\$ 15,028,860
Developer Equity:	1,502,886
Investor Equity:	13,525,974
Total Sources:	\$ 30,057,720

Uses of Funds:	
Land Acquisition Costs:	\$ 12,600,000
Construction Costs:	16,857,720
Replacement Reserves:	600,000
Total Uses:	\$ 30,057,720

From the assumptions above for Land and Construction Costs.

Extra amount to handle TI's, LCs, CapEx, etc. when tenants initially move in after construction is done.

Construction Loan is 50% * Total Uses; Developer Equity is 10% of the remaining 50%, and Investor Equity is 90% of the remaining 50%.

Simplified Model, Part 2 – Construction Projections

Next, you will project the Construction Costs over the 1-year period required to build this warehouse.

These numbers come from the case study instructions; in real life, you would look at the spending patterns of previous, similar developments, and use similar percentages in yours.

You initially draw on Equity to pay for these expenses, but you switch to the Construction Loan once the maximum amount of Equity has been drawn.

Lenders won't commit to a project until the equity investors have contributed a certain amount first – it's too risky to fund a loan when there's minimal-to-no Equity backing a development.

Here are the Development Cost projections in this model:

Debt and Equity Draws:	Units:	January	February	March	April	May	June	July	August	September	October	November	December	Total
Percentage of Land Acquisition Costs:	%	20.0%	-	-	-	-	80.0%	-	-	-	-	-	-	100.0%
Percentage of Construction Costs:	%	20.0%	10.0%	10.0%	5.0%	5.0%	5.0%	10.0%	10.0%	5.0%	10.0%	5.0%	5.0%	100.0%
Percentage of Replacement Reserves:	%	-	-	-	-	-	-	-	-	-	-	-	100.0%	100.0%
(+) Land Acquisition Costs:	\$	\$ 2,520,000	\$ -	\$ -	\$ -	\$ -	\$ 10,080,000	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 12,600,000
(+) Construction Costs:	\$	\$ 3,371,544	\$ 1,685,772	\$ 1,685,772	\$ 842,886	\$ 842,886	\$ 842,886	\$ 1,685,772	\$ 1,685,772	\$ 842,886	\$ 1,685,772	\$ 842,886	\$ 842,886	\$ 16,857,720
(+) Replacement Reserves:	\$	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 600,000	\$ 600,000
Total Development Costs:	\$	\$ 5,891,544	\$ 1,685,772	\$ 1,685,772	\$ 842,886	\$ 842,886	\$ 10,922,886	\$ 1,685,772	\$ 1,685,772	\$ 842,886	\$ 1,685,772	\$ 842,886	\$ 1,442,886	\$ 30,057,720

There's almost always an upfront deposit, followed by a final deposit, required for Land.

This Reserve is allocated to cover the capital costs associated with new tenants moving in in the first year.

And here are the Equity and Debt Draws:

	A	B	C	D	E	F	G	H	I	J	K	L	M
1													
2			Construction Period Projections										
3			(\$ in CAD as Stated)										
4													
5			Debt and Equity Draws:	Units:	January	February	March	April	May	June	July	August	
6													
7			Percentage of Land Acquisition Costs:	%	20.0%	-	-	-	-	80.0%	-	-	
8			Percentage of Construction Costs:	%	20.0%	10.0%	10.0%	5.0%	5.0%	5.0%	10.0%	10.0%	
9			Percentage of Replacement Reserves:	%	-	-	-	-	-	-	-	-	
10													
11			(+) Land Acquisition Costs:	\$	\$ 2,520,000	\$ -	\$ -	\$ -	\$ -	\$ 10,080,000	\$ -	\$ -	
12			(+) Construction Costs:	\$	\$ 3,371,544	\$ 1,685,772	\$ 1,685,772	\$ 842,886	\$ 842,886	\$ 842,886	\$ 1,685,772	\$ 1,685,772	
13			(+) Replacement Reserves:	\$	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	
14			Total Development Costs:	\$	\$ 5,891,544	\$ 1,685,772	\$ 1,685,772	\$ 842,886	\$ 842,886	\$ 10,922,886	\$ 1,685,772	\$ 1,685,772	
15													
16			Maximum Equity Draw:	\$	15,028,860								
17													
18			Equity Draw:	\$	\$ 5,891,544	\$ 1,685,772	\$ 1,685,772	\$ 842,886	\$ 842,886	=MIN(K14,Max_Equity-SUM(\$E18:J18))			
19			Construction Loan Draw:	\$	-	-	-	-	-	6,842,886	1,685,772	1,685,772	

The Equity Draw equals the minimum between the Development Costs and the Maximum Equity Draw minus Cumulative Equity Draws so far.

Construction Loan Draws cover everything we can't pay for with Equity, i.e. MAX(Development Costs - Equity Draw, 0).

Equity is simple to track because you update the Cumulative Equity Balance each month to reflect the Draws in the month.

Debt is trickier because you need to reflect **both** capitalized interest and capitalized loan fees – but you must base the interest on the *beginning balance* each month to avoid circular references:



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Debt and Equity Draws:	Units:	January	February	March	April	May	June	July
Equity Draw:	\$	\$ 5,891,544	\$ 1,685,772	\$ 1,685,772	\$ 842,886	\$ 842,886	\$ 4,080,000	\$ -
Construction Loan Draw:	\$	-	-	-	-	-	6,842,886	1,685,772
Cumulative Equity Balance:	\$	-	5,891,544	7,577,316	9,263,088	10,105,974	10,948,860	15,028,860
Beginning Debt Balance:	\$	-	-	-	-	-	=+J24+K19+K29	-
Ending Debt Balance:	\$	-	-	-	-	-	6,946,320	8,692,755
Monthly Interest Rate on Debt:	%	0.51%	0.51%	0.51%	0.51%	0.51%	0.51%	0.51%
Additions to Debt Balance:								
(+) Capitalized Loan Fees:	\$	-	-	-	-	-	68,429	16,858
(+) Capitalized Interest:	\$	-	-	-	-	-	35,005	43,806
Total Additions to Debt Balance:	\$	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 103,434	\$ 60,663

Beginning Debt Balance = Ending Debt Balance from Previous Month + Draw in This Month + Capitalized Loan Fees. This setup avoids circular references.

Debt and Equity Draws:	Units:	January	February	March	April	May	June	July
Equity Draw:	\$	\$ 5,891,544	\$ 1,685,772	\$ 1,685,772	\$ 842,886	\$ 842,886	\$ 4,080,000	\$ -
Construction Loan Draw:	\$	-	-	-	-	-	6,842,886	1,685,772
Cumulative Equity Balance:	\$	-	5,891,544	7,577,316	9,263,088	10,105,974	10,948,860	15,028,860
Beginning Debt Balance:	\$	-	-	-	-	-	6,911,315	8,648,949
Ending Debt Balance:	\$	-	-	-	-	-	6,946,320	8,692,755
Monthly Interest Rate on Debt:	%	0.51%	0.51%	0.51%	0.51%	0.51%	0.51%	0.51%
Additions to Debt Balance:								
(+) Capitalized Loan Fees:	\$	-	-	-	-	-	68,429	16,858
(+) Capitalized Interest:	\$	-	-	-	-	-	=+K23*K26	43,806
Total Additions to Debt Balance:	\$	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 103,434	\$ 60,663

And then the Capitalized Interest equals the Beginning Balance * Monthly Interest Rate - again, to avoid circular references.

Extend this projection until the end of the Development period to get the ending Debt and Equity balances.

They won't represent a 50/50 split *exactly* due to the capitalized loan fees and interest. You could attempt to fix this issue by making the beginning LTC calculation circular, but we don't think it's worth the effort and model instability to do that.

Simplified Model, Part 3 – Tenants and Pro-Forma Projections

With the Construction Projections finished, you can use individual tenants' lease terms to create a Pro-Forma.

We already covered this topic in [the section on Office, Retail, and Industrial properties](#), so we're not going to go through it in detail once again here.

The entire setup is the same: start with Base Rental Income, deduct Absorption & Turnover Vacancy in periods when tenants might cancel and leave space vacant, deduct Free Rent when new tenants move in, add Expense Reimbursements, and deduct General Vacancy to get to Effective Gross Income.



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Then, subtract Operating Expenses and Property Taxes, Management Fees, and Capital Cost Reserves to calculate the Net Operating Income (NOI).

The setup with the Reserves allocation and drawing on the Reserves to pay for CapEx, TIs, and LCs is also the same. Here's this property's Pro-Forma down to Adjusted NOI:

Property Pro-Forma:	Units:	Construction:	Operational Years:								Stabilized:						
		FY18	FY19	FY20	FY21	FY22	FY23	FY24	FY25	FY26							
Revenue:																	
(+) Base Rental Income:	\$	\$	2,450,270	\$	2,523,778	\$	2,599,491	\$	2,677,476	\$	2,757,800	\$	2,840,534	\$	2,925,750	\$	3,013,523
(-) Absorption & Turnover Vacancy:	\$		(768,712)		-		-		-		(351,484)		(178,230)		-		-
(-) Concessions & Free Rent:	\$		(520,482)		(263,924)		-		-		(410,065)		(207,935)		-		-
(+) Expense Reimbursements:	\$		489,253		736,514		758,610		781,368		694,677		776,598		853,822		879,436
Potential Gross Revenue:	\$		1,650,329		2,996,367		3,358,101		3,458,844		2,690,928		3,230,968		3,779,572		3,892,959
(-) General Vacancy:	\$		(120,111)		(123,715)		(127,426)		(131,249)		(135,186)		(139,242)		(143,419)		(147,722)
Effective Gross Income (EGI):	\$		1,530,217		2,872,653		3,230,675		3,327,595		2,555,742		3,091,726		3,636,153		3,745,237
Operating Expenses:																	
(-) Recoverable Expenses:	\$		(752,697)		(775,278)		(798,536)		(822,493)		(847,167)		(872,582)		(898,760)		(925,723)
(-) Management Fee:	\$		(45,907)		(86,180)		(96,920)		(99,828)		(76,672)		(92,752)		(109,085)		(112,357)
(-) CapEx, TI, and LC Reserves:	\$		(96,089)		(98,972)		(101,941)		(104,999)		(108,149)		(111,393)		(114,735)		(118,177)
Total Operating Expenses:	\$		(894,693)		(960,429)		(997,398)		(1,027,319)		(1,031,989)		(1,076,728)		(1,122,580)		(1,156,257)
Net Operating Income (NOI):	\$		635,525		1,912,223		2,233,277		2,300,275		1,523,753		2,014,999		2,513,573		2,588,980
NOI Margin:	%		41.5%		66.6%		69.1%		69.1%		59.6%		65.2%		69.1%		69.1%
(-) CapEx, TIs, and LCs:	\$		(508,265)		(243,508)		-		-		(417,778)		(200,469)		-		-
(+) Capital Costs Paid from Reserves:	\$		508,265		243,508		-		-		358,377		111,393		-		-
Adjusted Net Operating Income:	\$		635,525		1,912,223		2,233,277		2,300,275		1,464,352		1,925,923		2,513,573		2,588,980
Adjusted NOI Margin:	%		41.5%		66.6%		69.1%		69.1%		57.3%		62.3%		69.1%		69.1%

This property has two tenants, both on NNN leases, which explains the high Expense Reimbursements as well as the lower EGI in Years 5 and 6 when the tenants' leases expire.

The numbers for these tenants came from the case study instructions. In real life, you might look at tenants that occupy similar buildings in the area and use that data to come up with estimates for the numbers and the amount of time it might take to lease the property fully.

If you're modeling a multifamily or hotel development instead, your assumptions will be much higher-level, and you'll focus on the property-wide Occupancy Rate and Average Rent or ADR and the amount of time it will take the property to stabilize at those levels.

Simplified Model, Part 4 – Debt Service

To project the Debt service, you assume that the Construction Loan is refinanced with a "Permanent Loan" that will stay in place until the property is sold.

The tricky part is that you must pick an LTV for this Permanent Loan, which means you must estimate the property's value when the refinancing takes place.



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We do that here by assuming that Cap Rates fall slightly – meaning a higher valuation – before recovering to their initial levels:

Returns to All Equity Investors:	Units:	Construction:	Operational Years:							Stabilized:
		FY18	FY19	FY20	FY21	FY22	FY23	FY24	FY25	FY26
Forward NOI:	\$		1,912,223	2,233,277	2,300,275	1,523,753	2,014,999	2,513,573	2,588,980	
Applicable Cap Rate:	%		5.80%	5.70%	5.60%	5.50%	5.60%	5.70%	5.75%	
Implied Property Value:	\$		32,969,370	39,180,300	41,076,347	27,704,600	35,982,120	44,097,773	45,025,743	

But there's a problem with this set of numbers: the ~\$33 million in FY 19 represents the property's value *at the end of FY 19*, but the refinancing takes place *at the end of FY 18*!

When you value a property with a Cap Rate, it's always based on the property's forward NOI – so the property's value *at the end of FY 18* depends on its FY 19 NOI.

But that number here is very low – only around \$0.6 million – because the second tenant moves in in FY 20, so the property has not yet stabilized in FY 19.

This “pre-stabilization period” is always an issue with new property developments and the refinancing that takes place when construction ends.

We typically solve it by retrieving the property's value upon stabilization and then discount it back to its value at the time the refinancing takes place. Here's an example:

Permanent Loan Refinancing & Exit:		
Loan-to-Value (LTV) Ratio:	%	55.0%
Interest Rate - Annual:	%	4.75%
Issuance Fees:	%	1.00%
Amortization (Years):	# Years	30
Maturity:	Date	2028-12-31
Prepayment Penalty:	%	1.00%
Refinancing Date:	Date	2018-12-31
Discount Rate:		15.0%
Property Value 1 Year After Refi.	\$	\$ 32,969,370
Property Value @ Refinancing:	\$	28,669,018
Permanent Loan Amount:	\$	15,767,960
Property Selling Costs:	%	1.50%

As is usually the case, the Permanent Loan LTV is higher than the Construction Loan LTC (55% vs. 50%), the interest rate is lower (4.75% vs. 6.25%), the maturity period is longer (10 years), and the amortization period is much longer than the maturity.

We take the property's value at stabilization, 1 year after the refinancing, discount it back at a 15% discount rate, and then base the Permanent Loan on the LTV * that amount.

Once you have these assumptions, you can project the interest and principal repayments for this Permanent Loan using the IPMT and PPMT functions in Excel – or you can calculate the numbers manually, depending on the terms of the Debt.

There is only one tranche of Debt here (the Permanent Loan), the interest rate is fixed, there's no interest-only period, and there's no capitalized interest, so we use the built-in IPMT and PPMT functions in Excel:

Numerical Year:	Year	2018	2019	2020	2021	2022	2023	2024	2025	2026
Property Pro-Forma:	Units:	Construction:	Operational Years:							Stabilized:
		FY18	FY19	FY20	FY21	FY22	FY23	FY24	FY25	FY26
Adjusted Net Operating Income:	\$		635,525	1,912,223	2,233,277	2,300,275	1,464,352	1,925,923	2,513,573	2,588,980
Adjusted NOI Margin:	%		41.5%	66.6%	69.1%	69.1%	57.3%	62.3%	69.1%	69.1%
(-) Interest Expense on Permanent Loan:	\$		(748,978)	(737,212)	(724,887)	=IPMT(Perm_Loan_Ann_Interest,I104-\$F104,Perm_Loan_Amort_Period,Perm_Loan_Balance)				
(-) Permanent Loan Principal Repayment:	\$		(247,706)	(259,472)	(271,797)	(284,707)	(298,231)	(312,397)	(327,236)	
Cash Flow to Equity Investors:	\$		(361,159)	915,539	1,236,593	1,303,591	467,667	929,239	1,516,889	
Ending Permanent Loan Balance:	\$	15,767,960	15,520,254	15,260,782	14,988,985	14,704,277	14,406,046	14,093,649	13,766,414	
Debt Yield:	%		4.0%	12.1%	14.2%	14.6%	9.7%	12.8%	15.9%	
Interest Coverage Ratio - NOI:	x		0.85 x	2.59 x	3.08 x	3.23 x	2.18 x	2.94 x	3.75 x	
Interest Coverage Ratio - Adjusted NOI:	x		0.85 x	2.59 x	3.08 x	3.23 x	2.10 x	2.81 x	3.75 x	
Debt Service Coverage Ratio (DSCR) - NOI:	x		0.64 x	1.92 x	2.24 x	2.31 x	1.53 x	2.02 x	2.52 x	
Debt Service Coverage Ratio (DSCR) - Adj. NOI:	x		0.64 x	1.92 x	2.24 x	2.31 x	1.47 x	1.93 x	2.52 x	

The IPMT and PPMT functions ensure that the Interest Expense + Principal Repayment remain the same in each period.

Adjusted NOI minus Interest Expense minus Principal Repayments.

This must be repaid upon exit, so we track it.

Annual NOI / Initial Permanent Loan balance.

NOI or Adjusted NOI divided by Interest Expense.

NOI or Adjusted NOI divided by Interest + Principal Repayments.

The IPMT and PPMT functions accept the following inputs:

=IPMT(Annual Interest Rate, Year #, Total Years in Amortization Period, Initial Loan Balance)

These functions work *best* if all the conditions mentioned above are true: a fixed interest rate, no interest-only period, and no capitalized interest.

The Debt Yield (DY), Interest Coverage Ratio (ICR), and Debt Service Coverage Ratio (DSCR) measure **credit risk**.

The lower the Debt Yield, the less NOI there is to cover Interest Expense and Principal Repayments. Lenders like to see higher numbers for the Debt Yield because they indicate that there's more of a "cushion" in case things go wrong.

Lower ICR and DSCR figures also indicate more credit risk because they mean that there's less cash flow available to pay for Interest and Principal Repayments.

Lenders often set minimums for these, such as 2.00x for the ICR and 1.40x for the DSCR, and charge penalty fees or increase interest rates if these requirements are violated.

Simplified Model, Part 5 – Returns Calculations

Once you've projected the Debt Service, you can calculate the IRR to Equity Investors based on their initial contributions, the refinancing, the annual cash flows, and the eventual sale of the property.

You'll also have to factor in items such as the fees associated with the refinancing, the sale of excess land, costs associated with the sale of the property, the repayment of Debt upon exit, and any prepayment penalties associated with that.

Here's the calculation for this property with an assumed exit in FY 25:

Returns to All Equity Investors:	Units:	Construction:		Operational Years:						
		FY18	FY19	FY20	FY21	FY22	FY23	FY24	FY25	
Value of Excess Land:	\$	\$ 7,182,000	\$ 7,397,460	\$ 7,619,384	\$ 7,847,965	\$ 8,083,404	\$ 8,325,906	\$ 8,575,684	\$ 8,832,954	\$ 9,097,943
Annual Growth Rate in Land Value:	%	3.0%	3.0%	3.0%	3.0%	3.0%	3.0%	3.0%	3.0%	3.0%
Forward NOI:	\$		1,912,223	2,233,277	2,300,275	1,523,753	2,014,999	2,513,573	2,588,980	
Applicable Cap Rate:	%		5.80%	5.70%	5.60%	5.50%	5.60%	5.70%	5.75%	
Implied Property Value:	\$		32,969,370	39,180,300	41,076,347	27,704,600	35,982,120	44,097,773	45,025,743	
(-) Equity Draws:	\$		(15,028,860)							
(+) Permanent Loan Issued:	\$		15,767,960							
(-) Permanent Loan Financing Fees:	\$		(157,680)							
(-) Construction Loan Refinanced:	\$		(15,583,300)							
(+) Cash Flow to Equity Investors:	\$		(361,159)	915,539	1,236,593	1,303,591	467,667	929,239	1,516,889	
(+) Proceeds from Sale of Excess Land:	\$					8,325,906				
(+) Proceeds from Sale of Property:	\$								45,025,743	
(-) Selling Costs:	\$								(675,386)	
(-) Repayment of Permanent Loan:	\$								(13,766,414)	
(-) Prepayment Penalty on Permanent Loan:	\$								(137,664)	
Total Cash Flows to Equity Investors:	\$	(15,001,880)	(361,159)	915,539	1,236,593	9,629,498	467,667	929,239	31,963,169	
Internal Rate of Return (IRR):	%	20.2%								
Total Returns to Equity:	\$		44,807,525							
Invested Equity:	\$		15,028,860							
Cash-on-Cash Multiple:	x		3.0 x							
Annual Yield on Initial Investment:	%			(2.4%)	6.1%	8.2%	64.1%	3.1%	6.2%	212.7%

Refinancing upon construction completion doesn't provide *that much* of a benefit here.

Negative cash flows represent additional Equity Investment; positive cash flows are returns.

We sell the property at the end, pay ~1.5% of the selling price in fees, repay the Permanent Loan, and pay a prepayment penalty because this Loan only matures in FY 28.

Selling the unused land midway through is another way to boost returns, assuming it has appreciated in value.

Invested Equity is everything in the "Equity Draws" row at the top; Total Returns to Equity is everything below that.

Annual periods, so we use the standard IRR function.

Total Cash Flows to Equity Investors / Invested Equity.

We are simplifying this setup by ignoring the fact that the Equity Draws occur over several months of the first year, not all at the end. Also, this model is not flexible enough to support variable exit dates.

Past this point, we create a **waterfall returns schedule** to split up the cash flows to the Developers and Investors based on the overall Equity IRR.

We are not going to cover this schedule in detail here because it's explained in the [Waterfall Returns Schedule section](#) of this guide, but here is the basic idea, starting with Tier 1:



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The "Repayment" is the MIN between (90% * Total Cash Flows to Equity) and the (Beginning Balance + Returns Accrual) with a positive sign.

Essentially, this distributes cash flows that correspond to IRRs up to the 10% level, or if cash flow is negative, it makes the Ending Balance more negative so more is owed.

Waterfall Returns Schedule:		Units:	Construction:	Operational Years:						
			FY18	FY19	FY20	FY21	FY22	FY23	FY24	FY25
Tier 1 IRR - Up to 10.0%:										
Leveraged IRR to All Equity Investors:										
Beginning Balance:				(15,001,880)	(16,863,228)	(17,634,011)	(18,160,819)	(10,347,404)	(10,914,477)	(11,076,685)
Returns Accrual:	10.0%			(1,500,188)	(1,686,323)	(1,763,401)	(1,816,082)	(1,034,740)	(1,091,448)	(1,107,668)
Repayment:	100.0%			(361,159)	915,539	1,236,593	9,629,498	467,667	929,239	12,184,353
Ending Balance:				(15,001,880)	(16,863,228)	(17,634,011)	(18,160,819)	(10,347,404)	(10,914,477)	(11,076,685)
Limited Partners (LPs):										
Beginning Balance:				(13,501,692)	(15,176,905)	(15,870,610)	(16,344,737)	(9,312,663)	(9,823,029)	(9,969,016)
Returns Accrual:	10.0%			(1,350,169)	(1,517,691)	(1,587,061)	(1,634,474)	(931,266)	(982,303)	(996,902)
Repayment:	90.0%			=MIN(G\$168*SD190,-SUM(G188:G189))			8,666,548	420,901	836,315	10,965,918
Ending Balance:				(13,501,692)	(15,176,905)	(15,870,610)	(16,344,737)	(9,312,663)	(9,823,029)	(9,969,016)
Developer:										
Beginning Balance:				(1,500,188)	(1,686,323)	(1,763,401)	(1,816,082)	(1,034,740)	(1,091,448)	(1,107,668)
Returns Accrual:	10.0%			(150,019)	(168,632)	(176,340)	(181,608)	(103,474)	(109,145)	(110,767)
Repayment:	10.0%			(36,116)	91,554	123,659	962,950	46,767	92,924	1,218,435
Ending Balance:				(1,500,188)	(1,686,323)	(1,763,401)	(1,816,082)	(1,034,740)	(1,091,448)	(1,107,668)
Cash Flow Available for Tier 2 Distribution:										19,778,815

Returns Accrual = Beginning Balance * IRR Limit of 10% here.

Total Cash Flows to Equity minus Repayments in this tier.

We do the same thing in Tier 2, but now the distributions are based on a 20% IRR limit and 80% distributions to the Investors (Limited Partners) and 20% to the Developers.

We then "re-assemble" all the cash flows at the bottom and calculate the IRR and cash-on-cash multiple for each group:

In the earlier years, the positive cash flows represent an IRR that is below 10%, so nothing gets distributed in Tiers 2 and 3.

We only go above the 10% (and 20%) level at the end, upon exit, because only that cash flow distribution is sufficient to reach these levels.

Waterfall Returns Distributions by Investor Group and IRR Tier:										
Tier 1 IRR - Up to 10.0%:										
Limited Partners (LPs):	90.0%		(325,044)	823,985	1,112,934	8,666,548	420,901	836,315	10,965,918	
Developers:	10.0%		(36,116)	91,554	123,659	962,950	46,767	92,924	1,218,435	
Tier 2 IRR - Up to 20.0%:										
Limited Partners (LPs):	80.0%		-	-	-	-	-	-	15,502,287	
Developers:	20.0%		-	-	-	-	-	-	3,875,572	
Tier 3 IRR - 20.0% or Above:										
Limited Partners (LPs):	70.0%		-	-	-	-	-	-	280,670	
Developers:	30.0%		-	-	-	-	-	-	120,287	
Returns Analysis by Investor Group:										
Limited Partners (LP) - Leveraged Returns:										
Internal Rate of Return (IRR):	%		(13,501,692)	(325,044)	823,985	1,112,934	8,666,548	420,901	836,315	26,748,875
Cash-on-Cash Multiple:	x		19.3%							
Annual Yield on Initial Investment:	%		2.8 x	(2.4%)	6.1%	8.2%	64.2%	3.1%	6.2%	198.1%
Developers - Leveraged Returns:										
Internal Rate of Return (IRR):	%		(1,500,188)	(36,116)	91,554	123,659	962,950	46,767	92,924	5,214,294
Cash-on-Cash Multiple:	x		26.6%							
Annual Yield on Initial Investment:	%		4.3 x	(2.4%)	6.1%	8.2%	64.2%	3.1%	6.2%	347.6%

The IRR to the LPs is lower because of the "promote" that increased the distributions for Developers above the 10% IRR level.

Should We Pursue This Deal?

Whether or not we should pursue this deal depends on the IRR we're targeting.



It's difficult to give a definitive answer here because we have not examined the outcomes in different scenarios, such as longer/shorter construction periods, higher/lower construction costs, and Base/Upstate/Downside market environments.

But, for example, if we're targeting a 20% IRR, this deal seems like a "no" since the IRR to Investors is only 19%.

The large amount of excess land purchased in the beginning reduces the returns because the price of that land only appreciates by ~3% per year, and the Permanent Loan Refinancing takes place quite early; postponing it would allow for a higher LTV and more Debt.

The waterfall incentive structure here also works against us because a 10% IRR for Tier 1 is quite low for a new development; it would be more sensible to set the Tier 1 limit at 20% or 25% instead.

New developments are so risky that the targeted IRR is almost always well above 10%.

Some of the credit stats and ratios are also "iffy" in the first year because of the early refinancing problem described above.

So, we would not recommend this deal as it is presented here, though we would be open to it if some of the terms above changed and we could analyze the output in different scenarios.

More Complex Additions to Development Models

In this course, we also teach a more complex development model for an office property in London (100 Bishopsgate).

The basic concepts are the same, so we're not going to paste in screenshots showing each part of the process step-by-step.

Instead, we'll highlight just the added complexities below:

More Complex Model, Part 1 – Transaction Assumptions

More complex models tend to use multiple scenarios and calculate items like the Rentable vs. Gross Area in more sophisticated ways. Here are a few examples:



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100 Bishopsgate - Investment Analysis

(£ in GBP as Stated)

Development Overview:		Units:	Units:
Project Name:	Name	100 Bishopsgate	
Property Type:	Name	Office/Retail	
Location:	Name	London, U.K.	
Construction Start Date:	Date	2015-05-31	
Months in Year:	#	12	
Land Plot Size:	sq. m.	7,500 sq. m.	
Land Price per Square Metre:	£ / sq. m.	£ 20,000	
Land Acquisition Costs:	£	£ 150,000,000	
Floor Area Ratio (FAR):	#	12.0	
Gross Square Metres of Property:	sq. m.	90,000 sq. m.	
Rentable to Gross Area %:	%	93.0%	
Rentable Square Metres:	sq. m.	83,700 sq. m.	
Hard Costs per GSM:	£ / sq. m.	£ 8,000.00	
Total Hard Costs:	£	£ 720,000,000	
Soft Costs per GSM:	£ / sq. m.	£ 1,000.00	
Total Soft Costs:	£	£ 90,000,000	
Total Construction Costs:	£	£ 810,000,000	
Construction Completion Date:	Date	2019-04-30	
Operating Scenario:	Name	Base	
Developer Name:	Name	Great Portland Estates	
Investor Name:	Name	Brookfield	

Split into more categories (Hard Costs and Soft Costs) rather than just one.

All investment analysis is probabilistic, so you need multiple scenarios to make a decision.

Slightly more complex since it's a percentage of the Gross Square Meters, which is based on the allowable "Floor Area" on this plot of land.

The Construction Loan and Developer/Investor Equity assumptions are not much different; the Construction Loan stays outstanding longer because this property takes ~4 years to complete, but the interest is still capitalized during the development.

The waterfall returns assumptions are slightly different since it's a mixed IRR and multiple-based waterfall, but the Developer and Investor still contribute set percentages in the beginning.

The Operating Assumptions in different scenarios are as follows:

Operating Assumptions:	Units:	Construction Period:				Operational Period:				
		FY16	FY17	FY18	FY19	FY20	FY21	FY22	FY23	FY24
Numerical Year:	Year	2016	2017	2018	2019	2020	2021	2022	2023	2024
Numerical Year #:	#	1	2	3	4	5	6	7	8	9
Prime Office Yield:										
Upside	%	4.00%	4.25%	4.25%	3.80%	3.60%	3.90%	4.00%	4.10%	
Base	%	4.00%	4.25%	4.25%	4.40%	4.60%	4.70%	4.80%	4.90%	
Downside	%	4.00%	4.25%	4.25%	4.75%	5.30%	5.80%	5.60%	5.30%	
Selected Prime Office Yield:	%	4.00%	4.25%	4.25%	4.40%	4.60%	4.70%	4.80%	4.90%	
Change in Prime Office Market Rents (Starting Rates for Leases):										
Upside	%	10.00%	-	-	6.00%	5.00%	4.00%	3.50%	3.50%	3.50%
Base	%	10.00%	-	-	3.50%	3.50%	3.50%	3.00%	3.00%	3.00%
Downside	%	10.00%	-	-	(7.00%)	(3.00%)	(1.00%)	10.00%	5.00%	2.50%
Selected Change in Prime Office Market Rents:	%	10.00%	-	-	3.50%	3.50%	3.50%	3.00%	3.00%	3.00%
Fixed Annual Rental Escalations for New and Renewal Leases Starting in Specified Year:										
Upside	%	2.00%	2.00%	2.00%	2.50%	3.00%	3.00%	2.75%	2.75%	2.75%
Base	%	2.00%	2.00%	2.00%	2.25%	2.25%	2.50%	2.50%	2.50%	2.50%
Downside	%	2.00%	2.00%	2.00%	1.00%	1.00%	1.50%	1.75%	2.00%	2.25%
Selected Fixed Annual Rental Escalation:	%	2.00%	2.00%	2.00%	2.25%	2.25%	2.50%	2.50%	2.50%	2.50%
Prime Office Expense Growth Rate - City of London:										
Upside	%	5.00%	1.00%	1.00%	4.00%	3.00%	2.50%	2.50%	2.50%	2.50%
Base	%	5.00%	1.00%	1.00%	2.50%	2.50%	2.00%	2.00%	2.00%	2.00%
Downside	%	5.00%	1.00%	1.00%	(5.00%)	(2.00%)	(0.50%)	7.00%	3.00%	1.50%
Selected Prime Office Expense Growth Rate:	%	5.00%	1.00%	1.00%	2.50%	2.50%	2.50%	2.00%	2.00%	2.00%

Cap Rates, or "Yields," fall for a few years in the Base and Upside Cases before returning to historical levels, while they keep rising in the Downside Case, reflecting a weakening market.

Rental growth is moderate in the Base Case, strong in the Upside Case, and negative in the Downside Case, reflecting a recession followed by a recovery.

Fixed rental escalations are lower when there's a recession, but they're not *that* much different.

No such thing as a free lunch - if rental growth is stronger, expenses will also grow more quickly.

Finally, the assumptions for individual tenant leases, such as the Renewal Probability, Months of Downtime, Free Months of Rent, Tenant Improvements, and Leasing Commissions are as follows:



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Lease Terms, Renewals, and Capital Costs:		Selected:	Upside	Base	Downside
Renewal Probability:	%	50.0%	60.0%	50.0%	40.0%
# Months of Downtime for Non-Renewal:	#	9	6	9	12
New Tenants - # Months of Free Rent:	#	10	7	10	13
Renewal Tenants - # Months of Free Rent:	#	8	6	8	11
New - Tenant Improvements per RSM:	£ / sq. m.	£ 15.00	£ 10.00	£ 15.00	£ 20.00
Renewal - Tenant Improvements per RSM:	£ / sq. m.	7.00	5.00	7.00	10.00
New - Leasing Commissions % Lease Value:	%	2.5%	2.0%	2.5%	3.0%
Renewal - Leasing Commissions % Lease Value:	%	1.5%	1.0%	1.5%	2.0%
Initial Annual Expenses & Taxes per RSM:	£ / sq. m. / Yr	£ 320.00			
Initial Annual Replacement Reserves per RSM:	£ / sq. m. / Yr	£ 20.00			
Property Management Fees % EGI:	%	3.0%			
Lease Term for New Tenants (Years):	# Years	5			

It's harder to find new tenants and existing tenants are more likely to leave in weaker markets; tenants are more likely to stay in stronger markets.

In weaker markets, landlords and owners must offer tenants more concessions and incentives to move in; the opposite is true in stronger markets.

...and landlords and owners must also pay real estate brokers more to find and retain tenants in weaker markets, and less in stronger markets.

More Complex Model, Part 2 – Construction Projections

The Construction Projections are largely the same because the sponsors still draw on Equity to fund construction until the Maximum Draw is reached, at which point they switch to the Construction Loan.

Expenses may be split into more categories, but the *main difference* is that you'll often show the impact of the refinancing directly in these Construction Projections because of the scenarios and monthly numbers in the model.

With scenarios and monthly projections, it's harder to "separate" the Construction Projections from everything else because the development could end in different months, and once it finishes, the next phase needs to start *immediately in the next month*.

You will show the Debt Service for the Permanent Loans in the same spreadsheet for these reasons as well.

Here's the final month of the Construction Period here:

=IF(BA5=Refi_Date,-SUM(Monthly!BA26:BA29),0)

We check to see if this month represents the Refinancing Date when construction finishes - and if so, we assume full repayment of the Construction Loan.

As in the simplified model, Interest is still capitalized, based on the Beginning Construction Loan Balance, Loan Issuance Fees are capitalized, and the maximum amount of Equity is drawn first before the investors start drawing on the Construction Loan.

Monthly Projections - Construction and Lease-Up Period (£ in GBP as Stated)

Debt and Equity Draws:		Units:	Jan-19	Feb-19	Mar-19	Apr-19
Month #	#		45	46	47	48
Numerical Year:	Year		2019	2019	2019	2019
Numerical Year #:	#		4	4	4	4
Percentage of Land Acquisition Costs:	%		-	-	-	-
Percentage of Hard Costs:	%		0.3%	0.5%	0.5%	1.1%
Percentage of Soft Costs:	%		0.1%	0.7%	0.2%	3.5%
Percentage of Replacement Reserves:	%		-	-	-	100.0%
(+) Land Acquisition Costs:	£	£	-	-	-	-
(+) Hard Costs:	£	£	2,160,000	3,600,000	3,600,000	7,920,000
(+) Soft Costs:	£	£	90,000	630,000	180,000	3,150,000
(+) Replacement Reserves:	£	£	-	-	-	8,370,000
Total Development Costs:	£	£	2,250,000	4,230,000	3,780,000	19,440,000
Maximum Equity Draw:	£					
Equity Draw:	£					
Cumulative Equity Drawn:	£	£	415,237,056	415,237,056	415,237,056	415,237,056
Beginning Debt Balance:	£	£	569,981,463	574,576,138	581,189,333	587,374,970
(+) Construction Loan Draw:	£	£	2,250,000	4,230,000	3,780,000	19,440,000
(+) Capitalised Loan Fees:	£	£	22,500	42,300	37,800	194,400
(+) Capitalised Interest:	£	£	2,322,175	2,340,894	2,367,837	2,393,038
(-) Debt Principal Repayment:	£	£	-	-	-	(609,402,408)
Ending Debt Balance:	£	£	574,576,138	581,189,333	587,374,970	-
Monthly Interest Rate on Debt:	£	£	0.407%	0.407%	0.407%	0.407%

More Complex Model, Part 3 – Tenants and Pro-Forma Projections

We covered monthly lease and Pro-Forma projections in [the section of the guide dedicated to Office, Retail, and Industrial properties](#), so please refer to that for the details.

This property has 3 tenants, 2 with NNN leases with full Expense Reimbursements and 1 with a Gross Lease with no Expense Reimbursements, but a higher Rent per Square Meter.

The Pro-Forma looks “lumpier” when you model the cash flows on a monthly basis because the Concessions & Free Rent, Absorption & Turnover Vacancy, and other items stick out a lot more.

Here are the first few months after the end of the Construction Phase in this model:



Real Estate & REIT Modeling

Quick Reference Guide for Individual Properties

Property Pro-Forma:	Units:	May-19	Jun-19	Jul-19	Aug-19	Sep-19	Oct-19	Nov-19
Revenue:								
(+) Base Rental Income:	£	£ 5,192,421	£ 5,193,392	£ 5,194,367	£ 5,195,344	£ 5,196,324	£ 5,197,307	£ 5,198,293
(-) Absorption & Turnover Vacancy:	£	-	-	-	-	-	-	-
(-) Concessions & Free Rent:	£	(4,853,958)	(4,853,958)	(4,853,958)	(4,853,958)	(4,853,958)	(4,853,958)	(2,078,561)
(+) Expense Reimbursements:	£	1,698,484	1,701,636	1,704,794	1,707,958	1,711,128	1,714,304	1,717,485
Potential Gross Revenue:	£	2,036,947	2,041,071	2,045,203	2,049,344	2,053,494	2,057,653	4,837,217
(-) General Vacancy:	£	(338,463)	(339,435)	(340,409)	(341,386)	(342,367)	(343,349)	(344,335)
Effective Gross Income (EGI):	£	1,698,484	1,701,636	1,704,794	1,707,958	1,711,128	1,714,304	4,492,882
Operating Expenses:								
(-) Recoverable Expenses:	£	(2,426,405)	(2,430,908)	(2,435,420)	(2,439,940)	(2,444,468)	(2,449,000)	(2,453,550)
(-) Property Management Fees:	£	(50,955)	(51,049)	(51,144)	(51,239)	(51,334)	(51,429)	(134,786)
(-) CapEx, TI, and LC Reserves:	£	(151,650)	(151,932)	(152,214)	(152,496)	(152,779)	(153,063)	(153,347)
Total Operating Expenses:	£	(2,629,010)	(2,633,889)	(2,638,777)	(2,643,675)	(2,648,581)	(2,653,497)	(2,741,684)
Net Operating Income (NOI):	£	(930,526)	(932,253)	(933,984)	(935,717)	(937,454)	(939,193)	1,751,198
<i>NOI Margin:</i>	%	<i>(54.8%)</i>	<i>(54.8%)</i>	<i>(54.8%)</i>	<i>(54.8%)</i>	<i>(54.8%)</i>	<i>(54.8%)</i>	<i>39.0%</i>
(-) CapEx, TIs, and LCs:	£	(6,089,602)	-	-	-	-	-	-
(+) Capital Costs Paid from Reserves:	£	6,089,602	-	-	-	-	-	-
Adjusted Net Operating Income:	£	(930,526)	(932,253)	(933,984)	(935,717)	(937,454)	(939,193)	1,751,198
<i>Adjusted NOI Margin:</i>	%	<i>(54.8%)</i>	<i>(54.8%)</i>	<i>(54.8%)</i>	<i>(54.8%)</i>	<i>(54.8%)</i>	<i>(54.8%)</i>	<i>39.0%</i>

Two tenants move in in this first operational month, which results in a huge amount of TIs and LCs, as well as ~9 months of "Free Rent," which takes the Effective Gross Income down to much lower levels.

This is a monthly model, so we must divide the annual numbers by 12 to determine the appropriate values.

More Complex Model, Part 4 – Debt Service

The refinancing of the Construction Loan and new Permanent Loan are similar to the mechanics in the simplified model, but there are some significant differences.

First, there are two tranches of Debt: A **Senior Loan** and **Mezzanine**.

The Senior Loan is senior to the Mezzanine and is secured by collateral: the property itself. So, in a bankruptcy or other disaster, the Senior Lenders would be far more likely to recover their principal.

In exchange for this lower risk, the Senior Loan also offers a lower interest rate than the Mezzanine, and its principal amortizes over a long period.

The typical LTV for Senior Loans varies based on region, property type, and market conditions, but it's generally between 50% and 80%, and sometimes a bit lower if there's more risk.

If the investors want to go above that LTV range, they need **Mezzanine**, which is unsecured Debt that carries higher interest rates.

Mezzanine may add 10-20% to the LTV, perhaps taking it up to the 80-90% level.

Mezzanine does not amortize, its interest is often split between Cash and Accrued (also called PIK or "Paid-in-Kind"), and it tends to mature in less time, such as 2-5 years rather than 10+ years.

Its risk profile is closer to that of Equity, but in *most cases*, Mezzanine does not offer Equity ownership in properties (the screenshot below will contradict this statement):

Real Estate & REIT Modeling

Quick Reference Guide for Individual Properties

Permanent Loan Refinancing Assumptions:			
Refinancing Date:	Date	2019-04-30	
Refinancing Month #:	#	48	
Discount Rate:	%	20.0%	
Property Value 2 Years After Refi	£	£ 1,673,506,714	
Property Value @ Refinancing:	£	1,162,157,440	
Loan Issuance Fees:	%	1.00%	
Prepayment Penalty:	%	1.00%	
Senior Loan:			
Loan-to-Value (LTV) Ratio:	%	60.0%	
Senior Loan Amount:	£	697,294,464	
Interest Rate - Annual:	%	3.50%	
Interest Rate - Monthly:	%	0.287%	
Amortisation (Years):	# Years	30	
Tenor (Years):	# Years	10	
Maturity:	Date	2029-04-30	
Mezzanine:			
Loan-to-Value (LTV) Ratio:	%	10.0%	
Mezzanine Amount:	£	116,215,744	
Cash Interest Rate - Annual:	%	4.00%	
Cash Interest Rate - Monthly:	%	0.327%	
PIK Interest Rate - Annual:	%	4.00%	
PIK Interest Rate - Monthly:	%	0.327%	
Amortisation (Years):	# Years	N/A	
Tenor (Years):	# Years	5	
Maturity:	Date	2024-04-30	
Equity Percentage Upon Exit:	%	1.0%	

As in the simplified model, we need to move out one year past the refinancing and then discount the property's value back to get reasonable amounts for the Permanent Loan(s).

Very standard terms; we need the monthly interest rate because this is a monthly cash flow model. Since the interest rate is fixed and there's no interest-only period, we can use the IPMT and PPMT functions.

Most of these terms are fairly standard, but Mezzanine could easily be 100% Cash Interest rather than 50/50 Cash and PIK. It almost always has a fixed interest rate with no amortization.

The most unusual term here is the one at the bottom for 1% of the property's Equity upon exit - that is not common in real estate, but we're doing it here to demonstrate how it works in a model like this one.

These terms also create differences in the monthly Pro-Forma below the Adjusted NOI line:

Property Pro-Forma:	Units:	Apr-19	May-19	Jun-19	Jul-19	Aug-19	Sep-19	Oct-19	Nov-19	Dec-19	Jan-20
Adjusted Net Operating Income:	£	-	(930,526)	(932,253)	(933,984)	(935,717)	(937,454)	(939,193)	1,751,198	1,749,452	1,747,702
Adjusted NOI Margin:	%	N/A	(54.8%)	(54.8%)	(54.8%)	(54.8%)	(54.8%)	(54.8%)	39.0%	38.9%	38.8%
(-) Cash Interest Expense on Senior Loan:	£	-	(2,001,862)	(1,998,681)	(1,995,491)	(1,992,292)	(1,989,083)	(1,985,866)	(1,982,639)	(1,979,403)	(1,976,158)
(-) Cash Interest Expense on Mezzanine:	£	-	(380,460)	(381,706)	(382,955)	(384,209)	(385,467)	(386,729)	(387,995)	(389,265)	(390,539)
(-) Senior Loan Principal Repayment:	£	-	(1,107,964)	(1,111,145)	(1,114,335)	(1,117,534)	(1,120,742)	(1,123,960)	(1,127,186)	(1,130,422)	(1,133,668)
Cash Flow to Equity Investors:	£	-	(4,420,812)	(4,423,784)	(4,426,764)	(4,429,751)	(4,432,746)	(4,435,748)	(1,746,622)	(1,749,638)	(1,752,662)
PIK Interest on Mezzanine:	£	-	(380,460)	(381,706)	(382,955)	(384,209)	(385,467)	(386,729)	(387,995)	(389,265)	(390,539)
(+) Ending Senior Loan Balance:	£	697,294,464	696,186,500	695,075,356	693,961,021	692,843,488	691,722,746	690,598,786	689,471,600	688,341,178	687,207,510
(+) Ending Mezzanine Balance:	£	116,215,744	116,596,204	116,977,910	117,360,865	117,745,074	118,130,541	118,517,269	118,905,264	119,294,529	119,685,068
Total Permanent Loans:	£	813,510,208	812,782,704	812,053,266	811,321,886	810,588,562	809,853,286	809,116,055	808,376,864	807,635,706	806,892,578

The Senior Loan Principal Repayment also uses the standard PPMT function, but again, everything is converted to monthly numbers, and we need to check the refinancing date.

PIK Interest is based on a simple percentage times the beginning Mezzanine balance. It does NOT reduce Cash Flow to Equity Investors because it's non-cash!

The Senior Loan balance decreases in each period due to the amortization, while the Mezzanine balance increases because of the PIK Interest and lack of amortization.

The IPMT function still works, but we need to use the Monthly Interest Rate and convert all the years into months; we must also check to make sure we're past the refinancing date.

The Mezzanine Cash Interest is based on the Monthly Cash Interest Rate * Beginning balance since there's no amortization.

We could have skipped the IPMT and PPMT functions and calculated the Senior Loan Interest and Principal Repayments manually, but we chose to use the built-in functions.

More Complex Model, Part 5 – Returns Calculations

Many of the line items in the Returns Calculations stay the same, but everything is now on a monthly basis with variable exit dates, so we need to calculate the property's value *each month* and check the month against the exit date in many of the formulas.

Here's an early part of these Returns Calculations in this development model:

Returns to All Equity Investors:	Units:	May-15	Jun-15	Jul-15	Aug-15	Sep-15	Oct-15
Forward NOI:	£	-	-	-	-	-	-
Applicable Yield:	%	4.00%	4.00%	4.00%	4.00%	4.00%	4.00%
Implied Property Value:	£	-	-	-	-	-	-
(-) Equity Draws:	£	-£ 77,970,000	-£ 2,520,000	-£ 4,320,000	-£ 4,770,000	-£ 2,970,000	-£ 77,790,000
(+) Senior Loan Issued:	£	-	-	-	-	-	-
(+) Mezzanine Issued:	£	-	-	-	-	-	-
(-) Senior Loan Issuance Fees:	£	-	-	-	-	-	-
(-) Mezzanine Issuance Fees:	£	-	-	-	-	-	-
(-) Construction Loan Refinanced:	£	-	-	-	-	-	-
(+) Cash Flow to Equity Investors:	£	-	-	-	-	-	-
(+) Proceeds from Sale of Property:	£	-	-	-	-	-	-
(-) Selling Costs:	£	-	-	-	-	-	-
(+) Return of Replacement Reserves:	£	-	-	-	-	-	-
(-) Repayment of Senior Loan:	£	-	-	-	-	-	-
(-) Repayment of Mezzanine:	£	-	-	-	-	-	-
(-) Prepayment Penalty on Senior Loan:	£	-	-	-	-	-	-
(-) Prepayment Penalty on Mezzanine:	£	-	-	-	-	-	-
(-) Equity Granted to Mezzanine Investors:	£	-	-	-	-	-	-
Total Cash Flows to Equity Investors:	£	-£ 77,970,000	-£ 2,520,000	-£ 4,320,000	-£ 4,770,000	-£ 2,970,000	-£ 77,790,000
Internal Rate of Return (IRR):	%	19.0%					
Total Returns to Equity:	£	998,275,207					
Invested Equity:	£	448,644,011					
Cash-on-Cash Multiple:	x	2.23 x					

In the first period, all the cash flows are negative because they simply reflect Equity Draws.

XIRR since the schedule is monthly rather than annual.

Total Returns to Equity is the sum of all positive cash flows, and Invested Equity is the sum of all negative cash flows, with the signs flipped.

And here's the schedule when the refinancing takes place:

Returns to All Equity Investors:	Units:	Feb-19	Mar-19	Apr-19	May-19	Jun-19
Forward NOI:	£	3,401,379	7,161,777	10,920,415	15,713,561	20,506,474
Applicable Yield:	%	4.40%	4.40%	4.40%	4.60%	4.60%
Implied Property Value:	£	77,304,069	162,767,653	248,191,252	341,599,156	445,792,914
(-) Equity Draws:	£	-	-	-	-	-
(+) Senior Loan Issued:	£	-	-	697,294,464	-	-
(+) Mezzanine Issued:	£	-	-	116,215,744	-	-
(-) Senior Loan Issuance Fees:	£	-	-	(6,972,945)	-	-
(-) Mezzanine Issuance Fees:	£	-	-	(1,162,157)	-	-
(-) Construction Loan Refinanced:	£	-	-	(609,402,408)	-	-
(+) Cash Flow to Equity Investors:	£	-	-	-	(4,420,812)	(4,423,784)
(+) Proceeds from Sale of Property:	£	-	-	-	-	-
(-) Selling Costs:	£	-	-	-	-	-
(+) Return of Replacement Reserves:	£	-	-	-	-	-
(-) Repayment of Senior Loan:	£	-	-	-	-	-
(-) Repayment of Mezzanine:	£	-	-	-	-	-
(-) Prepayment Penalty on Senior Loan:	£	-	-	-	-	-
(-) Prepayment Penalty on Mezzanine:	£	-	-	-	-	-
(-) Equity Granted to Mezzanine Investors:	£	-	-	-	-	-
Total Cash Flows to Equity Investors:	£	£ -	£ -	£ 195,972,698	-£ 4,420,812	-£ 4,423,784
Internal Rate of Return (IRR):	%					

The property's value changes each month because the Forward NOI keeps changing.

The refinancing here generates a large positive cash flow because the Refinancing LTV is 70% and the property's value is much higher; the Construction Loan LTC was only 60% and the property's value was lower.

These initial cash flows are negative because of the high Concessions & Free Months of Rent in the beginning when the first two tenants move in; they turn positive once the Free-Rent period ends.

And here's the schedule at the end, upon exit:

Returns to All Equity Investors:	Units:	Jan-21	Feb-21	Mar-21	Apr-21	May-21	Jun-21
Forward NOI:	£	69,739,817	72,668,859	75,661,862	78,654,816	78,253,755	75,006,620
Applicable Yield:	%	4.70%	4.70%	4.70%	4.70%	4.80%	4.80%
Implied Property Value:	£	1,483,825,892	1,546,145,931	1,609,826,854	1,673,506,714	1,630,286,562	1,562,637,909
(-) Equity Draws:	£	-	-	-	-	-	-
(+) Senior Loan Issued:	£	-	-	-	-	-	-
(+) Mezzanine Issued:	£	-	-	-	-	-	-
(-) Senior Loan Issuance Fees:	£	-	-	-	-	-	-
(-) Mezzanine Issuance Fees:	£	-	-	-	-	-	-
(-) Construction Loan Refinanced:	£	-	-	-	-	-	-
(+) Cash Flow to Equity Investors:	£	330,838	327,515	324,185	320,845	425,818	-
(+) Proceeds from Sale of Property:	£	-	-	-	-	1,630,286,562	-
(-) Selling Costs:	£	-	-	-	-	(24,454,298)	-
(+) Return of Replacement Reserves:	£	-	-	-	-	2,196,105	-
(-) Repayment of Senior Loan:	£	-	-	-	-	(668,619,778)	-
(-) Repayment of Mezzanine:	£	-	-	-	-	(126,110,454)	-
(-) Prepayment Penalty on Senior Loan:	£	-	-	-	-	(6,686,198)	-
(-) Prepayment Penalty on Mezzanine:	£	-	-	-	-	(1,261,105)	-
(-) Equity Granted to Mezzanine Investors:	£	-	-	-	-	(8,057,767)	-
Total Cash Flows to Equity Investors:	£	330,838	327,515	324,185	320,845	797,718,886	-
Internal Rate of Return (IRR):	%						

The main "new" item here is the Equity Granted to Mezzanine Investors, which is based on the sum of everything above this bottom cell times the 1% equity they receive.

There are still the Proceeds from the Property Sale, Selling Costs, Repayment of the Permanent Loans, and Prepayment Penalties. We also assume that the new buyers must pay for the Replacement Reserves, so these count as a positive toward the cash flow.

We also calculate the Returns to Mezzanine Investors in this model since they receive a percentage of the Equity upon exit, and since the Accrued Interest may change the math slightly for them.

The original issuance of the Mezzanine counts as the "investment" here, and the issuance fees and cash interest count as positive cash flows.

The repayment of the Mezzanine principal, the prepayment penalty, and the Equity Granted all count toward the exit proceeds:

Returns to Mezzanine Investors:	Units:	Mar-19	Apr-19	May-19	Jun-19
(-) Mezzanine Issued:	£	-	-£ 116,215,744	-	-
(+) Cash Interest Received:	£	-	-	380,460	381,706
(+) Mezzanine Issuance Fees:	£	-	1,162,157	-	-
(+) Repayment of Mezzanine:	£	-	-	-	-
(+) Prepayment Penalty on Mezzanine:	£	-	-	-	-
(+) Equity Granted to Mezzanine Investors:	£	-	-	-	-
Total Cash Flows to Mezzanine Investors:	£	-£ 115,053,587	£ 380,460	£ 381,706	£ 381,706
Internal Rate of Return (IRR):	%	12.3%			
Total Returns:	£	145,324,036			
Invested Amount:	£	115,053,587			
Cash-on-Cash Multiple:	x	1.26 x			

Calculated with XIRR due to the monthly periods.

Mezzanine Issued upon refinancing counts as the initial investment, and Cash Interest and Issuance Fees are positive cash flows.

Returns to Mezzanine Investors:	Units:	Mar-21	Apr-21	May-21	Jun-21
(-) Mezzanine Issued:	£	£ -	£ -	£ -	£ -
(+) Cash Interest Received:	£	408,824	410,163	411,506	-
(+) Mezzanine Issuance Fees:	£	-	-	-	-
(+) Repayment of Mezzanine:	£	-	-	126,110,454	-
(+) Prepayment Penalty on Mezzanine:	£	-	-	1,261,105	-
(+) Equity Granted to Mezzanine Investors:	£	-	-	8,057,767	-
Total Cash Flows to Mezzanine Investors:	£	£ 408,824	£ 410,163	£ 135,840,831	£ -

The Equity Granted boosts the IRR significantly; without this item, the IRR would only be between 9% and 10%, just above the Total Interest Rate, primarily because of the Issuance and Repayment Fees.

By contrast, the PIK Interest makes a very small difference (~0.1%).

Finally, just as in the simplified model, there is also a waterfall returns schedule in this one.

The main difference here is that the cash flows are distributed proportionally (87.5% / 12.5%) up to a 15% IRR, but then between a 15% IRR and 2.0x multiple, the split changes to 77.5% / 22.5%, and then above a 2.0x multiple, it changes to 67.5% / 32.5%.

Beyond that, there's also a "Lookback" function that distributes proceeds back to the Investors if the Developer earns above a 20% IRR, but the Investors do not.

We do not cover the full details of this schedule here; for more, please see the section on [the Waterfall Returns Schedule](#).

Here is a summary of the output in the Base Case with an exit about 2 years after construction completion:



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Quick Reference Guide for Individual Properties

Returns Summary:	Units:	Construction Period:				Operational Period:									
		FY16	FY17	FY18	FY19	FY20	FY21	FY22	FY23						
Total Cash Flows to Mezzanine Investors:	£	£	-	£	-	£	115,053,587	£	4,648,630	£	4,834,575	£	135,840,831	£	-
Internal Rate of Return (IRR):	%		12.3%												
Total Returns:	£		145,324,036												
Invested Amount:	£		115,053,587												
Cash-on-Cash Multiple:	x		1.26 x												
Unleveraged Cash Flows:	£		(228,390,000)	(500,580,000)	(139,770,000)	(99,630,000)		9,074,516	46,221,168	1,611,975,519					
Internal Rate of Return (IRR):	%		12.9%												
Positive Cash Flows:	£		1,672,880,329												
Invested Amounts:	£		973,979,127												
Cash-on-Cash Multiple:	x		1.72 x												
Total Cash Flows to Equity Investors:	£		(228,390,000)	(186,847,056)	-	195,972,698	(32,892,019)	4,068,687	797,718,886						
Internal Rate of Return (IRR):	%		19.0%												
Positive Cash Flows:	£		998,275,207												
Invested Equity:	£		448,644,011												
Cash-on-Cash Multiple:	x		2.23 x												
Investor Cash Flow After Lookback:	£		(199,841,250)	(163,491,174)	-	171,476,110	(28,780,517)	3,560,101	736,884,854						
Internal Rate of Return (IRR):	%		20.0%												
Positive Cash Flows:	£		912,371,634												
Invested Equity:	£		392,563,510												
Cash-on-Cash Multiple:	x		2.32 x												
Developer Cash Flow After Lookback:	£		(28,548,750)	(23,355,882)	-	24,496,587	(4,111,502)	508,586	60,834,033						
Internal Rate of Return (IRR):	%		10.2%												
Positive Cash Flows:	£		85,903,573												
Invested Equity:	£		56,080,501												
Cash-on-Cash Multiple:	x		1.53 x												

Leverage significantly boosts all these metrics by reducing the upfront Equity Investment.

Initially, the Developers earn an IRR of 23% and the Investors earn an IRR of 18% - but the Lookback function changes that and redistributes the proceeds such that the Developers' IRR falls and the Investors' IRR rises.

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If you're building a monthly model, you'll always create an annual summary like the one above, which also shows the Pro-Forma, Debt Service, and Credit Stats and Ratios on an annual basis.

Should We Pursue This Deal?

In this deal, the case study prompt tells us that the Equity Investors are targeting a 20-25% IRR on property developments, with a 5% minimum IRR in Downside cases. The Mezzanine Investors are targeting a 10% IRR across all cases.

Based on that, this deal looks quite good for the Mezzanine Investors no matter what happens:

Sensitivity Analyses:											
Mezzanine IRR - Exit Cap Rate and Operating Scenario:											
Operating Scenario:		Exit Prime Office Yield (Exit Date of May 21):									
		4.00%	4.20%	4.40%	4.60%	4.80%	5.00%	5.20%	5.40%	5.60%	
Upside		12.3%	12.0%	11.7%	11.5%	11.2%	11.0%	10.9%	10.7%	10.5%	
Base		13.5%	13.2%	12.8%	12.6%	12.3%	12.0%	11.8%	11.6%	11.4%	
Downside		15.2%	14.7%	14.4%	14.0%	13.7%	13.4%	13.1%	12.9%	12.6%	
Mezzanine IRR - Exit Date and Operating Scenario:											
Operating Scenario:		Exit Date:									
		May-21	Jun-21	Jul-21	Aug-21	Sep-21	Oct-21	Nov-21	Dec-21	Jan-22	Apr-22
Upside		12.3%	11.9%	11.5%	11.2%	10.8%	10.5%	10.4%	10.3%	10.2%	9.9%
Base		12.3%	11.9%	11.5%	11.1%	10.8%	10.5%	10.2%	10.0%	9.8%	9.5%
Downside		12.6%	12.2%	11.7%	11.3%	11.0%	10.6%	10.3%	10.0%	9.7%	9.1%



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The returns fall a bit short of the 10% IRR with a late exit date in the Downside Case, but 9% is still close to 10%, and they don't even come close to losing money.

However, things look significantly worse for the Investors and Developers with later exit dates:

Investor IRR - Exit Date and Operating Scenario:

Operating Scenario:		Exit Date:											
		May-21	Jun-21	Jul-21	Aug-21	Sep-21	Oct-21	Nov-21	Dec-21	Jan-22	Feb-22	Mar-22	Apr-22
Upside		28.2%	26.8%	25.4%	24.0%	22.6%	21.2%	20.7%	20.3%	19.8%	19.3%	18.9%	18.4%
Base		20.0%	18.7%	17.3%	15.6%	13.8%	11.9%	10.0%	7.9%	6.7%	5.9%	5.1%	4.2%
Downside		7.1%	5.4%	3.6%	1.8%	(0.1%)	(2.2%)	(4.4%)	(6.8%)	0.0%	0.0%	0.0%	0.0%

Developer IRR - Exit Date and Operating Scenario:

Operating Scenario:		Exit Date:											
		May-21	Jun-21	Jul-21	Aug-21	Sep-21	Oct-21	Nov-21	Dec-21	Jan-22	Feb-22	Mar-22	Apr-22
Upside		34.6%	30.9%	27.1%	22.8%	18.0%	12.2%	10.2%	8.2%	6.1%	3.8%	1.4%	(0.7%)
Base		10.2%	3.5%	(4.4%)	(9.3%)	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Downside		0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%

So, we would **recommend against this deal** for the Equity Investors, while we would recommend it for the Mezzanine Investors.

For the deal to make sense for the Equity Investors, something would have to change to improve the property's performance in scenarios with later exit dates:

- Gradual sell-off of their stake and an earlier start date for this sale process; or
- A lower LTV for the Permanent Loan refinancing; or
- Longer-term leases that result in smoother EGI and NOI in exchange for lower rent or lower rental escalations; or
- A target of 5-10 tenants in the property rather than only 3 to provide better staggering for the lease expirations.

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Strategy #2: Develop a Property and Pre-Sell Individual Units ("Opportunistic")

We're not going to cover this strategy in detail because it's the same as [what was described above in the section on Condominiums](#).

The model there was a simplified one, but the more complex one in the course (for a pre-sold condo development in Brazil) uses the same principles.

There are still multiple phases, there are still Inflation Multipliers for Sales Prices and Expenses, a certain percentage of development costs are still paid each month, and Developer and Investor Equity still funds the project before the Construction Loan Draws.

We'll highlight below the **main differences and additions** to this model:

Addition #1 – Scenarios

Investing is always probabilistic, so, as with the other property and deal types, you *have to* consider different market scenarios to make a real decision.

The scenarios here differ regarding the units sold each month (“Sales Velocity”), the annual sales price inflation, and the annual expense inflation:

Assumption:		Selected Case:	Operating Cases:		
			Downside	Base	Upside
Sales Velocity per Month:	# Units	1.0	0.5	1.0	1.5
Annual Sales Price Inflation:	%	2.5%	1.0%	2.5%	4.0%
Annual Expense Inflation:	%	2.5%	2.0%	2.5%	3.0%

Faster sales, more price increases, but also faster expense growth in the Upside Case; the opposite in the Downside Case.

Addition #2 – Different Trigger Conditions for Debt/Equity Draws and Sale Phases

In this model, we don't set the Equity and Debt Draws based on the total amount of funding required and the targeted Loan to Cost (LTC) percentage.

Instead, **50%** of the total units in the development must be pre-sold before the sponsors can draw on the Construction Loan.

This assumption is often tied to the **region** of the property and the regulations and norms there.

In Brazil, Debt and Equity Draws are frequently based on the percentage of units pre-sold. But in other regions, they might be based on time (e.g., X months must pass to draw on the Construction Loan) or a percentage of total funding.

We do **not** assume in this model that Construction begins a fixed number of months after the beginning of a phase.

Instead, we assume that a certain percentage of units must be pre-sold in each phase for Construction to begin *in that phase*.

Then, once another percentage of units are pre-sold, the next pre-sales phase begins.

Here are the relevant assumptions:

Financial Assumptions:		Units:
Lot Price per Square Meter:	R\$ / sq. m.	R\$ 3,000
Lot Asking Price:	R\$	R\$ 16,007,500
Lot Upfront Payment %:	%	50.0%
Month # for Upfront Payment:	Month #	1
Month # for Remainder of Payment:	Month #	6
% of Units Pre-Sold Before Loan Draw:	%	50.0%
Construction Loan Interest Rate:	%	10.00%
Upfront Client Payment:	%	30.0%
Client Payment Upon Construction End:	%	30.0%
Client Payment Upon Building Move-In:	%	40.0%
% Units Pre-Sold for Construction to Begin:	%	30.0%
% Units Pre-Sold for Next Phase to Begin:	%	70.0%
# Months in Construction Period:	# Months	24
Condo Selling Price per Square Meter:	R\$ / sq. m.	R\$ 28,000
Hard Costs per Gross Square Meter:	R\$ / sq. m.	15,000
Soft Costs per Gross Square Meter:	R\$ / sq. m.	3,500
FF&E and Move-In Costs % Monthly Sales:	%	10.0%

Key difference - the Construction Loan Draws are based on the percentage of total units pre-sold rather than time or the percent of total funding.

Another key difference - the start of the Construction phase and the next phase in the process are based on the % of units pre-sold in THIS phase.

If Phase 1 has 10 units, then Construction will begin when 3 units are pre-sold, and Phase 2 will begin when 7 units are pre-sold.

You can see the impact in the Construction Timeline (the first ~10 months are shown below):

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Construction Timeline:	Units:	Jan 18	Feb 18	Mar 18	Apr 18	May 18	Jun 18	Jul 18	Aug 18	Sep 18	Oct 18
Absolute Year #:	#	1	1	1	1	1	1	1	1	1	1
Absolute Month #:	#	1	2	3	4	5	6	7	8	9	10
% of Total Building Units Sold:	%	3.0%	6.1%	9.1%	12.1%	15.2%	18.2%	21.2%	27.3%	33.3%	39.4%
Total Monthly Units Sold:	#	1.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0	2.0	2.0
Sales Price Inflation:	%	0.21%	0.21%	0.21%	0.21%	0.21%	0.21%	0.21%	0.21%	0.21%	0.21%
Sales Inflation Multiplier:	#	1.00	1.00	1.01	1.01	1.01	1.01	1.01	1.02	1.02	1.02
Expense Inflation:	%	0.21%	0.21%	0.21%	0.21%	0.21%	0.21%	0.21%	0.21%	0.21%	0.21%
Expense Inflation Multiplier:	#	1.00	1.00	1.01	1.01	1.01	1.01	1.01	1.02	1.02	1.02
% of Soft Costs Incurred:	%	7.0%	2.0%	3.0%	1.0%	2.0%	1.0%	8.0%	4.0%	2.0%	3.0%
Phase I:											
Monthly Units Sold:	#	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Cumulative Units Sold:	#	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0
Cumulative Percentage Sold:	%	10.0%	20.0%	30.0%	40.0%	50.0%	60.0%	70.0%	80.0%	90.0%	100.0%
Total Value of Condo Units Sold:	R\$	R\$ 15,852,587	R\$ 15,885,240	R\$ 15,917,961	R\$ 15,950,750	R\$ 15,983,606	R\$ 16,016,529	R\$ 16,049,521	R\$ 16,082,580	R\$ 16,115,708	R\$ 16,148,903
Status (1 = Pre, 2 = Constr, 3 = Post-Constr):	#	1	1	1	2	2	2	2	2	2	2
Phase II:											
Monthly Units Sold:	#								1.0	1.0	1.0
Cumulative Units Sold:	#								1.0	2.0	3.0
Cumulative Percentage Sold:	%								9.1%	18.2%	27.3%
Total Value of Condo Units Sold:	R\$								R\$ 16,424,157	R\$ 16,457,988	R\$ 16,491,889
Status (1 = Pre, 2 = Constr, 3 = Post-Constr):	#								1	1	1

Past 30% of Units Pre-Sold, the Construction Phase ("2") begins.

Once 70% of Units in Phase 1 are pre-sold, Phase 2 begins. Construction in Phase 2 begins once 30% are pre-sold.

The Initial, Construction-End, and Final Deposits are quite similar to those in the simplified case study, so relatively little changes on the revenue side ("Total Monthly Gross Sales").

The differences are more visible with the expenses:

Monthly Cash Flows:	Units:	Jan 18	Feb 18	Mar 18	Apr 18	May 18	Jun 18
Monthly Gross Sales:							
(+) Phase I - Initial Deposits:	R\$	R\$ 4,755,776	R\$ 4,765,572	R\$ 4,775,388	R\$ 4,785,225	R\$ 4,795,082	R\$ 4,804,959
(+) Phase I - Construction-End Deposits:	R\$	-	-	-	-	-	-
(+) Phase I - Final Deposits:	R\$	-	-	-	-	-	-
Phase I Total Sales:	R\$	4,755,776	4,765,572	4,775,388	4,785,225	4,795,082	4,804,959
(+) Phase II - Initial Deposits:	R\$	-	-	-	-	-	-
(+) Phase II - Construction-End Deposits:	R\$	-	-	-	-	-	-
(+) Phase II - Final Deposits:	R\$	-	-	-	-	-	-
Phase II Total Sales:	R\$	-	-	-	-	-	-
(+) Phase III - Initial Deposits:	R\$	-	-	-	-	-	-
(+) Phase III - Construction-End Deposits:	R\$	-	-	-	-	-	-
(+) Phase III - Final Deposits:	R\$	-	-	-	-	-	-
Phase III Total Sales:	R\$	-	-	-	-	-	-
Total Monthly Gross Sales:	R\$	4,755,776	4,765,572	4,775,388	4,785,225	4,795,082	4,804,959
Monthly Expenses:							
(-) Hard Costs:	R\$	-	-	-	(8,494,631)	(8,512,129)	(8,529,662)
(-) Soft Costs:	R\$	(5,239,888)	(1,500,195)	(2,254,927)	(753,191)	(1,509,484)	(756,297)
(-) FF&E and Move-In Costs:	R\$	(475,578)	(476,557)	(477,539)	(478,522)	(479,508)	(480,496)
Total Monthly Expenses:	R\$	(5,715,465)	(1,976,752)	(2,732,466)	(9,726,344)	(10,501,121)	(9,766,455)
(-) Land Acquisition Costs:	R\$	(8,003,750)	-	-	-	-	(8,003,750)
Gross Income:	R\$	(8,963,439)	2,788,820	2,042,922	(4,941,119)	(5,706,039)	(12,965,246)
(-) Cash Interest Expense:	R\$	-	-	-	-	-	-
Cash Flow After Interest, Before Draws:	R\$	-R\$ 8,963,439	R\$ 2,788,820	R\$ 2,042,922	-R\$ 4,941,119	-R\$ 5,706,039	-R\$ 12,965,246

Hard Costs begin as soon as Construction in Phase 1 begins and end once Phase 3 Construction ends.

Soft Costs, as always, follow a somewhat random pattern of percentages, and FF&E and Move-In Costs are linked to Monthly Gross Sales.

As always, there's an Initial Deposit and Final Deposit for the Land Acquisition.

The Debt and Equity schedule is almost the same, but the switch to Construction Loan Draws is based on the percentage of total units pre-sold instead.

Addition #3 – More Complex Waterfall

The waterfall returns schedule here is based on mixed multiple and IRR criteria, as well as a preferred return and catch-up return.

This setup is common for development deals because the investors' objective is to *avoid losing money* above all else – so a Preferred Return based on a 1x multiple or a low, positive IRR is normal.

The Catch-Up Return is distributed after the Preferred Return and goes to the Developers to “catch them up” to the 1x level.

If the Equity Multiple in the deal is above 1x, the proceeds are split according to the schedule below:

Units:			
Investor Equity Contribution:	%	80.0%	
Developer Equity Contribution:	%	20.0%	
Investor / Developer Returns Split:		Low:	High:
Limited Partners - Preferred Return:	x		1.00 x
Developers - Catch-Up Return:	x		1.00 x
IRR / Multiple Between:	%	1.00 x	20.0%
Limited Partners (Investors):	%		80.0%
Cyrela Brazil Realty (Developers):	%		20.0%
IRR / Multiple Between:	%	20.0%	3.00 x
Developer Promote:	%		10.0%
Limited Partners (Investors):	%		70.0%
Cyrela Brazil Realty (Developers):	%		30.0%
IRR / Multiple Between:	%	3.00 x	
Developer Promote:	%		10.0%
Limited Partners (Investors):	%		60.0%
Cyrela Brazil Realty (Developers):	%		40.0%

First, let's make sure that both groups earn their invested equity back...

Then, up to a 20% IRR, returns are split proportionally to Equity contributions.

If the IRR goes above 20%, then the Developer receives a higher percentage of the proceeds based on the equity multiple.

And here's the first part of the waterfall returns schedule so you can get a sense of the cash flows:

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Waterfall Returns Schedule:	Units:	Jan 18	Feb 18	Mar 18	Apr 18	May 18	Jun 18
Cash Flow to Equity Investors:							
(-) Invested Equity:	R\$	-R\$ 8,963,439	R\$ -	R\$ -	-R\$ 4,941,119	-R\$ 5,706,039	-R\$ 12,965,246
(+) Net Cash Flow After Debt Service:	R\$	-	2,788,820	2,042,922	-	-	-
Total Cash Flow to Equity Investors:	R\$	(8,963,439)	2,788,820	2,042,922	(4,941,119)	(5,706,039)	(12,965,246)
Investors - Preferred Return of 1.0x Invested Equity:							
(+) Beginning Balance:	R\$	-	7,170,752	4,381,931	2,339,009	6,291,904	10,856,736
(+) Investor Injections:	R\$	7,170,752	-	-	3,952,896	4,564,832	10,372,197
(+) Investor Accruals:	R\$	1.00 x	-	-	-	-	-
(-) Investor Preferred Distribution:	R\$	-	(2,788,820)	(2,042,922)	-	-	-
Ending Balance:	R\$	7,170,752	4,381,931	2,339,009	6,291,904	10,856,736	21,228,933
Cash Flow Avail. for Catch-Up Distributions:	R\$	-	-	-	-	-	-
Developers - Catch-Up Return of 1.0x Invested Equity:							
(+) Beginning Balance:	R\$	-	1,792,688	1,792,688	1,792,688	2,780,912	3,922,120
(+) Investor Injections:	R\$	1,792,688	-	-	988,224	1,141,208	2,593,049
(+) Investor Accruals:	R\$	1.00 x	-	-	-	-	-
(-) Developer Catch-Up Distribution:	R\$	-	-	-	-	-	-
Ending Balance:	R\$	1,792,688	1,792,688	1,792,688	2,780,912	3,922,120	6,515,169
Cash Flow Avail. for Tier 1 Distributions:	R\$	-	-	-	-	-	-
Tier 1 - 1.0x Multiple Up to 20.0% IRR:							
(+) Beginning Balance:	R\$	-	8,963,439	6,311,845	4,365,553	9,373,507	15,223,050
(+) Investor Injections:	R\$	8,963,439	-	-	4,941,119	5,706,039	12,965,246
(+) Investor Accruals:	R\$	20.0%	-	137,226	96,631	66,834	143,503
(-) Preferred and Catch-Up Distributions:	R\$	-	(2,788,820)	(2,042,922)	-	-	-
(-) Tier 1 Accrual Distribution:	R\$	-	-	-	-	-	-
Ending Balance:	R\$	8,963,439	6,311,845	4,365,553	9,373,507	15,223,050	28,421,353
Investor Cash Flow:	R\$	80.0%	-	-	-	-	-
Developer Cash Flow:	R\$	20.0%	-	-	-	-	-

Investors "take" everything up to a 1.0x return to earn this 1.0x Preferred Return. Once that is achieved, the Developers can earn some cash flows.

No cash flow is left to pay the Developers in these early months, so the amount they're owed for the Catch-Up Return keeps increasing.

This tier is a 1.0x multiple up to a 20% IRR, so we accrue 20% on a monthly basis and subtract the Preferred and Catch-Up Distributions, which represent 1.0x.

Addition #4 – Annual Summary and Sensitivities

You can use SUMIF functions based on the year numbers to create the annual summary:

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Annual Financial Summary:		Units:	FY 19	FY 20	FY 21	FY 22
Year Number:	#		1	2	3	4
Year End Date:	Date		Jan 19	Jan 20	Jan 21	Jan 22
Total Condo Units Sold:	#		15	16	2	-
Cumulative % Total Units Sold:	%		45.5%	93.9%	100.0%	100.0%
Total Gross Sales:	R\$		R\$ 72,738,953	R\$ 82,575,260	R\$ 113,403,159	R\$ 284,465,836
Expenses:						
(-) Hard Costs:	R\$		(77,084,630)	(105,025,495)	(107,651,133)	(45,645,116)
(-) Soft Costs:	R\$		(29,516,661)	(17,865,216)	(22,203,200)	(8,228,716)
(-) FF&E and Move-In Costs:	R\$		(7,273,895)	(8,257,526)	(11,340,316)	(28,446,584)
Total Expenses:	R\$		(113,875,186)	(131,148,237)	(141,194,649)	(82,320,416)
(-) Land Acquisition Costs:	R\$		(16,007,500)	-	-	-
Gross Income:	R\$		(57,143,733)	(48,572,977)	(27,791,490)	202,145,420
(-) Cash Interest Expense:	R\$		-	-	(1,256,618)	(1,532,507)
Cash Flow After Interest, Before Draws:	R\$		(57,143,733)	(48,572,977)	(29,048,108)	200,612,912
Equity & Debt Draws:						
(+) Developer Equity:	R\$		12,395,095	988,060	-	-
(+) Investor Equity:	R\$		49,580,380	3,952,239	-	-
(+) Construction Loan:	R\$		-	43,632,678	99,165,556	36,478,484
Total Equity & Debt Draws:	R\$		61,975,475	48,572,977	99,165,556	36,478,484
Construction Loan Repayments:	R\$		-	-	(70,117,448)	(120,918,609)
Cash Flow After Draws & Debt Repayment	R\$		R\$ 4,831,743	R\$ 0	R\$ -	R\$ 116,172,787
Unleveraged Cash Flows:	R\$		(57,143,733)	(48,572,977)	(27,791,490)	202,145,420
Internal Rate of Return (IRR):	%	17.2%				
Cash-on-Cash Multiple:	x	1.28 x				
Total Cash Flows to Equity Investors:	R\$		(57,143,733)	(4,940,299)	-	116,172,787
Internal Rate of Return (IRR):	%	20.9%				
Cash-on-Cash Multiple:	x	1.81 x				
Investor Cash Flow:	R\$		(44,748,638)	(3,952,239)	-	91,687,515
Internal Rate of Return (IRR):	%	21.2%				
Cash-on-Cash Multiple:	x	1.80 x				
Developer Cash Flow:	R\$		(12,395,095)	(988,060)	-	24,485,272
Internal Rate of Return (IRR):	%	19.9%				
Cash-on-Cash Multiple:	x	1.83 x				

You can also create sensitivity tables based on the scenarios and key assumptions, such as the Sales Velocity, Construction Months, Condo Selling Prices, and Hard Costs:



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Sensitivity Analyses:

Investors - Equity IRR and Scenario vs. Sales Velocity:

Scenario:		Sales Velocity per Month:										
		0.5	0.6	0.7	0.8	0.9	1.0	1.1	1.2	1.3	1.4	1.5
	Downside	10.2%	12.1%	13.2%	15.5%	17.9%	19.7%	21.5%	25.3%	28.4%	29.4%	38.4%
	Base	12.1%	14.0%	14.9%	17.1%	19.5%	21.2%	22.9%	26.6%	29.8%	30.8%	40.0%
	Upside	14.1%	15.8%	16.6%	18.6%	21.0%	22.6%	24.3%	28.0%	31.2%	32.2%	41.7%

Developers - Equity IRR and Scenario vs. Sales Velocity:

Scenario:		Sales Velocity per Month:										
		0.5	0.6	0.7	0.8	0.9	1.0	1.1	1.2	1.3	1.4	1.5
	Downside	8.6%	11.1%	12.3%	14.7%	16.7%	18.0%	20.9%	26.1%	30.0%	31.0%	38.5%
	Base	10.3%	12.7%	13.8%	16.3%	18.2%	19.9%	23.1%	28.1%	32.0%	32.9%	40.4%
	Upside	11.9%	14.3%	15.3%	17.8%	20.0%	22.0%	25.2%	30.1%	34.0%	34.8%	42.2%

Investors - Equity IRR and Scenario vs. Months in Construction Period:

Scenario:		Months in Construction Period:										
		20	21	22	23	24	25	26	27	28	29	30
	Downside	11.7%	11.3%	10.9%	10.5%	10.2%	9.9%	9.6%	9.4%	9.1%	8.9%	8.6%
	Base	22.8%	22.4%	22.0%	21.6%	21.2%	20.7%	20.3%	19.9%	19.4%	18.9%	18.4%
	Upside	42.4%	42.3%	42.0%	41.9%	41.7%	42.4%	43.3%	44.4%	45.8%	47.6%	49.8%

Developers - Equity IRR and Scenario vs. Months in Construction Period:

Scenario:		Months in Construction Period:										
		20	21	22	23	24	25	26	27	28	29	30
	Downside	8.8%	8.6%	8.7%	8.7%	8.6%	8.5%	8.4%	8.4%	8.2%	8.1%	7.9%
	Base	22.7%	22.0%	21.3%	20.6%	19.9%	19.2%	18.5%	18.0%	17.5%	17.1%	16.6%
	Upside	46.9%	45.8%	44.5%	43.4%	42.2%	40.9%	39.7%	38.5%	37.3%	36.0%	34.8%

Should We Pursue This Deal?

This decision depends on the IRRs our firm is targeting in different cases.

For example, if we're targeting a 20% IRR in the Base Case, a 35% IRR in the Upside Case, and we want to avoid losing money in the Downside Case, this deal would make sense because the IRR to the LPs is 21% in the Base Case, 42% in the Upside Case, and 10% in the Downside Case.

Even with a significantly slower Sales Velocity, such as 0.5 rather than 1.0 in the Base Case, or a much longer Construction Period, such as 30 months rather than 24 months, the deal is not a disaster.

And even if Condo Prices fell by 10%, the LPs would still not lose money in the Downside Case; if Hard Costs rose by 10-15%, the LPs would still avoid losing money.

In an Extreme Downside Case, where Condo Prices fall by 5% per year and Expenses also fall by 2% per year, the LPs would still earn a ~7% IRR.

And they would avoid losing money even when the initial Condo Price per square meter is R\$ 26,000 – R\$ 27,000 instead of R\$ 28,000.



Finally, the waterfall returns structure in this deal heavily favors the LPs because of the 1x Preferred Return. This provision limits the downside risk for the LPs, especially in more pessimistic scenarios.

So, assuming the IRRs here are within our targeted ranges, we would recommend this deal because the deal terms greatly limit risk for us (the Limited Partners).

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Strategy #3: Acquire an Existing Property and Then Sell It (“Core” and “Core-Plus”)

This strategy is the most basic and least risky one because **very little about the property changes**.

You use Debt and Equity to purchase a property based on a per-square-foot or per-square-meter figure, or a Cap Rate, you operate it for several years, and then you sell it to someone else.

If it's a “Core” deal, then nothing should change – perhaps just low-single-digit rent and expense increases and minor changes in the Occupancy Rate.

If it's a “Core-Plus” deal, then there will be minor changes. For example, if the In-Place Rents are a 10% discount to Market Rates, then the sponsor may upgrade the units to command higher rents.

“Upgrade” does **not** mean “full renovation,” which falls under [Strategy #4 below](#).

These minor upgrades might consist of improving the amenities or painting the interiors of units, but they would **not** be enough to move the property from “Class B” into “Class A” (for example).

In practice, there are relatively few true “Core” deals because most investors want to improve properties to boost their potential returns.

In the examples in this course, *something* always changes, even if it's fairly minor:

- The investors/owners offer the tenants lower Rental Escalations, such as 2% per year rather than 4%, in exchange for agreeing to longer-term leases, such as 10 years instead of 3-4 years.
- The investors/owners perform minor unit upgrades to boost In-Place Rents.
- The investors/owners increase Rents and accept a lower Occupancy Rate as the cost of higher Rents – under the logic that Effective Gross Income might be higher with the right numbers.



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Quick Reference Guide for Individual Properties

We'll start by looking at a simple acquisition of a multifamily property in Arizona and then move to a slightly more complex one for an office/retail property in New York:

Simplified Multifamily Acquisition Model, Step 1 – Property, Acquisition, and Exit Assumptions

Our starting assumptions are shown below:

Arcadia Gardens - Investment Analysis Model

(\$ USD in \$ as Stated, Unless Otherwise Noted)

Property Assumptions:		Units:		Units:
Building / Construction Name:	Name	Arcadia Gardens	Property Taxes % Property Value:	% 0.68%
Location:	Name	Phoenix, AZ	Property Management Fee % EGI:	% 3.00%
Months in Year:	#	12		
Number of Apartment Units:	#	76	Rentable Square Footage (RSF):	sq. ft. =+Apt_Units*RSF_per_Unit
Average Rentable Square Feet per Unit:	sq. ft.	573 sq. ft.	Rentable to Gross Square Feet Ratio:	% 80.0%
Parking Spots per Unit:	#	1.13	Gross Square Footage (GSF):	sq. ft. 54,435 sq. ft.
Parking Spots:	#	86		

Very standard assumptions for multifamily - we determine the space based on # of units and avg. size per unit.

The Acquisition and Exit assumptions are as follows:

Acquisition and Exit Assumptions:			Units:	Units:		
Acquisition Date:	Date	2017-12-31		Exit Date:	Date	2022-12-31
Acquisition Price:	\$	\$ 9,775,000		Exit Cap Rate:	%	6.00%
Implied Going-In Cap Rate:	%	5.80%		Exit Price:	\$	\$ 12,002,242
Acquisition Price per Unit:	\$ / Unit	\$ 128,618		Exit Price per Unit:	\$ / Unit	\$ 157,924
Acquisition Price per RSF:	\$ / sq. ft.	\$ 224.46		Exit Price per RSF:	\$ / sq. ft.	\$ 275.61
Acquisition Costs (% Gross Acquisition Price):	%	1.0%		Selling Costs (% Gross Sale Price):	%	2.0%
Loan Issuance Fees (% Senior Debt):	%	1.0%				
Acquisition Loan-to-Value (LTV) Ratio:	%	65.0%				
Loan Interest Rate:	%	5.0%				
Loan Amortization Period:	# Years	30				
Loan Maturity:	# Years	5				

Especially in Core deals, the Exit Cap Rate should be higher than the Going-In Cap Rate because older properties become less appealing vs. the competition over time; also, your analysis will be more conservative, and therefore more credible, with this assumption

Especially in Core deals, the Exit Cap Rate should be higher than the Going-In Cap Rate because older properties become less appealing vs. the competition over time; also, your analysis will be more conservative, and therefore more credible, with this assumption.

The LTV is sometimes a bit lower for Core deals since they're not quite as dependent on leverage.

Using these assumptions, we can create a Sources & Uses schedule that shows where the money is coming from and where it's going to:

Sources and Uses of Funds:

Sources of Funds:	
Senior Debt:	\$ 6,353,750
Investor Equity:	3,601,538
Total Sources:	\$ 9,955,288

Uses of Funds:	
Acquisition Price:	\$ 9,775,000
Acquisition Costs:	97,750
Loan Issuance Fees:	63,538
Replacement Reserves:	19,000
Total Uses:	\$ 9,955,288

"Total Sources" does not equal the Acquisition Price because of the fees and reserves; this is also why the Investor Equity is not 35% of total Sources.

Acquisition Costs are a % of the Acquisition Price, and Loan Issuance Fees are a % of the Debt used.

Reserves are based on a \$ per unit figure in the Operating Assumptions.

We allocate something to the Replacement Reserves under Uses because we "purchase" them with this property, which already has modest Reserves.

Simplified Multifamily Acquisition Model, Step 2 – Operating Assumptions

We covered this part in [the section on Multifamily properties above](#), so we're not going to repeat everything. As a reminder, here are the key points:

Operating Assumptions:	Units:	Historical:		Projected:					Stabilized Year:
		FY17	FY18	FY19	FY20	FY21	FY22		
Market Rent per RSF per Month:	\$ / RSF / Mo.	\$ 1.78	\$ 1.84	\$ 1.90	\$ 1.97	\$ 2.03	\$ 2.09	\$ 2.15	
In-Place Rent per RSF per Month:	\$ / RSF / Mo.	1.60	1.70	1.81	1.92	2.01	2.07	2.13	
Rental & Parking Income Growth Rate:	%		3.5%	3.5%	3.5%	3.0%	3.0%	3.0%	
In-Place Rent Discount to Market Rent:	%	10.0%	7.5%	5.0%	2.5%	1.0%	1.0%	1.0%	
Bad Debt & Concessions % Effective Rent:	%	(3.0%)	(3.0%)	(2.0%)	(2.0%)	(1.0%)	(1.0%)	(1.0%)	
Parking Fees per Spot per Month:	\$ / Spot / Mo.	50.00	51.75	53.56	55.44	57.10	58.81	60.58	
Utility Reimbursements % Utility Expense:	%	80.0%	83.0%	86.0%	89.0%	92.0%	95.0%	95.0%	
General Vacancy:	%	-	(1.0%)	(2.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)	
Insurance per Gross Square Foot per Year:	\$ / GSF / Yr.	0.40	0.41	0.42	0.44	0.45	0.46	0.48	
Utilities per Unit per Month:	\$ / Unit / Mo.	85.00	87.55	90.18	92.88	95.67	98.54	101.49	
Sales, Marketing & Administrative % EGI:	%	19.0%	19.2%	19.4%	19.6%	19.8%	20.0%	20.0%	
Replacement Reserves per Unit per Year:	\$ / Unit / Yr.	250.00	257.50	265.23	273.18	281.38	289.82	298.51	
Property Tax Growth Rate:	%		2.0%	2.0%	2.0%	2.0%	2.0%	2.0%	
Operating Expense Growth Rate:	%		3.0%	3.0%	3.0%	3.0%	3.0%	3.0%	
Capital Expenditures per Unit per Year:	\$ / Unit / Yr.	-	500.00	1,000.00	750.00	-	200.00	-	
Replacement Reserves:	\$	-	38,000	19,570	-	-	21,385	28,211	50,898

The basic idea here is that Market Rent will increase at a low single-digit rate, but the In-Place Rent Discount will drop, Concessions will fall, and Utility Reimbursements will increase as we spend CapEx on minor upgrades.

However, we'll also accept a higher Vacancy Rate in exchange for all of that.

Simplified Multifamily Acquisition Model, Step 3 – Property Pro-Forma

We covered this step in [the Multifamily section](#), so we're not going to repeat everything here. The parts we did *not* cover previously all relate to the Debt Service:



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Quick Reference Guide for Individual Properties

Property Pro-Forma:	Units:	Historical:	Projected:					Stabilized
		FY17	FY18	FY19	FY20	FY21	FY22	Year:
Revenue:								
(+) Base Rental Income:	\$	\$ 929,024	\$ 961,540	\$ 995,194	\$ 1,030,026	\$ 1,060,926	\$ 1,092,754	\$ 1,125,537
(-) Loss to Lease:	\$	(92,902)	(72,115)	(49,760)	(25,751)	(10,609)	(10,928)	(11,255)
(-) Bad Debt & Concessions:	\$	(25,084)	(26,683)	(18,909)	(20,085)	(10,503)	(10,818)	(11,143)
(+) Parking Income:	\$	51,528	53,331	55,198	57,130	58,844	60,609	62,428
(+) Utility Reimbursements:	\$	62,016	66,272	70,727	75,390	80,269	85,374	87,935
Potential Gross Revenue:	\$	924,582	982,345	1,052,451	1,116,710	1,178,927	1,216,991	1,253,501
(-) General Vacancy:	\$	-	(9,823)	(21,049)	(33,501)	(35,368)	(36,510)	(37,605)
Effective Gross Income (EGI):	\$	924,582	972,521	1,031,402	1,083,208	1,143,559	1,180,481	1,215,896
EGI Growth Rate:	%	N/A	5.2%	6.1%	5.0%	5.6%	3.2%	3.0%
Expenses:								
(-) Property Taxes:	\$	(66,470)	(67,799)	(69,155)	(70,538)	(71,949)	(73,388)	(74,856)
(-) Insurance:	\$	(21,774)	(22,427)	(23,100)	(23,793)	(24,507)	(25,242)	(25,999)
(-) Utilities:	\$	(77,520)	(79,846)	(82,241)	(84,708)	(87,249)	(89,867)	(92,563)
(-) Property Management Fee:	\$	(27,737)	(29,176)	(30,942)	(32,496)	(34,307)	(35,414)	(36,477)
(-) Sales, Marketing & Administrative:	\$	(175,671)	(186,724)	(200,092)	(212,309)	(226,425)	(236,096)	(243,179)
(-) Replacement Reserves:	\$	(19,000)	(19,570)	(20,157)	(20,762)	(21,385)	(22,026)	(22,687)
Total Expenses:	\$	(388,172)	(405,542)	(425,687)	(444,607)	(465,822)	(482,034)	(495,761)
Net Operating Income (NOI):	\$	536,410	566,980	605,714	638,602	677,738	698,447	720,135
NOI Margin:	%	58.0%	58.3%	58.7%	59.0%	59.3%	59.2%	59.2%
(-) Capital Expenditures:	\$	-	(38,000)	(76,000)	(57,000)	-	(15,200)	-
(+) Capital Costs Paid from Reserves:	\$	-	38,000	39,727	20,762	-	15,200	-
Adjusted Net Operating Income:	\$	536,410	566,980	569,441	602,364	677,738	698,447	720,135
Adjusted NOI Margin:	%	58.0%	58.3%	55.2%	55.6%	59.3%	59.2%	59.2%
(-) Interest Expense on Debt:	\$		(317,688)	(312,906)	(307,885)	(302,613)	(297,078)	
(-) Debt Principal Repayment:	\$		(95,633)	(100,415)	(105,435)	(110,707)	(116,243)	
Cash Flow to Equity Investors:	\$		153,659	156,121	189,043	264,417	285,127	
Ending Debt Balance:	\$	6,353,750	6,258,117	6,157,702	6,052,267	5,941,560	5,825,317	
Debt Yield:	%		8.9%	9.5%	10.1%	10.7%	11.0%	
Interest Coverage Ratio - NOI:	x		1.78 x	1.94 x	2.07 x	2.24 x	2.35 x	
Interest Coverage Ratio - Adjusted NOI:	x		1.78 x	1.82 x	1.96 x	2.24 x	2.35 x	
Debt Service Coverage Ratio (DSCR) - NOI:	x		1.37 x	1.47 x	1.55 x	1.64 x	1.69 x	
Debt Service Coverage Ratio (DSCR) - Adj. NOI:	x		1.37 x	1.38 x	1.46 x	1.64 x	1.69 x	

We use IPMT and PPMT because the interest rate is fixed, there's no Interest-Only period, and the terms are very standard.

Debt Yield stays relatively healthy, at or above the 9.0% figure lenders are seeking, and well above the Cap Rate.

Minimum ICR is 1.75x and Minimum DSCR is 1.30x, so these credit stats and ratios seem fine.

The Debt Yield equals the NOI in the current year divided by the *initial* Loan Balance (not reflecting any principal repayments), the Interest Coverage Ratio = NOI / Interest, and the Debt Service Coverage Ratio = NOI / (Interest + Principal Repayment).

For Core deals like this one, it's normal for the lenders to impose fairly strict requirements or "covenants" on the Debt, such as the ones above – or potentially even stricter ones.

Simplified Multifamily Acquisition Model, Step 4 – Returns Calculations

The calculations are straightforward for this simple model with a fixed exit date in Year 5.

We calculate the returns on both an unleveraged basis (as if no Debt at all were used, meaning no Debt Service and no principal repayment at the end, but a higher upfront purchase price) and a leveraged basis (the traditional method):

IRR Calculations:	Units:	Historical:	Projected:				
		FY17	FY18	FY19	FY20	FY21	FY22
Project Cash Flows (Unleveraged IRR):							
(+) Proceeds from Sale of Property:	\$		-	-	-	-	12,002,242
(+) Return of Replacement Reserve:	\$		-	-	-	-	28,211
(-) Selling Costs:	\$		-	-	-	-	(240,045)
(+) Adjusted NOI (Cash Flows to All Investors):	\$		566,980	569,441	602,364	677,738	698,447
(-) Property Acquisition Price:	\$	(9,891,750)	-	-	-	-	-
Total Project Cash Flows (Unleveraged IRR):	\$	\$ (9,891,750)	\$ 566,980	\$ 569,441	\$ 602,364	\$ 677,738	\$ 12,488,855
Unleveraged Internal Rate of Return (IRR):	%	9.41%					
Total Returns:	\$	14,905,377					
Total Initial Investment:	\$	9,891,750					
Cash-on-Cash Multiple:	x	1.51 x					
Cash Flow to Equity Investors (Leveraged IRR):							
(+) Proceeds from Sale of Property:	\$		-	-	-	-	12,002,242
(+) Return of Replacement Reserve:	\$		-	-	-	-	28,211
(-) Selling Costs:	\$		-	-	-	-	(240,045)
(+) Cash Flows to Equity Investors:	\$		153,659	156,121	189,043	264,417	285,127
(-) Debt Principal Repayment Upon Exit:	\$		-	-	-	-	(5,825,317)
(-) Initial Equity Investment:	\$	(3,601,538)	-	-	-	-	-
Total Cash Flows to Equity (Leveraged IRR):	\$	\$ (3,601,538)	\$ 153,659	\$ 156,121	\$ 189,043	\$ 264,417	\$ 6,250,218
Leveraged Internal Rate of Return (IRR):	%	15.23%					
Total Returns to Equity:	\$	7,013,457					
Invested Equity:	\$	3,601,538					
Cash-on-Cash Multiple:	x	1.95 x					

There are several differences here, but the much lower upfront price in the leveraged version explains the significantly higher IRR.

Based on the 6% Cap Rate at the top and the forward NOI.

Selling Costs are the same here, but in a more complex model, there might be a Prepayment Penalty for the Debt in the leveraged version.

There are several differences here, but the much lower upfront price in the leveraged version explains the significantly higher IRR.

Much lower cash flows and exit proceeds in the leveraged version due to the Debt Principal Repayment, but that makes a small difference next to the lower purchase price.

We calculate both returns to assess the impact of leverage – if the deal seems overly dependent on Debt, that could be a red flag since it also hurts us if the deal becomes a disaster.

Note that no refinancing takes place here because this is an acquisition of a stabilized property where relatively little changes.

That is in stark contrast to most development and renovation deals, where there's almost always a refinancing as the property's risk profile changes.

If we had reason to believe that interest rates or Cap Rates would fall significantly during this holding period, meaning that we'd benefit from refinancing, then we might build one in.

However, that assumption would be quite speculative for a Core or Core-Plus deal, and it might make our analysis tougher to defend.

Should We Pursue This Deal?



As always, the answer depends on the IRR our firm is targeting. For “Core” deals, an Equity IRR of 8-10% is often the target, so by that metric, this deal is a clear “Yes.”

We did not build in sensitivities or scenarios here, but if we were to assume a continued Rental Discount of 5.0% and Bad Debt/Concessions and Utility Reimbursement remaining at the same percentages, the IRR would only drop to ~11%.

The Exit Cap Rate of 6.0% vs. the 5.8% Going-In Cap Rate is also reasonable for an older property that has had some improvements and upgrades over time.

The main issue here is that **there no assumptions for Leasing Commissions (LCs) or Tenant Improvements (TIs), even though there will be significant turnover.**

These line items are less important for multifamily properties than they are for office/retail/industrial ones, but they still matter.

However, even if we “simulate” them by assuming ~3x higher CapEx, the IRR only drops to the 13-14% range, so this still looks like a reasonably good deal.

Slightly More Complex Core Property Deal (Office/Retail)

This next example is similar, in that relatively little changes over the holding period, but a few key differences make it more complex.

In this case, the NYC office/retail building we are considering acquiring has one retail tenant on a 10-year lease and two office tenants on 5-years leases that expire within the next few years.

These two office tenants occupy ~68% of the rentable square feet, so the lease expirations create a significant risk factor that would concern any potential acquirer.

We plan to negotiate with these two tenants and get them to sign 10-year leases in exchange for below-market rent escalations, which will reduce EGI but also make it more stable and predictable.

Despite the different property type, location, and strategy, *most of the steps* are similar to the ones in the example above, and we’ve covered [the Pro-Forma for office/retail properties in a previous section](#).

So, we’ll walk through each step in this process briefly and focus on the key differences:

Office/Retail Acquisition Model, Step 1 – Property, Acquisition, and Exit Assumptions



Real Estate & REIT Modeling

Quick Reference Guide for Individual Properties

Here are the main assumptions in this model:

Operating Assumptions:			Units:		Units:
Property Name:	Name	1201 Broadway	Real Estate Taxes % Property Value:	%	4.00%
Location:	Name	New York, NY	Real Estate Taxes Annual Growth Rate:	%	2.0%
Months in Year:	#	12	Replacement Reserve per RSF per Year:	\$ / sq. ft. / Yr	\$ 2.50
Rentable Square Feet (RSF):	sq. ft.	25,000 sq. ft.	Replacement Reserve Annual Growth Rate:	%	3.0%
Property Management Fees % EGI:	%	3.0%	Common Area Utilities per RSF per Year:	\$ / sq. ft. / Yr	\$ 3.00
Common Area Maintenance (CAM) per RSF per Year:	\$ / sq. ft. / Yr	\$ 5.00	Common Area Utilities Annual Growth Rate:	%	3.0%
Common Area Maintenance Annual Growth Rate:	%	3.5%			
Insurance per RSF per Year:	\$ / sq. ft. / Yr	\$ 2.00			
Insurance Annual Growth Rate:	%	2.5%			
Acquisition and Exit Assumptions:			Units:		Units:
Acquisition Date:	Date	2017-12-31	Exit Date:	Date	2022-12-31
Acquisition Price per Rentable Square Foot:	\$ / sq. ft.	\$ 1,000.00	Exit Cap Rate:	%	6.00%
Acquisition Price:	\$	\$ 25,000,000	Exit Price:	\$	\$ 35,684,042
Implied Going-In Cap Rate:	%	6.99%	Exit Price per RSF:	\$ / sq. ft.	\$ 1,427.36
Acquisition Costs (% Gross Acquisition Price):	%	1.00%	Selling Costs (% Gross Sale Price):	%	1.50%
Loan Issuance Fees (% Total Debt):	%	1.50%			
Senior Loan Loan-to-Value (LTV) Ratio:	%	50.0%	Mezzanine Loan-to-Value (LTV) Ratio:	%	10.0%
Senior Loan Interest Rate:	%	5.00%	Mezzanine Cash Interest Rate:	%	7.00%
Senior Loan Amortization Period:	# Years	25	Mezzanine PIK Interest Rate:	%	3.00%
Senior Loan Maturity:	# Years	5	Mezzanine Amortization Period:	# Years	N/A
			Mezzanine Maturity:	# Years	5
Operating Partner Equity Contribution:	%	10.0%			
Asset Management Fee % Initial Equity Invested:	%	1.5%			
Profit Split Below IRR Hurdle:	%	-			
IRR Hurdle (Based on Levered IRR to LPs):	%	10.0%			
Profit Split Beyond IRR Hurdle:	%	20.0%			

These are all fairly standard assumptions for office and retail properties - most items are based on \$ per rentable square foot figures and annual growth rates.

These assumptions could be problematic - going from a 7.0% Going-In Cap Rate to a 6.0% Exit Cap Rate is a big increase in value. But the property is older and not *that much* has changed!

Two tranches of Debt now; the Cash Interest here will end up being quite high as a result.

It's a similar waterfall structure, but the LPs get everything first, up to a 10% IRR; the Operating Partners just receive a small management fee.

The Sources & Uses schedule has nearly the same items as the one in the simplified case study, but the Equity is split between the Operating Partners (the ones doing the work in the deal) and the Limited Partners (the ones providing 90% of the Equity).

The Uses side still includes the Acquisition Price, Acquisition Fees, Loan Issuance Fees, and Replacement Reserves.

Office/Retail Acquisition Model, Step 2 – Operating Assumptions

Here, the “operating assumptions” consist of the cash flow projections for the tenants based on their leases (covered extensively in [the section on Office/Retail/Industrial properties](#)) as well as the numbers shown at the top of the screenshot above.

The main point is that there are **two significant lease expirations in Years 3 and 4**, which significantly reduce EGI, NOI, Adjusted NOI, and Cash Flow to Equity Investors in those years.



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Quick Reference Guide for Individual Properties

Therefore, if we negotiate for 10-year leases with the tenants, it is best to plan for an exit in Year 5 – one year *after* these two tenants have switched to longer-term leases, and a few years *before* the third tenant's lease expires.

Office/Retail Acquisition Model, Step 3 – Property Pro-Forma

Here's the Pro-Forma for this property, with a few new debt-related items highlighted:

Property Pro-Forma:	Units:	Historical:		Projected:					Stabilized
		FY17	FY18	FY19	FY20	FY21	FY22	Year:	
Revenue:									
(+) Base Rental Income:	\$	\$ 2,715,000	\$ 2,841,750	\$ 2,968,862	\$ 3,090,621	\$ 3,198,048	\$ 3,293,989	\$ 3,392,809	
(-) Absorption & Turnover Vacancy:	\$	-	-	-	(202,476)	(131,469)	-	-	
(-) Concessions & Free Rent:	\$	-	-	-	(438,697)	(284,850)	-	-	
(+) Expense Reimbursements:	\$	580,000	592,290	604,848	617,681	585,333	644,196	657,891	
Potential Gross Revenue:	\$	3,295,000	3,434,040	3,573,710	3,067,130	3,367,062	3,938,186	4,050,700	
(-) General Vacancy:	\$	(240,000)	(249,600)	(259,584)	(269,967)	(278,066)	(286,408)	(295,001)	
Effective Gross Income (EGI):	\$	3,055,000	3,184,440	3,314,126	2,797,163	3,088,996	3,651,777	3,755,700	
Expenses:									
(-) Property Management Fees:	\$	(91,650)	(95,533)	(99,424)	(83,915)	(92,670)	(109,553)	(112,671)	
(-) Common Area Maintenance (CAM):	\$	(125,000)	(129,375)	(133,903)	(138,590)	(143,440)	(148,461)	(153,657)	
(-) Common Area Utilities:	\$	(75,000)	(77,250)	(79,568)	(81,955)	(84,413)	(86,946)	(89,554)	
(-) Insurance:	\$	(50,000)	(51,250)	(52,531)	(53,845)	(55,191)	(56,570)	(57,985)	
(-) Real Estate & Property Taxes:	\$	(1,000,000)	(1,020,000)	(1,040,400)	(1,061,208)	(1,082,432)	(1,104,081)	(1,126,162)	
(-) CapEx, TI, and LC Reserves:	\$	(62,500)	(64,375)	(66,306)	(68,295)	(70,344)	(72,455)	(74,628)	
Total Expenses:	\$	(1,404,150)	(1,437,783)	(1,472,132)	(1,487,807)	(1,528,491)	(1,578,066)	(1,614,657)	
Net Operating Income (NOI):	\$	1,650,850	1,746,657	1,841,994	1,309,356	1,560,505	2,073,712	2,141,043	
<i>NOI Margin:</i>	%	54.0%	54.8%	55.6%	46.8%	50.5%	56.8%	57.0%	
(-) CapEx, TIs, and LCs:	\$	-	-	-	(266,974)	(175,934)	-	-	
(+) Capital Costs Paid from Reserves:	\$	-	-	-	266,974	127,347	-	-	
Adjusted Net Operating Income:	\$	1,650,850	1,746,657	1,841,994	1,309,356	1,511,918	2,073,712	2,141,043	
<i>Adjusted NOI Margin:</i>	%	54.0%	54.8%	55.6%	46.8%	48.9%	56.8%	57.0%	
(-) Cash Interest Expense on Senior Debt:	\$		(625,000)	(611,905)	(598,155)	(583,717)	(568,558)		
(-) Cash Interest Expense on Mezzanine:	\$		(175,000)	(180,250)	(185,658)	(191,227)	(196,964)		
(-) Senior Debt Principal Repayment:	\$		(261,906)	(275,001)	(288,751)	(303,189)	(318,348)		
Cash Flow to Equity Investors:	\$		684,751	774,838	236,792	433,786	989,842		
PIK Interest on Mezzanine:	\$		(75,000)	(77,250)	(79,568)	(81,955)	(84,413)		
(+) Ending Senior Debt Balance:	\$	12,500,000	12,238,094	11,963,093	11,674,342	11,371,154	11,052,806		
(+) Ending Mezzanine Balance:	\$	2,500,000	2,575,000	2,652,250	2,731,818	2,813,772	2,898,185		
Total Debt Balance:	\$	\$ 15,000,000	\$ 14,813,094	\$ 14,615,343	\$ 14,406,160	\$ 14,184,926	\$ 13,950,991		
Debt Yield:	%		11.6%	12.3%	8.7%	10.4%	13.8%		
<i>Interest Coverage Ratio - NOI:</i>	x		2.00 x	2.12 x	1.52 x	1.82 x	2.44 x		
<i>Interest Coverage Ratio - Adjusted NOI:</i>	x		2.00 x	2.12 x	1.52 x	1.76 x	2.44 x		
<i>Debt Service Coverage Ratio (DSCR) - NOI:</i>	x		1.54 x	1.61 x	1.14 x	1.35 x	1.78 x		
<i>Debt Service Coverage Ratio (DSCR) - Adjusted NOI:</i>	x		1.54 x	1.61 x	1.14 x	1.30 x	1.78 x		

IPMT and PPMT for the Senior Debt due to the fixed rate and no Interest-Only period; simple percentage times the Mezzanine Balance for the Cash Mezzanine Interest.

Simple fixed percentage times Mezzanine Balance; keeps increasing since it accrues to the principal.

Senior Debt keeps decreasing due to amortization; Mezzanine increases due to PIK Interest.

Min Debt Yield is 10%, Min ICR is 1.75x, and Min DSCR is 1.25x... so we have problems.

These credit stats are problematic because this *should* be a stabilized property where relatively little changes over the holding period.

However, as shown above, the lease expirations and the resulting possibility of vacant months can create problems for lenders, even in “Core” deals.

Office/Retail Acquisition Model, Step 4 – Returns Calculations

There’s little to say here because the Returns Calculations are almost the same as the ones in the simplified model. We left out the recovery of the Replacement Reserves at the end to be more conservative, but we could have easily changed that:

IRR Calculations:	Units:	Historical:	Projected:				
		FY17	FY18	FY19	FY20	FY21	FY22
Project Cash Flows (Unleveraged IRR):							
(+) Proceeds from Sale of Property:	\$		-	-	-	-	35,684,042
(-) Selling Costs:	\$		-	-	-	-	(535,261)
(+) Adjusted NOI (Cash Flows to All Investors):	\$		1,746,657	1,841,994	1,309,356	1,511,918	2,073,712
(-) Property Acquisition Price:	\$	(25,312,500)	-	-	-	-	-
Total Project Cash Flows (Unleveraged IRR):	\$	\$(25,312,500)	\$ 1,746,657	\$ 1,841,994	\$ 1,309,356	\$ 1,511,918	\$ 37,222,494
Unleveraged Internal Rate of Return (IRR):							
	%	12.7%					
Total Returns:	\$	43,632,419					
Total Initial Investment:	\$	25,312,500					
Cash-on-Cash Multiple:	x	1.7 x					
Cash Flow to Equity Investors (Leveraged IRR):							
(+) Proceeds from Sale of Property:	\$		-	-	-	-	35,684,042
(-) Selling Costs:	\$		-	-	-	-	(535,261)
(+) Cash Flows to Equity Investors:	\$		684,751	774,838	236,792	433,786	989,842
(-) Debt Principal Repayment Upon Exit:	\$		-	-	-	-	(13,950,991)
(-) Initial Equity Investment:	\$	(10,537,500)	-	-	-	-	-
Total Cash Flows to Equity (Leveraged IRR):	\$	\$(10,537,500)	\$ 684,751	\$ 774,838	\$ 236,792	\$ 433,786	\$ 22,187,633
Leveraged Internal Rate of Return (IRR):							
	%	19.6%					
Total Returns to Equity:	\$	24,317,800					
Invested Equity:	\$	10,537,500					
Cash-on-Cash Multiple:	x	2.3 x					

Based on a lower Exit Cap Rate of 6%, vs. 7% Entry. Red flags!

And even with these aggressive assumptions, we just barely reach a 20% IRR.

Based on a lower Exit Cap Rate of 6%, vs. 7% Entry. Red flags!

And even with these aggressive assumptions, we just barely reach a 20% IRR.

We also create a simple **waterfall schedule** to reflect the fact that the Limited Partners receive 100% of the positive cash flows up to a 10% IRR before the Operating Partner gets its share (other than a small annual fee of 1.5% * their initial Equity investment):



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Waterfall Returns Schedule:		Units:	Historical:	Projected:				
			FY17	FY18	FY19	FY20	FY21	FY22
(+) Total Cash Flow and Returns to Equity Investors:		\$		684,751	774,838	236,792	433,786	22,187,633
(-) Asset Management Fee to Operating Partner:		\$		(15,806)	(15,806)	(15,806)	(15,806)	(15,806)
Equity Cash Flow to Limited Partners:		\$		668,945	759,032	220,986	417,979	22,171,827
Tier 1 IRR - Up to 10.0% for JUST Limited Partners:								
Beginning Balance:			(9,483,750)	(9,763,180)	(9,980,466)	(10,757,527)	(11,415,300)	
Returns Accrual:		10.0%	(948,375)	(976,318)	(998,047)	(1,075,753)	(1,141,530)	
Repayment:		100.0%		668,945	759,032	220,986	417,979	12,556,830
Ending Balance:			(9,483,750)	(9,763,180)	(9,980,466)	(10,757,527)	(11,415,300)	-
Cash Flow Available for Tier 2 Distribution:		\$		-	-	-	-	9,614,997
Tier 2 IRR - Above 10.0% Limited Partners' IRR:								
Limited Partners (LPs):		80.0%		-	-	-	-	7,691,997
Operating Partner (OP):		20.0%		-	-	-	-	1,922,999
Returns Analysis by Investor Group:								
Limited Partners (LP) - Leveraged Returns:			(9,483,750)	668,945	759,032	220,986	417,979	20,248,827
Internal Rate of Return (IRR):		% 20.2%						
Cash-on-Cash Multiple:		x 2.4 x						
Annual Yield on Initial Investment:		%		7.1%	8.0%	2.3%	4.4%	213.5%
Operating Partner (OP) - Leveraged Returns:			(1,053,750)	15,806	15,806	15,806	15,806	1,938,806
Internal Rate of Return (IRR):		% 14.0%						
Cash-on-Cash Multiple:		x 1.9 x						
Annual Yield on Initial Investment:		%		1.5%	1.5%	1.5%	1.5%	184.0%

The OP gets barely any of the cash flow for the first 4 years due to this Preferred Return structure for the LPs. It's no surprise that the OP ends up with a much lower IRR as a result (time value of money).

These yields reflect the annual cash flow to equity, including the big hit in Years 3-4 due to lease expirations.

...while these reflect just the Asset Management Fee and the exit in Year 5.

If our target is a 15% IRR, this waterfall structure gives us two different stories about the deal's feasibility.

Should We Pursue This Deal?

No. As presented, this deal makes no sense for several reasons:

- The property cannot comply with the required Debt Yield, ICR, and DSCR at a 60% LTV.
- The 6.0% Exit Cap Rate seems unrealistically low. At a 7.0% Exit Cap Rate, matching the Going-In Cap Rate, the IRR to LPs would drop to ~16%.
- There's a significant lease-expiration risk since 68% of the space expires within the first 4 years.
- The Renewal Probability of 70% also seems too high for a scenario in which the tenants agree to much longer lease terms.
- A 10% Preferred Return for the LPs is unrealistic for a stabilized property; 5% might be acceptable.



- Reducing the LTV to 52% to improve the credit stats and ratios, changing the Exit Cap Rate to 7.0%, making the Preferred Return 5%, and reducing the Renewal Probability would reduce the LP IRR to 13-14%, putting it below the 15% target.
- And we haven't even considered the possibility of a recession or other market downturn in this holding period.

In short, the initial assumptions are too aggressive/unrealistic, and if we made them more realistic, the IRR would fall short of the 15-20% level.

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Strategy #4: Acquire an Existing Property, Renovate It, and Sell It ("Value-Added")

This strategy is similar to the one described above, and the basic steps in the process are the same: set up the property, acquisition, and exit assumptions; make operating assumptions; build a Pro-Forma including Debt Service, and calculate the returns.

The key difference is that **something significant** about the property changes during the holding period, and the investors **spend something** to enact this change. For example:

- **Hotels:** The investors might complete a renovation that boosts the hotel's Occupancy Rate and ADR significantly.
- **Multifamily:** The sponsors might complete a renovation that boosts the property from Class B to Class A, which will boost average rents significantly. They might also redevelop the ground floor and add retail units to diversify the income sources.
- **Office:** The sponsors might renovate a building with a 70% Occupancy Rate, attract 1-2 new tenants to boost the rate to 90%, and negotiate with the tenants to execute longer-term leases, such as 10 years rather than 3-4 years.
- **Retail:** The sponsors might renovate a shopping mall to attract higher-end luxury retailers and boost the average rent per square foot and Occupancy Rate; they might also negotiate new leases in which the tenants will owe a percentage of their gross monthly sales.

Regarding financial modeling, these differences translate into the following points:

- 1) Renovation/Redevelopment Costs** – These will reduce the Cash Flow to Equity investors, or you might assume a higher upfront purchase price to cover them, depending on the timing.
- 2) Penalty During the Renovation/Redevelopment Period** – For example, a hotel's Occupancy Rate might drop as rooms become unavailable due to the renovation.
- 3) Benefit Following the Renovation** – For example, the Occupancy Rate or average rent might increase once the renovation is done.
- 4) Permanent Loan Refinancing** – As with [Strategy #1 \(Develop a Property and Sell It Upon Completion\)](#), there is often a refinancing as the renovation finishes and the property stabilizes.
- 5) Exit Assumptions** – It is reasonable to assume a lower Cap Rate upon exit because the property should become more valuable. But you still need to check the numbers against historical data.

We'll walk through two renovation deals here, one for a hotel in Dubai and one for an office in Boston:

Hotel Acquisition & Renovation, Part 1 – Property, Acquisition, and Exit Assumptions

We covered the key hotel drivers in [the section on Hotels](#), so we'll focus on the points specific to the renovation here:

Acquisition, Financing, and Exit Assumptions:			Units:				Units:
Acquisition Financing:				Permanent Loan Refinancing:			
Acquisition Date:	Date	2016-12-31		Permanent Loan Refinancing Date:	Date	2018-12-31	
Going-In Cap Rate:	%	9.50%		Property Value @ Refinancing:	\$	\$ 274,186,597	
Assumed Growth of In-Place NOI w/o Renovation:	%	3.0%		Permanent Loan Issuance Fees:	%	1.00%	
Acquisition Price:	\$	\$ 227,965,466		Senior Loan Loan-to-Value (LTV) Ratio:	%	75.0%	
Acquisition Price per Key:	\$ / Key	\$ 368,876		Senior Loan Amount:	\$	\$ 205,639,948	
Acquisition Costs % Acquisition Price:	%	4.00%		Senior Loan Interest Rate:	%	3.50%	
Loan Issuance Fees % Total Debt:	%	1.50%		Senior Loan Amortization Period:	# Years	30	
Acquisition Loan Loan-to-Value (LTV) Ratio:	%	65.0%		Senior Loan Maturity:	# Years	10	
Acquisition Loan Amount:	\$	\$ 148,177,553		Mezzanine Loan-to-Value (LTV) Ratio:	%	10.0%	
Acquisition Loan PIK Interest Rate:	%	8.00%		Mezzanine Amount:	\$	\$ 27,418,660	
Acquisition Loan Amortization Period:	# Years	N/A		Mezzanine Cash Interest Rate:	%	4.00%	
Acquisition Loan Maturity:	# Years	10		Mezzanine PIK Interest Rate:	%	4.00%	
Exit Assumptions:				Mezzanine Amortization Period:	# Years	N/A	
Exit Date:	Date	2021-12-31		Mezzanine Maturity:	# Years	5	
Exit Capitalization Rate:	%	9.00%		The Exit Cap Rate and Date are both variable; the Exit Cap Rate is tied to the operating scenario (Independent vs. Franchise), with lower rates in the Franchise case. It's reasonable to assume a lower Cap Rate upon exit if the renovation was extensive.			
Exit Price:	\$	\$ 329,242,018					
Exit Price per Key:	\$ / Key	\$ 532,754					
Selling Costs (% Gross Sale Price):	%	2.00%					
Acquisition Sources and Uses of Funds:							
Sources of Funds:				Uses of Funds:			
Acquisition Loan:	\$	\$ 148,177,553		Acquisition Price:	\$	\$ 227,965,466	
Investor Equity:		91,129,195		Acquisition Costs:		9,118,619	
Total Sources:		\$ 239,306,747		Loan Issuance Fees:		2,222,663	
				Total Uses:		\$ 239,306,747	

"Forward NOI" is distorted here due to the renovation, so we must take the most recent year's NOI and assume a modest growth rate to get the Acquisition Price.

75% Permanent Loan LTV vs. 65% for Acquisition Loan.

Acquisition Loan is 100% PIK due to the property's negative cash flows while the renovation is taking place.

We're not assuming Reserves for the renovation costs, but that could be another approach.



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If we had allocated Reserves for the Renovation Costs, the Total Uses would have been higher, and we would have used more Investor Equity to fund the deal.

We do not do that here because it would decrease the IRR due to the time value of money; under our approach, the Renovation Costs turn Cash Flow to Equity negative, and Excel interprets that as an additional investment during the holding period.

Hotel Acquisition & Renovation, Part 2 – Operating Assumptions

We look at a total of **6 scenarios** in this model: Independent vs. Franchise for the hotel itself as well as Base, Downside, and Upside cases for the market conditions.

The Occupancy Rate and ADR are lower in the Independent scenario because the hotel won't benefit from brand-name recognition and a broader customer base, but the expenses, reserves, and fees will also be lower.

In the market scenarios, the Occupancy Rate and ADR are highest in the Upside Case and lowest in the Downside Case, and we assume that the hotel captures a certain percentage of the market-wide average figures.

Non-room revenue from food & beverages and other sources grows most slowly in the Downside Case and most quickly in the Upside Case, but expenses also grow the most quickly in the Upside Case and the most slowly in the Downside Case.

We also include a **renovation factor** in these assumptions because a certain percentage of rooms will be unavailable while the renovation is taking place:

Revenue and Expense Assumptions:	Units:	Historical:	Projected:					Stabilized:
		FY16	FY17	FY18	FY19	FY20	FY21	FY22
Renovation Factor:	%	-	25.0%	50.0%	-	-	-	-
Renovation Costs:	\$	\$ -	\$ 10,815,000	\$ 10,815,000	\$ -	\$ -	\$ -	\$ -
Days in Period:	# Days	366	365	365	365	366	365	365
Available Room Nights:	# Room Nights	226,188	225,570	225,570	225,570	226,188	225,570	225,570
Occupied Room Nights:	# Room Nights	176,427	135,004	87,860	177,862	179,424	180,005	181,076
F&B Revenue per Occupied Room Night:	\$ / Room Night	125.00	122.50	121.28	124.91	127.41	129.96	132.56
Other Revenue per Occupied Room Night:	\$ / Room Night	25.00	24.50	24.26	24.98	25.48	25.99	26.51
Utilities per Occupied Room Night:	\$ / Room Night	10.00	10.10	10.20	10.41	10.61	10.83	11.04
Insurance per Available Room:	\$ / Room / Yr.	1,500	1,515	1,530	1,561	1,592	1,624	1,656
Property Taxes per Available Room:	\$ / Room / Yr.	-	-	-	-	-	-	-
FF&E Reserves per Available Room:	\$ / Room / Yr.	2,100	2,121	2,142	2,185	2,229	2,273	2,319
Ground Lease:	\$ / Yr.	\$ 15,563,120	\$ 15,718,751	\$ 15,875,939	\$ 16,193,457	\$ 16,517,326	\$ 16,847,673	\$ 17,184,626

The Renovation reduces the Occupied Room Nights by 25% and 50% in the first two years. We could have also assumed a drop in the Available Room Nights, which then flowed through to the Occupied Room Nights.

These items are all fairly standard and are linked to Occupancy or Availability; there are no Property Taxes because it's Dubai, but there are higher-than-normal acquisition fees.

This is a major risk factor for the hotel because a "Ground Lease" means that it does not own the land on which the hotel operates. It's a huge fixed expense that could destroy cash flow in Downside scenarios.



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The rest of the assumptions are [in the More Complex example in the hotel section of the guide](#).

Hotel Acquisition & Renovation, Part 3 – Property Pro-Forma

You've seen [the basic Pro-Forma in the previous section](#), so we'll focus on the parts below the Net Operating Income line here.

Those parts include the Debt Service for the Acquisition Loan and Permanent Loans (Senior and Mezzanine), the Debt balances, and the credit stats and ratios.

There's also an Incentive Fee to Management and Renovation Costs, and we count both of them as "capital costs" that are deducted from NOI to calculate Adjusted NOI:

Property Pro-Forma:	Units:	Historical:	Projected:					Stabilized:
		FY16	FY17	FY18	FY19	FY20	FY21	FY22
Net Operating Income (NOI):	\$	21,025,941	15,026,797	2,876,539	25,773,540	27,588,558	28,764,927	29,631,782
NOI Margin:	%	23.8%	20.8%	6.2%	26.5%	27.2%	27.5%	27.6%
(-) Incentive Fee to Management:	\$	-	-	-	(1,903,588)	(2,194,005)	(2,360,657)	(2,457,276)
(-) Renovation Costs:	\$	-	(10,815,000)	(10,815,000)	-	-	-	-
Adjusted Net Operating Income (NOI):	\$	21,025,941	4,211,797	(7,938,461)	23,869,952	25,394,552	26,404,270	27,174,505
Adjusted NOI Margin:	%	23.8%	5.8%	(17.2%)	24.6%	25.0%	25.3%	25.3%
(-) Cash Interest Expense on Senior Loan:	\$	-	-	-	(7,197,398)	(7,057,975)	(6,913,672)	-
(-) Cash Interest Expense on Mezzanine:	\$	-	-	-	(1,096,746)	(1,140,616)	(1,186,241)	-
(-) Senior Loan Principal Repayment:	\$	-	-	-	(3,983,520)	(4,122,943)	(4,267,246)	-
Cash Flow to Equity Investors:	\$	-	4,211,797	(7,938,461)	11,592,288	13,073,018	14,037,111	-
PIK Interest on Acquisition Loan:	\$	-	(11,854,204)	(12,802,541)	-	-	-	-
PIK Interest on Mezzanine:	\$	-	-	-	(1,096,746)	(1,140,616)	(1,186,241)	-
(+) Ending Acquisition Loan Balance:	\$	148,177,553	160,031,757	-	-	-	-	-
(+) Ending Senior Loan Balance:	\$	-	-	205,639,948	201,656,428	197,533,485	193,266,239	-
(+) Ending Mezzanine Balance:	\$	-	-	27,418,660	28,515,406	29,656,022	30,842,263	-
Total Debt Balance:	\$	\$ 148,177,553	\$ 160,031,757	\$ 233,058,607	\$ 230,171,834	\$ 227,189,508	\$ 224,108,503	-
Debt Yield:	%	-	10.1%	1.9%	11.1%	11.8%	12.3%	-
Interest Coverage Ratio - NOI:	x	-	1.27 x	0.22 x	2.74 x	2.95 x	3.10 x	-
Interest Coverage Ratio - Adjusted NOI:	x	-	0.36 x	(0.62 x)	2.54 x	2.72 x	2.84 x	-
Debt Service Coverage Ratio (DSCR) - NOI:	x	-	1.27 x	0.22 x	1.93 x	2.05 x	2.12 x	-
Debt Service Coverage Ratio (DSCR) - Adjusted NOI:	x	-	0.36 x	(0.62 x)	1.78 x	1.89 x	1.95 x	-

Incentive Fees are not exactly "capital costs," but are typically shown below NOI because they're calculated based on NOI.

This is why the Acquisition Loan has 100% PIK Interest: The hotel's cash flow clearly cannot support Cash Interest during the renovation period.

IPMT and PPMT for the Senior Loan, but we check to make sure the refinancing has taken place and vary the Year # based on the refinancing date.

Refinancing replaces the Acquisition Loan with a Senior Loan and Mezzanine.

Min Debt Yield is 10%, ICR is 2.00x, and DSCR is 1.50x, so these #'s look fine in the Base Case.

Overall, we get numbers for the credit stats and ratios that indicate compliance with the lenders' requirements across all the cases.

That said, the outcomes in the Stabilized Year are quite different (everything below assumes the Independent operating scenario):

- **Base Case:** \$107 million revenue, \$30 million NOI, and \$27 million Adjusted NOI.



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- **Downside Case:** \$97 million revenue, \$25 million NOI, and \$24 million Adjusted NOI.
- **Upside Case:** \$116 million revenue, \$34 million NOI, and \$31 million Adjusted NOI.

These figures are reasonable; market conditions make a big impact on hotels, but you're still not going to see a *50% drop* in revenue if there's a recession.

But a 10-20% decline, as in the numbers above, is more plausible.

Hotel Acquisition & Renovation, Part 4 – Returns Calculations

There's no waterfall schedule here, so the only tricky part is building in support for variable exit dates and renovation end dates. To do that, you must do the following:

- 1) Check the current year against the Renovation End Date in the Permanent Loans Issued and Acquisition Loan Refinanced line items.
- 2) Only count Cash Flow to Equity Investors if the current year is the Exit Date or before the Exit Date; otherwise, it's 0.
- 3) Only count the Proceeds from the Property Sale and the Debt Repayment Upon Exit if the current year equals the Exit Date.

Returns to Equity Investors:	Units:	Historical:	Projected:				
		FY16	FY17	FY18	FY19	FY20	FY21
Forward NOI:	\$		\$ 2,876,539	\$ 25,773,540	\$ 27,588,558	\$ 28,764,927	\$ 29,631,782
Applicable Capitalization Rates:							
Independent	%	9.50%	9.70%	9.40%	9.25%	9.10%	9.00%
Franchise	%	9.50%	9.60%	9.25%	8.95%	8.70%	8.50%
Selected Capitalization Rate:	%	9.50%	9.70%	9.40%	9.25%	9.10%	9.00%
Implied Property Value:	\$		29,655,040	274,186,597	298,254,679	316,098,097	329,242,018
Cash Flow to Equity Investors (Leveraged IRR):							
(-) Investor Equity:	\$	(91,129,195)	-	-	-	-	-
(+) Senior Loan Issued:	\$	-	-	205,639,948	-	-	-
(+) Mezzanine Issued:	\$	-	-	27,418,660	-	-	-
(-) Permanent Loan Financing Fees:	\$	-	-	(2,330,586)	-	-	-
(-) Acquisition Loan Refinanced:	\$	-	-	(172,834,297)	-	-	-
(+) Cash Flow to Equity Investors:	\$	-	4,211,797	(7,938,461)	11,592,288	13,073,018	14,037,111
(+) Proceeds from Sale of Property:	\$	-	-	-	-	-	329,242,018
(-) Selling Costs:	\$	-	-	-	-	-	(6,584,840)
(-) Debt Repayment Upon Exit:	\$	-	-	-	-	-	(224,108,503)
Total Cash Flows to Equity (Leveraged IRR):	\$	\$ (91,129,195)	\$ 4,211,797	\$ 49,955,263	\$ 11,592,288	\$ 13,073,018	\$ 112,585,786
Internal Rate of Return (IRR):	%	21.9%					
Total Returns to Equity:	\$	191,418,152					
Invested Equity:	\$	91,129,195					
Cash-on-Cash Multiple:	x	2.1 x					
Annual Yield on Initial Investment:	%		4.6%	54.8%	12.7%	14.3%	123.5%

The property will be slightly more valuable in the Franchise Case because of the brand-name value and wider customer base. We don't factor in the market scenarios because in the Base, Downside, and Upside Case, there's a "return to market norms" by the final years (the most likely exit dates).

Big benefit to the refinancing because it is done at a higher LTV than the initial Acquisition Loan (75% vs. 65%) and the property's value is also higher.

A very good IRR for a renovation deal... but how will it change in different scenarios?

Should We Pursue This Deal?

Since this model includes multiple scenarios, it's much easier to assess the merits of this deal.

The case study instructions tell us the IRR targets are 10% in the Downside Case, 20% in the Base Case, and 30% in the Upside Case.

To assess the performance against these targets, we create several sensitivity tables:

Sensitivity Analysis - Equity IRR in Different Scenarios:

	Independent	Franchise
Base	21.9%	20.9%
Downside	10.4%	9.5%
Upside	35.0%	34.0%

At or above the targeted IRRs with a Year 5 Exit.

	Exit Date:		
	FY19	FY20	FY21
Base	14.4%	20.0%	21.9%
Downside	(1.9%)	6.1%	10.4%
Upside	32.1%	35.3%	35.0%

There's significant risk with a Year 3 exit, but Years 4-5 look reasonable.

	Exit Date:		
	FY19	FY20	FY21
Independent	14.4%	20.0%	21.9%
Franchise	11.1%	18.2%	20.9%

And it seems like the Independent option is slightly better due to the lower expenses.

We also created tables based on the LTV vs. Interest Rates, Going-In vs. Exit Cap Rates, and ADR vs. Occupancy Rate, but they all support the same conclusion: this deal meets or exceeds the targeted IRRs if we exit in Year 5.

Furthermore, the assumptions are plausible because the property's Occupancy Rate stays in the same range over time, the Cap Rates are in-line with market norms for 5-star hotels in Dubai, and the LTVs are also in-line with market figures.

The biggest issue is that the property's ADR jumps from 78% of the market-wide ADR to 90% as soon as the renovation begins, but that assumption alone doesn't make a huge difference since it affects the Cash Flows to Equity Investors only in the early period.

The main risk factors are less Exit Cap Rate compression than we assumed, lower-than-assumed ADR and Occupancy Rate capture, and the high fixed cost of the ground lease.



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We could mitigate risks these by refinancing at a higher LTV, or even using a higher LTV for the Acquisition Loan; we could also use 100% Debt to complete a larger, more expensive renovation.

To deal with the ground lease, we could negotiate with the landowner and offer a percentage of hotel revenue rather than a fixed annual payment so that the lease becomes a variable expense.

More Complex Acquisition & Renovation Model

Next, we'll look at a more complex acquisition & renovation model for a mixed office/retail property (45 Milk Street).

Office/Retail Acquisition & Renovation, Part 1 – Property, Acquisition, and Exit Assumptions

This deal uses a mixed monthly/annual model because some items, such as the tenant lease revenue and expenses, make more sense in a monthly context, while others, such as the Debt Service, can be projected on an annual basis.

There is no refinancing at a variable date in this model, either, which means that one of the main reasons to create a monthly Debt Schedule no longer applies.

Here are the starting assumptions:

45 Milk Street - Investment Analysis

(\$ in USD as Stated)

Acquisition Overview:			Units:	Operating Scenario:			Units:
Property Name:	Name	45 Milk Street		Operating Scenario:	Name	Base	
Property Type:	Name	Office/Retail		General Partners (GP) Name:	Name	HYM Investment	
Location:	Name	Boston, MA		Limited Partners (LP) Name:	Name	Limited Partners	
Acquisition Date:	Date	2018-12-31		GP Equity Contribution:	%	10.0%	
Months in Year:	#	12		LP Equity Contribution:	%	90.0%	
Gross Square Feet of Property:	sq. ft.	61,000 sq. ft.					
Rentable to Gross Area %:	%	90.0%					
Rentable Square Feet:	sq. ft.	54,900 sq. ft.					

You almost always need scenarios to make a real investment decision.

Fairly standard GP / LP Equity split, which will come up in the waterfall.

The rest of the assumptions are as follows:



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Acquisition, Financing, and Exit Assumptions: Units:

Acquisition & Exit Assumptions:		
Acquisition Price:	\$	\$ 20,500,000
Going-In Cap Rate:	%	8.02%
Acquisition Price per RSF:	\$ / sq. ft.	\$ 373.41
Acquisition Costs (% Gross Acquisition Price):	%	1.0%
Renovation Costs % Initial Acquisition Price:	%	15.0%
Renovation Costs:	\$	\$ 3,075,000
Renovation Costs per RSF:	\$ / sq. ft.	56.01
Additional Reserves % Initial Acquisition Price:	%	1.0%
Additional Reserves:	\$	\$ 205,000
Additional Reserves per RSF:	\$ / sq. ft.	3.73
Total Acquisition Costs:	\$	\$ 23,780,000
Total Acquisition Costs per RSF:	\$ / sq. ft.	433.15
Exit Date:	Date	2022-12-31
Exit Cap Rate:	%	7.00%
Exit Price:	\$	\$ 35,183,419
Exit Price per RSF:	\$ / sq. ft.	\$ 640.86
Selling Costs (% Gross Sale Price):	%	2.0%

Modest Cap Rate compression due to the renovation and, hopefully, higher Occupancy Rate by the end; the Renovation Costs and Reserves are added directly to the Acquisition Price here.

Financing Assumptions:		
Loan Issuance Fees:	%	1.00%
Prepayment Penalty:	%	1.00%
Senior Loan:		
Loan-to-Value (LTV) Ratio:	%	65.0%
Senior Loan Amount:	\$	15,457,000
Holdback:	\$	3,280,000
Annual Holdback Release:	\$ / sq. ft. / Yr.	\$ 35.00
LIBOR Spread:	%	3.00%
LIBOR Floor:	%	1.75%
Interest-Only Period (Years):	# Years	2
Amortization (Years):	# Years	30
Tenor (Years):	# Years	5
Maturity:	Date	2023-12-31
Mezzanine:		
Loan-to-Value (LTV) Ratio:	%	10.0%
Mezzanine Amount:	\$	2,378,000
Cash Interest Rate:	%	3.00%
PIK Interest Rate:	%	6.00%
Amortization (Years):	# Years	N/A
Tenor (Years):	# Years	5
Maturity:	Date	2023-12-31

There is no refinancing in this deal because the property's profile does not change *that much* even during the renovation - no additional space becomes vacant. Therefore, it's unnecessary to assume a higher-cost Acquisition Loan that gets refinanced with Permanent Loans later on.

The "Holdback" means that the lenders will not release ~21% of the Senior Loan upfront; it will be released as needed for the Renovation and Additional Reserves. However, the property owners will still pay interest on the full amount!

Nothing new with the Mezzanine, but the Senior Loan has a 2-year Interest-Only period and a floating interest rate, so we can't use IPMT and PPMT.

The Sources & Uses schedule is also different because the Senior Loan, Renovation Cost Reserves, and Replacement Reserves will **not** be fully drawn at the close.

This setup is common in renovation deals because lenders don't want to disburse the full amount of funding until they see clear progress on the property.

Not disbursing the full amount upfront helps them mitigate risk and potentially earn a higher IRR because of the time value of money.

When the acquisition initially closes, we owe only the Acquisition Price and associated fees:

Sources & Uses of Funds:

Sources of Funds:	Total:	Drawn at Close:
Senior Loan:	\$ 15,457,000	\$ 12,177,000
Mezzanine:	2,378,000	2,378,000
Investor Equity:	6,328,350	6,328,350
Total Sources:	\$ 24,163,350	\$ 20,883,350

Uses of Funds:	Total:	Drawn at Close:
Acquisition Price:	\$ 20,500,000	\$ 20,500,000
Acquisition Costs:	205,000	205,000
Loan Issuance Fees:	178,350	178,350
Renovation Cost (CapEx) Reserves:	3,075,000	-
Replacement Reserves:	205,000	-
Total Uses:	\$ 24,163,350	\$ 20,883,350

There is Cap Rate compression in each operating scenario, but the amount varies; the biggest difference is in the Upside Case, followed by the Base and Downside Cases:



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Returns to Equity Investors:	Units:	Historical:	Projected:			
		FY18	FY19	FY20	FY21	FY22
Forward NOI:	\$		\$ 1,746,814	\$ 1,838,542	\$ 2,328,801	\$ 2,462,839
Boston Financial District - Class-B Office/Retail Cap Rates:						
Upside	%	7.50%	7.25%	7.00%	6.85%	6.75%
Base	%	7.50%	7.25%	7.20%	7.10%	7.00%
Downside	%	7.50%	7.25%	8.00%	7.50%	7.25%
Selected Class-B Office/Retail Cap Rate:	%	7.50%	7.25%	7.20%	7.10%	7.00%
Implied Property Value:	\$		24,093,991	25,535,309	32,800,008	35,183,419

The renovation *a/ways* boosts the property's value by the last two years, but the extent of that boost varies; it's much weaker if the market experiences a downturn followed by a recovery.

Office/Retail Acquisition & Renovation, Part 2 – Operating Assumptions

We covered many of these operating assumptions in [the section on Office/Retail/Industrial properties](#), so please refer to that for the full details.

To summarize, there are 5 existing tenants and 2 potential tenants here with varied lease terms, and we consider 3 market scenarios that result in different numbers for the Renewal Probabilities, TIs, LCs, and other items:

Operating Assumptions:	Units:	Historical:	Projected:				Stabilized:
		FY18	FY19	FY20	FY21	FY22	FY23
Numerical Year:	Year	2018	2019	2020	2021	2022	2023
Numerical Year #:	#	1	2	3	4	5	6
Change in Boston Financial District Office Market Rents (Starting Rates for Leases):							
Upside	%	6.00%	4.00%	5.00%	4.00%	3.00%	3.00%
Base	%	6.00%	4.00%	3.00%	3.00%	2.50%	2.50%
Downside	%	6.00%	4.00%	(3.00%)	(1.00%)	6.00%	2.50%
Selected Change in Office Market Rents:	%	6.00%	4.00%	3.00%	3.00%	2.50%	2.50%
Fixed Annual Rental Escalations for New and Renewal Leases Starting in Specified Year:							
Upside	%	2.00%	3.00%	4.00%	3.00%	2.50%	2.50%
Base	%	2.00%	3.00%	2.50%	2.50%	2.00%	2.00%
Downside	%	2.00%	3.00%	1.00%	1.00%	1.50%	1.50%
Selected Fixed Annual Rental Escalation:	%	2.00%	3.00%	2.50%	2.50%	2.00%	2.00%
Office Expense Growth Rate - Boston Financial District:							
Upside	%	4.00%	3.00%	3.50%	3.00%	2.00%	2.00%
Base	%	4.00%	3.00%	2.00%	2.00%	1.50%	1.50%
Downside	%	4.00%	3.00%	(1.00%)	(0.50%)	4.00%	1.50%
Selected Office Expense Growth Rate:	%	4.00%	3.00%	2.00%	2.00%	1.50%	1.50%
Retail Sales Growth Rate - Boston Financial District:							
Upside	%	2.00%	3.00%	4.00%	3.00%	3.00%	2.50%
Base	%	2.00%	3.00%	3.00%	2.50%	2.50%	2.00%
Downside	%	2.00%	3.00%	(2.00%)	(1.00%)	5.00%	2.00%
Selected Retail Sales Growth Rate:	%	2.00%	3.00%	3.00%	2.50%	2.50%	2.00%



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Lease Terms, Capital Costs, and Operating Expenses:		Selected:	Upside	Base	Downside
Renewal Probability:	%	60.0%	70.0%	60.0%	50.0%
# Months of Downtime for Non-Renewal:	#	6	3	6	9
New Tenants - # Months of Free Rent:	#	3	2	3	4
Renewal Tenants - # Months of Free Rent:	#	2	1	2	3
New - Tenant Improvements per RSF:	\$ / sq. ft.	\$ 45.00	\$ 35.00	\$ 45.00	\$ 55.00
Renewal - Tenant Improvements per RSF:	\$ / sq. ft.	20.00	15.00	20.00	25.00
New - Leasing Commissions % Lease Value:	%	6.0%	5.0%	6.0%	7.0%
Renewal - Leasing Commissions % Lease Value:	%	3.0%	2.0%	3.0%	4.0%
Tenant #6 - Lease Start Date:	Date	2020-06-30	2020-01-31	2020-06-30	N/A
Tenant #7 - Lease Start Date:	Date	N/A	2021-01-31	N/A	N/A
Common Area Maintenance (CAM) / RSF / Year:	\$ / sq. ft. / Yr.	\$ 2.50			
Insurance / RSF / Year:	\$ / sq. ft. / Yr.	1.00			
Real Estate Taxes % Property Value:	%	2.55%			
Replacement Reserves / RSF / Year:	\$ / sq. ft. / Yr.	3.00			
Common Area Utilities / RSF / Year:	\$ / sq. ft. / Yr.	2.00			
Property Management Fees % EGI:	%	3.0%			
Lease Term for New Tenants:	# Years	5			

Of the existing tenants, 4 have NNN leases, and 1 has a NNN lease with Percentage Rent; of the potential tenants, 1 has a Full Service lease, and 1 has a NN lease with Percentage Rent.

The expiration dates are also different, with existing tenant leases ending in Years 1, 2, 3, and 4.

The status of the two new potential tenants (#6 and #7 in the screenshot above) varies based on the scenario, and they do not move in at all in the Downside Case, keeping the Occupancy Rate the same.

These assumptions create some “lumpiness” in the Pro-Forma, as you’ll see below.

Office/Retail Acquisition & Renovation, Part 3 – Property Pro-Forma

First, note that the TIs, LCs, CapEx, and Reserves work a bit differently because of the “Holdback,” which starts being released as soon as the Renovation begins.

The sponsors still pay for Capital Costs using the Reserves when it is possible to do so, but in addition to the normal monthly allocation, there is an *additional* allocation during the Renovation period to fund the Renovation Costs:



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Rent Roll:	Units:	Jun-19	Jul-19	Aug-19	Sep-19	Oct-19
Capital Costs and Replacement Reserve:						
Monthly Distribution of Renovation Costs:	%	-	6.0%	4.0%	8.0%	7.0%
Has Renovation Begun?	#	0	1	1	1	1
(-) Monthly Renovation Costs (CapEx):	\$	-	(184,500)	(123,000)	(246,000)	(215,250)
(-) Monthly Tenant Improvements (TIs):	\$	-	(69,707)	-	-	-
(-) Monthly Leasing Commissions (LCs):	\$	-	(26,174)	-	-	-
Total Monthly Capital Costs:	\$	\$ -	\$ (280,381)	\$ (123,000)	\$ (246,000)	\$ (215,250)
Release of TI/LC/CapEx Holdback:	\$	-	160,125	160,125	160,125	160,125
Replacement Reserve Balance:	\$	-	254,633	148,900	200,583	129,302

Start releasing the Holdback as soon as Renovation begins.

Reserve Balance = Old Balance + Normal Allocation This Month - Capital Costs Paid from Reserve + Monthly Release of Holdback.

The Monthly Pro-Forma takes the same shape as the Annual one, but we must divide by 12 for expenses that are quoted on an annual basis and factor in the monthly growth rates.

The items that contribute to Effective Gross Income – Base Rental Income, Loss to Lease, Absorption & Turnover Vacancy, Concessions & Free Rent, Expense Reimbursements, General Vacancy, and Percentage Rent – all come from the tenant-by-tenant schedules.

For reference, here's the Annual Pro-Forma down to the Adjusted NOI line:

Property Pro-Forma:	Units:	Historical:	Projected:					Stabilized:
		FY18	FY19	FY20	FY21	FY22	FY23	
Revenue:								
(+) Base Rental Income:	\$	\$ 2,805,084	\$ 2,942,642	\$ 3,044,289	\$ 3,135,618	\$ 3,221,150	\$ 3,301,678	
(-) Loss to Lease:	\$	(252,957)	(296,740)	(295,303)	(178,965)	(98,434)	(39,238)	
(-) Absorption & Turnover Vacancy:	\$	-	(55,228)	(48,771)	(252,211)	(63,248)	(79,547)	
(-) Concessions & Free Rent:	\$	(61,763)	(27,389)	(193,336)	(182,115)	(109,447)	(79,630)	
(+) Expense Reimbursements:	\$	695,031	701,479	721,686	675,469	745,972	752,454	
Potential Gross Revenue:	\$	3,185,396	3,264,764	3,228,565	3,197,795	3,695,993	3,855,718	
(-) General Vacancy:	\$	(578,017)	(606,363)	(441,442)	(266,052)	(273,310)	(280,142)	
(+) Percentage Rent:	\$	111,360	114,701	118,142	90,175	124,123	126,605	
Effective Gross Income (EGI):	\$	2,718,738	2,773,103	2,905,264	3,021,918	3,546,807	3,702,181	
EGI Growth Rate:	%	N/A	2.0%	4.8%	4.0%	17.4%	4.4%	
Operating Expenses:								
(-) Property Management Fees:	\$	(81,562)	(83,193)	(87,158)	(90,658)	(106,404)	(111,065)	
(-) Common Area Maintenance (CAM):	\$	(140,206)	(145,049)	(148,610)	(151,582)	(154,202)	(156,515)	
(-) Common Area Utilities:	\$	(112,165)	(116,039)	(118,888)	(121,266)	(123,362)	(125,212)	
(-) Insurance:	\$	(56,082)	(58,020)	(59,444)	(60,633)	(61,681)	(62,606)	
(-) Real Estate & Property Taxes:	\$	(534,008)	(552,455)	(566,018)	(577,338)	(587,315)	(596,125)	
(-) CapEx, TI, and LC Reserves:	\$	(168,247)	(174,059)	(178,332)	(181,899)	(185,042)	(187,818)	
Total Operating Expenses:	\$	(1,092,271)	(1,128,814)	(1,158,450)	(1,183,376)	(1,218,006)	(1,239,341)	
Net Operating Income (NOI):	\$	1,626,468	1,644,288	1,746,814	1,838,542	2,328,801	2,462,839	
NOI Margin:	%	59.8%	59.3%	60.1%	60.8%	65.7%	66.5%	
(-) Capital Expenditures (CapEx):	\$	-	(1,260,750)	(1,014,750)	(799,500)	-	-	
(-) Tenant Improvements (TIs):	\$	-	(69,707)	(553,404)	(527,757)	(386,765)	(234,219)	
(-) Leasing Commissions (LCs):	\$	-	(26,174)	(221,208)	(195,693)	(137,471)	(87,100)	
(+) Capital Costs Paid from Reserves:	\$	-	1,303,056	1,730,371	949,110	76,766	217,496	
Adjusted Net Operating Income:	\$	1,626,468	1,590,713	1,687,824	1,264,702	1,881,331	2,359,017	
Adjusted NOI Margin:	%	59.8%	57.4%	58.1%	41.9%	53.0%	63.7%	

This renovation aims to boost Rents up to market rates, bringing "Loss to Lease" close to 0, and also to reduce General Vacancy by finding new tenant(s); the lease expirations in Years 2 and 3 mean that the Potential Gross Revenue will not go straight up.

The CapEx here represents the Renovation Costs over the first 3 years, which are mostly, but not entirely, covered by the Reserves. The TIs and LCs follow the normal pattern for lease expirations and new tenants. As a result, Adjusted NOI is very "lumpy" and goes up and down in the first 3 years.



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Below that is the Debt Service for this property:

Property Pro-Forma:	Units:	Historical:	Projected:				Stabilized:
		FY18	FY19	FY20	FY21	FY22	FY23
Adjusted Net Operating Income:	\$	1,626,468	1,590,713	1,687,824	1,264,702	1,881,331	2,359,017
<i>Adjusted NOI Margin:</i>	%	59.8%	57.4%	58.1%	41.9%	53.0%	63.7%
(-) Cash Interest Expense on Senior Loan	\$		(911,963)	(958,334)	(1,004,705)	(1,016,040)	
(-) Cash Interest Expense on Mezzanine:	\$		(71,340)	(75,620)	(80,158)	(84,967)	
(-) Senior Loan Principal Repayment:	\$		-	-	(515,233)	(515,233)	
Cash Flow to Equity Investors:	\$		607,410	653,870	(335,394)	265,090	
PIK Interest on Mezzanine:	\$		(142,680)	(151,241)	(160,315)	(169,934)	
Additional Draw from Senior Loan TI/LC H	\$		960,750	1,921,500	397,750	-	
(+) Ending Senior Loan Balance:	\$	15,457,000	15,457,000	15,457,000	14,941,767	14,426,533	
(+) Ending Mezzanine Balance:	\$	2,378,000	2,520,680	2,671,921	2,832,236	3,002,170	
Ending Debt Balance:	\$	17,835,000	17,977,680	18,128,921	17,774,003	17,428,704	
LIBOR:	%	2.60%	2.90%	3.20%	3.50%	3.80%	
Interest Rate on Senior Loan:	%		5.90%	6.20%	6.50%	6.80%	
Senior Loan Balance - Basis to Lenders:	\$	12,177,000	13,137,750	15,059,250	14,941,767	14,426,533	
Replacement Reserves:	\$		-	369,461	-	108,276	
Credit Stats & Ratios for Permanent Loans:							
<i>Debt Yield - NOI:</i>	%	11.2%	10.5%	9.9%	10.3%	13.4%	
<i>Debt Yield - Adjusted NOI:</i>	%	11.2%	10.2%	9.5%	7.1%	10.8%	
<i>Cash Interest Coverage Ratio - NOI:</i>	x		1.67 x	1.69 x	1.69 x	2.12 x	
<i>Cash Interest Coverage Ratio - Adjusted</i>	x		1.62 x	1.63 x	1.17 x	1.71 x	
<i>Debt Service Coverage Ratio (DSCR) - NI</i>	x		1.67 x	1.69 x	1.15 x	1.44 x	
<i>Debt Service Coverage Ratio (DSCR) - Ac</i>	x		1.62 x	1.63 x	0.79 x	1.16 x	

These formulas are actually *easier*. For the Interest, we just multiply the rate by the previous period's ending balance. For the Principal Repayment, if we're past the Interest-Only Period, we divide the initial balance, including the Holdback, by the amortization period of 30 years.

PIK Interest is a simple fixed percentage times the previous Mezzanine balance; the Additional Loan Draw uses a SUMIF formula linked to the Monthly Holdback release...

...which is significant because the property owners must pay interest on the FULL Senior Loan amount, less principal repayments, even though they have only drawn a portion of it! The "Basis to Lenders" line reflects the initial balance, Holdback releases, and principal repayments over time.

Minimum Debt Yield of 7.0%, Min Cash ICR of 1.50x, and Min DSCR of 1.20x, so these numbers are "OK" but a bit iffy for the DSCR in Year 3.

You have to be careful with the Holdback because there's always confusion about how it works.

Both **Holdbacks** and **Earn-Outs** are mechanisms that allow lenders to distribute additional amounts to borrowers over time.

The difference is that with a Holdback, the property owners still pay Interest on the full amount of the loan – even including the portion that is “held back” and not distributed initially.

With an Earn-Out, the owners pay Interest only on the distributed amount.

People tend to mix up these terms, change the definitions, or make up their own definitions, so be careful and read the fine print before assuming anything.

Office/Retail Acquisition & Renovation, Part 4 – Returns Calculations

The basic setup is quite similar to the other Returns Calculations in this course, but there are a few differences to note:



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Returns to Equity Investors:	Units:	Historical:	Projected:			
		FY18	FY19	FY20	FY21	FY22
Forward NOI:	\$		\$ 1,746,814	\$ 1,838,542	\$ 2,328,801	\$ 2,462,839
Boston Financial District - Class-B Office/Retail Cap Rates:						
Upside	%	7.50%	7.25%	7.00%	6.85%	6.75%
Base	%	7.50%	7.25%	7.20%	7.10%	7.00%
Downside	%	7.50%	7.25%	8.00%	7.50%	7.25%
Selected Class-B Office/Retail Cap Rate:	%	7.50%	7.25%	7.20%	7.10%	7.00%
Implied Property Value:	\$		24,093,991	25,535,309	32,800,008	35,183,419
Cash Flow to Equity Investors:						
(-) Initial Equity Investment:	\$	(6,328,350)	-	-	-	-
(+) Proceeds from Sale of Property:	\$		-	-	-	35,183,419
(+) Return of Replacement Reserve:	\$		-	-	-	108,276
(-) Selling Costs:	\$		-	-	-	(703,668)
(+) Cash Flows to Equity Investors:	\$		607,410	653,870	(335,394)	265,090
(-) Senior Loan Repayment:	\$		-	-	-	(14,426,533)
(-) Mezzanine Repayment:	\$		-	-	-	(3,002,170)
(-) Prepayment Penalty on Senior Loan:	\$		-	-	-	(144,265)
(-) Prepayment Penalty on Mezzanine:	\$		-	-	-	(30,022)
Total Cash Flows to Equity Investors:	\$	\$ (6,328,350)	\$ 607,410	\$ 653,870	\$ (335,394)	\$ 17,250,126
Internal Rate of Return (IRR):						
	%	32.2%				
Total Returns to Equity:	\$	18,176,012				
Invested Equity:	\$	6,328,350				
Cash-on-Cash Multiple:	x	2.9 x				

Make sure we're in or before the Exit Date; otherwise, this is 0.

For all of these, we use MIN/MAX functions to ensure that we repay *only* what we can with the available proceeds - to handle cases in which the lenders do not get their full principal back.

The formula for the Senior Loan Repayment in Year 4, written out in words, is as follows:

=IF(Current Year = Exit Date, -MAX(0, MIN(Senior Loan Basis to Lenders in Year 4, Sum of Net Proceeds Above This Line))), 0)

In the screenshot above, the Net Proceeds far exceed the Senior Loan Basis to Lenders, so we repay everything.

But if the Net Proceeds were only \$15 million and the Senior Loan Basis to Lenders in Year 4 were \$20 million, we would repay only the \$15 million.

The MAX(0 in this formula ensures that if the Net Proceeds are negative (unlikely, but you never know), the entire inner term becomes 0. And the negative sign on the outside ensures that this repayment is shown as a negative.

Below this, we also calculate the Returns to Lenders. This topic is covered more in the section on [Credit Deals and Investments](#), but here's the basic idea for the Senior Lenders:

Returns to Senior Lenders:	Units:	Historical:	Projected:			
		FY18	FY19	FY20	FY21	FY22
(-) Senior Loan Issued:	\$	\$ (12,177,000)	(960,750)	(1,921,500)	(397,750)	-
(+) Cash Interest Received:	\$		911,963	958,334	1,004,705	1,016,040
(+) Senior Loan Principal Repayment:	\$		-	-	515,233	515,233
(+) Senior Loan Issuance Fees:	\$	154,570	-	-	-	-
(-) Repayment of Senior Loan:	\$		-	-	-	14,426,533
(+) Prepayment Penalty on Senior Loan:	\$		-	-	-	144,265
Total Cash Flows to Senior Lenders:	\$	\$ (12,022,430)	\$ (48,787)	\$ (963,166)	\$ 1,122,188	\$ 16,102,072
Internal Rate of Return (IRR):	%	7.6%				
Recovery:	%	100.0%				
Total Returns:	\$	19,646,877				
Invested Amount:	\$	15,457,000				
Cash-on-Cash Multiple:	x	1.3 x				

Initial issuance and holdbacks released; this release over time reduces risk and boosts the IRR because of the time value of money.

Link to the Pro-Forma, but only count these if the year is before the Exit Date.

"Recovery" measures the total principal repaid vs. the total principal distributed over time. Should be 100% or the lenders will be in trouble.

In this model, there's also a waterfall schedule with a Preferred Return of 8% for the LPs, a Catch-Up Return of 20% for the GPs, and an 80% / 20% split between the LPs and GPs thereafter.

The LPs contribute 90% of the Equity in this deal, so this schedule provides a big boost for the GPs... *if* the deal performs well enough to beat this 8% Preferred Return (the effective "hurdle rate").

We'll address these terms, the formulas, and a more complex version of this schedule in [the section on Waterfall Returns Schedules](#), but the schedule output is shown below:

Equity Investors - Waterfall Returns Sched	Units:	Historical:	Projected:			
		FY18	FY19	FY20	FY21	FY22
Cash Flow to Equity Investors:						
(-) Invested Equity:	\$	\$ (6,328,350)	\$ -	\$ -	\$ (335,394)	\$ -
(+) Net Cash Flow After Debt Service:	\$	-	607,410	653,870	-	17,250,126
Total Cash Flow to Equity Investors:	\$	(6,328,350)	607,410	653,870	(335,394)	17,250,126
Limited Partners - Preferred Return of 8.0%:						
(+) Beginning Balance:	\$	-	5,695,515	5,543,746	5,333,376	6,061,901
(+) Investor Injections:	\$	5,695,515	-	-	301,855	-
(+) Investor Accruals:	\$	8.0%	455,641	443,500	426,670	484,952
(-) Preferred Distribution:	\$		(607,410)	(653,870)	-	(6,546,853)
Ending Balance:	\$	5,695,515	5,543,746	5,333,376	6,061,901	-
Cash Flow Available for Catch-Up Distrib	\$		-	-	-	10,703,274
General Partners - Catch-Up Return of 20.0%:						
(+) Beginning Balance:	\$	-	632,835	759,402	911,282	1,127,078
(+) Investor Injections:	\$	632,835	-	-	33,539	-
(+) Investor Accruals:	\$	20.0%	126,567	151,880	182,256	225,416
(-) Catch-Up Distribution:	\$		-	-	-	(1,352,494)
Ending Balance:	\$	632,835	759,402	911,282	1,127,078	-
Remaining Cash Flow Available for Distr	\$		-	-	-	9,350,780
Cash Flow Split for Remaining Available Distributions:						
LP Cash Flow:	\$	80.0%	-	-	-	7,480,624
GP Cash Flow:	\$	20.0%	-	-	-	1,870,156
Remaining Cash to Distribute:	\$		-	-	-	-

Cash Flows to Equity Investors are more than enough to achieve an 8% IRR for the LPs in the first 2 years, so the Investor Accruals are less than the absolute value of the Preferred Distributions.

Since the LPs get all the cash flows first, there's nothing for the GPs until the Exit in Year 4; at that point, the Equity IRR clearly exceeds 8%, so more proceeds are available for distribution.

We know for sure that the Preferred and Catch-Up Returns are satisfied only at the end, upon exit - so at that stage, the 80% / 20% split for the LPs and GPs kicks in, and the remaining cash flows are distributed like that.



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This schedule is a bit odd because in Years 1 and 2, the LPs earn *more than* 8% * Initial Investment since they receive *all* the Cash Flows to Equity Investors.

The Preferred Return works based on the *ending balance* upon exit and ensures that the LPs earn at least 8% by then; terms for waterfalls vary widely, so it's different but not completely unexpected.

We could change the formulas around and do a check that limits the Preferred Distribution to the Investor Accrual in each year prior to exit – if that's what the investor agreement states.

The final results in the Base Case look like this:

Returns Summary:	Units:		Historical:	Projected:				
			FY18	FY19	FY20	FY21	FY22	
Total Cash Flows to Equity Investors:	\$		(6,328,350)	607,410	653,870	(335,394)	17,250,126	
Internal Rate of Return (IRR):	%	32.2%						
Positive Cash Flows:	\$	18,511,406						
Invested Equity:	\$	6,663,744						
Cash-on-Cash Multiple:	x	2.8 x						
Limited Partner Cash Flows:	\$		(5,695,515)	607,410	653,870	(301,855)	14,027,477	
Internal Rate of Return (IRR):	%	29.6%						
Positive Cash Flows:	\$	15,288,756						
Invested Equity:	\$	5,997,370						
Cash-on-Cash Multiple:	x	2.5 x						
General Partner Cash Flows:	\$		(632,835)	-	-	(33,539)	3,222,650	
Internal Rate of Return (IRR):	%	49.6%						
Positive Cash Flows:	\$	3,222,650						
Invested Equity:	\$	666,374						
Cash-on-Cash Multiple:	x	4.8 x						

The waterfall returns hurt the LPs a little but benefit the GPs significantly.

Should We Pursue This Deal?

This is a risky deal with a wide range of outcomes and very different targeted IRRs as well: our firm is aiming for a 20% IRR in the Base Case, a 40% IRR in the Upside Case, and it wants to avoid losing money (i.e., at least a 0% IRR) in the Downside Case.

We use a variety of sensitivity tables to make a recommendation:

General Partners - IRR vs. Year 4 Exit Cap Rate and Market Scenario:

Market Scenario:		Year 4 Exit Cap Rate:							
		6.25%	6.50%	6.75%	7.00%	7.25%	7.50%	7.75%	8.00%
	Upside	75.5%	72.8%	70.1%	67.6%	65.1%	62.7%	60.4%	58.1%
	Base	58.5%	55.5%	52.5%	49.6%	46.8%	43.9%	41.1%	38.3%
	Downside	24.9%	20.3%	(10.0%)	#NUM!	#NUM!	#NUM!	#NUM!	#NUM!

Limited Partners - IRR vs. Year 4 Exit Cap Rate and Market Scenario:

Market Scenario:		Year 4 Exit Cap Rate:							
		6.25%	6.50%	6.75%	7.00%	7.25%	7.50%	7.75%	8.00%
	Upside	49.9%	47.8%	45.8%	43.8%	41.9%	40.0%	38.2%	36.4%
	Base	36.3%	34.0%	31.8%	29.6%	27.4%	25.3%	23.2%	21.1%
	Downside	11.4%	8.2%	8.0%	6.3%	2.7%	(1.1%)	(5.1%)	(9.4%)

The LPs mostly avoid losing money in the Downside Case, but the GPs do not because of the 8% Preferred Return, which means they receive nothing when the Total Equity IRR is below 8%.

We can also look at assumptions like the Acquisition Price vs. the Year 4 Exit Cap Rate and the Acquisition Price vs. the Market Scenario:

General Partners - IRR vs. Acquisition Price and Market Scenario:

Market Scenario:		Acquisition Price:							
		\$ 17,500,000	\$ 18,500,000	\$ 19,500,000	\$ 20,500,000	\$ 21,500,000	\$ 22,500,000	\$ 23,500,000	\$ 24,500,000
	Upside	83.4%	78.8%	74.4%	70.1%	66.0%	61.8%	57.8%	53.8%
	Base	64.9%	59.8%	54.7%	49.6%	44.5%	39.4%	34.1%	28.6%
	Downside	31.0%	23.3%	(17.9%)	#NUM!	#NUM!	#NUM!	#NUM!	#NUM!

Limited Partners - IRR vs. Acquisition Price and Market Scenario:

Market Scenario:		Acquisition Price:							
		\$ 17,500,000	\$ 18,500,000	\$ 19,500,000	\$ 20,500,000	\$ 21,500,000	\$ 22,500,000	\$ 23,500,000	\$ 24,500,000
	Upside	58.3%	54.0%	49.7%	45.8%	42.0%	38.4%	34.9%	31.6%
	Base	42.8%	38.3%	33.9%	29.6%	25.4%	21.4%	17.5%	13.6%
	Downside	16.2%	10.4%	8.0%	2.7%	(4.4%)	(12.2%)	(21.1%)	(32.5%)

The conclusion is clear just from glancing at these tables: **The Preferred Return structure means that the GPs lose money when the Total Equity IRR is below 8%, so we recommend against the deal.**

The numbers look OK in the Base and Upside Cases, but the Downside Case presents too much risk to justify this deal, at least from the perspective of the GPs.

Also, the Downside Case in this model is not *that* extreme: the property's occupancy rate stays the same, at 83%, there's a mild recession over 2 years, and the Loss to Lease nearly disappears because the existing tenants renew their leases at market rates. The Effective Gross Income still grows by 9% over 5 years, and the NOI still grows by 10%.



In an “Extreme Downside” Case, where all the existing tenants cancel (i.e., Renewal Probability of 0%), both Equity Investor groups would earn negative IRRs, and the GPs would lose everything.

On the other hand, we *would* recommend this deal for both the Senior Lenders and the Mezzanine Investors because the Senior Lenders are likely to earn a 7-8% IRR in any operational scenario, and the Mezzanine Investors are likely to earn a 9-10% IRR in any scenario.

Even with worse numbers for the Renewal Probability, the Downtime Months, and the Exit Cap Rate, the IRRs for the lenders hold up relatively well.

We might change our mind about the Equity Investors if the Senior Loan LTV were lower (~50% rather than 65%), if the Preferred Return were lower (~5%) or it applied equally to the LPs and GPs, or if the Renovation Costs or Acquisition Price were lower.

Risk factors for the lenders include rising Cap Rates, declining market rents, longer periods of downtime between tenants, WeWork as an “anchor tenant,” and leases for 35% of rentable square feet expiring in Year 3.

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Valuation of Individual Properties and Other Analyses

When you analyze entire companies, **valuation** is often your key concern because a company’s “intrinsic value” vs. its market value determines whether or not the company is a good potential investment, whether it should raise Debt or Equity, and more.

By contrast, valuation is *somewhat* less important for real estate – at least for individual properties – because it’s not necessarily the primary goal of an analysis.

As a real estate investor, you aim to earn a certain IRR and money-on-money multiple over a specified timeframe, such as 3-5 years.

Valuation plays a role because if a property’s asking price is too high, it will be very difficult to earn an acceptable IRR.

But valuation is more of a *supplemental* analysis, and it’s not nearly as rigorous for properties as it can be for normal companies.



For example, with real estate, you don't have to spend hours combing through filings looking for non-recurring charges, strange dilutive securities, and other oddities buried in the footnotes.

Instead, the challenge is usually **finding relevant market data** for your property and making sure that its valuation is not wildly off the mark.

You can use **3 basic methodologies** to value properties:

- 1) **"Multiples"** – But instead of EV / EBITDA or EV / Revenue, you use **Cap Rates** (also called "Yields") or metrics such as \$ per unit, \$ per square foot, or \$ per square meter.
- 2) **DCF** – The Discounted Cash Flow analysis can be useful when the property's risk and cash flow profiles change substantially, and Cap Rates don't produce meaningful numbers as a result.
- 3) **Replacement Cost** – How much would it cost to construct this same property today, and how does that number compare with the property's asking price?

Let's go through each of these in more detail:

"Multiples" AKA Cap Rates or \$ per Unit / SF / SM Figures

We covered the basic idea behind Cap Rates [in the first section of this guide](#); here, we'll give you a quick refresher and expand on the fundamentals.

The Cap Rate is defined as Net Operating Income / Property Price. Sometimes, you solve for the Property Price, and other times, you solve for the Cap Rate.

Lower Cap Rates mean that the property is more expensive or more highly valued by the market; a 5% Cap Rate corresponds to a $1 / 5\% = 20x$ multiple, while a 10% Cap Rate corresponds to a $1 / 10\% = 10x$ multiple.

Just as valuation multiples are abbreviated ways to express the Cash Flow, Cash Flow Growth Rate, and Discount Rate of companies (since $\text{Company Value} = \text{Cash Flow} / (\text{Discount Rate} - \text{Cash Flow Growth Rate})$), Cap Rates serve a similar function for properties.

If two properties have similar Cash Flows and Discount Rates because they are in similar locations and have similar financial characteristics, the one that is growing at a faster rate should have a *lower* Cap Rate, indicating a higher valuation.

In practice, however, Cap Rates tend to tell you more about *the location* and *the type of property* in that location.



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For example, Class A multifamily properties in Santa Monica in Los Angeles are likely to sell for similar Cap Rates, just as Class B offices in downtown Cleveland are likely to sell for similar Cap Rates.

Here's how you might use Cap Rates in real life to value a property:

Comparable Property Sales for The Lyric (215 10th Ave E)

Property Name:	# Units:	# Square Feet:	Avg. Unit Size in SF:	Year Built:	Sale Date:	Sale Price:	Price per Unit:	Price per SF:	Cap Rate:
Griffis Belltown	233	178,392	766	1991	2016-01-21	\$ 90,750,000	\$ 389,485	\$ 508.71	3.90%
Packard Building	61	53,679	880	2010	2016-02-29	25,866,000	424,033	481.86	4.00%
Cornelius	137	62,380	455	1925	2016-03-16	29,950,000	218,613	480.12	5.40%
Whitworth	54	40,297	746	1927	2016-07-22	18,230,000	337,593	452.39	4.20%
Rivet	131	134,900	1,030	2014	2016-08-01	54,750,000	417,939	405.86	4.40%
Walton Lofts	137	98,875	722	2015	2016-08-08	76,675,000	559,672	775.47	4.60%
Cue	90	59,391	660	2015	2016-08-12	39,663,000	440,700	667.83	4.50%
Anthem on 12th	120	73,813	615	2015	2016-09-15	35,034,877	291,957	474.64	5.20%
JUXT	361	343,203	951	2016	2016-10-05	151,400,000	419,391	441.14	4.60%
Summit	52	31,602	608	1949	2016-12-30	17,000,000	326,923	537.94	4.30%
Union Bay	73	86,525	1,185	1994	2017-01-12	24,000,000	328,767	277.38	4.60%
Radius	282	225,600	800	2015	2017-02-03	141,000,000	500,000	625.00	4.00%
Helix-Ellipse	150	121,000	807	2006	2017-08-15	53,961,430	359,743	445.96	4.50%
Bridges @ 11th	184	159,716	868	2015	2017-10-12	64,400,000	350,000	403.22	4.40%
8th & Republican	211	280,000	1,327	2016	2017-10-30	101,300,000	480,095	361.79	4.10%
Sherwood Apartments	49	32,398	661	1912	2018-03-02	13,995,000	285,612	431.97	3.60%
Median:	134	92,700	783	2012	2016-09-25	\$ 46,812,215	\$ 374,614	\$ 463.52	4.40%

If our property is truly comparable to these, then purchasing it for a 4.2% - 4.6% Cap Rate seems reasonable. On the other hand, a 3.5% Cap Rate would be too expensive, and a 6.0% Cap Rate would be "too good to be true."

For this analysis to be meaningful, our property can't be *too different* from these median stats. For example, if our property's asking price is \$75 million and it has 100,000 square feet, that's within range. An asking price of \$500 million with 600,000 square feet is not.

Not exactly "valuations," but useful quick checks.

You don't usually set up a football field graph or back into the "implied value" for the property across different methodologies; instead, the goal is to verify or establish reasonable ranges for the Going-In Cap Rate, Price per Unit, and Price per Square Foot.

The Price per Unit figures are useful for multifamily properties but not for the other property types for obvious reasons, though you could use Price per Room or "Price per Key" for hotels.

Price per Unit is a "rough" metric that makes sense mostly if the units are comparable in size and type. For example, if the properties above have primarily 1-bedroom units with 700 – 800 square feet of space, the average Price per Unit figure of \$374,614 is useful.



But if 30% of the units are studios, 30% are 1-bedroom, 20% are 2-bedroom, and 20% are 3-bedroom, this metric is far less useful because the square footage varies too much.

Price per Square Foot or Square Meter is also a “rough” metric because the size of a property only tells you so much about its income potential and profitability.

For example, Multifamily Properties A and B could both have 100,000 rentable square feet, but Multifamily A could have an Occupancy Rate of 95% with an NOI of \$5.0 million, while Multifamily B could have an Occupancy Rate of 85% with an NOI of \$4.4 million.

Looking at just the Price per Square Foot would give a misleading impression because Multifamily A's NOI is significantly higher – so it *should* have a higher Price per Square Foot.

These Price per Square Foot and Price per Square Meter metrics are arguably more useful for single-family owner-occupied homes because there's nothing else to go on there.

The tricky part about valuing properties with Cap Rates, or at least determining whether or not an asking price is reasonable with Cap Rates, is **getting good data**.

It's easy to find Cap Rates for office properties in the City of London or Manhattan, but it's much harder in Small Town A, Saskatchewan.

Cap Rate and NOI calculations also tend to be inconsistent, which adds to the confusion:

- 1) **Do You Use the Forward or Trailing NOI?** It should be the forward, projected NOI because that's what investors are paying to earn, but the calculations vary.
- 2) **Which Expenses Go into NOI?** For example, there's controversy over whether or not you should count the TI/LC/CapEx Reserves as expenses that reduce NOI. We always do so to reflect the true costs of running and maintaining the property, but other people argue that they should be shown below NOI instead.
- 3) **Stabilized or Historical, Extended NOI?** Do you use the “stabilized” NOI from after the property has reached its targeted Occupancy Rate, or do you extend the historical NOI, or do you do something else?

Because of these issues, it's almost impossible to get “clean data” when you're working with Cap Rates.

So, you use them as a rough valuation measure, and in investment analyses with acquisitions and exits based on Cap Rates, you always sensitize the Cap Rates.



Selecting Cap Rates for use in your analysis depends on the market and where you are in the cycle.

Ideally, you will review the historical data and create normal, optimistic, and pessimistic scenarios, with different Cap Rate progressions in each one.

For example, if Cap Rates increased to 7.0% in the last recession, the long-term average is 5.5%, and they're currently at 6.0%, then in the pessimistic case, you might assume an increase to 7.0% in the first two years, followed by an eventual decline to between 5.5% and 6.0% over time.

Real estate is a cyclical industry, and Cap Rates must reflect that cyclical nature.

If the property changes significantly, as in a renovation deal, you will often select an Exit Cap Rate that is less than the Going-In Cap Rate, under the logic that the property becomes more appealing during the holding period.

It's harder to justify such an assumption in a Core or Core-Plus deal because little about the property changes, so Cap Rates there are more likely to reflect simple market trends.

DCF Analysis

The Discounted Cash Flow (DCF) Analysis applies to individual properties just as it applies to normal companies.

You project the Unlevered Free Cash Flow from the property over 5-10 years, calculate the Terminal Value with a Terminal Multiple (Terminal Cap Rate here) or a Terminal FCF Growth Rate, discount the Terminal Value to Present Value, discount the Cash Flows to Present Value, and add them up.

Key differences for properties include:

1. **Cost of Equity and WACC** – Instead of un-levering and re-levering Beta to calculate Cost of Equity, you will usually base it on the sponsor's targeted Equity IRR. So, it might be in the 5-10% range for stabilized properties; for Value-Added deals, it might be in the 15-20% range.

The Cost of Debt will be based on the interest rates on the Debt used to fund the acquisition, and WACC should equal $(LTV * \text{Weighted Average Interest Rate on Debt}) + (1 - LTV) * \text{Cost of Equity}$.

2. **Changing Discount Rates** – If it's a Core or Core-Plus deal, you can probably assume the same Discount Rate for the entire projection period because very little changes.



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But if it's a Value-Added or Opportunistic deal where the property changes significantly, you should use a higher Discount Rate in the initial period and then reduce it over time as the property stabilizes.

- Unlevered Free Cash Flows** – These are simply the Adjusted NOI on a property's Pro-Forma. You can link in the first few years from your Pro-Forma and then project the ones in the years after that based on a simple growth rate.
- Terminal Value** – The calculation via the Perpetuity Growth Method is the same, but with the "Multiples Method," you normally apply a Cap Rate to the NOI in Year 6 or Year 11 (i.e., one year after the explicit forecast period). This Cap Rate is often based on a premium to the Exit Cap Rate in your returns model, meaning the property becomes less valuable over time.

Properties that stay the same become less competitive relative to newer properties over time, so Cap Rates tend to rise in the absence of major market changes.

- Implied Property Value** – Similar to the approach used for private companies, you do not back into the Implied Equity Value or Implied Share Price because they do not apply to properties. Instead, you stop at the Implied Total Property Value, which is the "Implied Enterprise Value for a Property."

You then compare this number to the asking price of the property to determine whether or not that price is reasonable.

Here's a short example for The Lyric, a multifamily property in Seattle used in one of the case studies:

DCF - Unleveraged Cash Flow Projections:	Units:	Historical:	Projected:										Stabilized
		FY18	FY19	FY20	FY21	FY22	FY23	FY24	FY25	FY26	FY27	FY28	Year
Net Operating Income (NOI):	\$	\$ 4,441,188	\$ 4,703,157	\$ 5,106,361	\$ 5,511,234	\$ 5,872,721	\$ 6,117,944	\$ 6,348,531	\$ 6,570,729	\$ 6,767,851	\$ 6,937,048	\$ 7,075,789	\$ 7,181,925
NOI Growth Rate:	%	N/A	5.9%	8.6%	7.9%	6.6%	4.2%	3.8%	3.5%	3.0%	2.5%	2.0%	1.5%
Unleveraged Cash Flow (Adjusted NOI):	\$	4,441,188	4,703,157	5,106,361	5,434,465	5,872,721	6,113,099	6,348,531	6,472,168	6,666,334	6,832,992	6,969,652	7,074,196
Adjusted NOI % NOI:	%	100.0%	100.0%	100.0%	98.6%	100.0%	99.9%	100.0%	98.5%	98.5%	98.5%	98.5%	98.5%
Present Value of Cash Flows:	\$		4,554,168	4,636,286	4,626,522	4,687,878	$=+K377/((1+Discount_Rate)^{K383})$			4,113,207	3,953,153	3,780,793	
Normal Discount Period:	Year Frac.		1.000	2.000	3.000	4.000	5.000	6.000	7.000	8.000	9.000	10.000	11.000
Mid-Year Discount Period:	Year Frac.		0.500	1.500	2.500	3.500	4.500	5.500	6.500	7.500	8.500	9.500	10.500

The historical year and first 6 projected years come from our Pro-Forma; after that, we use low-single-digit growth rates to project the rest.

This is a multifamily property, so Adjusted NOI and NOI are quite close due to the lack of huge TIs/LCs. And this is a "core-plus" deal, so CapEx isn't that big.

If this were a Value-Added or Opportunistic deal, the Discount Rate would change over time, and we'd have to use a different formula to calculate the PV of Cash Flows.

However, even those types of properties should stabilize over time, so the last few years would use similar assumptions for the growth rates, margins, and Stabilized Terminal Year.

Here are the Discount Rate and Terminal Value assumptions and Implied Value calculation in this analysis:

Discounted Cash Flow (DCF) Analysis - Assumptions and Output:

Cost of Equity:
Discount Rate (WACC):

15.00%
6.65%

The 15% Cost of Equity is based on the annualized rate of return the equity investors are targeting, not "Beta" or the ERP; WACC is $LTV * \text{Weighted Average Debt Interest Rate} + (1 - LTV) * \text{Cost of Equity}$.

Terminal Value - Multiples Method:

Terminal Cap Rate - Spread Over Year 5 Rate:

0.50%

Baseline Terminal Cap Rate:

5.25%

Baseline Terminal Value: \$ 136,798,578

Implied Terminal Cash Flow Growth Rate: 1.31%

(+) PV of Terminal Value: \$ 71,857,646

(+) Sum of PV of Cash Flows: 43,641,882

Implied Total Property Value: \$ 115,499,528

% Implied Value from PV of Terminal Value: 62.2%

Terminal Value - Perpetuity Growth Method:

Baseline Terminal Cash Flow Growth Rate:

1.31%

Baseline Terminal Value:

\$ 136,798,578

Implied Terminal Cap Rate:

5.25%

(+) PV of Terminal Value: \$ 71,857,646

(+) Sum of PV of Cash Flows: 43,641,882

Implied Total Property Value: \$ 115,499,528

% Implied Value from PV of Terminal Value: 62.2%

Cash flows for a stabilized property in a major urban area should *not* be growing much into the future. The Terminal Value uses the familiar formula, i.e., Terminal Year Adjusted NOI / (Discount Rate - Terminal Growth Rate), adjusted for the mid-year convention.

Premium / (Discount) to Asking Price: 10.5%

Premium / (Discount) to Asking Price: 10.5%

These results indicate that the Asking Price of ~\$104 million is a modest bargain and that the property is a bit undervalued at that price; on the other hand, it's *not* the deal of the century.

We can sensitize these results based on the Market Scenario, Discount Rate, Terminal Cap Rate, and Terminal Cash Flow Growth Rate, but we stick to the basics here:

		Discount Rate:							
		5.75%	6.00%	6.25%	6.50%	6.75%	7.00%	7.25%	7.50%
Terminal Cap Rate (Applied to Year 11 Stabilized NOI):	7.00%	\$ 104,134,128	\$ 102,244,409	\$ 100,398,898	\$ 98,596,422	\$ 96,835,839	\$ 95,116,044	\$ 93,435,960	\$ 91,794,546
	6.75%	106,306,706	104,366,287	102,471,375	100,620,760	98,813,265	97,047,751	95,323,109	93,638,264
	6.50%	108,646,405	106,651,386	104,703,273	102,800,816	100,942,801	99,128,051	97,355,423	95,623,806
	6.25%	111,173,280	109,119,294	107,113,723	105,155,277	103,242,700	101,374,775	99,550,322	97,768,191
	6.00%	113,910,728	111,792,860	109,725,044	107,705,942	105,734,256	103,808,726	101,928,129	100,091,276
	5.75%	116,886,214	114,698,911	112,563,437	110,478,405	108,442,470	106,454,325	104,512,702	102,616,368
	5.50%	120,132,200	117,869,148	115,659,864	113,502,910	111,396,886	109,340,433	107,332,236	105,371,013
	5.25%	123,687,327	121,341,312	119,051,190	116,815,463	114,632,674	112,501,409	110,420,297	108,388,006
	5.00%	127,597,967	125,160,692	122,781,649	120,459,271	118,192,040	115,978,482	113,817,164	111,706,698

Most regions of this table are fairly close to the \$104 million asking price, so the property seems appropriately-valued-to-modestly-undervalued depending on the assumptions used.

You can certainly build a DCF to value a property, but in many cases, it doesn't give you more useful information than the Cap Rate.



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Here, for example, the median Cap Rate for recent multifamily sales in this area is 4.40%, and we're acquiring the property for a 4.50% Going-In Cap Rate.

Just based on those two numbers, we could conclude that this property is appropriately-valued-to-modestly-undervalued, which is what the DCF tells us.

The DCF is arguably *more useful* when the property's cash flows change significantly during the holding period because of a renovation or redevelopment.

There, Cap Rates are less reliable because the property does not generate a stable stream of cash flows initially.

Also, the Discount Rate will change over time as the property stabilizes, and a simple Cap Rate applied to the Year 1 NOI cannot capture that dynamic.

If you build a DCF for this type of property, you still need to select a Terminal Cap Rate or Terminal Cash Flow Growth Rate to calculate the Terminal Value at the end.

To do that, you will have to look at comparable stabilized properties in the area and base your assumptions on recent transactions.

Replacement Cost Analysis

If Cap Rates and the DCF Analysis give you a "rough idea" of a property's value, the Replacement Cost Analysis produces even rougher values.

The idea is to estimate the cost of developing the entire property again, from the Land Acquisition Costs to the Hard Costs, Soft Costs, Financing Costs, and Profits for the Developers, and compare this total to the property's asking price.

But if you ask 10 different developers for their estimates, you'll get 10 very different answers.

Real estate developments almost always get delayed and go over-budget, so the *actual costs* in real life will almost certainly be different from these 10 estimates as well.

As a result, the Replacement Cost Analysis is used as more of a "sanity check" for the other methodologies and less of a strict valuation methodology by itself.

Here's an example, once again based on The Lyric, the multifamily property in Seattle:

Replacement Cost Analysis:

Category / Line Item:		Units:	Rentable SF:		
The Lyric (215 10th Ave E) - Property Statistics:		234	186,215		
Expense Category:	% Driver:	Total in \$ as Stated:	% Total: Dev Costs:	\$ per Unit:	\$ per Rentable SF:
Land Acquisition Costs:					
(+) Land Purchase:		\$ 22,000,000		\$ 94,017	\$ 118.14
(+) Taxes & Fees:		1,700,000		7,265	9.13
Total Land Costs:		23,700,000	24.1%	101,282	127.27
Hard Costs:					
(+) Excavation & Construction:		51,035,700		218,101	274.07
(+) Contingency:	3.00%	1,531,071		6,543	8.22
(+) General Contractor Fee:	5.00%	2,551,785		10,905	13.70
Total Hard Costs:		55,118,556	56.0%	235,549	295.99
Soft Costs:					
(+) Architectural & Engineering:		2,495,150		10,663	13.40
(+) Real Estate Taxes:		600,000		2,564	3.22
(+) Office and Common Area FF&E:		748,000		3,197	4.02
(+) Engineering:		745,123		3,184	4.00
(+) Startup Expenses:		345,178		1,475	1.85
(+) Legal & Closing:		415,415		1,775	2.23
(+) Impact & Permit Fees:		3,589,012		15,338	19.27
(+) Contingency:		1,500,000		6,410	8.06
(+) Development Fee:		2,415,975		10,325	12.97
Total Soft Costs:		12,853,853	13.1%	54,931	69.03
Financing Costs:		6,782,895	6.9%	28,987	36.43
Total Development Costs:		98,455,304	100.0%	420,749	528.72
Development Profit (20%):	20.00%	19,691,061		84,150	105.74
Total Replacement Cost:		\$ 118,146,365		\$504,899	\$ 634.46

Fair amount above the ~\$104 million asking price, indicating that the price is reasonable, if not a modest discount.

Fair amount above the ~\$104 million asking price, indicating that the price is reasonable, if not a modest discount.

You cannot "calculate" these numbers by yourself. You have to speak with developers, look at recently completed projects in the market, and do outside research to get a sense of what's appropriate.

Here, the Replacement Cost Analysis confirms what we already know: the property is likely to be modestly undervalued at an asking price of ~\$104 million.

If this methodology produced a value of, say, \$50 million or \$150 million, that would be more noteworthy and might require us to revisit the other valuation methodologies.

Analysis of Comparable Properties

Just as you can create a set of comparable public companies when you're valuing a normal company, you can also create a set of comparable properties when you're valuing a property.



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However, this set has no “valuation multiples” because individual properties do not trade on the public markets, so they have no trading multiples.

Instead, the purpose is to determine whether or not the assumptions used to create your Pro-Forma are reasonable. Here’s an example:

Comparable Properties for The Lyric (215 10th Ave E)

Property Name:	Address:	Neighborhood:	Year Built:	# Units:	# Square Feet:	Avg. Unit Size in SF:	In-Place Rent per Unit:	In-Place Rent per SF:	Effective Rent per Unit:	Effective Rent per SF:	Occupancy Rate:	% Concessions:	Comments:
Joule	523 Broadway Avenue E	Capitol Hill	2010	295	222,430	754	\$ 2,600	\$ 3.45	\$ 2,555	\$ 3.39	99%	2%	Similar amenities; 5 levels of underground parking; 26,000 retail SF.
The Broadway Building	1641 Nagle Place	Capitol Hill	2010	94	61,100	650	2,310	3.55	2,250	3.46	95%	3%	Inferior amenities; mixed office/retail space.
Citizen	1222 E Madison Street	Capitol Hill	2012	107	64,414	602	1,850	3.07	1,825	3.03	95%	1%	Inferior amenities; 59 parking spots; 6,000 retail SF.
Chloe	1408 E Union Street	Capitol Hill	2011	117	80,561	689	2,500	3.63	2,465	3.58	92%	1%	Similar amenities; inferior location.
Pearl	1530 15th Avenue	Capitol Hill	2008	80	48,160	602	2,105	3.50	2,065	3.43	95%	2%	Inferior amenities; similar location.
Packard	1530 12th Avenue	Capitol Hill	2009	61	45,124	740	2,434	3.29	2,378	3.21	93%	2%	Similar amenities; lacks 2BR units.
Terraviva Luxury	1615 Belmont Avenue	Capitol Hill	2012	108	76,140	705	2,395	3.40	2,345	3.33	99%	2%	Similar amenities; similar location.
REO Flats	1525 14th Avenue	Capitol Hill	2014	108	60,800	563	2,295	4.08	2,250	4.00	97%	2%	Similar amenities; similar location.
Sunset Electric	1111 E Pine Street	Capitol Hill	2014	92	71,213	774	2,550	3.29	2,495	3.22	95%	2%	Similar amenities; similar location.
Gatsby	1145 10th Avenue E	Capitol Hill	2014	70	56,035	801	2,750	3.44	2,700	3.37	98%	2%	Superior amenities; inferior location.
Total or Average:			2011	1,132	785,977	694	\$ 2,407	\$ 3.47	\$ 2,362	\$ 3.40	96%	2%	
Median:			2012	101	62,757	697	\$ 2,415	\$ 3.44	\$ 2,362	\$ 3.38	95%	2%	

Each property is unique, so these comments can add "color" and explain why other properties in the market are higher or lower quality even if the numbers look similar.

These numbers tell us whether or not our Pro-Forma assumptions are reasonable. For example, if our In-Place Rent per Square Foot is \$5.00, that's not reasonable for this market. But if it's in the \$3.00 - \$3.50 range, that makes more sense. The same goes for the Occupancy Rate and Concessions (i.e., Free Rent Months, Free Utilities, etc.).

For other property types, such as hotels or industrial/retail/office properties, you might see metrics such as the Average ADR and Occupancy Rate or the Average Lease Expiration Date, Free Months of Rent, TIs, LCs, and In-Place vs. Market Rents.

As always, the trick is *finding* the proper data in the first place. There’s nothing difficult about using an AVERAGE or MEDIAN function once you’ve collected the data from brokers or local developers.

Rent Roll Analysis

Finally, you’ll often analyze the “Rent Roll” of a property to determine how much **lease-expiration risk** there is.

This analysis is more meaningful if there are *relatively* few tenants; there’s no point doing it for a large multifamily property with 500 tenants where everyone has very similar 1-year leases.

So, it's most common to complete this process for office, retail, and industrial properties with a few tenants up to several dozen tenants.

Here is an example from a deprecated version of one of the case studies in this course (we skipped it in the current version due to lack of time and a lower tenant count):

Key Tenant Schedule:

Tenant:	Rentable Square Feet:	Percent of Building:	Lease Expiration
Suffolk Construction	9,197.4	8.0%	2017-10-31
Keurig Business Unit	8,869.0	7.7%	2018-06-30
Eaton Vance	8,507.9	7.4%	2017-03-31
Google	8,291.8	7.2%	2021-03-31
HarborOne Credit Union	8,173.8	7.1%	2017-09-30
Vistaprint	7,859.7	6.9%	2016-06-30
Massachusetts Medical Society	6,462.3	5.6%	2020-03-31
Cubist Pharmaceuticals	5,815.8	5.1%	2022-12-31
Hill Holiday	5,467.9	4.8%	2020-11-30
Athenahealth	4,637.9	4.0%	2020-06-30
HubSpot	4,257.3	3.7%	2019-07-31
Rockland Trust	3,500.5	3.1%	2019-06-30
Consigli Construction	3,474.4	3.0%	2016-02-29
Vacant	30,152.4	26.3%	N/A
Total:	114,668.0	100.0%	

We've sorted the tenants by # Rentable Square Feet occupied; this is a value-added deal because over 25% of the property is currently vacant, which we're aiming to reduce with a renovation.

These percentages help us see that there is significant lease-expiration risk in Years 2-4 because over 50% of in-place revenue, and over 40% of potential revenue, expires in those years. This is a big risk factor! Any lender looking at this deal would look at the worst-case scenario of leases in these years expiring and it taking a very long time to find new tenants, if at all.

Lease Expiration Schedule:

Year of Lease Expiration:	Rentable Square Feet:	Percent of Building:	# Tenants:	Annual In-Place Revenue @ Current Rates:	% Total In-Place Revenue:	Annual Potential Revenue @ Market Rates:	% Total Potential Revenue:	Year End Date:	Calendar Year:
Vacant	30,152.4	26.3%	8	\$ -	-	\$ 1,145,791	26.3%		1900
Year 1	-	-	0	-	-	-	-	2015-12-31	2015
Year 2	11,334.1	9.9%	2	410,351	13.2%	430,695	9.9%	2016-12-31	2016
Year 3	25,879.0	22.6%	3	949,361	30.6%	983,403	22.6%	2017-12-31	2017
Year 4	8,869.0	7.7%	1	332,586	10.7%	337,020	7.7%	2018-12-31	2018
Year 5	7,757.8	6.8%	2	292,754	9.4%	294,798	6.8%	2019-12-31	2019
Year 6	16,568.1	14.4%	3	603,077	19.4%	629,587	14.4%	2020-12-31	2020
Year 7	8,291.8	7.2%	1	310,113	10.0%	315,088	7.2%	2021-12-31	2021
Year 8	5,815.8	5.1%	1	205,299	6.6%	221,002	5.1%	2022-12-31	2022
Year 9	-	-	0	-	-	-	-	2023-12-31	2023
Year 10	-	-	0	-	-	-	-	2024-12-31	2024
Total:	114,668.0	100.0%	21	\$ 3,103,541	100.0%	\$ 4,357,384	100.0%		

It's important to calculate the percentages for *both* In-Place Revenue and Potential Revenue because in Value-Added deals, sponsors always try to present numbers that are rosier than reality.

For example, the 26% vacant space here is counted in the % Total Potential Revenue column, which makes the lease-expiration risk in Years 2-4 seem less serious.



But the % Total In-Place Revenue column makes it clear that lease expiration is a big risk factor since leases corresponding to over 50% of current rental income expire in Years 2-4.

You could also calculate a Weighted Average Lease Expiration (WALE) number or conduct other analysis around the expected TIs, LCs, and Concessions when these leases expire, but the rental income figures are the most important ones.

A lender would not necessarily look at the results above and turn this deal down, but it might request more safeguards, such as Holdbacks or Earn-Outs, and it might also push for a higher interest rate in exchange for this elevated risk.

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The Waterfall Returns Schedule

A “waterfall schedule” means that if the deal’s total Equity IRR or Equity Multiple are above certain thresholds, then the proceeds will *not* be split proportionally among different investor groups.

To illustrate, let’s say that the 3rd Party Investors contribute 80% of the Equity to fund an office development, and the Developers contribute the remaining 20%.

Normally, the 3rd Party Investors would receive 80% of the positive cash flows and exit proceeds in this deal, and the Developers would receive 20%.

But if there’s a “Waterfall” or “Promote” structure in place, this split will differ if the deal performs well enough.

For example, the terms might state that the proceeds are distributed proportionately up to a 20% Equity IRR – but if the Equity IRR is above 20%, then the Developers receive an additional 10%.

And if the Equity IRR goes above 30%, the Developers receive another 10%. So:

- **Proceeds Up to a 20% IRR:** 80% Investor / 20% Developer Split
- **Proceeds Between a 20% and 30% IRR:** 70% Investor / 30% Developer Split
- **Proceeds Above a 30% IRR:** 60% Investor / 40% Developer Split



Creating a schedule to support this distribution is trickier than it seems because these splits apply *only to the proceeds in the level*.

For example, let's say that there's an initial Equity Investment of \$10 million in Year 0 to fund this Development.

There are no additional Equity Investments and no positive or negative cash flows for Equity Investors until the end, in Year 5, when they sell the stabilized, finished property for \$50 million.

This result represents an IRR of 38% since $(\$50 / \$10)^{(1 / 5)} - 1 = 38\%$.

So, all 3 tiers will be "activated" in this deal, **but to do that, you have to determine the amount of proceeds that *corresponds* to each IRR level.**

For a simple scenario like this one, it's not difficult because you can rearrange the terms and use algebra to solve for the proceeds:

$$30\% = (x / 10)^{(1 / 5)} - 1$$

$$130\% = (x / 10)^{(1 / 5)}$$

$$130\% ^ 5 = x / 10$$

$$3.71293 = x / 10$$

$$37.1293 = x$$

And sure enough, \$37.1293 is the amount of proceeds in Year 5 that corresponds to a 30% IRR. A similar process for the 20% IRR level produces \$24.7927 for the amount there.

Therefore, to distribute the proceeds in this simple waterfall, you do the following:

- **Proceeds Up to a 20% IRR** = \$24.7927, so 80% of that amount goes to the Investors (\$19.8341), and 20% goes to the Developers (\$4.9585).
- **Proceeds Between a 20% and 30% IRR** = \$37.1293 – \$24.7927 = \$12.3367, so 70% of that amount goes to the Investors (\$8.6357) and 30% goes to the Developers (\$3.7010).
- **Proceeds Above a 30% IRR** = \$50.0000 – \$37.1293 = \$12.8707, so 60% of that amount goes to the Investors (\$7.7224) and 40% goes to the Developers (\$5.1483).



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As a result of these distributions, the Investors' IRR of 35% is slightly below the total Equity IRR of 38%, and the Developers' IRR of 47% is far above that total Equity IRR.

Here it is in Excel, starting with Tier 1, which is the "proportional level" where Investors and Developers earn proceeds based on their initial Equity contributions:

Waterfall Distribution in a Real Estate Deal

(\$ in Millions)

Waterfall Returns Distribution - Assumptions:

Initial Amount Invested (Total Equity):	\$ 10
Developer Contributions/Ownership %:	20%
Net Proceeds at End:	\$ 50

Level 1 of Waterfall Distribution Calculation:

IRR Up To: 20%

Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
\$ 10	\$ 12	\$ 14	\$ 17	\$ 21	\$ 25

25 <-- Distribute and Split This Amount in This Tier
25 <-- Amount of Proceeds Left to Distribute

Investor Percentage: 80%
Developer Percentage: 20%

20
5

If the total Equity IRR is at or below 20%, it stops here. But in this case, the total Equity IRR is above 20%, so the schedule continues. The splits happen in Tiers 2 and 3:

Level 2 of Waterfall Distribution Calculation:

IRR Through: 30%

Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
\$ 10	\$ 13	\$ 17	\$ 22	\$ 29	\$ 37

Based on the \$50 in Net Proceeds minus the \$25 distributed in Tier 1 minus the \$12 distributed in this tier.

12 <-- Distribute and Split This Amount in This Tier
13 <-- Amount of Proceeds Left to Distribute

Investor Percentage: 70%
Developer Percentage: 30%

9
4

Level 3 of Waterfall Distribution Calculation:

IRR Above: 30%

13 <-- Distribute and Split This Amount in This Tier
- <-- Amount of Proceeds Left to Distribute

Investor Percentage: 60%
Developer Percentage: 40%

8
5

And then we sum up everything at the end to determine the IRR to each group:

IRR Calculations:

Overall Deal IRR:

Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
\$ (10)	\$ -	\$ -	\$ -	\$ -	\$ 50

5-Year IRR:

38.0%

IRR Only to the Investors:

Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
\$ (8)	\$ -	\$ -	\$ -	\$ -	\$ 36

5-Year IRR:

35.2%

IRR Only to the Developers:

Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
\$ (2)	\$ -	\$ -	\$ -	\$ -	\$ 14

5-Year IRR:

47.2%

This deal performed very well, so the "Developer Promotes" boost the Developers' IRR to 47%, well above the total equity IRR of 38% while the Investors' IRR decreases by ~3% as a result of these same Promotes.

Why Do Investors and Developers Agree to These Terms?

The short answer is **incentives**. Offering Developers "Promotes" (i.e., 30% of cash flows rather than 20%) costs the Investors very little because they contribute the vast majority of Equity in deals.

But these Promotes provide a big incentive for the Developers doing the work because they encourage the Developers to finish on-time and on-budget and earn a significantly higher IRR in the process.

It's similar to a CEO who earns \$10 million per year at a large company giving up, say, \$100,000 of his compensation to reward a junior-level employee who performed some heroic task for the company.

For the CEO, it's no big deal because he/she still earns \$9.9 million, but it might double the employee's pay, so it's a massive bonus for him/her.

The difference is that these "Promotes" are used in almost any real estate deal with multiple investor groups, while CEOs hardly ever give away a portion of their compensation to lowly employees.

What Makes Waterfall Schedules More Complex?



The example above is simple because there's only one initial investment in Year 0 and only one positive cash flow at the end in Year 5, and the returns split is based on simple IRR tiers.

But in real life, many factors make these waterfall schedules more difficult to interpret and build:

- 1) **Timing:** What if there's a mix of positive and negative cash flows throughout the holding period, and they arrive in random months rather than following a clear pattern?
- 2) **Thresholds:** What if the waterfall is based on a mix of IRR-based tiers and multiple-based tiers, such as 80% / 20% up to a 20% IRR, 70% / 30% up to a 3x multiple, and 60% / 40% beyond a 3x multiple? Or what if the waterfall is based purely on multiples?
- 3) **Which IRR or Multiple?** Are the Promotes based on the total Equity IRR or multiple, or *just* the IRR or multiple for the Developers or Investors?
- 4) **Preferred and Catch-Up Returns:** Does one investor group receive all the cash flows before the other groups, up to a certain IRR or multiple ("Preferred Return")? If so, are the other groups then "caught up," until they reach that same IRR or multiple? Or, is this Preferred Return based on simultaneous distributions of cash flows to both groups with the same priority?
- 5) **Lookback and Clawback Provisions:** If one group, such as the Investors, does *not* achieve its targeted IRR, but another group does, are the proceeds redistributed at the end of the project so that the Investors *do* achieve this targeted IRR, even if it hurts the other group?

There are other variations and complexities, but the ones above are the most important in financial modeling and case studies.

The first one – **Timing** – almost always comes up in real life and dictates the basic requirements for any "real" waterfall structure in a financial model.

The second one – **Thresholds** – comes up a fair amount as well, but it's not that difficult to implement. As you'll see, multiple-based tiers are *easier* than IRR-based ones.

The third point – **Which IRR or Multiple?** – we deliberately avoid in most of this course because it tends to cause a massive amount of confusion. The first example in the section below uses it, but after that, we switch to thresholds based on the total Equity IRRs and multiples.

The fourth point – **Preferred and Catch-Up Returns** – is fairly simple to implement but can make a dramatic impact on the returns, especially in Downside Cases when the IRR is low but still positive.



The fifth point – **Lookback and Clawback Provisions** – is the most challenging point to implement because it requires VBA. It's still not "hard," but it requires more than just Excel formulas.

Below, we examine waterfall schedules for different deal types and highlight the more complex features in each category:

Waterfall Schedules for Acquisition and Renovation Deals (Preferred Return, Catch-Up Return, and Management Fee)

The first example here is for a relatively simple deal: the acquisition of an NY office/retail property with 3 main tenants and a small amount of vacant space.

The Limited Partners contribute 90% of the Equity in this deal, and the Operating Partner contributes the remaining 10%.

The Limited Partners in this deal receive a **10% Preferred Return based on *their own* IRR**, meaning that all cash flows corresponding to the LP IRR up to 10% go to them *first*.

Then, if the LP IRR is above 10%, the proceeds corresponding to the LP IRR above 10% are distributed 80% / 20% to the Limited Partners and Operating Partner.

That 80% / 20% split represents a "Promote" because the Limited Partners contributes 90% of the Equity, so they *should* receive 90% of the proceeds.

However, they only receive 80% of the proceeds and willingly give up 10% to incentivize the Operating Partner to perform.

The Operating Partner also receives a "Management Fee" each year of 1.5% * Initial Equity Invested.

This fee is *not* a Preferred Return because it doesn't accrue on a compounded basis in the same way that IRR does; it's just a fixed annual fee.

Here's the first part of the waterfall returns schedule where we deduct this Management Fee:

Waterfall Returns Schedule:	Units:	Historical:	Projected:				
		FY17	FY18	FY19	FY20	FY21	FY22
(+) Total Cash Flow and Returns to Equity Investors:	\$		684,751	774,838	236,792	433,786	22,187,633
(-) Asset Management Fee to Operating Partner:	\$		<u>=-OP_Equity*Asset_Mgmt_Fee</u>		(15,806)	(15,806)	(15,806)
Equity Cash Flow to Limited Partners:	\$		668,945	759,038	220,986	417,979	22,171,827



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This fee and the 10% Preferred Return to the LPs have the same “rank,” but since the fee is so low, we’re ignoring this fact and deducting the fee first. Here’s the next part:

			Historical:			Projected:		
			FY17	FY18	FY19	FY20	FY21	FY22
Waterfall Returns Schedule:	Units:							
(+) Total Cash Flow and Returns to Equity Investors:	\$			684,751	774,838	236,792	433,786	22,187,633
(-) Asset Management Fee to Operating Partner:	\$			(15,806)	(15,806)	(15,806)	(15,806)	(15,806)
Equity Cash Flow to Limited Partners:	\$			668,945	759,032	220,986	417,979	22,171,827
Tier 1 IRR - Up to 10.0% for JUST Limited Partners:								
Beginning Balance:				(9,483,750)	(9,763,180)	(9,980,466)	(10,757,527)	(11,415,300)
Returns Accrual:	10.0%			(948,375)	(976,318)	(998,047)	(1,075,753)	(1,141,530)
Repayment:	100.0%			668,945	=MIN(\$D205*H\$200,-SUM(H203+H204))			12,556,830
Ending Balance:				(9,483,750)	(9,763,180)	(9,980,466)	(10,757,527)	(11,415,300)
Cash Flow Available for Tier 2 Distribution:	\$				-	-	-	9,614,997

The LPs' initial equity investment.

Returns Accrual represents the *additional amount* the LPs are owed each year. It's based on the Beginning Balance * 10%, and the Beginning Balance becomes more negative when these returns accrue and less negative when cash flows are distributed.

"Repayment" represents the cash flows distributed to LPs. It's the MIN between 100% of the Equity Cash Flow to LPs and the Beginning Balance + Returns Accrual This Year. Either distribute all the cash flows we can, or the entire remaining owed balance.

This idea is the core concept behind all waterfall schedules: track the Equity Invested, increase the amount owed by 10% per year (or whatever the IRR is in this tier), and reduce it whenever there's a positive cash flow we can distribute.

Each year, we distribute all the positive cash flows, or, if the positive cash flows exceed the remaining owed balance, we distribute the entire remaining owed balance.

You see this condition in Year 5 (FY 22) above when the exit takes place, and the “Ending Balance” goes to 0.

Then, if there are *more* cash flows left after reaching this 10% IRR, we move to the next tier:



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Waterfall Returns Schedule:		Units:	Historical:	Projected:				
			FY17	FY18	FY19	FY20	FY21	FY22
(+) Total Cash Flow and Returns to Equity Investors:	\$			684,751	774,838	236,792	433,786	22,187,633
(-) Asset Management Fee to Operating Partner:	\$			(15,806)	(15,806)	(15,806)	(15,806)	(15,806)
Equity Cash Flow to Limited Partners:	\$			668,945	759,032	220,986	417,979	22,171,827
Tier 1 IRR - Up to 10.0% for JUST Limited Partners:								
Beginning Balance:				(9,483,750)	(9,763,180)	(9,980,466)	(10,757,527)	(11,415,300)
Returns Accrual:	10.0%			(948,375)	(976,318)	(998,047)	(1,075,753)	(1,141,530)
Repayment:	100.0%			668,945	759,032	220,986	417,979	12,556,830
Ending Balance:			(9,483,750)	(9,763,180)	(9,980,466)	(10,757,527)	(11,415,300)	-
Cash Flow Available for Tier 2 Distribution:	\$			-	-	-	-	9,614,997
Tier 2 IRR - Above 10.0% Limited Partners' IRR:								
Limited Partners (LPs):	80.0%			-	-	-	-	7,691,997
Operating Partner (OP):	20.0%			-	-	-	-	1,922,999

This second tier is simple because there are no thresholds or limits – it's a simple 80% / 20% split of all the cash flows above the 10% LP IRR.

We then “reassemble” these cash flows and add up the totals for the LPs and OP:

Waterfall Returns Schedule:		Units:	Historical:	Projected:				
			FY17	FY18	FY19	FY20	FY21	FY22
(+) Total Cash Flow and Returns to Equity Investors:	\$			684,751	774,838	236,792	433,786	22,187,633
(-) Asset Management Fee to Operating Partner:	\$			(15,806)	(15,806)	(15,806)	(15,806)	(15,806)
Equity Cash Flow to Limited Partners:	\$			668,945	759,032	220,986	417,979	22,171,827
Tier 1 IRR - Up to 10.0% for JUST Limited Partners:								
Beginning Balance:				(9,483,750)	(9,763,180)	(9,980,466)	(10,757,527)	(11,415,300)
Returns Accrual:	10.0%			(948,375)	(976,318)	(998,047)	(1,075,753)	(1,141,530)
Repayment:	100.0%			668,945	759,032	220,986	417,979	12,556,830
Ending Balance:			(9,483,750)	(9,763,180)	(9,980,466)	(10,757,527)	(11,415,300)	-
Cash Flow Available for Tier 2 Distribution:	\$			-	-	-	-	9,614,997
Tier 2 IRR - Above 10.0% Limited Partners' IRR:								
Limited Partners (LPs):	80.0%			-	-	-	-	7,691,997
Operating Partner (OP):	20.0%			-	-	-	-	1,922,999
Returns Analysis by Investor Group:								
Limited Partners (LP) - Leveraged Returns:								
Internal Rate of Return (IRR):	%	20.2%	(9,483,750)	668,945	759,032	220,986	417,979	20,248,827
Cash-on-Cash Multiple:	x	2.4 x						
Annual Yield on Initial Investment:	%			7.1%	8.0%	2.3%	4.4%	213.5%
Operating Partner (OP) - Leveraged Returns:								
Internal Rate of Return (IRR):	%	14.0%	(1,053,750)	15,806	15,806	15,806	15,806	1,938,806
Cash-on-Cash Multiple:	x	1.9 x						
Annual Yield on Initial Investment:	%			1.5%	1.5%	1.5%	1.5%	184.0%

The overall equity IRR was 19.6%, so the waterfall here hurt the OP and benefited the LPs.

The overall equity IRR was 19.6%, so the waterfall here hurt the OP and benefited the LPs.



This waterfall structure is quite strange because:

- A **10% Preferred Return** for a stabilized property in a mature market is extremely high. This number should be lower, or it should be a Preferred Return to both the LPs and the OP, distributed at the same time.
- Earning a **~20% IRR** on a stabilized property in a market like New York is also odd, bordering on “too good to be true” or at least “you should re-think your assumptions.”
- The **benefits** here go to the LPs, but a waterfall returns structure usually incentivizes the Developers, Operating Partner, or the others who do the work in the deal.

Now let's look at another waterfall schedule for the acquisition & renovation of an office/retail property in Boston (45 Milk Street).

The one used in the case study in this course features an 8% Preferred Return for the LPs, a 20% Catch-Up Return for the GPs, and then an 80 / 20 split of the remaining cash flow between the LPs and GPs.

That is fairly simple, so we are going to look at a modified, more complex version here with the following terms:

- **10% Preferred Return to LPs:** The Limited Partners receive ALL cash flows first, before the General Partners, until the cash flows are sufficient to give them a 10% IRR.
- **10% Catch-Up Return to GPs:** Then, the General Partners receive ALL cash flows next, ahead of the Limited Partners, until the cash flows are sufficient for them to achieve a 10% IRR.
- **Tier 1 Cash Flow Split:** Between a 10% and 20% Equity IRR, the LPs receive 90% of the cash flows and the GPs receive 10% because the LPs contributed 90% of the Equity and the GPs contributed 10%. If the IRR is less than 20%, we stop there. If it's above 20%, we continue.
- **Tier 2 Cash Flow Split:** Between a 20% and 30% Equity IRR, the LPs receive 80% of the cash flows and the GPs receive 20%. If the IRR is less than 30%, we stop there. If it's above 30%, we continue.
- **Tier 3 Cash Flow Split:** Above a 30% Equity IRR, the LPs receive 70% of the cash flows and the GPs receive 30%.



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This arrangement is more equitable for both investor groups, but as you'll see, the LPs are still at an advantage because **the cash flows go to them first**.

Here's the first part of the schedule in the Base Case (the signs for Accruals and Distributions are the opposite of those in the example above):

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Equity Investors - Waterfall Returns Schedule:	Units:	Historical:	Projected:			
		FY18	FY19	FY20	FY21	FY22
Cash Flow to Equity Investors:						
(-) Invested Equity:	\$	\$ (6,328,350)	\$ -	\$ -	\$ (335,394)	\$ -
(+) Net Cash Flow After Debt Service:	\$	-	607,410	653,870	-	17,250,126
Total Cash Flow to Equity Investors:	\$	(6,328,350)	607,410	653,870	(335,394)	17,250,126
Limited Partners - Preferred Return of 10.0%:						
(+) Beginning Balance:	\$	-	5,695,515	5,657,656	5,569,552	6,428,362
(+) Investor Injections:	\$	5,695,515	-	-	301,855	-
(+) Investor Accruals:	\$	10.0%	569,552	565,766	556,955	642,836
(-) Preferred Distribution:	\$		(607,410)	=MIN(0,MAX(-(H272+H273+H274),-H269))		
Ending Balance:	\$	5,695,515	5,657,656	5,569,552	6,428,362	-
Cash Flow Available for Catch-Up Distributions:	\$		-	-	-	10,178,928
General Partners - Catch-Up Return of 10.0%:						
(+) Beginning Balance:	\$	-	632,835	696,119	765,730	875,843
(+) Investor Injections:	\$	632,835	-	-	33,539	-
(+) Investor Accruals:	\$	10.0%	63,284	69,612	76,573	87,584
(-) Catch-Up Distribution:	\$		-	-	-	(963,427)
Ending Balance:	\$	632,835	696,119	765,730	875,843	-
Cash Flow Available for Tier 1 Distributions:	\$		-	-	-	9,215,501

Annual Invested Equity * % LP Contribution.

10% * Beginning Balance.

Distribute either Total Cash Flow to Equity, or just the remaining owed balance.

Annual
Invested
Equity * % LP
Contribution.

10% * Beginning
Balance.

Distribute either
Total Cash Flow
to Equity, or just
the remaining
owed balance.

The same logic as the Preferred Distribution, but based on row 278 instead of 269. The GPs are "caught up," but are still at a disadvantage due to the time value of money.

We need to include the Investor Injections to account for the possibility of additional Equity Investments even after the initial acquisition. We did **not** do this in the simplified example above.

The Preferred and Catch-Up Distribution formulas may look complicated, but the logic is simple: distribute all the cash flows, or if the total cash flows exceed the remaining owed balance, distribute the remaining owed balance.

This schedule illustrates one of the problems with Preferred and Catch-Up Returns as well: it seems like the LPs and GPs get similar treatment, **but the LPs get their cash flows much earlier**.

This is especially the case because we allow the Preferred Distribution to exceed the LP Investor Accruals, meaning that in Year 1, for example, the LPs earn *more than* 10% * Beginning Balance.

As a result, the LPs are at an advantage because of the time value of money, and the GPs are at a disadvantage.

To earn a higher IRR than the LPs, they'll need the deal to perform very well and achieve the threshold IRRs in Tiers 1-3.

Here's the next part for Tier 1 of the schedule:

	A	B	C	D	E	F	G	H	I	J
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Top part is similar, but now Investor Injections reflect *all* Invested Equity from both LPs and GPs.

Subtract Pref. and Catch-Up Distrib. so this *only* reflects the IRR between 10% and 20%, not 0% and 20%.

And then, once again, we distribute the lesser of the Cash Flows Available for Tier 1 Distribution and the Total Remaining Owed Balance. Here, the IRR is above 20%, so in the exit year, we distribute the Total Remaining Owed Balance.

Simple 90 / 10 split, and we have more to distribute in the next tier.

Tier 2 is nearly the same, but now we must subtract the Preferred and Catch-Up Distributions *and* the Tier 1 Distributions so that this one represents only the Equity IRR between 20% and 30%:

	A	B	C	D	E	F	G	H	I	J
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The same as in Tier 1, but with 30% rather than 20%.

Direct links to the previous tiers.

Familiar formula, but the "Remaining Owed Balance" term now subtracts distributions in previous tiers.

Ending Balance of 0 in Tier 2 indicates that the Equity IRR *is* above 30%, which is confirmed by the Tier 3 numbers below.



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Finally, we reassemble the cash flows to calculate the IRR and cash-on-cash multiple to the Limited Partners and General Partners:

Returns Summary:	Units:	Historical:		Projected:			
			FY18	FY19	FY20	FY21	FY22
Total Cash Flows to Equity Investors:	\$		(7,717,800)	255,707	426,290	244,305	17,173,058
Internal Rate of Return (IRR):	%	24.7%					
Positive Cash Flows:	\$	18,099,359					
Invested Equity:	\$	7,717,800					
Cash-on-Cash Multiple:	x	2.3 x					
Limited Partner Cash Flows:	\$		(6,946,020)	255,707	426,290	244,305	15,091,431
Internal Rate of Return (IRR):	%	24.2%					
Positive Cash Flows:	\$	16,017,733					
Invested Equity:	\$	6,946,020					
Cash-on-Cash Multiple:	x	2.3 x					
General Partner Cash Flows:	\$		(771,780)	-	-	-	2,081,626
Internal Rate of Return (IRR):	%	28.2%					
Positive Cash Flows:	\$	2,081,626					
Invested Equity:	\$	771,780					
Cash-on-Cash Multiple:	x	2.7 x					

The Preferred Return hurts the GPs, but the 10% Promote above the 20% IRR level more than makes up for the difference.

The GPs benefit from the 10% Promote in Tier 2, and they earn a higher IRR than the LPs despite the Preferred and Catch-Up Return structure. In the Downside Case, however, their returns are disastrous:

Returns Summary:	Units:	Historical:		Projected:			
			FY18	FY19	FY20	FY21	FY22
Total Cash Flows to Equity Investors:	\$		(6,328,350)	577,414	557,125	(914,060)	5,964,909
Internal Rate of Return (IRR):	%	(0.6%)					
Positive Cash Flows:	\$	7,099,448					
Invested Equity:	\$	7,242,410					
Cash-on-Cash Multiple:	x	1.0 x					
Limited Partner Cash Flows:	\$		(5,695,515)	577,414	557,125	(822,654)	5,964,909
Internal Rate of Return (IRR):	%	2.7%					
Positive Cash Flows:	\$	7,099,448					
Invested Equity:	\$	6,518,169					
Cash-on-Cash Multiple:	x	1.1 x					
General Partner Cash Flows:	\$		(632,835)	-	-	(91,406)	-
Internal Rate of Return (IRR):	%	#NUM!					
Positive Cash Flows:	\$	-					
Invested Equity:	\$	724,241					
Cash-on-Cash Multiple:	x	-					

We never reach the 10% IRR level required for the Preferred Return, so the GPs get... pretty much nothing.



This outcome illustrates a key risk of *sequential, prioritized* Preferred and Catch-Up Returns – even if the deal achieves a neutral-to-positive Equity IRR, individual investor groups such as the GPs or Developers may end up losing money if they have unfavorable terms.

Waterfall Schedules for Development Deals with a Sale of the Entire Property at the End (Mixed IRR/Multiple-Based Tiers, Monthly Schedule, and Lookback Function)

The basic waterfall returns structure is nearly the same for development deals.

The **main difference** is that the cash flows tend to be more irregular, especially with office, retail, and multifamily properties that take more time to construct (compared with industrial properties, which are often simple and fast).

For example, the cash flows might be negative initially, move closer to neutral as construction begins, turn positive briefly if there's a sale of excess land, turn negative again as tenants begin moving in, and then turn positive near the end of the holding period into the exit date.

To start, we'll look at a waterfall schedule for an industrial development where the Developers contribute 10% of the Equity and receive a 10% promote for an Equity IRR above 10% and another 10% promote for an Equity IRR above 20%.

The development takes a year to finish and another 1-2 years to stabilize, resulting in negative Cash Flow to Equity in Year 1. Here's Tier 1 of the waterfall schedule:



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	A	B	C	D	F	G	H	I	J	K	L	M
168			Total Cash Flows to Equity Investors:	\$	(15,001,880)	(361,159)	915,539	1,236,593	9,629,498	467,667	929,239	31,963,169
169												
176												
177												
178			Waterfall Returns Schedule:	Units:	Construction:					Operational Years:		
179					FY18	FY19	FY20	FY21	FY22	FY23	FY24	FY25
180			Tier 1 IRR - Up to 10.0%:									
181			Leveraged IRR to All Equity Investors:									
182			Beginning Balance:			(15,001,880)	(16,863,228)	(17,634,011)	(18,160,819)	(10,347,404)	(10,914,477)	(11,076,685)
183			Returns Accrual:	10.0%		(1,500,188)	(1,686,323)	(1,763,401)	(1,816,082)	(1,034,740)	(1,091,448)	(1,107,668)
184			Repayment:	100.0%		(361,159)	915,539	1,236,593	9,629,498	467,667	929,239	12,184,353
185			Ending Balance:			(15,001,880)	(16,863,228)	(17,634,011)	(18,160,819)	(10,347,404)	(10,914,477)	(11,076,685)
186												
187			Limited Partners (LPs):									
188			Beginning Balance:			(13,501,692)	(15,176,905)	(15,870,610)	(16,344,737)	(9,312,663)	(9,823,029)	(9,969,016)
189			Returns Accrual:	10.0%		(1,350,169)	(1,517,691)	(1,587,061)	(1,634,474)	(931,266)	(982,303)	(996,902)
190			Repayment:	90.0%		(325,044)	823,985	1,112,934	8,666,548	420,901	836,315	10,965,918
191			Ending Balance:			(13,501,692)	(15,176,905)	(15,870,610)	(16,344,737)	(9,312,663)	(9,823,029)	(9,969,016)
192												
193			Developer:									
194			Beginning Balance:			(1,500,188)	(1,686,323)	(1,763,401)	(1,816,082)	(1,034,740)	(1,091,448)	(1,107,668)
195			Returns Accrual:	10.0%		(150,019)	(168,632)	(176,340)	(181,608)	(103,474)	(109,145)	(110,767)
196			Repayment:	10.0%		(36,116)	=MIN(H\$168*\$D196,-SUM(H194:H195))			46,767	92,924	1,218,435
197			Ending Balance:			(1,500,188)	(1,686,323)	(1,763,401)	(1,816,082)	(1,034,740)	(1,091,448)	(1,107,668)
198												
199			Cash Flow Available for Tier 2 Distribution:									19,778,815

Similar setup, but we combine Injections and Distributions into the single "Repayment" line item.

If Cash Flow to Equity is positive, we distribute either the appropriate percentage of it or the entire Remaining Owed Balance; if it's negative, "Repayment" is also negative and it simply increases the Remaining Owed Balance.

We don't separate Injections and Distributions into separate line items here because of the issue described above with the cash flows potentially flipping from negative to positive to negative throughout the holding period.

We prefer to simplify the setup and let the single "Repayment" formula handle everything.

Tier 2, with an IRR limit of 20%, is similar, and Tier 3, for IRRs over 30%, is nearly the same as the setup described above for acquisition deals.

One *difference* in this schedule is that Tier 2 does **not** just represent the IRR between 10% and 20%. It represents everything from 0% up to 20%, so we do **not** subtract the Tier 1 Distributions here.

In a simple model like this one, we can easily adjust the "cash flow reassembly" formulas to reflect this change:



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	A	B	C	D	F	G	H	I	J	K	L	M
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This Tier 2 is "up to 20%," so we do not subtract any Tier 1 distributions.

... which means we must take Cash Flow Available for Tier 2 and subtract Cash Flow Available for Tier 3 to calculate the distributions for Tier 2.

We get a fairly standard result: an overall Equity IRR of ~20%, with the LPs at ~19% and the Developers at ~27% due to this waterfall structure:

Returns Analysis by Investor Group:

Returns Analysis by Investor Group:									
Limited Partners (LP) - Leveraged Returns:									
Internal Rate of Return (IRR):	\$	(13,501,692)	(325,044)	823,985	1,112,934	8,666,548	420,901	836,315	26,748,875
Cash-on-Cash Multiple:	%	19.3%							
Annual Yield on Initial Investment:	x	2.8 x							
	%		(2.4%)	6.1%	8.2%	64.2%	3.1%	6.2%	198.1%
Developers - Leveraged Returns:									
Internal Rate of Return (IRR):	\$	(1,500,188)	(36,116)	91,554	123,659	962,950	46,767	92,924	5,214,294
Cash-on-Cash Multiple:	%	26.6%							
Annual Yield on Initial Investment:	x	4.3 x							
	%		(2.4%)	6.1%	8.2%	64.2%	3.1%	6.2%	347.6%

These results are expected because the Developers receive significant promotes at the 10% IRR and 20% IRR levels.

If this deal had performed much worse and the Equity IRR had been below 10%, then both investor groups would have earned the same IRR.

We'll now look at 3 ways to make this development waterfall schedule more complex: monthly accruals, injections, and distributions; mixed IRR and multiple-based tiers; and the Lookback provision.

The first complexity – a **monthly schedule** – does not change that much. The main difference is that we must now track Investor Accruals using a formula that converts the annual IRR into a monthly one:

$$\text{Monthly Investor Accruals} = \text{Beginning Balance} * ((1 + \text{Annual IRR Limit})^{(1 / 12)} - 1)$$

Here it is in Excel, based on a monthly office development model with proportional distributions (87.5% / 12.5%) up to a 15% Equity IRR:

	A	B	C	D	E	F	G	H	I
227									
228			Waterfall Returns Schedule:	Units:		May-15	Jun-15	Jul-15	Aug-15
229									
230			Cash Flow to Equity Investors:						
231			(-) Invested Equity:	£		-£ 77,970,000	-£ 2,520,000	-£ 4,320,000	-£ 4,770,000
232			(+) Net Cash Flow After Debt Service:	£		-	-	-	-
233			Total Cash Flow to Equity Investors:	£		(77,970,000)	(2,520,000)	(4,320,000)	(4,770,000)
234									
235			Tier 1 - Up to 15.0% IRR:						
236			(+) Beginning Balance:	£		-	77,970,000	81,403,412	86,677,046
237			(+) Investor Injections:	£		77,970,000	2,520,000	4,320,000	4,770,000
238			(+) Investor Accruals:	£	15.0%	-	=G236*((1+\$E238)^(1/Months)-1)		
239			(-) Tier 1 Accrual Distribution:	£		-	-	-	-
240			Ending Balance:	£		77,970,000	81,403,412	86,677,046	92,462,461
241									
242			Investor Cash Flow:	£	87.5%	(68,223,750)	(2,205,000)	(3,780,000)	(4,173,750)
243			Developer Cash Flow:	£	12.5%	(9,746,250)	(315,000)	(540,000)	(596,250)
244									
245			Cash Flow Available for Tier 2 Distribution:	£		-	-	-	-

Top row is only for negative Cash Flows to Equity; bottom row is only for positive ones.

Same idea as usual - Investor Injections link directly to "invested Equity." Accruals are calculated on a monthly basis. Tier 1 Accrual Distributions are based on the typical MIN/MAX formula - distribute all the positive cash flow we can, or the total Remaining Owed Balance.

This process of splitting up cash flows is also slightly different in this model - see the explanation in text below.

The "Investor Cash Flow" formula in the first column here (cell F242) is the following:

$$=\text{MIN}(-(\text{F237}+\text{F239}), \text{F236}+\text{F238}) * \$\text{E242}$$

Written out in words, that is:

$$=\text{MIN}(-(\text{Investor Injections} + \text{Tier 1 Accrual Distribution}), \text{Beginning Balance} + \text{Investor Accruals}) * \text{Investor's \% of Cash Flow in This Tier}$$

This formula enforces the same constraint as in the Tier 1 Accrual Distributions: distribute the lesser of (Injections This Month + Accruals) or the Remaining Owed Balance for This 15% IRR.

Then, we divide by the Investor's % of Cash Flow in This Tier to gross this up to 100% of total Cash Flow, and we multiply by the Developer's % of Cash Flows in This Tier to determine the Developer Cash Flow.



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If Cash Flow to Equity is positive this month, Cash Flow Available for Tier 2 Distribution = Cash Flow to Equity – Investor Cash Flow – Developer Cash Flow.

If Cash Flow to Equity is negative, nothing is available for Tier 2, so Cash Flow Available for Tier 2 Distribution is 0.

This distribution into Tier 2 only “activates” in the exit month because of the large cash proceeds from the property sale at that time.

Moving down, Tier 2 in this schedule is based on an **Equity multiple** rather than IRR, which makes the schedule *easier* to set up:

	A	B	C	D	E	F	G	H
227								
228			Waterfall Returns Schedule:	Units:		May-15	Jun-15	Jul-15
229			Cash Flow Available for Tier 2 Distribution:	£		-	-	-
245								
246								
247			Tier 2 - 15.0% IRR to 2.0x Multiple:					
248			(+) Beginning Balance:	£		-	155,940,000	160,980,000
249			(+) Investor Injections:	£		77,970,000	2,520,000	4,320,000
250			(+) Investor Accruals:	£	2.00 x	77,970,000	2,520,000	=+H249*(\$E250-1)
251			(-) Tier 1 Accrual Distribution:	£		-	-	-
252			(-) Tier 2 Accrual Distribution:	£		-	-	-
253			Ending Balance:	£		155,940,000	160,980,000	169,620,000
254								
255			Investor Cash Flow:	£	77.5%	-	-	-
256			Developer Cash Flow:	£	22.5%	-	-	-
257								
258			Cash Flow Available for Tier 3 Distribution:	£		-	-	-
259								
260			Tier 3 - Above 2.0x Multiple:					
261								
262			Investor Cash Flow:	£	67.5%	-	-	-
263			Developer Cash Flow:	£	32.5%	-	-	-
264								
265			Remaining Cash to Distribute:	£		£	- £	- £

Accruals here simply equal the Investor Injections in this month * (Multiple - 1). So, if the investors contribute \$100, and it's a 3x multiple, the Accruals will be \$100 * (3 - 1) = \$200, and the Ending Balance will increase by \$100 + \$200 = \$300 total.

The other formulas are similar or the same - Investor Injections and Tier 1 Accrual Distributions are direct links, and the Tier 2 Accrual Distribution formula uses a nearly identical MIN/MAX function.

This type of mixed IRR/multiple-based waterfall is common in development deals because **the timing is more uncertain** than in Core, Core-Plus, or Value-Added deals.

As always, we reassemble the cash flows at the bottom; the overall Equity IRR is ~19%, and the Equity multiple is 2.23x, so all 3 IRR tiers get activated, and the Developers do quite well due to the Promotes:

Returns Summary:	Units:	May-15	Jun-15	Jul-15
Unleveraged Cash Flows:	£	(77,970,000)	(2,520,000)	(4,320,000)
Internal Rate of Return (IRR):	%	12.9%		
Positive Cash Flows:	£	1,672,880,329		
Invested Amounts:	£	973,979,127		
Cash-on-Cash Multiple:	x	1.72 x		
Total Cash Flows to Equity Investors:	£	(77,970,000)	(2,520,000)	(4,320,000)
Internal Rate of Return (IRR):	%	19.0%		
Positive Cash Flows:	£	998,275,207		
Invested Equity:	£	448,644,011		
Cash-on-Cash Multiple:	x	2.23 x		
Investor Cash Flow Before Lookback:	£	(68,223,750)	(2,205,000)	(3,780,000)
Internal Rate of Return (IRR):	%	18.3%		
Positive Cash Flows:	£	847,968,250		
Invested Equity:	£	392,563,510		
Cash-on-Cash Multiple:	x	2.16 x		
Developer Cash Flow Before Lookback:	£	(9,746,250)	(315,000)	(540,000)
Internal Rate of Return (IRR):	%	23.4%		
Positive Cash Flows:	£	150,306,957		
Invested Equity:	£	56,080,501		
Cash-on-Cash Multiple:	x	2.68 x		

Calculated with XIRR due to monthly schedule; SUMIFs used for other parts.

The waterfall helps the Developers quite a bit here because the IRR is above 15% and the Equity Multiple is above the 2x threshold.

We're not quite done yet because we still have to factor in the **Lookback**, which states the following:

"At the end of the deal, if the Developer has achieved an IRR above 20%, but the Investor's IRR is below 20%, the Developer will give up its proceeds to the Investor until the Investor achieves a 20% IRR."

If you're wondering how this could happen, just look at the screenshot above: The Developer Promotes result in the Developers earning a 23% IRR, while the Investors earn only an 18% IRR.

If the overall Equity IRR is in the 15-20% range, this scenario could easily happen because the Developers will earn a higher IRR than the Investors.

Investors typically negotiate Lookback provisions **to protect themselves**, especially when the waterfall uses more complex criteria and they are more concerned about Downside scenarios.

Here's what happens with the Lookback provision in place:

Returns Summary:		Units:	Apr-21	May-21
Total Cash Flows to Equity Investors:	£		320,845	797,718,886
Internal Rate of Return (IRR):	%	19.0%		
Positive Cash Flows:	£	998,275,207		
Invested Equity:	£	448,644,011		
Cash-on-Cash Multiple:	x	2.23 x		
Investor Cash Flow Before Lookback:	£		280,739	672,481,469
Internal Rate of Return (IRR):	%	18.3%		
Positive Cash Flows:	£	847,968,250		
Invested Equity:	£	392,563,510		
Cash-on-Cash Multiple:	x	2.16 x		
Developer Cash Flow Before Lookback:	£		40,106	125,237,417
Internal Rate of Return (IRR):	%	23.4%		
Positive Cash Flows:	£	150,306,957		
Invested Equity:	£	56,080,501		
Cash-on-Cash Multiple:	x	2.68 x		
Lookback Required:	£			64,403,385
Ctrl + Shift + L to activate Lookback VBA function.				
Investor Cash Flow After Lookback:	£		280,739	736,884,854
Internal Rate of Return (IRR):	%	20.0%		
Positive Cash Flows:	£	912,371,634		
Invested Equity:	£	392,563,510		
Cash-on-Cash Multiple:	x	2.32 x		
Developer Cash Flow After Lookback:	£		40,106	60,834,033
Internal Rate of Return (IRR):	%	10.2%		
Positive Cash Flows:	£	85,903,573		
Invested Equity:	£	56,080,501		
Cash-on-Cash Multiple:	x	1.53 x		

Investor IRR is below 20%, but Developer IRR is above 20% due to the waterfall, so the Lookback is activated.

We literally "take away" ~£64 million in proceeds from the Developer and distribute them to the Investors instead, so that the Investors' IRR equals exactly 20%. This is very bad for the Developers!

The Excel formulas for this Lookback are easy: subtract it from the Developer Cash Flow and add it to the Investor Cash Flow.

The hard part is **calculating the Lookback amount and determining the proper date**.

We need VBA to do that because we must apply the Goal Seek function to a different cell depending on the exit date, and then we must insert the correct value elsewhere in the spreadsheet.

Here's the function:



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```
Sub Lookback()  
'  
' Macro1 Macro  
'  
' Keyboard Shortcut: Ctrl+Shift+L  
'  
Dim LookbackAmount  
Dim DevProceedsOnExit  
Dim LookbackCell  
Dim DevProceedsCell  
  
[Lookback_Range].ClearContents  
  
If [Investor_IRR].Value < [Investor_Lookback_IRR].Value And [Developer_IRR].Value > [Investor_Lookback_IRR].Value Then  
  
Set LookbackCell = WorksheetFunction.Index([Returns_Full_Range], [Returns_Full_Range].Rows.Count, WorksheetFunction.Match([Exit_Date], [Returns_Date_Range], 0))  
Set DevProceedsCell = WorksheetFunction.Index([Returns_Full_Range], [Returns_Full_Range].Rows.Count - 6, WorksheetFunction.Match([Exit_Date], [Returns_Date_Range], 0))  
  
[Investor_IRR].GoalSeek _  
Goal:=[Investor_Lookback_IRR].Value, _  
ChangingCell:=LookbackCell  
  
LookbackAmount = LookbackCell.Value  
DevProceedsOnExit = DevProceedsCell.Value  
  
If (LookbackAmount > DevProceedsOnExit) Then  
LookbackCell.Value = DevProceedsOnExit  
End If  
  
End If  
End Sub
```

The first few lines declare the variables we'll be using (LookbackAmount, DevProceedsOnExit, LookbackCell, and DevProceedsCell).

Here are the explanations for the rest:

[Lookback_Range].ClearContents – This deletes the existing contents of the Lookback cell in case we change the assumptions and want to re-run this function or refresh the results.

If [Investor_IRR].Value < [Investor_Lookback_IRR].Value And [Developer_IRR].Value > [Investor_Lookback_IRR].Value Then

This IF statement checks to see if the Investor IRR is below 20% and the Developer IRR is above 20%. If so, we proceed with the rest of the function.

Set LookbackCell = WorksheetFunction.Index([Returns_Full_Range], [Returns_Full_Range].Rows.Count, WorksheetFunction.Match([Exit_Date], [Returns_Date_Range], 0))

Set DevProceedsCell = WorksheetFunction.Index([Returns_Full_Range], [Returns_Full_Range].Rows.Count - 6, WorksheetFunction.Match([Exit_Date], [Returns_Date_Range], 0))

This next part uses INDEX/MATCH functions to find the appropriate cell locations for the Developer Proceeds received upon exit and the Lookback Amount, which will be 6 rows below the Developer Proceeds.



```
[Investor_IRR].GoalSeek _  
  Goal:=[Investor_Lookback_IRR].Value, _  
  ChangingCell:=LookbackCell
```

Then, we use the Goal Seek function to set the Investor IRR to 20% and solve for the Exit Proceeds that will produce that 20% IRR.

We already have the locations of these cells, so we then retrieve the output of this Goal Seek:

```
LookbackAmount = LookbackCell.Value  
DevProceedsOnExit = DevProceedsCell.Value
```

This “LookbackAmount” is the amount of proceeds the Investors *should* receive to earn a 20% IRR. Just one small problem: **that amount might exceed the total that actually goes to the Developers upon exit (i.e., DevProceedsOnExit)!**

If we need £200 million to boost the Investors’ IRR to 20%, but the Developers only receive £150 million upon exit, we obviously cannot redistribute all £200 million. The next part handles that:

```
If (LookbackAmount > DevProceedsOnExit) Then  
  LookbackCell.Value = DevProceedsOnExit  
End If
```

So, in the example directly above, we would “override” the Goal Seek function and set the LookbackCell’s value to £150 million instead. We distribute all the Developers’ proceeds, and that’s all we can do.

On the other hand, if the Developers had received £250 million upon exit, this IF statement would be false, and we would do nothing here – £200 million of the Developers would simply be redistributed, following the Goal Seek.

The Lookback function is interesting, but it’s in the “Optional” category for waterfall schedules.

Waterfall Schedules for Pre-Sold Condo Development Deals

This section will be brief because we’ve covered the main idea in the previous sections on [condo developments](#) and [development & pre-sale strategies](#).

Also, the waterfall returns schedules in the pre-sold condo models in this course are nearly identical to the ones in the development examples above.

The main difference is that the cash flows are even more irregular because in most pre-sold deals, there isn't a single "exit date," but rather a series of fluctuating positive cash flows that arrive as units are pre-sold and deposits are collected.

For your reference, here's the top part of the waterfall in the simplified condo model:

Waterfall Returns Schedule:	Units:	Jan 15	Feb 15	Mar 15	Apr 15
Cash Flow to Equity Investors:					
(-) Invested Equity:	HK\$	\$ (754,775,012)	\$ (6,449,172)	\$ (3,242,818)	\$ (3,261,154)
(+) Net Cash Flow After Debt Service:	HK\$	-	-	-	-
Total Cash Flow to Equity Investors:	HK\$	(754,775,012)	(6,449,172)	(3,242,818)	(3,261,154)
Tier 1 - Up to 20.0% IRR:					
(+) Beginning Balance:	HK\$	-	754,775,012	772,779,390	787,853,051
(+) Investor Injections:	HK\$	754,775,012	6,449,172	3,242,818	3,261,154
(+) Investor Accruals:	HK\$	20.0%	11,555,206	11,830,843	12,061,613
(-) Tier 1 Accrual Distribution:	HK\$	-	=MIN(0,MAX(-(G121+G122+G123),-G118))	-	-
Ending Balance:	HK\$	754,775,012	772,779,390	787,853,051	803,175,818
Investor Cash Flow:	HK\$	80.0%	(603,820,009)	(5,159,338)	(2,594,255)
Developer Cash Flow:	HK\$	20.0%	(150,955,002)	(1,289,834)	(648,564)
Cash Flow Available for Tier 2 Distribution:	HK\$	-	-	-	-

Almost the same setup as the Development waterfall above - Injections are directly linked to Invested Equity, Accruals are calculated on a monthly basis, and the Tier 1 Accrual Distribution is the lesser of Total Cash Flow to Equity and the Remaining Owed Balance.

Tier 2 is based on a 3x Equity multiple threshold, which is nearly the same as in the development examples above, and Tier 3 is for everything over a 3x Equity multiple.

In the more complex pre-sold condo model, the waterfall also includes a Preferred Return for the Investors, a Catch-Up Return for the Developers, and several tiers of Promotes.

These Preferred and Catch-Up Returns are based on **1x Equity multiples** instead of IRRs, but they work the same way as the Preferred and Catch-Up Returns in the acquisition/renovation example:

Waterfall Returns Schedule:	Units:	Jan 18	Feb 18	Mar 18	Apr 18	May 18
Cash Flow to Equity Investors:						
(-) Invested Equity:	R\$	-R\$ 8,963,439	R\$ -	R\$ -	-R\$ 4,941,119	-R\$ 5,706,039
(+) Net Cash Flow After Debt Service:	R\$	-	2,788,820	2,042,922	-	-
Total Cash Flow to Equity Investors:	R\$	(8,963,439)	2,788,820	2,042,922	(4,941,119)	(5,706,039)
Investors - Preferred Return of 1.0x Invested Equity:						
(+) Beginning Balance:	R\$	-	7,170,752	4,381,931	2,339,009	6,291,904
(+) Investor Injections:	R\$	7,170,752	-	-	3,952,896	4,564,832
(+) Investor Accruals:	R\$	1.00 x	-	-	-	-
(-) Investor Preferred Distribution:	R\$	-	(2,788,820)	(2,042,922)	=MIN(0,MAX(-(1141+1142+1143),-1138))	-
Ending Balance:	R\$	7,170,752	4,381,931	2,339,009	6,291,904	10,856,736
Cash Flow Avail. for Catch-Up Distributions:	R\$	-	-	-	-	-
Developers - Catch-Up Return of 1.0x Invested Equity:						
(+) Beginning Balance:	R\$	-	1,792,688	1,792,688	1,792,688	2,780,912
(+) Investor Injections:	R\$	1,792,688	-	-	988,224	1,141,208
(+) Investor Accruals:	R\$	1.00 x	-	-	-	-
(-) Developer Catch-Up Distribution:	R\$	-	-	-	-	-
Ending Balance:	R\$	1,792,688	1,792,688	1,792,688	2,780,912	3,922,120
Cash Flow Avail. for Tier 1 Distributions:	R\$	-	-	-	-	-

Accruals are still based on Injections This Month * (Equity Multiple - 1), so for a 1.0x multiple, the Accruals are always 0, and the Ending Balance simply increases by the Investor Injections each month and declines whenever there are Preferred Distributions.

Put simply, any positive cash flow generated will go to the Investors FIRST until they earn this 1.0x multiple of their invested equity.

Then, if the cash flows take us beyond this 1.0x equity multiple, the Developers are "caught up" - and then we move into the next tiers.

We use tiers that represent ranges *in between* specific IRRs or multiples here, so we must subtract the Preferred and Catch-Up Distributions and Tier 1 Accrual Distributions in the subsequent tiers.

Here's the waterfall schedule in the exit month:

Waterfall Returns Schedule:	Units:	Sep 21	Oct 21	Nov 21
Tier 1 - 1.0x Multiple Up to 20.0% IRR:				
(+) Beginning Balance:	R\$	108,279,499	109,937,201	111,620,281
(+) Investor Injections:	R\$	-	-	-
(+) Investor Accruals:	R\$	20.0%	1,657,702	1,683,080
(-) Preferred and Catch-Up Distributions:	R\$	-	-	(62,084,032)
(-) Tier 1 Accrual Distribution:	R\$	-	-	(51,245,096)
Ending Balance:	R\$	109,937,201	111,620,281	-
Investor Cash Flow:	R\$	80.0%	-	40,996,077
Developer Cash Flow:	R\$	20.0%	-	10,249,019
Cash Flow Avail. for Tier 2 Distributions:	R\$	-	-	2,843,659
Tier 2 - 20.0% IRR to 3.0x Multiple:				
(+) Beginning Balance:	R\$	195,915,582	195,915,582	195,915,582
(+) Investor Injections:	R\$	-	-	-
(+) Investor Accruals:	R\$	3.00 x	-	-
(-) Preferred and Catch-Up Distributions:	R\$	-	-	(62,084,032)
(-) Tier 1 Accrual Distribution:	R\$	-	-	(51,245,096)
(-) Tier 2 Accrual Distribution:	R\$	-	-	(2,843,659)
Ending Balance:	R\$	195,915,582	195,915,582	79,742,795
Investor Cash Flow:	R\$	70.0%	-	1,990,561
Developer Cash Flow:	R\$	30.0%	-	853,098
Cash Flow Avail. for Tier 3 Distributions:	R\$	-	-	-
Tier 3 - Above 3.0x Multiple:				
Investor Cash Flow:	R\$	60.0%	-	-
Developer Cash Flow:	R\$	40.0%	-	-

This tier is for a 1.0x multiple *up to* a 20% IRR, so we must subtract the Preferred and Catch-Up Distributions.

And this tier is for the returns *in between* a 20% IRR and a 3.0x multiple, so we must subtract the Preferred and Catch-Up Distributions and the Tier 1 Accrual Distributions.

There is a positive Ending Balance in Tier 2, indicating that the equity multiple has not reached 3x. The IRR is only slightly above 20% here.

In this deal, the waterfall structure gives a slight advantage to the Investors because of the Preferred Return in the beginning.

The Developer Promotes are not enough to compensate for that because the overall Equity IRR is not high enough:

Returns Summary:		Units:	Oct 21	Nov 21
Unleveraged Cash Flows:	R\$		-	190,917,238
Internal Rate of Return (IRR):	%	17.2%		
Positive Cash Flows:	R\$	314,829,713		
Invested Amounts:	R\$	246,192,494		
Cash-on-Cash Multiple:	x	1.28 x		
Total Cash Flows to Equity Investors:	R\$		-	116,172,787
Internal Rate of Return (IRR):	%	20.9%		
Positive Cash Flows:	R\$	121,004,530		
Invested Equity:	R\$	66,915,775		
Cash-on-Cash Multiple:	x	1.81 x		
Investor Cash Flow:	R\$		-	91,687,515
Internal Rate of Return (IRR):	%	21.2%		
Positive Cash Flows:	R\$	96,519,258		
Invested Equity:	R\$	53,532,620		
Cash-on-Cash Multiple:	x	1.80 x		
Developer Cash Flow:	R\$		-	24,485,272
Internal Rate of Return (IRR):	%	19.9%		
Positive Cash Flows:	R\$	24,485,272		
Invested Equity:	R\$	13,383,155		
Cash-on-Cash Multiple:	x	1.83 x		

The total equity IRR here is not high enough for the Developers to benefit much from the incentive structure; they only get a Promote when the IRR exceeds 20%, which barely happens. As a result, the Preferred Return up to a 1.0x equity multiple benefits the Investors more and boosts their IRR.

That's it for our coverage of waterfall returns schedules.

We have not taught every possible combination of assumptions and conditions, but we have covered the variations most likely to come up in case studies, modeling tests, and even on the job.

If you understand the main points in this section, you should be well-prepared for interview questions and modeling tasks related to waterfall schedules.

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Credit Analysis and Investments

Credit analysis is all about **analyzing the worst-case scenario** and assessing the chances of losing money if there's a severe market downturn or something goes disastrously wrong with the property.

Lenders have extremely limited upside because they earn a fixed, or nearly fixed, interest rate regardless of the property's performance.

If the property's NOI increases by 3x, Cap Rates go from 6% to 4%, and the Equity Investors earn a 50% IRR, the Senior Lenders who funded the deal at a 5% fixed interest rate will earn... a 5% IRR.

On the other hand, if the property's NOI falls by 20%, Cap rates rise from 5% to 7%, and the Equity Investors earn a 2% IRR, the Senior Lenders will earn... that same 5% IRR.

Other terms could boost lenders' returns slightly above the interest rate (e.g., origination fees, prepayment penalties, and small Equity grants), but most do not change the picture *that much*.

The lenders would earn substantially less than the interest rate only if the deal were a complete disaster and they were not fully repaid at the end.

For example, suppose that the Senior Lenders fund \$100 million of a \$150 million property purchase and earn 5% fixed interest per year.

The Going-In Cap Rate is 6%, so there is \$9 million of NOI to cover the \$5 million of interest per year.

However, this property's NOI declines by 6% each year; it falls from \$9 million in Year 1 to ~\$6.6 million in Year 6.

At the same time, the market also softens, and Cap Rates for similar properties rise to 8%.

Therefore, when the property is sold in Year 5, the exit price is \$6.6 million / 8% = \$82.6 million.

Warning signs should already be flashing because **that is less than the initial \$100 million Senior Loan balance**.

To be fair, we should probably factor in some amortization; let's say there's a 2-year interest-only period followed by a 30-year amortization period.

At that rate, the Senior Loan balance will be down to \$90 million by the end of Year 5 – but there's only \$82.6 million in proceeds, so the Senior Lenders get \$82.6 million rather than \$90 million.



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As a result, their IRR is **3.69%** rather than the ~5.00% they were expecting (initial investment of \$100 million, two \$5 million cash inflows, 2 cash inflows of \$8.3333 million, and in Year 5, the \$8.3333 million + \$82.6 million).

This scenario is not even a real “disaster,” but it is a disappointing outcome for the Senior Lenders because the IRR is well below their targeted level.

And if there are other lenders in the deal, such as Mezzanine Investors, they’ll get \$0 at the end, meaning they lose almost everything in the deal.

Your goal in credit analysis is to assess the chances of such scenarios happening and then make an investment recommendation based on the likelihood of avoiding losses.

To do this, you often look at historical market data such as **peak-to-trough rents** for similar properties in the area to determine how bad things could get.

In recessions, rents fall, vacancy rates rise, concessions increase, and Tenant Improvements and Leasing Commissions both increase as it becomes more difficult to attract tenants.

But operating expenses often stay about the same or even decrease since it’s more difficult for providers and employees to find work as well.

You must consider *all* these factors together and build them into discrete scenarios because everything is interconnected in real estate. For example, it wouldn’t make sense to assume that the vacancy rate rises while TIs and LCs decline.

The percentages here matter more than the absolute numbers. For example, if average rents for 1-bedroom units in Class-A multifamily properties in your area declined from \$2,500 to \$2,300 in the last recession, 10-15 years ago, you wouldn’t use that same \$2,300 rate today because of inflation.

However, you might use the 8.0% decrease in your analysis. So, if current rates are \$3,400 per month, perhaps you would assume a decrease to \$3,128 in a Downside scenario.

Sometimes, you will create multiple scenarios based not entirely on market conditions, but also on whether or not a specific strategy succeeds.

For example, let’s say that current rents in an office property are a 20% discount to market rates, and the occupancy rate of the property is only 75% vs. an average of 90% in the area. A sponsor plans to acquire the property, renovate it, and take the rents and occupancy rates up to market rates.



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In the Downside Case there, you might assume “the status quo”: a continued 20% discount to market rents and the same 75% occupancy rate, along with slower growth in market rents and higher concessions, TIs, and LCs.

In the Base Case, you might assume an improvement, such as a 10% discount to market rents and an 85% occupancy rate.

And in the Extreme Downside Case, you might assume that things get *worse* following the renovation: perhaps a 25% discount to market rents and a 60% occupancy rate because of the loss of a key tenant.

There is no universal rule for creating these cases because it depends on the deal type, the strategy, the location, and recent market history, but the approaches described above are representative.

In the credit case study in this course, we analyze a multifamily property in Seattle that the sponsor wants to acquire for a very low Cap Rate (4.50%) using high leverage (85% LTV).

Furthermore, the sponsor wants to use a Senior Loan, Mezzanine, and Preferred Equity in the capital structure, with progressively higher interest rates and PIK interest.

Without even opening Excel, you should be **quite skeptical** of this deal because the Going-In Cap Rate is very low, indicating a high valuation, and the LTV of 85% is also very high, indicating more downside risk for both the Equity Investors and the Lenders.

Remember that additional leverage does not “boost returns”; it *amplifies* returns, so that good results become even better, and disasters become apocalypses.

You already know what a multifamily Pro-Forma looks like, so we will focus on the **scenarios** and **credit analysis** here. To start with, here are the Debt terms (purchase price of ~\$104 million):



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Term:	Senior Loan:	Mezzanine:	Preferred Equity:
Loan-to-Value (LTV)	65.0%	10.0%	10.0%
Cash Interest Rate	L + 2.00%; LIBOR Floor of 2.00%	4.00%	N/A
PIK Interest Rate	N/A	4.00%	10.00%
Issuance Fees	1.00%	1.00%	1.00%
Interest-Only Period	2 years	Entire Period	Entire Period
Amortization Period	30 years	N/A	N/A
Maturity	10 years	5 years	5 years
Prepayment Penalty	1.5%	1.5%	1.5%
Equity Percentage Upon Exit	N/A	N/A	5.0%

Assume that LIBOR rises from 1.90% in FY 19 to 2.40% in FY 23.

Your firm is seeking a minimum Debt Yield of 6.0%, Cash Interest Coverage Ratio of 1.50x, and Debt Service Coverage Ratio (DSCR) of 1.20x.

The sponsor plans to complete minor renovations (enough to make this a “Core-Plus” deal) and reduce the in-place rental discount to market rent at the same time.

To model this plan, we focus on the Base, Downside, and Extreme Downside Cases; there’s no point looking at an Upside Case because there’s nothing in it for most of the lenders:

Operating Assumptions:	Units:	Historical:	FY19	FY20	FY21	FY22	FY23	Stabilized Year:
Property Management Fee % EGI:	%	3.0%						
Market Rent per Unit per Month:	\$ / RSF / Mo.	\$ 2,500	\$ 2,625	\$ 2,756	\$ 2,880	\$ 3,010	\$ 3,130	\$ 3,240
In-Place Rent per Unit per Month:	\$ / RSF / Mo.	2,300	2,428	2,618	2,808	2,980	3,099	3,207
In-Place Rent Discount to Market Rent:	%	8.0%	7.5%	5.0%	2.5%	1.0%	1.0%	1.0%
Parking Fees per Spot per Month:	\$ / Spot / Mo.	150.00	157.50	165.38	172.82	180.59	187.82	194.39
Utility Reimbursements % Utility Expense:	%	85.0%	86.0%	87.0%	88.0%	89.0%	90.0%	90.0%
Income Growth Rate:								
Base	%	5.0%	5.0%	5.0%	4.5%	4.5%	4.0%	3.5%
Downside	%	5.0%	5.0%	(3.0%)	(1.0%)	6.0%	4.5%	3.5%
Extreme Downside	%	5.0%	5.0%	(6.0%)	(3.0%)	5.0%	4.0%	3.0%
Selected Income Growth Rate:	%	5.0%	5.0%	5.0%	4.5%	4.5%	4.0%	3.5%
General Vacancy:								
Base	%	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)
Downside	%	(3.0%)	(3.0%)	(6.0%)	(4.0%)	(3.5%)	(3.0%)	(3.0%)
Extreme Downside	%	(3.0%)	(3.0%)	(8.0%)	(6.0%)	(4.0%)	(3.0%)	(3.0%)
Selected General Vacancy:	%	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)

Rental declines were 3-5% in the last two recessions, so we assume slightly worse numbers in the Extreme Downside Case; 2-year recession followed by a recovery.

The Vacancy Rate rose to 7-8% in the past two recessions, so we use that in the Extreme Downside Case; we move closer to the 4-5% market average over time but stay slightly below it.

And here’s the next segment:



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Operating Assumptions:	Units:	Historical:	Projected:					Stabilized
		FY18	FY19	FY20	FY21	FY22	FY23	Year:
Bad Debt & Concessions % In-Place Rent:								
Base	%	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)
Downside	%	(3.0%)	(3.0%)	(6.0%)	(5.0%)	(4.0%)	(3.0%)	(3.0%)
Extreme Downside	%	(3.0%)	(3.0%)	(8.0%)	(6.0%)	(5.0%)	(4.0%)	(3.0%)
Selected Bad Debt & Concessions %:	%	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)	(3.0%)
Expense Growth Rate:								
Base	%	4.0%	4.0%	4.0%	3.5%	3.5%	3.0%	2.5%
Downside	%	4.0%	4.0%	(2.0%)	(1.0%)	4.0%	3.0%	2.5%
Extreme Downside	%	4.0%	4.0%	(3.0%)	(2.0%)	3.5%	3.0%	2.5%
Selected Expense Growth Rate:	%	4.0%	4.0%	4.0%	3.5%	3.5%	3.0%	2.5%
Insurance per Gross Square Foot per Year:	\$ / GSF / Yr.	0.55	0.57	0.59	0.62	0.64	0.66	0.67
Utilities per Unit per Month:	\$ / Unit / Mo.	130.00	135.20	140.61	145.53	150.62	155.14	159.02
Replacement Reserves per Unit per Year:	\$ / Unit / Yr.	400.00	416.00	432.64	447.78	463.45	477.36	489.29
Sales, Marketing & Administrative % EGI:	%	10.0%	10.1%	10.2%	10.3%	10.4%	10.5%	10.5%
% Apartment Unit Turnover:	%	10.0%	12.0%	14.0%	16.0%	18.0%	20.0%	20.0%

This percentage moves in-line with the Vacancy Rate; more concessions and flaky tenants in recessions.

More mild version of the Income Growth Rate assumptions; expenses may fall in recessions, but not *that* much.

Far too low! Unit Turnover is usually more like 40-50% for multifamily, but we're just following the instructions here. It should increase as rents increase, but the starting level should be much higher.

And the last segment:

Tenant Improvement (TI) Growth Rate:								
Base	%	4.0%	4.0%	4.0%	3.5%	3.5%	3.0%	2.5%
Downside	%	4.0%	4.0%	10.0%	10.0%	(8.0%)	(4.0%)	2.5%
Extreme Downside	%	4.0%	4.0%	15.0%	10.0%	(12.0%)	(7.0%)	2.5%
Selected TI Growth Rate:	%	4.0%	4.0%	4.0%	3.5%	3.5%	3.0%	2.5%
Tenant Improvements (TIs) per Unit Leased:	\$ / Unit / Yr.	800.00	832.00	865.28	895.56	926.91	954.72	978.58
Leasing Commissions % Effective Rent:								
Base	%	3.0%	3.0%	3.0%	3.0%	3.0%	3.0%	3.0%
Downside	%	3.0%	3.0%	8.0%	6.0%	5.0%	4.0%	3.0%
Extreme Downside	%	3.0%	3.0%	12.0%	8.0%	6.0%	4.5%	3.0%
Selected Leasing Commission %:	%	3.0%	3.0%	3.0%	3.0%	3.0%	3.0%	3.0%
Capital Expenditures per Unit per Year:	\$ / Unit / Yr.	-	1,000.00	1,500.00	750.00	-	200.00	-
Replacement Reserves:	\$	-	556,086	372,258	64,195	-	25,579	16,279

More incentives required to find tenants in a downturn; we should also check the per-unit figures against the historical ones in previous recessions.

LCs also increase in recessions because brokers must be paid more to find tenants; this is based on Effective Rent so it includes both new and renewal tenants.

Some of these assumptions are a bit questionable (see the notes above), but this is the general outline for scenarios in credit analysis.

Here's the Pro-Forma down to NOI in the Base Case:



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Property Pro-Forma:	Units:	Historical:		Projected:				Stabilized	
		FY18	FY19	FY20	FY21	FY22	FY23	Year:	
Revenue:									
(+) Base Rental Income:	\$	\$ 7,020,000	\$ 7,371,000	\$ 7,739,550	\$ 8,087,830	\$ 8,451,782	\$ 8,789,853	\$ 9,097,498	
(-) Loss to Lease:	\$	(561,600)	(552,825)	(386,978)	(202,196)	(84,518)	(87,899)	(90,975)	
(-) Bad Debt & Concessions:	\$	(193,752)	(204,545)	(220,577)	(236,569)	(251,018)	(261,059)	(270,196)	
(+) Parking Income:	\$	648,648	681,080	715,134	747,315	780,945	812,182	840,609	
(+) Utility Reimbursements:	\$	310,284	326,492	343,500	359,609	376,424	392,074	401,875	
Potential Gross Revenue:	\$	7,223,580	7,621,202	8,190,629	8,755,989	9,273,615	9,645,152	9,978,812	
(-) General Vacancy:	\$	(216,707)	(228,636)	(245,719)	(262,680)	(278,208)	(289,355)	(299,364)	
Effective Gross Income (EGI):	\$	7,006,873	7,392,566	7,944,911	8,493,309	8,995,407	9,355,798	9,679,447	
EGI Growth Rate:	%	N/A	5.5%	7.5%	6.9%	5.9%	4.0%	3.5%	
Expenses:									
(-) Property Taxes:	\$	(1,068,128)	(1,110,853)	(1,155,287)	(1,195,722)	(1,237,572)	(1,274,700)	(1,306,567)	
(-) Insurance:	\$	(128,023)	(133,144)	(138,469)	(143,316)	(148,332)	(152,782)	(156,601)	
(-) Utilities:	\$	(365,040)	(379,642)	(394,827)	(408,646)	(422,949)	(435,637)	(446,528)	
(-) Property Management Fee:	\$	(210,206)	(221,777)	(238,347)	(254,799)	(269,862)	(280,674)	(290,383)	
(-) Sales, Marketing & Administrative:	\$	(700,687)	(746,649)	(810,381)	(874,811)	(935,522)	(982,359)	(1,016,342)	
(-) Replacement Reserves:	\$	(93,600)	(97,344)	(101,238)	(104,781)	(108,448)	(111,702)	(114,494)	
Total Expenses:	\$	(2,565,684)	(2,689,408)	(2,838,550)	(2,982,076)	(3,122,686)	(3,237,853)	(3,330,917)	
Net Operating Income (NOI):	\$	4,441,188	4,703,157	5,106,361	5,511,234	5,872,721	6,117,944	6,348,531	
NOI Margin:	%	63.4%	63.6%	64.3%	64.9%	65.3%	65.4%	65.6%	

In the Extreme Downside Case, the EGI Growth Rate and NOI Margin in the Stabilized Year are not much different, but the absolute numbers are lower by nearly 20%:

Property Pro-Forma:	Units:	Historical:		Projected:				Stabilized							
		FY18	FY19	FY20	FY21	FY22	FY23	Year:							
Revenue:															
(+) Base Rental Income:	\$	\$	7,020,000	\$	7,371,000	\$	6,928,740	\$	6,720,878	\$	7,056,922	\$	7,339,199	\$	7,559,375
(-) Loss to Lease:	\$		(561,600)		(552,825)		(346,437)		(168,022)		(70,569)		(73,392)		(75,594)
(-) Bad Debt & Concessions:	\$		(193,752)		(204,545)		(526,584)		(393,171)		(349,318)		(290,632)		(224,513)
(+) Parking Income:	\$		648,648		681,080		640,216		621,009		652,060		678,142		698,486
(+) Utility Reimbursements:	\$		310,284		326,492		320,380		317,581		332,431		346,252		354,908
Potential Gross Revenue:	\$		7,223,580		7,621,202		7,016,314		7,098,274		7,621,526		7,999,568		8,312,661
(-) General Vacancy:	\$		(216,707)		(228,636)		(561,305)		(425,896)		(304,861)		(239,987)		(249,380)
Effective Gross Income (EGI):	\$		7,006,873		7,392,566		6,455,009		6,672,378		7,316,665		7,759,581		8,063,282
EGI Growth Rate:	%		N/A		5.5%		(12.7%)		3.4%		9.7%		6.1%		3.9%
Expenses:															
(-) Property Taxes:	\$		(1,068,128)		(1,110,853)		(1,077,527)		(1,055,977)		(1,092,936)		(1,125,724)		(1,153,867)
(-) Insurance:	\$		(128,023)		(133,144)		(129,149)		(126,566)		(130,996)		(134,926)		(138,299)
(-) Utilities:	\$		(365,040)		(379,642)		(368,252)		(360,887)		(373,518)		(384,724)		(394,342)
(-) Property Management Fee:	\$		(210,206)		(221,777)		(193,650)		(200,171)		(219,500)		(232,787)		(241,898)
(-) Sales, Marketing & Administrative:	\$		(700,687)		(746,649)		(658,411)		(687,255)		(760,933)		(814,756)		(846,645)
(-) Replacement Reserves:	\$		(93,600)		(97,344)		(94,424)		(92,535)		(95,774)		(98,647)		(101,113)
Total Expenses:	\$		(2,565,684)		(2,689,408)		(2,521,414)		(2,523,392)		(2,673,658)		(2,791,565)		(2,876,165)
Net Operating Income (NOI):	\$		4,441,188		4,703,157		3,933,595		4,148,986		4,643,007		4,968,016		5,187,117
NOI Margin:	%		63.4%		63.6%		60.9%		62.2%		63.5%		64.0%		64.3%

We then set up the Debt Schedule and calculate the returns to Equity Investors, Senior Lenders, Mezzanine Investors, and Preferred Investors:



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Property Pro-Forma:	Units:	Historical:	Projected:				
		FY18	FY19	FY20	FY21	FY22	FY23
Adjusted Net Operating Income:	\$	4,441,188	4,703,157	5,106,361	5,434,465	5,872,721	6,113,099
<i>Adjusted NOI Margin:</i>	%	63.4%	63.6%	64.3%	64.0%	65.3%	65.3%
(-) Cash Interest Expense on Senior Loan:	\$		(2,717,380)	(2,751,347)	(2,853,249)	(2,823,811)	(2,789,843)
(-) Cash Interest Expense on Mezzanine:	\$		(418,058)	(434,781)	(452,172)	(470,259)	(489,069)
(-) Senior Loan Principal Repayment:	\$		-	-	(2,264,483)	(2,264,483)	(2,264,483)
Cash Flow to Equity Investors:	\$		1,567,719	1,920,233	(135,439)	314,168	569,704
PIK Interest on Mezzanine:	\$		(418,058)	(434,781)	(452,172)	(470,259)	(489,069)
PIK Interest on Preferred Equity:	\$		(1,045,146)	(1,149,661)	(1,264,627)	(1,391,089)	(1,530,198)
(+) Ending Senior Loan Balance:	\$	\$ 67,934,496	\$ 67,934,496	\$ 67,934,496	\$ 65,670,013	\$ 63,405,530	\$ 61,141,046
(+) Ending Mezzanine Balance:	\$	10,451,461	10,869,519	11,304,300	11,756,472	12,226,731	12,715,800
(+) Ending Preferred Equity Balance:	\$	10,451,461	11,496,607	12,646,268	13,910,894	15,301,984	16,832,182
Ending Debt Balance:	\$	88,837,418	90,300,622	91,885,064	91,337,379	90,934,245	90,689,029
LIBOR:	%	1.70%	1.90%	2.05%	2.20%	2.30%	2.40%
Interest Rate on Senior Loan:	%		4.00%	4.05%	4.20%	4.30%	4.40%
<i>Debt Yield - NOI:</i>	%		5.3%	5.7%	6.2%	6.6%	6.9%
<i>Debt Yield - Adjusted NOI:</i>	%		5.3%	5.7%	6.1%	6.6%	6.9%
<i>Cash Interest Coverage Ratio - NOI:</i>	x		1.50 x	1.60 x	1.67 x	1.78 x	1.87 x
<i>Cash Interest Coverage Ratio - Adjusted NOI:</i>	x		1.50 x	1.60 x	1.64 x	1.78 x	1.86 x
<i>Debt Service Coverage Ratio (DSCR) - NOI:</i>	x		1.50 x	1.60 x	0.99 x	1.06 x	1.10 x
<i>Debt Service Coverage Ratio (DSCR) - Adj. NOI:</i>	x		1.50 x	1.60 x	0.98 x	1.06 x	1.10 x

Based on the Interest Rates * beginning Balances; Principal Repayment is 0 for the Interest-Only years and starts afterward.

The Mezzanine and Preferred keep increasing due to the PIK Interest, while the Senior Loan decreases because of the amortization.

Minimum Debt Yield is 6.0%, minimum Cash ICR is 1.50x, and minimum DSCR is 1.20x, so we have some problems.



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Returns to Equity Investors:	Units:	Historical:	Projected:				
		FY18	FY19	FY20	FY21	FY22	FY23
Forward NOI:	\$		\$ 5,106,361	\$ 5,511,234	\$ 5,872,721	\$ 6,117,944	\$ 6,348,531
Applicable Capitalization Rates:							
Base	%	4.50%	4.55%	4.60%	4.65%	4.70%	4.75%
Downside	%	4.50%	4.55%	6.00%	5.50%	5.00%	5.00%
Extreme Downside	%	4.50%	4.55%	6.50%	6.00%	5.50%	5.25%
Selected Capitalization Rate:	%	4.50%	4.55%	4.60%	4.65%	4.70%	4.75%
Implied Property Value:	\$		112,227,709	119,809,433	126,295,071	130,169,027	133,653,280
Cash Flow to Equity Investors (Leveraged IRR):							
(-) Initial Equity Investment:	\$	(18,110,712)	-	-	-	-	-
(+) Proceeds from Sale of Property:	\$		-	-	-	-	133,653,280
(+) Return of Replacement Reserve:	\$		-	-	-	-	-
(-) Selling Costs:	\$		-	-	-	-	(2,673,066)
(+) Cash Flows to Equity Investors:	\$		1,567,719	1,920,233	(135,439)	314,168	569,704
(-) Senior Loan Repayment:	\$		-	-	-	-	(61,141,046)
(-) Mezzanine Repayment:	\$		-	-	-	-	(12,715,800)
(-) Preferred Equity Repayment:	\$		-	-	-	-	(16,832,182)
(-) Prepayment Penalty on Senior Loan:	\$		-	-	-	-	(917,116)
(-) Prepayment Penalty on Mezzanine:	\$		-	-	-	-	-
(-) Prepayment Penalty on Preferred Equity:	\$		-	-	-	-	-
(-) Equity Granted to Preferred Investors:	\$		-	-	-	-	(1,997,189)
Total Cash Flows to Equity (Leveraged IRR):	\$	\$ (18,110,712)	\$ 1,567,719	\$ 1,920,233	\$ (135,439)	\$ 314,168	\$ 37,946,585
Internal Rate of Return (IRR):	%		19.8%				
Total Returns to Equity:	\$		41,613,266				
Invested Equity:	\$		18,110,712				
Cash-on-Cash Multiple:	x		2.3 x				

Exit Cap Rates reflect a property that becomes significantly less valuable in the pessimistic cases.

Fairly standard returns schedule, but upon exit, we *only* repay the Debt to the extent that it is possible to do so with the net proceeds available. Junior lenders could take a haircut!

Returns to Senior Lenders:	Units:	Historical:	Projected:				
		FY18	FY19	FY20	FY21	FY22	FY23
(-) Senior Loan Issued:	\$	\$ (67,934,496)	-	-	-	-	-
(+) Cash Interest Received:	\$		2,717,380	2,751,347	2,853,249	2,823,811	2,789,843
(+) Senior Loan Principal Repayment:	\$		-	-	2,264,483	2,264,483	2,264,483
(+) Senior Loan Issuance Fees:	\$	679,345	-	-	-	-	-
(+) Repayment of Senior Loan:	\$		-	-	-	-	61,141,046
(+) Prepayment Penalty on Senior Loan:	\$		-	-	-	-	917,116
Total Cash Flows to Senior Lenders:	\$	\$ (67,255,151)	2,717,380	2,751,347	5,117,732	5,088,294	67,112,489
Internal Rate of Return (IRR):	%		4.7%				
Recovery:	%		100.0%				
Total Returns:	\$		83,466,586				
Invested Amount:	\$		67,934,496				
Cash-on-Cash Multiple:	x		1.2 x				

Cash Interest increases and then decreases due to rising interest rates and principal repayments; prepayment penalty is incurred because the maturity is 10 years.

IRR is above the initial 4.0% interest rate due to the rising interest rate over time, issuance fee, and prepayment penalty.

We can then do the same thing for the Mezzanine Investors and Preferred Investors, and the IRRs for those are 8.2% and 12.7%, respectively, in the Base Case.

You can see how dramatically the seniority of the Debt affects its IRR when you compare the returns for Mezzanine and Preferred Investors in more pessimistic cases.

In the screenshots above, we assumed a Year 5 Exit in the Base Case. But if we assume a Year 4 Exit in the Extreme Downside Case, here are the numbers:

Returns to Mezzanine Investors:	Units:	Historical:	Projected:				
		FY18	FY19	FY20	FY21	FY22	FY23
(-) Mezzanine Issued:	\$	\$ (10,451,461)	-	-	-	-	-
(+) Cash Interest Received:	\$		418,058	434,781	452,172	470,259	-
(+) Mezzanine Issuance Fees:	\$	104,515	-	-	-	-	-
(+) Repayment of Mezzanine:	\$	-	-	-	-	12,226,731	-
(+) Prepayment Penalty on Mezzanine:	\$	-	-	-	-	-	-
Total Cash Flows to Mezzanine Investors:	\$	\$ (10,346,946)	418,058	434,781	452,172	12,696,990	-
Internal Rate of Return (IRR):	%	8.3%					
Recovery:	%	100.0%					
Total Returns:	\$	14,106,516					
Invested Amount:	\$	10,451,461					
Cash-on-Cash Multiple:	x	1.3 x					

The Mezzanine Investors receive back their full proceeds, but there is not enough left to pay for the prepayment penalty.

So, while the Mezzanine Investors do recover 100% of their initial investment, their IRR is a bit lower because there is no prepayment penalty.

Returns to Preferred Investors:	Units:	Historical:	Projected:				
		FY18	FY19	FY20	FY21	FY22	FY23
(-) Preferred Issued:	\$	\$ (10,451,461)	-	-	-	-	-
(+) Preferred Issuance Fees:	\$	104,515	-	-	-	-	-
(+) Repayment of Preferred:	\$	-	-	-	-	11,958,289	-
(+) Prepayment Penalty on Preferred:	\$	-	-	-	-	-	-
(+) Equity Granted to Preferred:	\$	-	-	-	-	-	-
Total Cash Flows to Preferred Investors:	\$	\$ (10,346,946)	\$ -	\$ -	\$ -	\$ 11,958,289	\$ -
Internal Rate of Return (IRR):	%	3.7%					
Recovery:	%	78.1%					
Total Returns:	\$	12,062,804					
Invested Amount:	\$	10,451,461					
Cash-on-Cash Multiple:	x	1.2 x					

The Preferred Investors are owed back \$15.3 million by Year 4, but they only receive back ~ \$12 million because there aren't enough exit proceeds to pay for the rest.

And so the IRR ends up being far lower than the stated 10% interest rate - and the lack of a prepayment penalty and equity don't help.

To make an investment decision for each tranche of Debt, we create sensitivity tables that show the impact of different Exit Dates, Scenarios and Exit Cap Rates.

Should We Do This Deal?

From an Equity perspective, this deal is horrible because the IRR could easily turn negative in the Extreme Downside Case and even certain Downside Cases.

And even in the Base Case, there's far too much risk to justify the potential returns, primarily because the 85% LTV is abnormally high.

As lenders, though, we don't necessarily care about that – if we earn close to *our* targeted IRRs even in very pessimistic cases, the deal could work.

Here are the numbers for the Exit Date vs. the Market Scenario for each lender group:

Sensitivity Analyses:

Senior Lenders - IRR vs. Exit Date and Market Scenario:

	Exit Date:		
	FY21	FY22	FY23
Base	4.9%	4.7%	4.7%
Downside	4.4%	4.7%	4.7%
Extreme Downside	4.4%	4.4%	4.7%

Senior Lenders are pretty safe no matter what happens, but their IRRs are also the lowest.

Mezzanine Investors - IRR vs. Exit Date and Market Scenario:

	Exit Date:		
	FY21	FY22	FY23
Base	8.9%	8.7%	8.2%
Downside	8.4%	8.7%	8.2%
Extreme Downside	(1.7%)	8.3%	8.2%

The Mezzanine Investors are also relatively safe except for one case, and their IRRs are significantly higher.

Preferred Investors - IRR vs. Exit Date and Market Scenario:

	Exit Date:		
	FY21	FY22	FY23
Base	14.8%	13.7%	12.7%
Downside	5.3%	11.9%	11.4%
Extreme Downside	#NUM!	3.7%	10.5%

And the Preferred Investors are... not so safe. There's a significant chance of the IRR falling short here.

We can also look at the **Year 5 Exit Cap Rate vs. the Market Scenario** (i.e., assume a fixed Exit Date and let the Exit Cap Rate vary).

For reference, the Going-In Cap Rate was 4.5%, and the range for similar multifamily property sales in the same area over the past ~7 years was 4.5% – 5.5%.

The maximum Cap Rates in the past two recessions were 5.8% and 7.3%.

Even with extremely pessimistic assumptions, such as a Year 5 Exit at a 7.0% Cap Rate in the Extreme Downside Case, the Senior Lenders have a near-0 chance of losing money.

Therefore, we focus on the other lenders here:



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Mezzanine Investors - IRR vs. Year 5 Exit Cap Rate and Market Scenario:

Market Scenario:	Year 5 Exit Cap Rate:								
	5.00%	5.25%	5.50%	5.75%	6.00%	6.25%	6.50%	6.75%	7.00%
Base	8.2%	8.2%	8.2%	8.2%	8.2%	8.2%	8.2%	8.2%	8.2%
Downside	8.2%	8.2%	8.2%	8.2%	8.2%	8.2%	8.2%	8.2%	8.2%
Extreme Downside	8.2%	8.2%	8.2%	8.2%	8.2%	8.2%	8.2%	8.2%	5.2%

Preferred Investors - IRR vs. Year 5 Exit Cap Rate and Market Scenario:

Market Scenario:	Year 5 Exit Cap Rate:								
	5.00%	5.25%	5.50%	5.75%	6.00%	6.25%	6.50%	6.75%	7.00%
Base	12.3%	12.0%	11.6%	11.3%	11.0%	10.8%	10.5%	10.3%	8.5%
Downside	11.4%	11.1%	10.8%	10.5%	10.2%	6.9%	1.3%	(5.3%)	(13.7%)
Extreme Downside	10.8%	10.5%	10.2%	6.1%	(0.2%)	(7.9%)	(18.5%)	(39.7%)	#NUM!

Even if Cap Rates rise to their highest-ever level, the Mezzanine Investors still earn a positive IRR with a Year 5 Exit in the Extreme Downside Case.

By contrast, it would take far less for the Preferred Investors to lose money - just a modest Cap Rate increase in-line with the one in the most recent recession.

Our views on this deal would be the following:

First, it seems far too risky for the Equity Investors, so we would **not** recommend it for them under any circumstances.

On the lending side, the **Mezzanine Investors** seem to perform the best in terms of IRR (8-9% range) and the ability to avoid losing money. Even if Cap Rates increase to nearly their all-time highs, the Mezzanine Investors still earn a positive IRR (though the Recovery drops below 100%).

The Senior Loan is safer than the Mezzanine, but its IRR is also ~50% lower than that of the Mezzanine; for slightly more risk, we could earn significantly more with the Mezzanine.

In some cases, the Preferred Investors earn the highest IRR, but the numbers are quite bad in the Extreme Downside Case, especially when Cap Rates rise to the levels seen in previous recessions.

The fact that the Preferred Equity has 100% PIK Interest and 0% Cash Interest also makes it far riskier than the other Debt because the returns depend completely on the exit.

In short, the Preferred Equity is far riskier than the Mezzanine, but it does not offer IRRs that are far higher; we might earn 10-12% vs. 8-9% on the Mezzanine.

By contrast, the Mezzanine offers IRRs that are twice as high as those of the Senior Loan, but risk that is only incrementally higher.

There are some risks associated with the Mezzanine, such as an early Exit Date, a double-dip recession with rising Cap Rates and falling rents, but we can mitigate these risks by tweaking the terms of the Debt (e.g., by requesting a higher Prepayment Penalty in Year 3 in exchange for no penalty after that).



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If we could invest in only one tranche of Debt, we would pick the Mezzanine, but the Senior Loan would also be acceptable.

We would avoid the Preferred Equity, at least with the terms described here, because we do not believe its IRR justifies the higher risk and the very real chance of losing money.

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Real Estate Interview Questions & Answers

Interview questions about real estate span a wide range of topics, but you're most likely to receive questions about fundamental concepts such as the different property types, NOI and Cap Rates, and how Debt and Equity are used to finance deals.

If interviewers want to test your ability to assess a deal, they will give you a case study or modeling test rather than asking detailed math-based questions.

Similarly, if they want to test your ability to enter formulas and calculate line items such as rent and concessions, they will give you an Excel test.

We do include a few technical and math-based questions here just so you're prepared for all possibilities, but you should **focus** on the higher-level, conceptual questions.

High-Level Concepts in Real Estate

These are the most important interview questions in this guide because they're the most likely to come up in real life. Also, if you can't answer these basic questions correctly, interviewers will conclude that you don't know much about real estate.

1. Why would you invest in real estate?

As an asset class, real estate combines many elements of fixed income and equities and is somewhere in between them regarding risk/volatility and potential returns.

You could invest in properties with risk and potential returns in-line with investment-grade corporate bonds, high-yield bonds, blue-chip equities, growth equities, or ones with even higher risk and potential returns than growth equities.

Real estate investments could provide anything from a stable, tax-advantaged income stream (due to the Depreciation deduction for individuals) to the potential for high capital appreciation.

There are also many ways to invest in real estate, from individual properties and loans to REITs, other real estate companies, and even crowdfunding. And given its long history, real estate is likely to be around for centuries to come.

2. Explain the main property types and how they differ from each other.



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Quick Reference Guide for Individual Properties

The main types are office, industrial, retail, and multifamily properties; others include condominiums, hotels, and variants like data centers and healthcare properties.

Office, industrial, and retail properties have businesses as tenants and offer long-term leases of 5-10 years. The lease terms are highly variable and often include different rental rates, rental escalations, free months of rent, expense reimbursements, and tenant improvements.

Industrial properties can be constructed more quickly and tend to have fewer tenants, while office and retail properties take more time and money and tend to have more tenants.

Multifamily properties have individuals as tenants and offer short-term leases of 1 year in most cases, with lease terms that are very similar except for the rent, which depends on each tenant's space.

Hotels are even shorter-term than multifamily, with guests that stay for an average of a few days and who pay similar rates for similar room sizes.

Finally, condominiums are different from everything else because they are sold to individuals rather than rented out, and developers aim to pre-sell them during the construction period to reduce risk.

3. What about other property types beyond these main categories?

Other property types are similar to the main categories described in the previous question but are sometimes a “mix” of types.

To decide on the closest match, think about the number of tenants, the average lease length, the similarity of lease terms, and the granularity of the revenue and expenses.

For example, student housing is nearly the same as multifamily because leases are typically for a year or less, there are dozens or hundreds of students in each building, and the lease terms are very similar.

Data centers are closer to industrial/office/retail properties because there tend to be fewer tenants, leases are longer-term (e.g., 3-7 years), and individual lease terms vary significantly.

And healthcare properties may be closer to multifamily or industrial/office/retail properties. For example, nursing homes have dozens or hundreds of tenants (patients) with similar terms, while the corporate tenants in pharmaceutical labs have longer-term leases with highly variable terms.

4. From an investor's perspective, which property type should be valued most highly / lowly?

Hotels tend to be valued at the highest Cap Rates, i.e., the lowest valuations, because they do not have long-term leases and, therefore, do not have much revenue visibility.



Multifamily properties are often valued at the lowest Cap Rates – despite the 1-year leases, they are more resistant to recessions because people always need somewhere to live, and fewer people will buy homes in downturns.

There's also less risk because the lease terms for a single property are very similar, there tend to be dozens or hundreds of tenants in most properties, and capital costs tend to be lower than for office, industrial, and retail properties.

Office, industrial, and retail properties are somewhere in between these two. The long-term leases offer high revenue visibility, but they also tend to have fewer tenants, which creates more risk, as do the highly variable lease terms and higher capital costs.

Within that category, industrial properties sometimes sell for higher Cap Rates because they are faster and cheaper to construct and often have fewer tenants; single-tenant properties are quite common in the industrial sector.

5. What are the main strategies that sponsors use to invest in real estate, and what are the trade-offs of those strategies?

In terms of equity investments in real estate, the main strategies are "Core" (buy an existing, stabilized property, change very little, and sell it again), "Core-Plus" (similar but make minor upgrades), "Value-Added" (acquire an existing property, renovate or greatly improve it, and then sell it again), and "Opportunistic" (develop or re-develop a property and then sell it).

Core offers the lowest risk and potential returns, Core-Plus is slightly higher, Value-Added is higher, and Opportunistic offers the highest risk and potential returns.

No strategy is "the best" because it depends on the investor's goals and the rest of his/her portfolio; a well-diversified real estate investor should ideally use a mix of all these strategies.

6. Why are some combinations of strategies and properties, such as "Core" hotel deals, not common?

Some combinations do not make sense for risk/potential return or practical reasons. For example, Core hotel deals where the sponsor buys the property and does nothing to improve it are not common because hotels are already quite risky as-is.

Typically, sponsors must do something to tilt the odds in their favor and boost the potential returns to acceptable levels, which means Opportunistic and Value-Added deals are more viable than Core or Core-Plus ones.



It's also rare to develop condominiums and then attempt to sell them upon completion because doing it that way creates significantly more risk than pre-selling them as the construction is taking place.

Finally, sponsors hardly ever pre-sell non-condo properties, such as multifamily or office buildings, before development is complete because the new potential owners want to see a stabilized occupancy rate and financial performance first.

7. What is a property's Net Operating Income, and why is it important?

Net Operating Income, or NOI, represents the property's cash flow from operations on a capital structure-neutral basis before *most* of the capital costs ("most" because the Reserves are sometimes deducted and sometimes not deducted).

It equals, roughly, the property's revenue minus operating expenses and property taxes, and it excludes interest expense, debt principal repayments, and capital costs such as Tenant Improvements, Leasing Commissions, and Capital Expenditures.

NOI may *partially* reflect these capital costs if you deduct the Reserves when calculating it.

NOI is important because it lets us compare different properties and value them.

8. What is the "Cap Rate," and how do you use it in real estate?

The Cap Rate, or Capitalization Rate, equals the stabilized forward NOI of a property divided by its price (either the asking price or the actual sale price). For example, if the property generates \$5 million in NOI next year and its asking price is \$100 million, then the Cap Rate is 5%.

The Cap Rate is the reciprocal of a valuation multiple; this 5% Cap Rate corresponds to a $1 / 5\%$, or 20x, multiple.

You use the Cap Rate to determine the purchase price and exit price of a property in investment analysis.

Typically, you calculate the property's NOI, select a range of Cap Rates from market data for similar properties in the area, and apply those rates to estimate this property's value.

Cap Rates are central to most property deals and investment analyses because you always need to assume an exit value.

9. Why do nearly all real estate deals involve Debt?



Real estate investors use Debt for the same reason that other investors use Debt: **to amplify their returns** (i.e., positive returns become more positive and negative returns become even more negative).

Leverage also makes it possible to diversify by acquiring and developing more properties since investors are not required to pay the full, upfront price using Cash.

Normal companies do not always use Debt to make acquisitions, but that is because they often plan to hold the acquired companies indefinitely.

By contrast, real estate investors always plan to **sell** their assets, so leverage makes a huge difference.

10. What are the most common types of Debt in real estate, and how do they differ from each other?

The most common types of Debt are Construction Loans, Bridge/Temporary Loans, and Permanent Loans.

Construction Loans are used to fund new developments, and interest rates are relatively high due to the high risk of new developments; interest is capitalized during the development, and the loan is refinanced and replaced with a Permanent Loan once construction ends and the property stabilizes.

Bridge/Temporary Loans are used for many acquisition and renovation deals, and interest rates tend to be higher than the ones on Permanent Loans due to the added risk; these loans are refinanced once the renovation is done and the property stabilizes.

Permanent Loans are split into Senior Loans and Mezzanine, and interest rates on the Senior Loans tend to be lower than the rates on the other loan categories because of the reduced risk. These loans are used primarily for stabilized, developed properties that are not changing much. The maturity of these loans tends to be longer than the maturity of the others as well.

Mezzanine is used similarly, but it's junior, unsecured Debt with higher interest rates, no principal amortization, and shorter maturity.

11. How can property acquisitions use 60% or 70% leverage? Private equity firms do not use that much leverage for normal companies in leveraged buyouts.

This high leverage works in real estate because real estate loans often have interest-only periods without any principal amortization. And the loans often amortize over 20-30 years instead of 5-10 years, even if the loan maturity is only 5-10 years.



As a result, the Debt Service on these loans, relative to the property's cash flow, tends to be less than the Debt Service on, say, Term Loans for normal companies relative to their cash flows, assuming the same leverage ratios for both.

12. What's the benefit of refinancing loans in real estate?

Refinancing boosts Equity Investors' potential returns in real estate deals because in a refinancing, the sponsor repays existing Debt with new Debt, and the Net Proceeds get distributed to the sponsor.

For example, if the property is worth \$1 million and the initial loan is for a 75% LTV, a refinancing in 2 years at the same 75% LTV but with a property value of \$1.2 million means that the new loan will be \$900K.

The original loan was only for \$750K, so the extra \$150K in Net Proceeds goes to the Equity Investors.

Sometimes sponsors also refinance to take advantage of lower rates if interest rates have been falling.

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Leases, the Pro-Forma, and Projections

These questions delve into the "nitty gritty" of specific line items and how to move from individual tenant leases to the Pro-Forma.

You don't necessarily need to be able to calculate all the numbers in your head, but you do need to be familiar with the basic concepts and terms like "Tenant Improvements" and "Leasing Commissions."

1. Walk me through a property Pro-Forma and explain what it means.

First, clarify which *type* of Pro-Forma the interviewer wants because they differ based on the property type. Assuming it is an office, industrial, or retail Pro-Forma:

You start at the top with the Base Rental Income, which represents the total potential rental income *if the property were 100% occupied at market rental rates*.

Then, you adjust this for items such as the Absorption & Turnover Vacancy, Concessions & Free Rent, Expense Reimbursements, Loss to Lease, General Vacancy, and Percentage Rent, all of which bridge the gap between potential income and actual income, known as Effective Gross Income or EGI.



Then, you show Operating Expenses such as Management Fees, Maintenance & Repairs, Utilities, Insurance, Property Taxes, and the Capital Cost Reserves. EGI minus Operating Expenses equals Net Operating Income or NOI.

You then deduct capital costs, such as Capital Expenditures, Tenant Improvements, and Leasing Commissions, to calculate Adjusted NOI, and then you deduct Debt Service to calculate Cash Flow to Equity Investors.

The Pro-Forma is a combined Income Statement and Cash Flow Statement for a property that shows its historical and projected cash flow and ability to service Debt; the NOI line item also tells you what the property might be worth upon exit.

2. Why don't we need the 3 full financial statements for properties?

Because properties are much simpler than normal companies, and their financials tend to be based on cash accounting rather than accrual accounting.

Also, Working Capital is less significant, you typically ignore corporate-level taxes, and “other activities” that represent many of the line items in Cash Flow from Investing and Financing for normal companies do not exist for properties.

Finally, you can handle Debt and Equity Issuances and Debt Service directly on the Pro-Forma and in the Returns Schedule rather than showing them on a separate Cash Flow Statement.

3. When would you model individual tenant leases separately to create a Pro-Forma?

Modeling individual tenant leases is most useful when the terms of those leases vary significantly, as in most office, industrial, and retail properties, and when the number of tenants in the property is relatively low (e.g., 5-10 tenants rather than 50-100).

4. What are the key lease terms and assumptions you use for modeling purposes?

For modeling purposes, you need to calculate the Base Rental Income, Loss to Lease (if applicable), Absorption & Turnover Vacancy, Concessions & Free Rent, Expense Reimbursements, and Percentage Rent (if applicable) for each tenant.

On the expense side, you need the Tenant Improvements, Leasing Commissions, and Capital Expenditures (CapEx), if any.

For the property as a whole, you need the General Vacancy to determine the potential rental income lost to permanently vacant space.



The Base Rental Income is based on the Market Rent for the tenant's space, and Loss to Lease is based on the difference between the tenant's In-Place Rent and Market Rent.

The Absorption & Turnover Vacancy is based on the number of months it will take to find a new tenant if an existing one does not renew, and Concessions & Free Rent is determined by the number of months of free rent given to new and renewal tenants as incentives.

Expense Reimbursements is based on the proportional share of operating expenses and property taxes for which the tenant is responsible, and Percentage Rent for retail tenants is based on a percentage of tenants' monthly retail sales.

Tenant Improvements refer to customizations made to a space to suit a tenant's business, and they're specified in the lease agreement on a per-square-foot or per-square-meter basis.

Leasing Commissions are not part of tenants' leases, but they are paid to brokers as a percentage of the total lease value in exchange for finding tenants and closing deals with them.

5. What's the difference between Triple Net (NNN), Double Net (NN), Single Net (N), and Full Service or "Gross" Leases?

These lease types refer to the expenses for which the tenant is responsible. If the tenant is responsible for fewer expenses, it will tend to pay higher rent, and if it is responsible for more expenses, it will tend to pay lower rent. Here's a chart:

Lease Type:	Abbreviation	Tenant Pays Its Proportional Share of...	Implications in Model
Full Service or Gross Lease	FS	Rent.	The highest rent, but no expense reimbursements from this tenant.
Single Net Lease	N	Rent + Property Taxes.	Lower rent and moderate expense reimbursements.
Double Net Lease	NN	Rent + Property Taxes + Insurance.	Even lower rent and more substantial expense reimbursements.
Triple Net Lease	NNN	Rent + Property Taxes + Insurance + Maintenance/Utilities.	The lowest rent and the highest expense reimbursements.



“Proportional Share” means that if the tenant occupies 20% of the rentable square feet of the building, it will be responsible for 20% of the total property taxes (for example).

6. Walk me through the calculations for Effective Gross Income (EGI) for an office/retail/industrial property.

You start with Base Rental Income, which represents the potential rental income if the property were 100% occupied and all the tenants were paying market rates for their spaces.

Then, you deduct the Loss to Lease for tenants that are paying below-market rates, the Absorption & Turnover Vacancy for spaces where leases have expired and where the search for a new tenant is underway, and Concessions & Free Rent for the free months of rent given to new/renewal tenants as incentives.

Then, you add Expense Reimbursements for tenants that are responsible for a portion of the property’s operating expenses and property taxes. This takes you to Potential Gross Revenue or Potential Gross Income.

Finally, you deduct General Vacancy for spaces that are permanently vacant (not just temporarily vacant due to a lease expiration) and add Percentage Rent for retail tenants, which results in Effective Gross Income.

7. Tenant A occupies 5,000 square feet of an office building and pays rent of \$50 per square foot per year vs. a market rate of \$60 per square foot per year.

Tenant A is on a 5-year lease and receives 3 months of free rent upon move-in. This property is 25,000 square feet and has total operating expenses and property taxes of \$500,000 per year, and this tenant is on a Triple Net Lease.

Walk me through the calculation of the Year 1 Effective Gross Income from Tenant A, assuming a January 1 move-in.

The Base Rental Income is $\$60 * 5,000 = \$300,000$. Then, you deduct $5,000 * (\$60 - \$50) = \$50,000$ for the Loss to Lease line item, so you’re at \$250,000 now.

In Year 1, this tenant receives 3 months of free rent, which is $1/4$ of the year, or \$62,500 of this tenant’s in-place rent ($\$250,000 * 25\% = \$62,500$).

So, instead of paying \$250,000, this tenant only pays $\$250,000 - \$62,500 = \$187,500$.



Also, this tenant is responsible for 20% of the \$500,000 in property taxes + operating expenses, or \$100,000, since it occupies 20% of the building's space and is on a Triple Net Lease.

Therefore, the Year 1 EGI from just Tenant A is \$287,500. There is no information about the Percentage Rent or General Vacancy, so this is the best answer you can give.

8. Now assume the lease expires on December 31 of Year 5. By Year 6, the market rent is \$70 per square foot per year, and Tenant A's in-place rent is \$60 per square foot per year. Property-wide expenses have also increased to \$600,000 per year.

There is a 70% probability that Tenant A will renew its lease and a 30% probability that it will cancel, and we will need to find a new tenant.

If Tenant A renews, it will receive 4 months of free rent. If Tenant A does not renew, it will take 6 months to find a new tenant, and the new tenant will receive 6 months of free rent as a sign-up incentive.

Walk me through the EGI calculation for Tenant A in Year 6. You may write down the numbers and do the math on paper.

We'll divide this into the Renewal Case and the Non-Renewal Case and probability-weight each of them.

First, the Renewal Case: If Tenant A renews, Year 6 Base Rental Income is $\$70 * 5,000 = \$350,000$, and the Loss to Lease is $(\$70 - \$60) * 5,000 = \$50,000$, so its in-place rent is \$300,000.

It receives 4 months of free rent, so the Concessions & Free Rent line item is $\$300,000 * 1/3 = \$100,000$, reducing income to \$200,000.

Tenant A also reimburses 20% of the \$600,000 in operating expenses + property taxes, for \$120,000 total. Therefore, the Tenant A EGI is \$320,000.

In the Non-Renewal Case, the Year 6 Base Rental Income is still \$350,000, the Loss to Lease is still \$50,000, and the in-place rent is still \$300,000.

There will be 6 months of downtime and 6 months of free rent, so the Absorption & Turnover Vacancy will be \$150,000, and the Concessions & Free Rent will be \$150,000.

You deduct both of those items, so the rental income goes from \$300,000 to \$0 after factoring them in.



But the tenant will still pay for Expense Reimbursements for the 6 months of the year in which it occupies the space. So, $\$600,000 * 50\% * 20\% = \$60,000$.

Therefore, the EGI in this Non-Renewal Case is \$60,000.

Combining the cases, the Year 6 EGI is $\$320,000 * 70\% + \$60,000 * 30\% = \$242,000$.

9. How is “General Vacancy” different from “Absorption & Turnover Vacancy” on the Pro-Forma?

General Vacancy represents potential rental income lost because of space that is permanently vacant and which is not expected to be filled during the holding period – for example, a small suite that has been empty for a long time and is unlikely to be filled anytime soon.

Absorption & Turnover Vacancy is for space that is temporarily vacant because an existing tenant’s lease has expired, the tenant has not renewed, and the search for a new tenant is underway. Sometimes this line item is also used in property developments when a new tenant has signed and committed to a lease, but the lease has not yet begun.

10. How do you project the most common expenses in a Pro-Forma?

Maintenance, Utilities, and Insurance are typically projected on a per-square-foot or per-square-meter basis, and sometimes there is a starting value with an annual growth rate in-line with inflation. Other expenses such as Staff Payroll and Sales & Marketing also fall into this category.

Management Fees are usually a percentage of Effective Gross Income.

Property Taxes are initially a percentage of the property's assessed value or purchase price and grow by a modest percentage each year (unless the property's value changes dramatically).

Finally, you also forecast allocations to the CapEx, TI, and LC Reserve on a per-square-foot or per-square-meter basis.

11. Can you explain the concept of Replacement Reserves and why they appear on the Pro-Forma?

Replacement Reserves exist to “smooth out” a property’s cash flow over time so that the fluctuations are smaller when there is a significant amount of spending on CapEx, Tenant Improvements, or Leasing Commissions in a given year.

The property owners allocate a small amount to the Reserves each year in anticipation of future costs, under the assumption that capital costs will be highest when leases expire.



Then, the owners draw on these Reserves to pay for these capital costs in future years, and if the Reserves are insufficient to meet their needs, they use the property's cash flow.

If the Reserves are completely sufficient for capital costs, NOI and Adjusted NOI will be the same; if they are not, then Adjusted NOI will be lower than NOI.

12. Should the Reserves affect the property's NOI?

There is a lot of debate about this point, and many people claim that allocations to the Reserves should be *below* the Net Operating Income line, under the logic that these allocations are non-cash expenses.

On the other hand, you could argue that Reserves should reduce NOI because the NOI must reflect the true, ongoing costs of maintaining and operating the property. Without the Reserves, nothing else in NOI reflects the capital costs.

We agree with the second line of reasoning, so in all the models in this course, Reserves always count as a deduction when calculating NOI.

But in real life, you will see many variations because not everyone uses this logic. Sponsors often try to claim higher NOI figures by excluding the Reserves, while lenders tend to be more conservative and include the Reserves to get lower NOI figures.

13. How do NOI, Adjusted NOI, and Cash Flow to Equity differ?

Net Operating Income, or NOI, represents the property's cash flow from operations on a capital structure-neutral basis before *most* of the capital costs ("most" because the Reserves are sometimes deducted and sometimes not deducted).

Adjusted NOI equals NOI minus Net Capital Costs (i.e., TIs, LCs, and CapEx netted against the Reserves used to cover them). So, it's after the full operational expenses and capital costs, but before Debt Service.

Cash Flow to Equity equals Adjusted NOI minus Cash Debt Service (Cash Interest and Debt Principal Repayments). It's the "bottom line" because it represents what's truly available for distribution to the Equity Investors.

14. What does "Loss to Lease" mean, and how do you calculate it?

Loss to Lease represents the difference between market rents and in-place rents, and it's a direct deduction from Base Rental Income when you calculate Effective Gross Income (EGI).



You could calculate it by looking at each tenant's in-place rent and comparing it to the market rent for the same space, multiplying the tenant's space by that differential, and then aggregating everything on the Pro-Forma.

You could also take the average in-place rent per square foot or square meter, compare it to the market rent per square foot or per square meter, and then multiply that differential by the total rentable square feet.

15. What are the advantages and disadvantages of a retail lease with a Percentage Rent term?

For tenants, Percentage Rent makes rent into more of a variable expense, which reduces the risk in recessions but also costs tenants more when the economy is strong and monthly sales are high.

Retailers often agree to this term if they are worried about their fixed expenses when business takes a turn for the worse.

For owners and sponsors, leases with Percentage Rent increase the risk but also the potential returns because they mean lower income in downturns but higher income in periods of strong growth.

No owner would ever want to use significant Percentage Rent for all the tenants, but it can be a useful tool to incentivize tenants and negotiate other lease terms.

16. Tenant B currently occupies 100,000 square feet in a 300,000 square-foot industrial warehouse.

At the end of Year 2, its lease expires. If it renews, the new lease will be for 6 years; if we find a new tenant instead, the lease will be for 5 years.

The rental rate in both cases will be \$10.00 per square foot per year, with an escalation of 3% per year.

Tenant Improvements will be \$1.00 per square foot if Tenant B renews and \$1.50 per square foot for a new tenant.

Leasing Commissions will be 1.0% of total lease value if Tenant B renews and 3.0% of total lease value for a new tenant.

The Renewal Probability is 60%. Calculate the TIs and LCs in Year 3, and show your work on paper. Feel free to use approximations and round your numbers.

First, consider the Renewal Case: The TIs will be $\$1.00 \times 100,000 = \$100,000$, and the LCs will be based on the total lease value.



For a 6-year lease at \$10.00 per square foot per year, that is $\$10.00 * 100,000 * 6 = \6 million. But we have to factor in the escalation as well: $1.03^6 = 1.19$, which means that the rent per square foot per year will be \$11.9 by Year 6 of the lease. You could approximate that as “about \$12.0.”

We can then say that the “average” rent over that 6-year time frame is \$11.00, so $\$11.00 * 100,000 * 6 = \6.6 million (in Excel, it’s closer to \$6.5 million, slightly under this estimate).

The LCs will be 1.0% of that, or \$66,000.

Now, consider the Non-Renewal Case: The TIs will be $\$1.50 * 100,000 = \$150,000$.

The LCs will be based on a 5-year lease at \$10.00 per square foot per year. Factoring in the escalation, the rate by the end will be $\$10.00 * 1.03^5 = \11.6 (or you could round this to “about \$11.5”).

We can then say that the “average” rent over that 5-year period is \$10.80, so the approximate total lease value is $\$10.80 * 100,000 * 5 = \5.4 million (in Excel, it’s closer to \$5.3 million).

The LCs are 3.0% of that, and $\$5.4 \text{ million} * 3\% = \$162,000$.

Putting this together, the TIs in Year 3 are $\$100,000 * 60\% + \$150,000 * 40\% = \$120,000$.

The LCs in Year 3 are $\$66,000 * 60\% + \$162,000 * 40\% = \$104,400$.

17. Explain the Expense Reimbursement calculation and how it factors into the Pro-Forma.

The Expense Reimbursement calculation offsets some or all of the property’s operating expenses + property taxes, depending on tenants’ lease types and their responsibilities.

To calculate this item, you start by calculating all the expenses for the entire property each year based on the assumptions and drivers there.

Then, you go through each tenant and assume that it pays for its proportional share of the expenses that it owes (e.g., just Property Taxes, Property Taxes + Insurance, etc.).

If there’s a lease expiration, you must consider both the Renewal and Non-Renewal cases and assume that there are no Expense Reimbursements when the space is vacant, and the owners are searching for a new tenant in the Non-Renewal Case. There are full Expense Reimbursements in the Renewal case.

Then, you probability-weight each case, add the numbers together, and aggregate all the tenants. Expense Reimbursements always increase the property’s EGI.



18. Why do you probability-weight numbers such as the Free Months of Rent and Absorption & Turnover Vacancy in a Pro-Forma?

Because you don't know whether or not existing tenants will renew their leases or cancel; cancellation creates the need to find new tenants to fill their spots.

And if there are multiple lease expirations per suite in the holding period, you can very rapidly get many cases in the model (e.g., Renewal → Renewal, Renewal → Non-Renewal, Non-Renewal → Renewal, and Non-Renewal → Non-Renewal).

You need to consider all these cases separately and probability-weight them because the numbers for the Free Months of Rent, TIs, and LCs will differ, and there will be a period of downtime with no Rent or Expense Reimbursements if you must search for a new tenant.

19. What are the advantages and disadvantages of monthly or quarterly models rather than annual ones when building a property Pro-Forma?

An annual model is fine for coming up with a "rough view" about the viability of a deal. It's also fine if you are under time pressure because annual models are simpler and faster to complete.

However, annual models are also less accurate than monthly or quarterly ones because tenant leases often start in the middle of the year, not December 31 or January 1, the Downtime Months between tenants may span across two calendar years, the Free Months of Rent could also span over two calendar years, and the lease itself might not last for a whole number of years.

In short, monthly or quarterly models produce more accurate results, but they also take more time and effort to set up, and the formulas are more complex.

20. Do you need to use ARGUS for office/retail/industrial properties with multiple tenants? When might Excel be sufficient?

If there's a certain amount of complexity, such as 30 tenants each with different lease terms, and a holding period that includes multiple lease expirations per tenant, you pretty much need ARGUS.

However, if you limit the assumptions to one lease expiration per tenant in the holding period, and there are relatively few tenants (e.g., 3-5 rather than 30), Excel is sufficient.

Also, if the lease terms are all very similar, Excel may suffice because you could copy and paste around the formulas in that case (but you still need to assume only one lease expiration per tenant in the holding period).



21. How do you set up the Excel formulas for items such as the Free Months of Rent, Tenant Improvements, and Leasing Commissions in a monthly Pro-Forma?

The main idea is the same: divide these items into a Renewal case, a Non-Renewal case, and an “Initial” case if the model assumes that brand-new tenants move into currently vacant space.

The difference is that in the Non-Renewal Case, you will assume a certain number of Downtime Months and check to make sure that you’re past that downtime period (i.e., Lease Expiration Date + Downtime Months) before projecting these figures. If you’re still in the Downtime Period, they are 0.

In the Renewal Case, by contrast, you can project these items starting in the month immediately following the Lease Expiration Date because there’s no interruption in between tenants.

The Tenant Improvement *calculation* will be similar because it’s always a lump-sum tied to a specific amount per square foot or square meter, but the Free Months of Rent will be a direct deduction each month instead of a calculation based on a fraction of the year such as 1/2 or 3/4.

The Leasing Commissions will also be similar, but it can be trickier to calculate the Total Lease Value if rent increases by something *other* than a constant annual percentage.

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Multifamily, Student Housing, and Hotels

With these property types, you will generally see a “higher-level” Pro-Forma that is *not* linked to individual tenants and leases in the same way that a Pro-Forma for an industrial, retail, or office property is.

As a result, there’s less room for detailed questions about specific line items, but you still need to understand the fundamentals – especially how these properties compare and contrast with others.

1. Walk me through a multifamily Pro-Forma and explain how it differs from an industrial, office, or retail Pro-Forma.

The basic shape of the Pro-Forma is the same: you start with Base Rental Income and adjust for items such as the Loss to Lease, Concessions, and General Vacancy, and then you deduct Operating Expenses and Property Taxes to calculate NOI, subtract capital costs to calculate Adjusted NOI, and subtract Debt Service to calculate Cash Flow to Equity.



The main difference is that you tend **not** to project individual tenants in detail since the lease terms are similar and there are so many tenants.

Turnover tends to be high (50%+) for multifamily properties, which can make them more expensive to operate in terms of staffing and sales & marketing.

Capital costs such as TIs and LCs are smaller and less significant for multifamily properties since small, individual tenants on largely identical leases have much less bargaining power than a large company renting a huge percentage of space in an office property (for example).

You usually won't see Absorption & Turnover Vacancy as a separate line item because the owners can find new tenants relatively quickly, and there may be a few new items, such as "Bad Debt" (i.e., tenants not paying), Parking Income, Laundry Income, and Utility Reimbursements.

2. How does a multifamily Pro-Forma change in different market scenarios, such as Base, Upside, and Downside cases?

The Upside Case tends to have the highest Rent and Expense Growth, the lowest Vacancy Rate, the lowest Bad Debt & Concessions, and the lowest Tenant Improvements and Leasing Commissions because it's easier to attract tenants in strong markets.

The Downside Case will be the opposite, with the lowest Rent and Expense Growth, the highest Vacancy Rate, the highest Bad Debt & Concessions, and the highest TIs and LCs.

The Base Case will be in between these two extremes, with moderate growth rates and percentages for many of these items.

3. How might a student housing Pro-Forma differ from a multifamily one?

The basic shape will be the same because there are dozens or hundreds of tenants, all with very similar short-term leases, but student housing is often viewed as "countercyclical" because people tend to go back to school during recessions.

So, the Upside, Base, and Downside Cases might not differ as substantially as they would for a multifamily Pro-Forma.

4. What are the strengths and weaknesses of multifamily properties as an asset class?

Multifamily properties usually have 1-year leases, so rents and occupancy rates adjust quickly, which is beneficial in strong markets and a risk factor in weak markets and downturns.



There's far less risk of an important "anchor tenant" leaving because there are often dozens or hundreds of individual tenants, all with very similar lease terms, so you do not need to read the fine print as closely as you do in office, industrial, or retail deals. Capital costs also tend to be lower.

Also, in major cities, there are usually limits to rent and occupancy rate declines because people always need somewhere to live regardless of economic conditions.

One weakness is that individuals tend to be less reliable than companies, so there is more risk of tenants failing to pay their rent on time.

Also, multifamily properties tend to have high turnover, which creates more staffing needs and the need for ongoing sales & marketing spending.

5. What makes hotels different from other property types?

Hotels are the closest to "normal companies" because they do not have "leases" at all – just short-term guests staying for a few days at a time.

As a result, you can project revenue and expenses for hotels based on the Average Room Rate, Occupancy Rate, Number of Rooms, and Per-Room Costs.

Because of this lack of long-term leases, hotels are more closely correlated with economic conditions than the other property types; in recessions, hotel revenue tends to plummet as business and leisure travel falls, and in strong markets, hotel revenue jumps up as travel increases.

The shape of a hotel Pro-Forma is also closer to that of a normal company's Income Statement.

6. Walk me through a hotel Pro-Forma.

Revenue at the top is split into Room Revenue, Food & Beverage, and "Other," which includes fees from Parking, Telecom Services, and Events.

Then, there are Departmental Expenses, similar to COGS for a normal company, that correspond to specific revenue categories such as Room Revenue.

Revenue – Departmental Expenses = Gross Operating Income.

Next, there are "Undistributed Expenses" for items that do not correspond to a specific revenue category, such as Sales & Marketing and Repairs & Maintenance; these are variable expenses that depend on the Occupancy Rate.



Gross Operating Income – Undistributed Expenses = House Profit.

Finally, you subtract Fixed Expenses, such as Insurance, Property Taxes, Reserves, and the Ground Lease, to calculate NOI, and you subtract Capital Costs to calculate the Adjusted NOI.

7. What do the ADR and RevPAR for a hotel mean, and why are they important?

The ADR is the Average Daily Rate, which tells you how much the hotel earns from each occupied room each night, on average. It's important because it tells you how the hotel is positioned in its market and the types of guests it attracts, as well as the hotel's revenue potential.

RevPAR is Revenue per Available Room and equals the ADR * the Occupancy Rate. It's useful because it tells you the income-earning potential of the hotel, *factoring in the Occupancy Rate*.

8. What are the advantages and disadvantages of a hotel's main sources of revenue?

Room Revenue tends to have the highest margins because it does not cost much to clean and maintain rooms; Room Revenue also has *a bit* more visibility than revenue in other categories, but it still tends to drop substantially in recessions.

Food & Beverage Revenue almost always has lower margins and tends to drop even more in recessions as hotel guests cut back on their spending.

Finally, Other Revenue often has the lowest margins, depending on what's in it (hotels often lose money with Telecom Services, but earn something with Parking and Events).

The stability/visibility of this last one varies depending on the items within it, but some revenue sources, such as fees from managed apartment units in the building, are more predictable than others (e.g., Parking Fees).

9. How do you project the key expenses for a hotel?

Typically, you make Departmental Expenses percentages of revenue since there are direct links and associated costs for each category.

Undistributed Expenses can be percentages of Revenue as well, except for Utilities, which are based on the Occupied Room Nights (there's no electricity or water usage in a vacant room).

You can project Fixed Expenses on a per-room-night or per-available-room basis, or use simple percentage growth rates depending on the item.



10. What are the biggest risk factors in hotel acquisition and development deals?

The biggest risk factor for hotels is the one described above: they're highly sensitive to the overall business environment, and their revenue tends to decline sharply in recessions.

Many hotels also have high Fixed Expenses, which is bad news when revenue falls due to a declining ADR or Occupancy Rate. For example, an expensive Ground Lease could cut into the hotel's NOI quite heavily, and the hotel would not be able to do much about it since it's a long-term lease.

Finally, hotels are more dependent on **the human element** (i.e., the management team) than other property types. It's hard to screw up if you own an office building where all the tenants are Fortune 500 companies that have signed 10-year leases, but it's easy to run a hotel into the ground if you annoy the guests and they start leaving bad reviews online.

11. What might change on a hotel Pro-Forma in different market scenarios?

In the Upside Case, the Occupancy Rate, ADR, and Revenue and Expense Growth will tend to be highest, and these figures will be the lowest in the Downside Case, with the Base Case in between.

Total revenue growth over the holding period should be highest in the Upside Case, and the ending NOI and Adjusted NOI margins should be the highest as well; the Downside Case should be the opposite.

You should also see more of a difference between the scenarios than you would with other property types, especially if you project the performance over 5-10 years.

Incentive Fees to Management and other bonuses are far less likely in the Downside Case (and far more likely in the Upside Case) because they're often based on NOI Achieved vs. Targeted NOI.

12. What are the advantages and disadvantages of a hotel operating independently vs. joining a franchise?

If a hotel joins a franchise, it will be more expensive to operate, and it will owe fees to the franchisor, but it may also be able to boost its Occupancy Rate and ADR – the association with this larger brand and access to the brand's customer base should boost the hotel's business.

If the hotel operates independently, its expenses and obligations will be lower, but it will not receive a boost from another company's brand name and customer base in the same way, and it will have to develop more of its own reputation.



The hotel's revenue is likely to be higher when it's part of a franchise, but its NOI margins could easily be lower (and vice versa for the independent case).

13. What are the strengths and weaknesses of hotels as an asset class?

Hotels are almost always valued at higher Cap Rates, meaning lower valuations, than the other property types because they're significantly riskier due to the lack of revenue visibility.

The lack of long-term leases, reliance on the management team, and possibility of high Fixed Expenses create quite a lot of risk – but they also create an opportunity if the hotel's team can perform well and build a solid reputation in a desirable location.

Hotels are more like normal companies than traditional multifamily, industrial, office, or retail properties, and their risk and potential returns reflect that.

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Development Deals

These questions tend to be higher-level because it's difficult to do mental math for development deals, given the timing of the cash flows and the model complexity.

You should be familiar with key concepts such as the major cost categories, why and how Construction Loans get refinanced, and the exit assumptions.

1. Walk me through a property development model (i.e., one where a new property is constructed and eventually sold).

You start by setting up assumptions for the total amount of land to purchase and the square feet or square meters of space that translates into, as well as the construction costs and the amount of Debt and Equity to use (for example, Debt might fund 50% of the total costs).

Then, you project the construction costs and initially draw on Equity to pay for them. Once you reach the maximum allowable Equity, you switch to the Construction Loan and assume that interest and loan fees are capitalized when the property's cash flow is negative due to the construction.

Once the construction period finishes, you assume that the Construction Loan gets refinanced with a Permanent Loan, and you project the "lease-up period," during which individual tenants move into the property.



You create the standard Pro-Forma for this lease-up period as the property stabilizes, and you project Debt Service on the Permanent Loan to calculate the Cash Flow to Equity Investors.

Then, you assume the property is sold based on its forward, stabilized NOI and a range of Cap Rates taken from historical and market data.

You calculate the IRR to Equity Investors based on their Equity contributions during the construction period, the refinancing, the annual cash flows, and the sale of the property and repayment of Debt.

You may also factor in items such as the sale of excess land, refinancing and exit fees, and prepayment penalties on the Debt, and you may create a “waterfall returns” schedule to split up the cash flows for the Investors and Developers.

2. What’s the relationship between the Land Plot Size, the FAR, and Gross and Rentable Area in a development?

The Land Plot Size is the number of acres, hectares, square feet, or square meters that you purchase in the beginning.

The FAR, or Floor Area Ratio, refers to the ratio of Gross Area to Land Plot Size allowed by the local jurisdiction, such as the city that governs property developments. The FAR times the Land Plot Size gives you the Gross Area of the property.

For example, if the Land Plot Size is 10,000 square meters and the FAR is 10, then the Gross Area of the property can be up to 100,000 square meters.

The Rentable Area is a smaller percentage of this Gross Area because every building has some space that is *not* rentable (e.g., space for elevators and staircases, the front lobby, and hallways).

These distinctions matter because some expenses will be linked to the Land Plot Size, and others will be linked to the Gross or Rentable Area.

3. What are the main categories of construction costs, and how do you project them?

The main categories are Land Acquisition Costs, Hard Costs, and Soft Costs.

You could also include Replacement Reserves, FF&E (Furniture, Fixtures, and Equipment), Tenant Improvements (TIs), and Leasing Commissions (LCs) in this list, but these are incurred when the development is complete or nearing completion, and they’re *not specific to developments*.



Land Acquisition Costs are what they sound like; you assume a \$ per acre, hectare, square foot, or square meter figure, and then project an initial deposit in the first month and then a final deposit for the rest within the first year.

Hard Costs represent the material and labor costs for constructing the building, and they are normally distributed according to a bell curve or straight-line pattern.

Soft Costs represent design, accounting, legal, architectural, and permit fees, and they are incurred somewhat “randomly” over the construction period; you might look at recent, similar developments to come up with the appropriate monthly percentages.

4. Why might you assume the allocation of a Replacement Reserve in the final months of construction?

You might do this because you assume that new tenants start to move in following the end of construction, and Tenant Improvements (TIs) and Leasing Commissions (LCs) will be incurred as you find and sign these tenants.

However, the property’s cash flow will be insufficient to pay for these capital costs initially, so it’s more sensible to put in place a Reserve funded by Debt or Equity to cover these early capital costs.

5. Why do you assume that the Construction Loan fees and interest are capitalized? Couldn’t you assume funding for an Interest Reserve in the beginning?

You assume that fees and interest on the Construction Loan are capitalized because the property does not generate the positive cash flow required to pay for these when it is still under development.

You could assume an Interest Reserve in the beginning, but that will reduce the IRR to Equity Investors if Equity funds it (since the upfront contribution will be higher).

And if this Interest Reserve is funded by Debt, the ending Debt balance and interest will be higher because the initial Debt balance starts out at a non-zero number.

So, on balance, you almost always assume capitalized interest and fees on Construction Loans in development deals.

6. How do you estimate the new property's value upon refinancing if it hasn't stabilized yet?

The easiest method is to move forward to the property’s stabilization 1-2 years after the refinancing and then discount its value based on an appropriate Discount Rate in this period – which should be lower than the rate during development but higher than the rate for stabilized properties.



For example, if the property's forward NOI will be \$5 million upon stabilization in two years, an appropriate Cap Rate is 6%, and the sponsor's Discount Rate during development is 20%, with a 10% Discount Rate for stabilized properties, you might pick 15% as the Discount Rate.

Then, $\$5 \text{ million} / 6\% = \83.3 million . $\$83.3 \text{ million} / ((1 + 15\%)^2) = \63.0 million . So, you would use this \$63.0 million figure and an appropriate LTV to calculate the size of the Permanent Loan when the refinancing takes place.

7. If it's too difficult to estimate the new property's value when the Construction Loan is refinanced, what's another approach to sizing the Permanent Loan(s)?

Another approach is to calculate the credit stats and ratios following the refinancing and "size" the Permanent Loan to those.

For example, if the lenders want a Debt Service Coverage Ratio (DSCR) of at least 1.30x and an Interest Coverage Ratio (ICR) of at least 2.00x, then you might size the Permanent Loan by working backward from those.

Continuing with this example, if the NOI upon stabilization is \$5 million, then the Interest Expense could be at most \$2.5 million to support a 2.00x ICR. But you might aim for something a bit lower than that, such as \$2.2 or \$2.3 million, to provide more of a cushion.

If prevailing market interest rates are 4% for these types of loans, then you might conclude that the Permanent Loan should be between \$55.0 and \$57.5 million, since $\$2.2 \text{ million} / 4\% = \55.0 million and $\$2.3 \text{ million} / 4\% = \57.5 million .

8. How does the "lease-up" period in a development Pro-Forma differ from the Pro-Forma for an existing, stabilized property?

The main difference is that there will be far more **variability** in the Effective Gross Income (EGI) and capital costs such as Tenant Improvements and Leasing Commissions. The operating expenses and property taxes should not change that much because they are paid regardless of the property's occupancy.

As a simple example, if two big tenants have committed to moving in toward the end of Year 1 in this period, you might see a significant Absorption & Turnover Vacancy or General Vacancy in Year 1 since these tenants have not yet moved in, and the Expense Reimbursements will be lower than normal.



Then, in Year 2, you'll see significant Concessions & Free Rent but lower numbers or even 0's for the other items. The TIs and LCs will also be significant during these tenants' move-in dates in Year 1 and will decrease after that.

9. What are the major components that go into the Equity Returns calculation for a new property development?

The main components are the Equity Draws or Contributions from the Investors and Developers during the construction phase, the Net Proceeds from the Refinancing, the Cash Flow to Equity Investors as the property stabilizes, and the Net Exit Proceeds at the end (the property's selling price minus fees and Debt Repayment).

Other components might include the Return of the Replacement Reserves at the end, prepayment penalties on the Permanent Loans, Equity Granted if any of the loans include it, and the Sale of Excess Land if the development turns out not to need the entire Land Plot.

10. How does the Sale of Excess Land affect the Equity IRR and Multiple in a development deal?

In a purely mechanical sense, positive cash flow from an item like this midway through the holding period will boost the Equity IRR and Multiple compared with *not* selling the excess land at all.

However, the real question is, "Would we have been better off buying *less land* in the beginning?"

The answer depends on the land-price appreciation vs. the Equity IRR during the holding period. For example, if land prices increase 3% per year over the holding period, but the Equity IRR is 15%, we would have been better off buying less land in the beginning.

On the other hand, if land prices increase by 15% per year, but the Equity IRR is 10%, then the excess land purchase and sale helped us.

11. Why are there multiple types of Equity in most development deals?

There are almost always multiple Equity Investors in development deals because one group (the "Investors" or "3rd Party Investors" or "Sponsor" or "Limited Partners") contributes most of the Equity but is not actively involved, while the other group (the "Developer" or "General Partners") contributes a smaller portion of the Equity and does the work.

This is how the industry works: investment firms have the capital to invest, but they do not want to spend time on the day-to-day business of developing and running properties.



So, they bring Developers on board to do the work, but they want the Developers to have some “skin in the game,” so they require the Developers to contribute Equity to fund part of the process as well.

12. What are the key assumptions that you might sensitize in a development deal?

You might create sensitivity tables for the length of the construction period, the construction costs, the time to stabilization, the Exit Cap Rate, the initial LTC and the LTV at refinancing (and other Debt terms), and the market scenario.

Of these, the construction costs, market scenario, and Exit Cap Rate are likely to make the biggest difference because the returns depend mostly on the upfront costs and Equity contributions and the Net Equity Proceeds at the end.

13. How are monthly and annual development models different?

First, note that almost all development models include at least a *partial* monthly schedule because construction spending differs depending on the month, and it’s important to record the timing of Equity contributions, especially if the construction period lasts more than one year.

Monthly vs. annual projections for the lease-up period and exit do not make a huge difference, but you will have to convert all the Rent, Expense, Concession, Vacancy, and other assumptions to a monthly basis and go into more granular detail for a monthly model.

Besides the additional detail, one benefit of a monthly model is that you can be more flexible with the refinancing and exit dates, so it is often easier to create sensitivities and examine different cases.

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Pre-Sold Condo Development Deals

Pre-sold condo developments are a more specialized topic, so you should focus on the previous section to learn about the most common development-related interview questions.

The questions here are **unlikely** to come up if the firm you’re interviewing with only does acquisition or renovation deals; they’re more likely at development-focused firms, but even there, they’re less likely than the questions in the previous section.

1. Walk me through the process of building a pre-sold condo development model.



First, you make assumptions for the size of the development, the number of units that will be sold, and key revenue and expense items like the selling price per square meter or square foot, the construction costs, the deposit payment structure, the sales velocity per month, and the sales and expense inflation.

Then, you create a timeline for the pre-construction phase, construction phase, and post-construction phase, and you split the pre-sales of condo units into different phases, which may be triggered by different conditions and which may take different amounts of time to finish.

You then forecast the initial and final deposits for the pre-sales in each phase, as well as the deposits collected during construction.

Then, you forecast the main construction expenses, such as the Hard Costs, Soft Costs, FF&E and Move-In Costs, and Land Acquisition Costs, and you calculate the Gross Income based on sales minus expenses.

You draw on Equity and Debt to fund the development, capitalize interest and loan fees when cash flow is insufficient to cover them, and track the Equity and Debt balances over time.

Then, you calculate the IRR and multiples at the end based on the Equity contributions and eventual positive cash flows from the development.

2. How do the risk and potential returns of pre-sold condos compare to those of other property types and strategies?

Pre-sold condo developments are development deals, so the risk and potential returns are higher than those of stabilized property acquisitions or renovation deals for any property.

However, they're less risky than developing single-family owned homes and attempting to sell them at the end, and they're also less risky than developing brand-new properties, leasing them up, and waiting until stabilization to sell them.

3. How do you determine the Debt and Equity draws in a pre-sold condo development model?

You typically base the Debt and Equity Draws on a percentage of the total funding required for the deal, such as 50% or 60% Debt, and you draw on Equity first and then switch to Debt.

To do this, you may have to make the calculation circular, depending on how the model is set up and the level of precision you need.

Another approach is to link the Debt and Equity Draws to the percentage of units pre-sold and require Equity up to a certain percentage of total units pre-sold and then switch to Debt after that.



4. Why are condominium pre-sales typically split into multiple phases?

Splitting the pre-sales into multiple phases allows the Developer and Investor to “test the market” and assess demand before committing to construction for all the units.

If everything goes well in the first phase – when they attempt to sell smaller units or a lower number of units – they’ll plan to sell more units, at a greater average size, in the next phase.

Doing it this way also allows the Developer and Investor to adjust midway through the project if something dramatic happens, such as a market downturn or major competitive development starting next door.

5. What are the trade-offs of different Debt/Equity and Sales Phase triggers in a pre-sold condo development?

On the Debt and Equity side, the main advantage of linking Draws to the percentage of units pre-sold is that the lenders can be certain the development is closer to “fully sold” before committing funding.

But the disadvantage is that if expenses rise by significantly more than forecast over the course of the development, the lenders may end up having to pay for a higher percentage of the total costs, which increases their risk.

If you link Debt and Equity Draws to the percentage of total funding instead, it’s the opposite: lenders lock in the total costs for which they are responsible, but they assume the additional risk if the development is delayed or never “fully sold.”

With the Sales Phase triggers, the main options are to base the phases on a fixed month number, such as 30 months before the end of development, or to start the next phase after a certain number or percentage of units in the current phase are sold.

With fixed months, there’s more visibility regarding the construction costs, but you also run the risk of starting construction without knowing if there’s sufficient demand.

With the staggered approach, there’s less visibility with the construction costs and timing, so the project could easily be delayed, but you could match sales to construction costs more effectively.

6. What are the main expenses in a condo development, and how do you project them?



The main ones are Hard Costs (buying the materials and paying for the labor), Soft Costs (legal and architectural fees, permits, and other expenses owed to service providers), FF&E and Move-In Costs (FF&E is “Furniture, Fixtures & Equipment”), and Land Acquisition Costs.

Hard Costs and Soft Costs tend to be distributed irregularly, so you might use hard-coded monthly percentages based on the spending patterns from prior projects, or you might use a Normal (or other) Distribution.

FF&E and Move-In Costs should be linked to the buyer deposits because they are only incurred once the new owners purchase their units and move in.

Land Acquisition Costs are normally paid in the first month of the project with an initial deposit, and then again with a final deposit sometime within the first year (around the time construction begins).

7. Explain the overall pattern to Equity and Debt Draws and Debt Repayments in a pre-sold condo development.

It varies based on the trigger conditions, but in the earlier months, Gross Income tends to be negative, creating the need to draw on Equity, followed by Debt.

In later months, as the deposits arrive, the positive cash flows from those deposits are used to repay the Construction Loan.

In most models, there will be little available for distribution to the Equity Investors until the final deposits arrive, which usually happens upon construction completion.

If the deposits follow a different pattern, such as more gradual payments over time, then the cash flows may not be quite as lopsided, and distributions might start a bit earlier.

8. What’s the point of calculating both the leveraged and unleveraged IRR and multiple in a condo development deal?

The unleveraged metrics are based strictly on Gross Income, before Interest and Debt and Equity Draws, while the leveraged ones are based on the Equity contributions and positive Cash Flows to Equity.

The purpose is the same as in any other real estate deal: to assess the deal’s dependency on leverage. Leverage increases both risk and potential returns, so you must assess how big a role it plays in any development deal.



If a development only “works” with a certain amount of leverage, that could be a negative sign and a reason to turn down the deal – because the results will also be much worse if the deal does not go as planned.

9. Why are waterfall returns schedules often used in pre-sold condo development models?

Waterfall schedules are common in all development deals, including ones for pre-sold condos, because they incentivize the Developers to perform and earn higher returns for everyone without “costing” the Investors much.

Since development deals have higher risk and higher potential returns than acquisitions of stabilized properties, waterfall schedules can be used to ensure that each investor group earns back at least 1x its Invested Equity before splitting up the returns (for example).

Also, construction delays and cost overruns are big risk factors in development deals, and waterfall schedules can be used to mitigate these risks by incentivizing the Developers to finish more quickly and closer to the targeted budget.

10. How would you make an investment decision as a potential Equity Investor in a pre-sold condo development, and what are the key risk factors?

You would create scenarios for different outcomes, such as Downside, Base, and Upside Cases, and then assess whether or not you could reasonably earn your targeted IRR in each case.

For example, you might want to earn at least a 1.2x multiple in the Downside Case, while you might target a 20% IRR in the Base Case and a 30% IRR in the Upside case.

To assess whether or not you can reach those targets, you might create sensitivity tables based on those scenarios as well as unit selling prices, construction costs, sales velocity, and sales price and expense inflation, which are some of the key risk factors in these deals.

If these targets seem feasible and there are no huge red flags, then you might choose to invest in the development.

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Acquisition & Renovation Deals



Acquisitions are one of the main types of real estate deals, so this section is just as important as the one on developments.

Renovations are similar to acquisitions, so we group them together here and cover the specific differences between them in these questions.

1. Walk me through a stabilized property acquisition model.

You start by assuming a purchase price for the property based on a Cap Rate or per-square-foot or per-square-meter figure, and you use certain percentages of Debt and Equity to fund the deal.

You then make assumptions for the property's revenue and expenses, sometimes projecting individual tenant leases (for office/retail/industrial properties) and sometimes using higher-level assumptions such as the average rent or ADR (multifamily and hotels).

You forecast the Pro-Forma over several years, project the Debt Service, and you assume an exit in the future based on a Cap Rate and the property's stabilized forward NOI.

Finally, you calculate the returns based on the initial Equity contribution, the Cash Flows to Equity, and the Net Proceeds after Debt repayment upon exit.

2. What's the difference between "Core" and "Core-Plus" acquisition deals?

In a "Core" deal, nothing about the property changes – the tenants or types of tenants, leases, rent, and expenses all follow familiar trends.

In a "Core-Plus" deal, there may be minor changes to improve the property. For example, the sponsor may get tenants to sign longer-term leases in exchange for lower rent escalations, or it may perform small upgrades to units in a multifamily property to boost rents slightly.

3. How does a "Value-Added" acquisition & renovation deal differ from a stabilized acquisition, and what are the key assumptions?

Acquisition & renovation deals are similar to stabilized acquisition deals and follow the same basic outline and set of modeling steps.

The difference is that you must assume something for the **renovation costs**, usually incurred early in the holding period, and you usually assume a **penalty** while the renovation is taking place, such as reduced room availability for a hotel or reduced occupancy for a multifamily.



Also, there will almost always be a **refinancing** because the risk profile of the property changes and the lenders that funded the acquisition or bridge loan don't necessarily want to stay on once the renovation is complete.

Once the renovation is finished, you then assume some **benefit** from it, such as an increased occupancy rate or higher average rent.

Finally, it may be reasonable to assume an Exit Cap Rate that's lower than the Going-In Cap Rate, especially if the property had serious problems before the renovation began.

4. How do you set up the exit assumptions in an acquisition or renovation deal, and how can you check whether or not they're reasonable?

You almost always base the exit price on the property's stabilized forward NOI (from one year past the exit date) divided by a selected Cap Rate.

The Cap Rate should be based on market data and the operating scenario, with higher rates in Downside cases and lower rates in Base and Upside cases.

If the property does not change at all during the holding period, it's difficult to justify a *lower* Exit Cap rate, especially over longer holding periods. Over time, the property becomes less competitive with newer ones.

You can check the reasonability of these assumptions by looking at long-term Cap Rate trends in the market and also calculating the annualized NOI growth rate over the holding period.

5. Suppose that you're comparing the Equity Returns for a stabilized acquisition deal and a renovation deal. How will the **shape of these returns differ?**

In a renovation deal, more of the returns are likely to come from capital appreciation (i.e., the Exit Price exceeding the Purchase Price by a wide margin), since you often assume Cap Rate compression as the property improves.

Some portion is also likely to come from the refinancing that takes place once the renovation is done; by contrast, this refinancing is less common in stabilized acquisition deals.

A smaller percentage of the returns will come from the property's cash flows during the holding period because of the greater percentages from the items above and the fact that the cash flows may fall to lower levels when the renovation is taking place.

6. What are the key risk factors in acquisition and renovation deals?



For stabilized acquisitions, the main risk factors are Cap Rate trends and the stability of the cash flows. If the occupancy rate falls or rents fall, the property could suddenly become “un-stabilized,” and the returns from cash flows will drop.

Cap Rates are always a big risk factor because even if the property performs as expected, it will be nearly impossible to realize an acceptable IRR if Cap Rates rise significantly over the holding period.

For renovation deals, these risk factors also apply, but you could add the risk of the renovation going poorly and not producing the desired boost to rent and occupancy rates, and the risk of the refinancing not producing the expected results due to a market downturn or construction delay.

Renovation deals are even more susceptible to the risk of Cap Rates increasing because a higher percentage of the returns comes from capital appreciation. Even if the renovation goes well, the deal might fall apart if there's a market downturn afterward.

7. How do you make an investment decision in acquisition/renovation deals?

You create different scenarios for the market and deal-specific performance, calculate the IRR and multiple in each one, and assess how well they meet your targeted returns.

For example, if the targeted multiple in the Downside Case is 1.0x, the targeted Base Case IRR is 10%, and the targeted Upside Case IRR is 20%, and the deal produces results of 1.1x, 13%, and 25%, respectively, it is a positive deal upon first glance.

Then, you verify that the assumptions are realistic; if they are not (e.g., rental growth of 10% per year when the average is 3-4%), then you adjust them and see if the new results still meet these targets.

You'll also do qualitative research on the ground because real estate is a hyper-local industry – if you physically inspect a building, you will find information that never appears in online databases or the official documents.

8. What does the “renovation factor” mean, and how does it affect the cash flows and returns?

The renovation factor is most relevant for hotels and multifamily properties, or others with large numbers of tenants and higher-level projections.

It refers to the “penalty” for completing the renovation, such as a certain percentage of rooms not being available while the construction is taking place.



By itself, this renovation factor always reduces the cash flows and returns, especially since much of the capital spending for the renovation takes place during this period as well.

9. How do renovation deals differ for different property types?

The mechanics are similar, but the “penalty” and “benefits” differ based on the property type. Hotels and multifamily properties might see lower occupancy rates, or a portion of the units might be unavailable. When the renovation is done, these units will become available again, and occupancy rates and per-unit rates might increase.

With office, retail, and industrial properties, there’s often less of a “penalty” because the building itself might already have significant vacant space and below-market rents – so, the renovation requires capital costs, but existing tenants do not leave.

Instead, the goal is to use the renovation to attract new tenants, boost the occupancy rate, and boost rents to market rates when new tenants move in or existing tenants’ leases come up for renewal.

10. In acquisition and renovation deals, how far apart should the different cases or scenarios be?

It depends on the property type, the strategy, the location, and the exact plan, so there is no specific answer to this question.

In general, though, the cases should be **further apart** in renovation deals than in stabilized acquisition deals because renovation deals have higher risk and higher potential returns. Significant changes to the property mean that there’s more potential upside but also more room for disaster.

So, for example, if the IRRs are 10%, 12%, and 15% in the Downside, Base, and Upside Cases in a renovation deal, that is **unrealistic** and reflects scenarios that are not different enough.

The differences between the scenarios also depend on the property type – for example, you would expect to see bigger differences for hotels than you would for office or retail properties because of the long-term leases and annual rental escalations for those properties.

11. How can lenders mitigate some of the risk in renovation deals?

One method is to use a **Holdback** or **Earnout** to avoid distributing the full loan proceeds upfront when the acquisition first takes place. The funds “held back” are distributed over time as certain benchmarks are achieved, which reduces the risk of lenders losing money if the deal goes wrong.

Another method is to negotiate for an **Interest Reserve**, which sets aside extra funds to cover interest payments if the property’s cash flows might be insufficient to pay for interest in the future.



On that note, the lenders could also shift some of the Interest from Cash to PIK or back depending on their key concerns (PIK Interest if the cash flows are potentially unstable, or Cash Interest if the deal's returns are too dependent on the exit).

Finally, lenders could also link the interest rates and other terms to the property's performance. For example, they could require higher interest rates if the property's occupancy rate falls below a certain level or if rents decline.

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Valuation

Valuation is typically *not* the main point of real estate models, so while the concepts here are good to know, they're less important than the questions about development and acquisition/renovation deals.

Part of the issue is that real estate valuation is simple compared with valuation for normal companies, and the main challenges are **finding the right data** and making sure the numbers are comparable in the first place – but it's difficult to assess your abilities to do those with interview questions.

1. What are the 3 main valuation methodologies for properties?

You can use Cap Rates ("Capitalization Rates," also known as Yields) or metrics such as per-square-foot or per-square-meter figures, the DCF ("Discounted Cash Flow Analysis"), and the Replacement Cost methodology.

You divide the property's stabilized forward Net Operating Income by the selected Cap Rate to value it like that, or you can multiply its square feet or square meters by the per-square-foot or per-square-meter figures for comparable properties.

With the DCF, you project the property's Unlevered Free Cash Flow, or Adjusted NOI, calculate the Terminal Value, discount everything back to Present Value based on the Discount Rate, and add up the PV of the Terminal Value and the PV of the UFCFs.

With the Replacement Cost methodology, you estimate the cost of building the entire property from the ground up today and compare that to the property's asking price.



2. You are valuing an office property, and current market Cap Rates of 6.00% – 6.50% suggest a valuation of approximately \$90 – \$100 million. The DCF produces a valuation of \$80 – \$90 million, and the Replacement Cost methodology produces estimates of \$70 – \$85 million.

What do these numbers tell you about the property and market?

These numbers tell you that prevailing Cap Rates in the market are lower than those implied by the property/market fundamentals, meaning that valuations might be elevated and the market might be at a high point in the cycle.

These results don't necessarily mean that the property is a "bad deal" at a price of \$90 – \$100 million, but they do suggest that it is at least slightly overvalued at that price and that there's a risk of a market downturn.

3. Why is it tricky to use Cap Rates to value properties, in practice?

First, people tend to calculate Net Operating Income (NOI) differently, especially when it comes to items like TI/LC/CapEx Reserves and other capital costs. There is also the question of whether you should use the property's historical or stabilized forward NOI, and what "stabilized" means.

Second, selecting the appropriate range of Cap Rates is trickier than it seems because the property sales you select must be comparable regarding location, quality (e.g., Class A vs. Class B), amenities, average rents, and so on.

Finally, there might also be extremely limited Cap Rate data for your market. You'll never have trouble finding stats for New York or London, but it's much harder to find numbers for Neepawa, Manitoba (for example).

4. How would you select the appropriate Exit Cap Rate in a Core vs. Value-Added vs. Opportunistic deal?

In Core and Core-Plus deals, you usually select Cap Rates based on market data, the scenario, and how rates have changed in past cycles.

For example, in the Downside Case, you might assume that Cap Rates increase up to the level they reached in the last recession before moving back to the average over the past 10-15 years; the Base Case might have more of an even progression toward the average, and the Upside Case might see a decline before moving back to the historical average.

The property should almost always have an Exit Cap Rate that's **higher** than its Going-In Cap Rate because it becomes less appealing over time as newer, more competitive properties enter the market.



The process for a Value-Added deal is similar, but you can more easily justify a **lower** Exit Cap Rate since the property changes significantly. Even if the market softens or goes through a downturn, the whole point of a renovation is to make the property more valuable.

Finally, in Opportunistic deals (developments and some redevelopments), there isn't a real "Going-In Cap Rate" in the same way, so you should pick an Exit Cap Rate based on market data for similar properties and trends in past cycles, as explained above.

5. Why might industrial properties be valued at higher Cap Rates than office properties?

First, industrial properties are typically faster, easier, and cheaper to develop than office properties, with construction periods commonly of ~1 year rather than 2-4 years. These differences make industrial properties such as warehouses less defensible and easier to replace.

Second, industrial properties often have fewer tenants than office properties, with single-tenant buildings fairly common – and fewer tenants means more risk.

These are broad generalizations because industrial properties are not *always* valued at higher Cap Rates – if an office property has similar characteristics to the ones above, it could easily be valued at about the same level as industrial properties.

6. Walk me through a DCF for a property and explain how it's different from a DCF for a normal company.

You start by projecting the Unlevered Free Cash Flow for the property, known as Adjusted NOI in real estate, over 5-10 years, and then you calculate the Terminal Value using a Terminal Multiple or Cap Rate, or by using the Terminal Growth Rate method.

You then discount the Terminal Value to its Present Value and discount the Unlevered FCF to its Present Value, add them up, and compare the total to the asking price of the property.

Unlike in a DCF for a normal company, Cost of Equity is not based on re-levered Beta from the comparable public companies, but is instead linked to the sponsor's targeted Equity IRR in the deal; WACC is based on the Cost of Equity times the percentage Equity used to fund the deal plus the Cost of Debt times the percentage Debt.

The Discount Rate may change over time, especially in Value-Added and Opportunistic deals, so you will have to account for that with a Cumulative Discount Factor.



Finally, if you calculate the Terminal Value using a Terminal Cap Rate, you often assume a **premium** to the Exit Cap Rate in the model, under the assumption that the property will be less competitive over time if nothing changes.

7. When is a property DCF most useful and least useful?

A property DCF tends to be **least useful** for stabilized acquisition deals (Core and Core-Plus), where the property's cash flows increase only modestly over time and where the Terminal Value is still based on a Cap Rate assumption. In those cases, the DCF might confirm what you already know.

It tends to be more useful in deals with unstable cash flows, such as Opportunistic and Value-Added ones; with those, traditional Cap Rate-based valuation might not work well since it's difficult to determine the "stabilized NOI."

You still need to pick a range of Exit Cap Rates at the end of the holding period for those deals, but the DCF might be useful for establishing an *upfront* purchase price.

8. Walk me through a Replacement Cost Analysis and explain when it's useful.

In a Replacement Cost Analysis, you estimate the total costs required to reconstruct a property today and compare that total to the property's asking price.

You start by estimating the Hard Costs, Soft Costs, Land Acquisition Costs, and anything else required to build the property, and then you make the Financing Costs a percentage of that total based on the expected construction period, current interest rates, and the Debt/Equity split.

Then, you assume that the Developers will take a certain percentage (usually around 20%), add this portion to the total costs, and compare that to the property's asking price.

The Replacement Cost Analysis is most useful as a "sanity check" for the other valuation methodologies because these cost estimates are so imprecise – ask 10 different Developers, and you'll get 10 different answers.

It also tends to be more useful for stabilized acquisitions because with other deal types, the property will change significantly, so it's even more difficult to make reasonable estimates.

9. When you're analyzing comparable properties, which metrics do you look at, and what's the purpose of the analysis?



Depending on the property type, you might look at the In-Place Rent per Square Foot or per Unit as well as the Market Rent and Effective Rent. You'll also examine the Occupancy Rate, Concessions & Bad Debt, # of Units or Amount of Space, and the year the property was built.

For office/retail/industrial properties, you might also look at the average lease length and TIs, LCs, and other capital costs per square foot or square meter.

For hotels, you will look at the ADR, Occupancy Rate, percentage revenue from different sources, margins, and incentive fees paid to management.

Unlike comparable public companies for normal companies, the purpose is **not** to value the property but instead to determine whether or not the assumptions used in your Pro-Forma are reasonable.

For example, if your assumed Occupancy Rate is far above the rate for any other similar property in the local market, you might have to assume a lower number.

10. How might you analyze a property's Rent Roll and draw conclusions from it?

First, you'll retrieve a list of the tenants, along with their lease start dates and expiration dates, rentable square feet, and current rental rates.

Then, you'll create rows for the future years and show the rentable square feet, in-place revenue, and "potential revenue" (at market rates) that are expiring each year in both dollars (or other currency) and percentages.

You might also show supplemental information, such as the expected TIs and LCs to be incurred each year if all the tenants with lease expirations fail to renew their leases.

This analysis helps you determine **the risk of lease expirations**, so it is most useful for properties with a relatively low number of tenants instead of, say, a multifamily property with 500 individual tenants.

If it suggests too much risk, then lenders and sponsors might request additional safeguards in the deal terms.

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Waterfall Returns



The waterfall returns schedule is a more advanced topic. If they want to test your understanding in-depth, they will give you a modeling test and ask you to build or complete this schedule.

We do not delve into numerical questions and answers here because they're unlikely to come up – you need to know the basic ideas, but would not be expected to do the mental math required to set up an entire schedule.

1. Explain the waterfall returns schedule and why it is common in real estate.

A waterfall returns schedule allows the Equity Proceeds from the deal to be split up in a way that is **not** proportional to the Equity contributed by each group.

For example, if the Investors contribute 80% and the Developers contribute 20%, normally the Investors would earn 80% of the Equity Proceeds at the end, and the Developers would earn 20%.

A waterfall schedule allows that to vary so that the Developers end up with 25% or 30% of the Equity Proceeds if the deal performs well enough.

Often, these bonuses or “Promotes” are based on the overall Equity IRR.

For example, the Investors and Developers might earn proportionally to their Equity contributed up to a 20% IRR, but then the Developers might receive an extra 10% for the returns between a 20% and 30% IRR, and another 10% for the returns above that 30% IRR.

The Developers earn a significantly higher IRR if the deal performs well enough, but it doesn't “cost” the Investors much since they contribute the majority of the Equity and still earn most of the returns.

These Promotes provide a big incentive for the Developers because they encourage the Developers to finish on time and on budget.

2. Can you explain the basic logic required to set up a waterfall returns schedule?

For a simple schedule based on Equity IRRs, like the one described above, you track the Equity Invested in each period and increase the “amount owed” to investors based on the IRR of the given tier.

For example, if Tier 1 represents everything up to a 20% IRR, and the Equity Invested in Year 1 is \$1,000, you increase the “amount owed” to \$1,200 in Year 2 to reflect this 20% annualized return. You keep doing that based on the starting balance in each year, and you add the Equity Invested to the balance in each year.



You reduce the “amount owed” whenever there is a positive cash flow to distribute, which usually happens toward the end of the project.

Each year, you distribute all the positive cash flows, or, if the positive cash flows exceed the remaining “amount owed” for the tier, you distribute the remaining “amount owed.”

You split these distributions according to the Investor / Developer split in this tier, which is often proportional to Equity Invested in Tier 1.

Then, you move to the next tier and do the same thing, but if this tier is **only** for the IRR between 20% and 30% (for example), then you increase the “amount owed” by 30% per year.

You need to subtract the amount distributed in the previous tier(s) as well as distributions in this tier (once again, equal to the lesser of the positive cash flows or the remaining “amount owed”).

You then split up the distributions according to the percentages in this tier, such as 70% for the Investors and 30% for the Developers.

You keep doing this for all the tiers and then reassemble the cash flows at the end to calculate each group’s overall IRR and multiple.

3. The basic concept doesn’t sound that complicated. What makes waterfall schedules hard?

The basic concept is not difficult, but waterfall schedules can get more complex because of timing, mixed IRR and multiple-based thresholds, *which* IRR or multiple you use in each tier, Preferred and Catch-Up Returns, and Lookback and Clawback provisions.

The timing adds complexities because you often build monthly schedules, and the property’s cash flows often switch from negative to positive and back throughout the holding period.

Also, you do not always use the overall Equity IRR or Multiple in each tier – sometimes you might use the Equity IRR or Multiple for *just* the LPs or GPs or Developers or some other group, which also makes the calculations more complex.

Preferred and Catch-Up Returns make the calculations more complicated by requiring *all* the cash flows in certain tiers to be distributed to just one group before other groups receive anything.

And Lookback and Clawback provisions make schedules more complex by requiring a *redistribution* of the proceeds if one group achieves a specific IRR, but the other group does not.



4. Why are Preferred and Catch-Up Returns common in waterfalls for renovation deals, and what do they mean?

Preferred Returns give one group, such as the Investors or Limited Partners, 100% of the positive cash flows from the property until they reach a specific Equity IRR or Multiple, such as 10% or 1.0x.

Then, the other group(s) may receive Catch-Up Returns that “catch them up” to that same Equity IRR or Multiple, which means that the other group(s) will receive 100% of the next available positive cash flows up to that level.

These terms are common in many real estate deals, but they’re especially common in renovation deals because Investors and LPs want to *avoid losing money* no matter what happens – and a Preferred Return based on a 1.0x Equity Multiple is one way to tip the odds in their favor.

These terms are not as common in Core or Core-Plus deals because the risk and potential returns are lower, so investors do not necessarily need them as safeguards in the same way.

5. Walk me through a waterfall schedule with Preferred and Catch-Up Returns and traditional IRR-based tiers after that.

First, you track the Net Cash Flow to Equity in each period and count the negatives as Investor Injections.

For the Preferred Return, you make the “amount owed” increase by the Investor Injections for just the group that receives this Preferred Return, and you multiply the beginning balance by the IRR percentage for this tier (so, a 10% increase if it’s a 10% Preferred Return).

Then, you distribute the lesser of the positive Net Cash Flow to Equity and the “amount owed” at the end of the period.

At the bottom, you subtract the Preferred Distribution in each period from the Net Cash Flow to Equity to calculate the Cash Flow Available for Catch-up Distributions.

Then, you do the same thing for the Catch-Up Return, but you distribute the lesser of the Cash Flow Available for Catch-Up Distribution and the “amount owed” to this specific investor group.

The Cash Flow Available for Tier 1 Distributions then equals the Cash Flow Available for Catch-Up Distributions minus the Catch-Up Distributions.



The rest of the schedule is the same as usual, but in each tier, you'll have to subtract these Preferred and Catch-Up Distributions to capture the proceeds for just the IRR between, say, 10% and 20% if that's Tier 1 of this schedule and the Preferred and Catch-Up Returns are both 10%.

The other difference is that the traditional IRR tiers are usually based on total Investor Injections instead of the ones for just a single group such as the GPs or LPs. And then you divide the accrual distribution in each tier according to the split between investor groups in that tier.

6. Could a waterfall schedule ever *hurt* the Developers and benefit the Investors? If so, how?

Yes. Although waterfall schedules are normally used to incentivize the Developers to perform and earn a higher IRR, certain terms could also hurt the Developers.

For example, if the Investors receive a Preferred Return in the deal and there's no Catch-Up Return for the Developers, the Developers would lose all their money if the Equity IRR is below the Preferred Return.

This scenario is **unlikely** since few, if any, Developers would agree to such terms, but Developers could also be hurt if there are Lookbacks or Clawbacks that redistribute the Exit Proceeds based on the IRR to each group at the end.

7. What does the "Accrual Distribution" line item in each tier of a waterfall schedule mean, and how do you calculate it?

The "Accrual Distribution" line refers to the amount of positive cash flow available for distribution to investor groups in a given tier.

You may calculate it differently depending on the waterfall setup, but broadly speaking, it equals the lesser of the positive cash flows available for distribution in the year and the "amount owed" to investor groups in the year.

For example, if a tier is based on an Equity IRR up to 15%, then the Accrual Distribution line will be 0 in each period where Net Cash Flow to Equity is negative.

When Net Cash Flow to Equity is positive but under the "amount owed" in the period – for example, \$1 million vs. \$10 million owed – this line item will be \$1 million.

And if Net Cash Flow to Equity ever exceeds this "amount owed," which usually only happens at the end when the property is sold, then this line item will reflect the "amount owed" in this tier.



So, if the proceeds from the sale of the property after fees and Debt repayment are \$20 million, and the investors are owed \$10 million to earn a 15% IRR, the Accrual Distribution will be \$10 million.

Then, to calculate the amount available for distribution in the next tier, you subtract this \$10 million Accrual Distribution from the Net Cash Flow to Equity.

8. What's different about waterfalls for development deals?

The mechanics of all waterfall schedules are similar, so the formulas in a waterfall for a development deal are not necessarily that different.

One difference is that it's more important to build a **monthly schedule** rather than an annual one because development spending changes significantly from month to month; another difference is that tiers that use a mix of Equity IRR and Equity Multiples are more common since the timing of developments can be more unpredictable.

Finally, terms such as Lookbacks and Clawbacks are arguably more common because Developers could earn a much higher IRR than Investors if the deal performs well enough, which might trigger the conditions for these terms.

9. How are monthly and annual waterfall models different?

The main difference is that the "Investor Accruals" formula is different. In annual models, it can simply increase the beginning "amount owed" in the period by the IRR percentage in the tier because the IRR is already an annual figure.

But in monthly models, you must convert this IRR to a monthly percentage by raising it to the power of $(1 / 12)$ and subtracting 1.

For example, if it's a 15% IRR tier, you would use:

$$\text{Investor Accruals} = \text{Beginning Amount Owed} * ((1 + 15\%) ^ (1 / 12) - 1)$$

to calculate the accrual on a monthly basis.

10. If the waterfall schedule has a tier based on an Equity Multiple rather than the Equity IRR, how does that affect the calculations?

Almost everything stays the same, but the Investor Accruals line item changes since you no longer have to factor in the time value of money as you do with IRR.



When the tier is based on an Equity Multiple, the formula is:

Investor Accruals = Investor Injections in This Period * (Equity Multiple – 1)

So, if it's a 3x Equity Multiple, and the Investor Injections are \$10 million in this period, the Investor Accruals will be \$10 million * (3 – 1) = \$20 million, and the Ending “amount owed” will increase by \$30 million total.

11. What is a “Lookback” in a waterfall schedule, and what conditions might activate it?

A “Lookback” redistributes Equity Proceeds at the end of the holding period based on the IRRs achieved by each investor group.

Unlike in a waterfall schedule, this redistribution happens only at the end – not to all the positive cash flows in between.

For example, a Lookback in a development deal might state that if the Investors earn less than a 20% IRR, but the Developers earn over a 20% IRR, then the Developers must give up their proceeds to the Investors until the Investors reach that 20% IRR.

This Lookback would activate if the Investors earned an 18% IRR, but the Developers earned a 21% IRR.

The Developers would then give up the lesser of their Equity Proceeds at the end and the amount required for the Investors to achieve the 20% IRR.

Depending on the numbers and Equity contributed by each group, this term could make a huge impact on the results – for example, the Developers might fall from a 21% IRR to a 1% IRR in an example like the one above if they contributed only ~10% of the Equity.

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Credit Analysis

Questions on credit analysis are quite specialized and are unlikely unless you interview with a real estate lending firm or some other firm that invests in credit.

Conceptual knowledge is important in this question category, but you may also be asked to approximate figures such as the IRR or Recovery based on assumptions that the interviewers provide.



1. What is the basic idea behind credit analysis for real estate loans?

The goal in credit analysis is to model the impact of extremely negative outcomes and then make an investment recommendation based on the likelihood of avoiding losses.

To do this, you create scenarios for the property and market, focusing on the worst-case outcomes (e.g., a major recession or the complete failure to execute a certain strategy for the property).

You look at historical market data such as peak-to-trough rents for similar properties in the area to determine how key assumptions might change in the course of a cycle, and you use these numbers in the scenarios you create.

If it seems like the lenders will recover 100% of their principal and the property will be able to service the Debt even when its cash flows worsen significantly, you might recommend the investment.

In some cases, you might also recommend the deal even if this does not happen – if, for example, the Recovery is less than 100%, but the IRR is still positive, meaning that the lenders avoid a loss.

2. What changes in the financial profile of a property when there's a recession in a Downside or Extreme Downside Case?

In recessions, Rents fall, Vacancy Rates rise, Concessions increase, and Tenant Improvements and Leasing Commissions both increase as it becomes more difficult to attract tenants.

Operating expenses often stay about the same or even decrease since it's more difficult for providers and employees to find work as well.

Cap Rates also increase in recessions as properties become less valuable, which could affect the refinancing in a development or renovation deal and the Exit Price for all deal types.

3. How might you determine the appropriate numbers to use in the Downside or Extreme Downside Case of a credit analysis?

You might go back to the past few downturns in this area and use the *percentage changes* from them.

Percentages matter more than absolute numbers because inflation tends to increase prices over time.

So, for example, if the average rent per unit fell from \$2,500 to \$2,000 in the last recession 10 years ago, you shouldn't necessarily use that same \$2,000 figure today – but you might use the 20% decline and apply it to today's average rent per unit.



Similarly, it's better to use the *percentage increases* for items like TIs and LCs rather than the absolute amounts.

4. What are the main differences between Senior Loans and Mezzanine, and why do many deals use both?

Senior Loans are secured Debt where the property acts as collateral, meaning that lenders can foreclose on it and recoup some of their losses if the deal goes wrong.

Senior Loans tend to have the lowest interest rates, which may be either fixed or floating, because they're the most senior form of Debt, and they often have amortization periods that far exceed their maturities (e.g., 30-year amortization vs. 10-year maturity).

Senior Loans are used to fund property acquisitions up to a certain LTV that lenders will accept, such as 60% or 70%. If the sponsor wants to go beyond that, it will have to use Mezzanine, which is unsecured Debt that is junior to Senior Loans.

Mezzanine has higher interest rates to compensate for the additional risk, this interest can be paid in cash or accrue to the loan principal (or a mix of both), rates are often fixed, there's rarely any amortization, and the maturity is almost always shorter than the maturity of Senior Loans.

Mezzanine can take the LTV up to a higher level, such as 80% or 90% total, so the sponsor might use it if it wants higher leverage (beyond the level that just the senior lenders are comfortable with).

5. How does Preferred Equity compare with Senior Loans and Mezzanine?

Preferred Equity is unsecured Debt that is even more junior than Mezzanine and has even higher interest rates as a result.

Although it has "Equity" in its name, Preferred Equity works more like Debt, with fixed payments based on the coupon rate. These payments often accrue to the loan principal instead of being paid out in cash, but the treatment varies based on the property type and strategy.

Preferred Equity may also have a warrant or Equity option attached so that the investors receive a small percentage of the Equity Proceeds upon exit (Mezzanine may also have this).

In short, Preferred Equity is the closest to Common Equity out of all the forms of Debt used in real estate, but it's still Debt – albeit a risky, high-interest-rate form of Debt.

6. How can you determine the appropriate Loan-to-Value (LTV) or Loan-to-Cost (LTC) Ratio to use in a deal?



Typically, you look at the LTV or LTC for similar, recent deals in the market and use something in that range. These numbers are not difficult to establish because there are usually industry-wide norms for these numbers, such as 50% for the LTC of new office developments or 70% for the maximum LTV of stabilized multifamily acquisitions.

(These are just examples – do **not** take them literally as the acceptable numbers since they change over time.)

Another approach is to size the Debt based on the credit stats and ratios the lender is seeking, such as a minimum DSCR of 1.2x and a minimum ICR of 2.0x.

The problem with that approach is that these ratios change in different cases, so you'll have to use the most pessimistic case to determine the proper amount – and that might mean that too little leverage in other cases.

7. How do you calculate the Debt Yield (DY), Interest Coverage Ratio (ICR), and Debt Service Coverage Ratio (DSCR), and what do they tell you about a credit deal?

The Debt Yield is based on $\text{NOI} / \text{Initial Debt Balance}$, the ICR equals $\text{NOI} / \text{Interest Expense}$, and the Debt Service Coverage Ratio equals $\text{NOI} / (\text{Interest Expense} + \text{Debt Principal Repayments})$. There are other variations as well, such as versions that use the Adjusted NOI or just the Cash Interest Expense.

Broadly, these metrics tell you how risky a deal is from a lending perspective. The lower the ICR and DSCR, the closer the property comes to *not* being able to service its Debt obligations, and the higher the chances of something going wrong in pessimistic scenarios.

The Debt Yield is similar, but you can also compare it to prevailing Cap Rates since they both incorporate NOI.

If Cap Rates ever rise *above* the Debt Yield, the lenders are in trouble because it means the loans may be worth more than the property (but it depends on the loan amortization, accrued interest, and Holdbacks and other special terms as well).

8. What are the advantages and disadvantages of Accrued or Capitalized Interest, also known as Paid-in-Kind or PIK Interest?

From the perspective of the sponsor, PIK Interest is beneficial because it allows them to avoid Cash Interest payments, thereby making it easier to service the Debt, which is important if cash flows ever decline, or there's a renovation that temporarily depresses cash flows.



However, it also means that there will be more Debt to repay upon exit, which could reduce the sponsor's IRR, especially in pessimistic cases.

From the lenders' perspective, PIK Interest helps if the lenders believe that the property will have trouble servicing its Debt fully with cash. It can't "miss" an interest payment if the interest is capitalized instead.

But PIK Interest also makes the Debt's risk profile closer to that of Equity since the IRR is more dependent on the exit (and entirely dependent on the exit if it's 100% PIK Interest).

However, despite that added risk, the *potential returns* are still nearly the same because the lenders do not receive any Equity upon exit.

9. Why might a lender negotiate for a "Holdback" when issuing a loan, and how would it work mechanically?

A lender might negotiate for a Holdback when issuing a loan to mitigate risk, especially when the property's cash flows might become unstable, or if the property will require high capital costs for a specific period.

With a Holdback, the lender does not distribute the full amount of loan proceeds upfront but instead releases only a portion of them (e.g., \$10 million on a \$15 million loan).

However, the sponsor still pays interest *on the full amount* – the \$15 million – before it is fully distributed. Any amortization is also based on the full loan amount.

The proceeds are released over time as certain milestones are met, or as capital costs such as TIs, LCs, and CapEx are incurred.

A Holdback boosts the lender's IRR because a portion of its investments occurs later in the holding period.

10. How do you calculate the IRR and Recovery for a real estate loan?

For the IRR, the amounts invested – the negatives in Excel – are based on the initial issuance and the subsequent release of Holdbacks or Earnouts (if applicable).

Then, the positive cash flows are based on the cash interest earned, the principal repayments (if applicable), the issuance and prepayment penalty fees (if applicable), and the loan repayment upon maturity or exit.



PIK Interest does not factor in directly because it simply increases the loan balance that must be repaid at the end; it generates neither positive nor negative cash flows in between. If there's an Equity Grant or Options, you will also factor that in at the end.

You can then use the IRR or XIRR functions to calculate the IRR.

For the Recovery, you look at the total principal repayments plus repayment upon maturity or exit and divide that amount by the total loan proceeds issued, including the initial issuance, any Holdback or Earnout releases, and any balance increases because of PIK Interest.

11. A \$100 million Senior Loan has a 4% fixed interest rate, a 2-year interest-only period, 20-year amortization and 10-year maturity, and a 1.0% issuance fee and 1.0% prepayment penalty fee.

What is the approximate IRR to lenders if the Senior Loan remains in place for 5 years and is repaid when the property is sold at the end of Year 5?

To approximate this IRR, we can say that the 4% fixed interest rate generates a 4% IRR, and the 1.0% issuance fee adds approximately 0.2% per year over 5 years (since $1.0\% / 5 = 0.2\%$).

Then, the 1.0% prepayment penalty fee will add a bit less than that to the IRR because 3 years of amortization have already taken place; we could say it's about $0.9\% / 5 = 0.18\%$ per year.

So, our guesstimate is "just under 4.4%" for the IRR here.

In Excel, it is exactly 4.39% (assuming a \$1 million issuance fee for a net initial investment of \$99 million, \$4 million in interest per year for the first 3 years and slightly less in the last 2 years due to amortization, \$5 million in principal prepayment in each of the last 3 years, an \$85 million repayment upon maturity, and a \$0.85 million prepayment fee).

12. A \$20 million Mezzanine issuance has 5% Cash Interest and 5% PIK Interest, no amortization, a 5-year maturity, and a 1.0% issuance fee and 1.0% prepayment penalty fee.

What is the approximate IRR to lenders if the Mezzanine remains in place for 5 years and is repaid when the property is sold at the end of Year 5?

There is no prepayment penalty fee here because the 5-year maturity is the same as the exit date.

So, we can approximate the IRR by saying that the total Interest is 10% per year, and the 1.0% issuance fee distributed over 5 years is 0.2% per year. Therefore, the IRR should be "approximately 10.2%."

In Excel, it's 10.24%, almost exactly what we predicted.



If this were 10% Cash Interest instead, the IRR would barely change (10.27% rather than 10.24%).

It would increase slightly because money today is worth more than money tomorrow, and 10% Cash Interest would give us more of the returns upfront.

13. A \$10 million Preferred Equity issuance has 10% PIK Interest and 2% Cash Interest, no amortization, a 5-year maturity, a 1.0% issuance fee, a 1.0% prepayment penalty fee, and receives 2.0% of the property's Net Equity Proceeds upon exit.

Assume the property's forward NOI upon exit is \$15 million and that the property is sold for a 6% Cap Rate; due to debt repayment and fees, the Net Equity Proceeds represent ~40% of this amount.

What is the approximate IRR to the Preferred Investors if this issuance stays in place for 3 years and is repaid when the property is sold at the end of Year 3?

The Cash Interest and PIK Interest together produce a ~12% IRR; the issuance fee and prepayment penalty fee together add about $2.0\% / 3 = 0.67\%$ per year, but the prepayment penalty will be higher by the end because of the PIK Interest, which increases the balance to about \$13 million by Year 3. So, we can round this up to ~0.80% per year.

The property is sold for \$15 million / 6%, or \$250 million, and $40\% * \$250 \text{ million} = \100 million . Therefore, \$2 million goes to the Preferred Investors at the end.

\$2 million is 20% of \$10 million, and over 3 years, that represents an annualized return of approximately $20\% / 3 = 6.7\%$. But it's a bit less than that due to the compounding, so we'll say this contributes "between 5% and 6%" to the IRR.

Therefore, the IRR is approximately $12\% + 0.80\% + \text{"between 5% and 6%,"}$ which we'd say is "just above 18%."

It's 17.89% in Excel, so our estimate is decently close. It's off by more than usual here because of the Equity at the end and our imprecise estimate of its IRR contribution.

14. Suppose that you analyze the IRR and Recovery for a Mezzanine issuance that your firm might invest in. The Recovery is 100% even in the Extreme Downside Case, but the IRR is in the 7-8% range, while your firm targets an IRR of 10%.

How might you negotiate with the sponsors to earn a higher IRR here?



Since the Recovery is 100% and the IRR remains positive even in the Extreme Downside Case, this deal is probably *not* a disaster – i.e., the Net Equity Proceeds at the end are likely sufficient to produce a positive IRR for the Equity Investors.

Therefore, we could negotiate for a percentage of the Net Equity Proceeds upon exit in exchange for a lower interest rate, lower issuance fee, or lower prepayment penalty.

We could also negotiate for a higher percentage of PIK Interest so that the sponsors pay less in Cash Interest, in exchange for a higher overall rate, which will also boost the IRR.

PIK Interest doesn't present that much of a risk here since the Recovery is 100% even in the Extreme Downside Case.

The key is to realize that there's little chance of a disastrous exit here, so you can boost the IRR by shifting more of the returns to that exit.

15. How does credit analysis differ for different deal types, such as Core vs. Value-Added vs. Opportunistic ones?

"Core" deals follow fairly standard guidelines, so you'll evaluate the property's performance and Debt Service in different cases, focusing on the Recovery, IRR, and credit stats and ratios to ensure that there's a low chance of losing money.

Value-Added deals are a bit different because the initial Acquisition Loan or Bridge Loan there is often refinanced and replaced with a Permanent Loan once the renovation is done.

With those, you might pay less attention to the credit stats and ratios in the initial period and more attention to the IRR since the lenders initially fund a higher-risk, higher-potential-return loan.

After the refinancing is finished, the analysis would be very similar to that of a Core Deal.

Opportunistic deals are similar since the Construction Loan will be refinanced once the development is finished and the property begins to stabilize, so you'll focus more on the IRR for the initial lenders and more on traditional credit stats and ratios after the refinancing takes place.

16. How might the financial covenants on the loans be different for these different deal types?

You will see traditional covenants, such as a minimum DSCR, ICR, and Debt Yield in Core and Core-Plus deals because the property barely changes, and its cash flows should not fluctuate much during the holding period.



These also apply to the permanent loans that are put in place after the refinancing in Value-Added and Opportunistic deals.

When the renovation or development is taking place in the first phases of those deals, the financial covenants are likely to be much lighter – if they exist at all.

The property may not even have positive cash flows in these periods, so it is unrealistic to require a minimum DSCR or ICR (for example).

So, the only financial covenants in these periods are likely to be a maximum LTV or LTC on the loans themselves.

17. Suppose that you've run the numbers and found that there is a significant chance of losing money on a Mezzanine issuance if the sponsor sells the property in Year 3 or 4 rather than waiting until Year 5 or beyond to sell. How could you mitigate this risk?

You could negotiate a higher prepayment penalty fee in Years 3 and 4 in exchange for a lower interest rate, a lower issuance fee, a higher percentage of PIK Interest, or something else that might benefit the sponsor.

Another option might be to extend the loan maturity and remove the prepayment penalty *after* Years 3-4 in exchange for a higher penalty fee *in* Years 3-4.

18. Why might lenders negotiate for a higher issuance fee in exchange for a lower prepayment penalty fee?

Lenders might do this if the key risk is not an early exit, but the property's cash flows in the holding period being "spotty," resulting in an inability to service the Debt in full.

A higher issuance fee in the beginning would mitigate some of the risks of missed or partial interest payments since the lenders would receive more money upfront.

19. You've built a cash flow model for a property and concluded that the main risk is that the DSCR and ICR will fall to very low levels if the occupancy rate and average rents decline in a recession.

How might you mitigate this risk and ensure that the property can properly service its Debt?

If the main risk is that the property will not be able to service its Debt, i.e., pay the full interest expense and principal repayments, then there are several options:



- 1) Offer a longer interest-only period in exchange for a higher interest rate so that the total payments in the first few years are lower, but the overall IRR increases.
- 2) Offer to make a percentage of the Interest Paid-in-Kind in exchange for a higher overall rate so that the sponsor pays less in Cash Interest during this possible recession.
- 3) Ask for a higher issuance fee in exchange for a lower interest rate or a longer amortization schedule so that the Debt Service is lower, but the lenders receive more in upfront Cash.

20. You've analyzed a potential Mezzanine investment and concluded that if Cap Rates increase from 5.0% to 6.5%, as they did in the last recession, then your firm could potentially lose money (Recovery of ~50%).

How could you mitigate this risk?

If the returns in any deal depend too much on the exit, the usual answer is: "Shift the returns to the holding period instead." So:

- 1) Reduce the percentage of PIK Interest (if applicable) and negotiate for a higher percentage of Cash Interest, perhaps in exchange for a lower overall rate.
- 2) Negotiate for a higher issuance fee in exchange for a lower overall rate, a longer maturity, or no prepayment penalty fee.
- 3) If the issuance has an Equity option attached (rare, but possible), negotiate for its removal in exchange for a higher overall interest rate, higher Cash Interest, or a higher issuance fee.

Negotiating for a higher prepayment penalty fee would **not** make sense in this case because there's a decent chance of the Recovery being well under 100%, which means the Equity Proceeds upon exit won't be sufficient to pay for this penalty fee.

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